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SANTHI SCHOOL OF YOGA & VEDANTA STUDIES



SANTHI YOGA SCHOOL

# 200-Hour Yoga Teacher Training Manual

*Tradition, Anatomy & the Art of Teaching*

PHILOSOPHY · ASANA · ANATOMY · PRĀṆĀYĀMA · MEDITATION · TEACHING METHODOLOGY

*Kerala, India*

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# 200-Hour Yoga Teacher Training Manual

*Tradition, Anatomy & the Art of Teaching*

Section I · Yoga Philosophy, Tradition & Practice

Section II · Asana, Pranayama, Meditation & Anatomy

Section III · Teaching, Professional Practice & Graduation

Santhi School of Yoga & Vedanta Studies

Chirattapalam, Fort Kochi, Kerala 682001, India

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**Santhi Yoga School – 200-Hour Yoga Teacher Training Manual** First edition, 2026.

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**How claims are made in this book.** Traditional teachings are presented as traditional teachings. Where a modern scientific claim is made, a source is given, and where the research is limited or conflicting we say so. Descriptions of the effects of postures and breathing practices “according to tradition” are not claims that those effects have been demonstrated scientifically.

**Lineage and certification.** Santhi Yoga School teaches classical Haṭha Yoga. The school is independent and is not affiliated with any other yoga organisation. The school is not a Yoga Alliance Registered Yoga School; graduates receive the school's own 200-hour certificate.

**Illustrations and photographs.** All diagrams and posture illustrations were drawn specifically for this manual. Anatomical drawings are schematic teaching illustrations, simplified for clarity; they are not to scale. Photographs are of Santhi Yoga School classes and trainings in Fort Kochi and are © Santhi Yoga School.

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ॐ सह नावतु । सह नौ भुनक्तु ।  
सह वीर्यं करवावहै ।  
तेजस्वि नावधीतमस्तु मा विद्विषावहै ।  
ॐ शान्तिः शान्तिः शान्तिः ॥

*oṃ saha nāv avatu · saha nau bhunaktu  
saha vīryaṃ karavāvahai  
tejasvi nāv adhītam astu · mā vidviṣāvahai  
oṃ śāntiḥ śāntiḥ śāntiḥ*

May it protect us both, teacher and student.  
May it nourish us both.  
May we work together with energy.  
May our study be bright and effective.  
May we not turn against each other.  
Om — peace, peace, peace.

PEACE INVOCATION OF THE TAITTIRĪYA AND KATHA UPANIṢADS,  
TRADITIONALLY CHANTED AT THE BEGINNING OF STUDY

# Welcome

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You have come to Kerala to learn to teach yoga. Before you learn to teach it, this course asks you to practise it more carefully, study it more honestly and understand it more deeply than you have before. The two cannot be separated. A teacher can only give what they have digested.

This manual is the written companion to twenty-eight days in Fort Kochi. It is not a substitute for the practice hall, the teacher in front of you or the discipline of getting up before sunrise to sit and breathe. What it can do is hold the material steady, so that you can return to it in the weeks of the course and in the years afterwards, when you are teaching your own students and a question arises that you have not met before.

We have tried to write the book we wished we had when we were training. It honours the classical tradition of Haṭha Yoga — practical, joyful and disciplined — and it takes the modern sciences of anatomy, physiology and learning seriously. Where the two speak in different languages, we have tried to let each speak in its own voice rather than forcing one to say what only the other can know. A cakra is not a nerve plexus, and a nerve plexus is not a cakra; a good teacher can talk about both without confusing them.

Three requests as you begin:

1. **Practise first.** Read a chapter after you have practised what it describes, not instead of practising it.
2. **Question everything, respectfully.** The tradition itself asks for enquiry (*vicāra*). If something in this manual does not match your experience, bring it to class.
3. **Remember who the practice is for.** Every chapter ends with teaching exercises because the purpose of this training is not your own perfect pose; it is your future students' safety, understanding and freedom.

We are glad you are here.

*Master Aksharanand A S*

*and the teaching team, Santhi Yoga School, Fort Kochi*

# How to Use This Manual

This manual has three sections. **Section I** covers the philosophy, history and inner practices of yoga. **Section II**, the largest, covers the body and the practices that work through it: anatomy, biomechanics, the twelve basic postures, safety, prāṇāyāma and meditation. **Section III** covers the craft and ethics of teaching, and ends with the practicum and final examination. Appendices provide glossaries, indexes, a cue library, sample classes, assessment forms and the full reference list.

Every chapter opens with **learning objectives** and closes with a **chapter review**: key takeaways, reflection questions, teacher-practice exercises, a knowledge check (answers are printed upside down beneath it) and a teaching assignment. The assignments build towards your practicum, so do them in order.

## Reading the evidence labels

Yoga is an ancient tradition studied today by historians, philosophers, physiologists and clinicians — and each speaks with a different kind of authority. To keep them from blurring, statements that could be misread are tagged:

| Label                   | Meaning                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HISTORICAL</b>       | Historical evidence — what texts, inscriptions and archaeology show about when and where something appeared.                                                    |
| <b>TRADITIONAL</b>      | Traditional teaching — what the yoga tradition teaches, as the tradition teaches it. It is presented faithfully and with respect; it is not a scientific claim. |
| <b>INTERPRETIVE</b>     | Interpretation — a philosophical or teacher's reading of a text or practice; reasonable people differ.                                                          |
| <b>SCIENTIFIC</b>       | Modern scientific understanding — what current research suggests, with a citation and an honest statement of its strength.                                      |
| <b>CLINICAL CAUTION</b> | Clinical caution — a safety point where expert consensus, physiology or case reports advise care.                                                               |

A traditional teaching is not “wrong” because science has not measured it, and a scientific finding is not “spiritual” because it concerns yoga. We ask you to hold both honestly — and to teach your students to do the same.

## The boxes

### ◆ KEY CONCEPT

Key concepts summarise an idea you must be able to explain to a student in your own words.

 **TEACHER TIP**

Teacher tips pass on practical wisdom from the teaching hall — what works, what to watch for, what to say.

 **SAFETY FIRST**

Safety boxes flag contraindications, precautions and moments when a student should stop or be referred on.

 **PRACTICAL EXERCISE**

Practical exercises are done on the mat or with a partner during the course.

 **WHAT THE EVIDENCE SAYS**

Evidence boxes summarise what research does and does not show, with citations.

 **TRADITIONAL TEACHING**

Traditional teaching boxes present a teaching as the tradition gives it.

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## Citations

Sources are cited as (Author, year) and listed in full in **Appendix N**. Yoga texts are cited by chapter and verse — for example YS 2.46 for *Yoga Sūtra* book 2, sūtra 46, and HYP 1.17 for *Haṭha Yoga Pradīpikā* chapter 1, verse 17. Verse numbering of the *Haṭha Yoga Pradīpikā* differs slightly between editions; this manual follows Akers (2002).

# The Programme at a Glance

The Santhi 200-hour teacher training is a residential course of twenty-eight days. The table below shows how the two hundred contact and study hours are distributed. It is indicative: the exact timetable for your batch is confirmed on arrival.

**Table 0.1** Indicative distribution of the 200 hours

| Area                                                                            | Hours      | Main chapters               |
|---------------------------------------------------------------------------------|------------|-----------------------------|
| Techniques, training and practice — āsana, prāṇāyāma, meditation, mantra, kriyā | 85         | 14, 23–24, 27–28            |
| Anatomy, physiology and biomechanics                                            | 25         | 16–22, 25–26                |
| Yoga philosophy, lifestyle and ethics                                           | 30         | 1–15, 30                    |
| Teaching methodology                                                            | 25         | 29–39                       |
| Practicum — practice teaching, observation and feedback                         | 20         | 40, Appendices G–I          |
| Self-study, assignments and examinations                                        | 15         | Chapter reviews, Appendix K |
| <b>Total</b>                                                                    | <b>200</b> |                             |

**Table 0.2** A sample day (timings vary by batch and season)

| Time      | Session                                                                              |
|-----------|--------------------------------------------------------------------------------------|
| Dawn      | Silent sitting · prāṇāyāma · morning Hatha practice                                  |
| Morning   | Breakfast · philosophy and theory · āsana workshop (alignment, anatomy, adjustments) |
| Midday    | Lunch · rest, reading and assignments                                                |
| Afternoon | Teaching methodology · practice teaching in small groups                             |
| Evening   | Slower, grounding practice · meditation, chanting and reflection                     |

## A four-week study plan

| Week | Focus                                                             | Read                        | Practicum milestone                                                        |
|------|-------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------|
| 1    | Foundations: what yoga is; the body's language; the breath        | Ch. 1–6, 16–18, 21, 23      | Teach one pose to a partner using the cue structure ( <b>Figure 33.1</b> ) |
| 2    | The classical path; the twelve postures; muscles and biomechanics | Ch. 7–9, 19–20, 24–25       | Teach Sūrya Namaskār and two basic postures to a small group               |
| 3    | Inner practices; safety; prāṇāyāma; meditation; sequencing        | Ch. 10–14, 22, 26–28, 32–35 | Teach a 30-minute class; peer and teacher feedback                         |
| 4    | Teaching as a profession; practicum and examinations              | Ch. 15, 29–31, 36–40        | Final teaching examination and written paper                               |



 **KEEP A PRACTICE JOURNAL**

Use the daily practice journal in **Appendix J** from the first morning. Teachers who have recorded their own learning — what was hard, what changed, what a cue did to their body — are far better at understanding their students.

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## SECTION I

# Yoga Philosophy, Tradition & Practice

Before yoga was a set of postures it was a way of understanding the mind and a path towards freedom. This section traces that path from the Vedic hymns and the Upaniṣads, through the Bhagavad Gītā and Patañjali's Yoga Sūtras, to the Haṭha texts and the practice as it is taught today — and it introduces the inner practices, the subtle-body maps and the responsibilities that a teacher carries.

- 1 What Is Yoga? Meaning, Purpose and How to Study It
- 2 Origins and Historical Development of Yoga
- 3 The Vedic Roots and the Upaniṣads
- 4 The Bhagavad Gītā
- 5 Classical Yoga — Sāṃkhya and the Yoga Sūtras of Patañjali
- 6 The Eight Limbs I — Yamas and Niyamas
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## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER

## 1

# What Is Yoga? Meaning, Purpose and How to Study It

*“When the five senses are still, together with the mind, and the intellect does not stir — that, they say, is the highest state. This firm holding of the senses is what they call yoga.” — Kaṭha Upaniṣad 2.3.10–11*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ explain the meanings of the word *yoga* and give three classical definitions with their sources;
- ◆ describe the traditional purpose of yoga and distinguish it from the purposes that bring most modern students to class;
- ◆ distinguish historical evidence, traditional teaching, interpretation and scientific evidence, and explain why a teacher must not confuse them;
- ◆ judge the strength of a research claim using a simple hierarchy of evidence;
- ◆ describe how the course is structured around practice, study and teaching.

## 1.1 The word “yoga”

The Sanskrit word *yoga* comes from the verbal root *yuj*. Traditional Sanskrit grammar actually lists two roots of that form: *yujir yoge*, “to yoke, to join”, and *yuj samādhau*, “to concentrate, to be absorbed” (Bryant, 2009). Both meanings are alive in the tradition. The first gives the image, familiar from the Upaniṣads, of harnessing the senses like horses to a chariot so that they pull in one direction. The second points to the meditative absorption (*samādhi*) that Patañjali places at the summit of practice. The popular translation “union” is not wrong, but it is only half of the story, and in classical Sāṃkhya-Yoga the goal is better described as *separation* — the discrimination of pure awareness from everything it observes (Chapter 5).

In ordinary Sanskrit, *yoga* also means simply a method, a means, an application or a connection: astronomers speak of the *yoga* of two planets, physicians of the *yoga* of medicines. When the texts use the word for a spiritual discipline, the everyday sense of “a disciplined method” is never far away.



## 1.2 Classical definitions

No single definition covers everything that has been called yoga over three thousand years. It is more useful to know several and to know where each comes from.

**Table 1.1** Five classical definitions of yoga

| Source                | Definition                                                                          | Emphasis                                                |
|-----------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------|
| Kaṭha Upaniṣad 2.3.11 | "The firm holding of the senses" (sthīrām indriya-dhāraṇām)                         | restraint of the senses; the oldest explicit definition |
| Bhagavad Gītā 2.48    | "Evenness of mind is called yoga" (samatvaṃ yoga ucyate)                            | equanimity in success and failure                       |
| Bhagavad Gītā 2.50    | "Yoga is skill in action" (yogaḥ karmasu kauśalam)                                  | acting without attachment to results                    |
| Bhagavad Gītā 6.23    | "The severing of the connection with suffering is known as yoga"                    | freedom from suffering                                  |
| Yoga Sūtra 1.2        | "Yoga is the stilling of the fluctuations of the mind" (yogaś citta-vṛtti-nirodhaḥ) | the mind itself; the basis of classical Yoga            |

### YOGA SŪTRA 1.2

योगश्चित्तवृत्तिनिरोधः

*yogaś citta-vṛtti-nirodhaḥ*

Yoga is the stilling of the fluctuations of the mind.

Notice what these definitions have in common. None of them mentions posture, flexibility or health. All of them describe a change in the relationship between a person and their own mind and senses. The *Haṭha Yoga Pradīpikā*, the most influential text of physical yoga, agrees: it opens by presenting Haṭha Yoga as "a stairway" for those who wish to climb to Rāja Yoga, the yoga of meditation (HYP 1.1-2) (Akers, 2002). Postures were, from the start of the Haṭha tradition, a means rather than an end.

#### ◆ YOGA AS MEANS AND AS GOAL

The word *yoga* refers both to a set of **methods** (ethical discipline, posture, breath, concentration) and to the **state** those methods aim at (a still, clear mind; freedom). A good teacher keeps both in view: the methods are taught in detail, but always as steps towards something larger than the method.

## 1.3 The purpose of yoga

### The traditional answer

Every classical school of Indian philosophy begins from the fact of suffering (*duḥkha*) and asks how it can be ended. Patañjali puts it with characteristic economy: *heyam duḥkham anāgatam* — "suffering that has not yet come is to be avoided" (YS 2.16). The traditional purpose of yoga is therefore liberation: *kaivalya*, the "aloneness" or independence of pure awareness in Patañjali's system, or *mokṣa*, release, in Vedānta **TRADITIONAL**. Health, calm and strength are welcomed along the way as supports for the jour-

ney, not as its destination. The *Haṭha Yoga Pradīpikā* says that success comes to the one who practises, “whether young, old, very old, sick or weak” (HYP 1.64) — a statement about perseverance, not a promise of physical results (Akers, 2002).

## Why students actually come

Surveys of modern practitioners consistently find that most people begin yoga for exercise, flexibility and stress relief; a spiritual motive often grows later, if at all (Penman et al., 2012). There is nothing wrong with this. Haṭha Yoga has always begun with the body: exercise, breathing, relaxation, sensible diet and a positive, steady mind form a practical path that anyone can follow (Chapter 9). But a teacher who knows only the modern purposes teaches a thinner practice, and a teacher who scorns them loses the students who most need to begin.

### Q MEETING STUDENTS WHERE THEY ARE

A student who comes for a stiff back and leaves ten weeks later sleeping better, a little kinder to themselves and curious about the breath has been well taught. You do not need to lecture on *kaivalya* in a beginners’ class — but you should know why the relaxation at the end matters more than the headstand.

## 1.4 What yoga is — and is not

Yoga developed within the religious and philosophical cultures of South Asia; Hindu, Buddhist and Jain traditions all have yoga practices and yoga literature **HISTORICAL** (Samuel, 2008). It is therefore neither true that yoga is “just exercise” nor that practising yoga requires adopting a particular religion. People of many faiths, and of none, practise it seriously. The tradition taught here, in which this school teaches, has always welcomed students of every background while chanting in Sanskrit and honouring its Hindu roots. As a teacher you should be able to do both: speak about the tradition knowledgeably and respectfully, and never pressure a student to believe anything.

### ② “YOGA IS FIVE THOUSAND YEARS OLD.”

You will hear this claim in many classes. The evidence does not support it. The earliest texts that describe yoga as a practice date from the second half of the first millennium BCE, and interpretations of older Indus Valley seals as depicting yogis are speculative **HISTORICAL** (Samuel, 2008). The traditions that became yoga are certainly ancient, and many practitioners regard the teachings as timeless — but that is a traditional claim **TRADITIONAL**, not a historical date. Chapter 2 sets out what the evidence does show.

## 1.5 Four kinds of knowledge

Throughout this manual you will see small labels in the text. They exist because four different kinds of statement are routinely mixed together in yoga teaching, and much confusion — and some harm — comes from mixing them.

1. **Historical evidence** **HISTORICAL** tells us when and where a text, practice or idea appeared. It comes from manuscripts, inscriptions, archaeology and the methods of historians and philologists. It can be revised when new manuscripts are found.

2. **Traditional teaching** **TRADITIONAL** is what a living tradition teaches about practice and its effects, handed down through texts and teachers. It deserves to be transmitted accurately and respectfully. It is not made true or false by laboratory experiment; it is a different kind of knowledge, validated within the tradition by practice and experience.
3. **Interpretation** **INTERPRETIVE** is a reading of a text or a practice by a commentator or teacher. Commentaries on the Yoga Sūtras disagree with one another; so do modern teachers. Interpretation should be labelled as such.
4. **Scientific evidence** **SCIENTIFIC** comes from systematic observation and measurement — anatomy, physiology, clinical trials. It is always provisional and must be cited so others can check it.

#### ◆ THE RULE OF LABELS

When you teach, say which kind of statement you are making. “In the tradition, twists are said to massage the abdominal organs” is honest. “Twists detoxify your liver” is a scientific claim that has no evidence behind it and should not be made.

## 1.6 How to judge a research claim

Much of Section II discusses research. You do not need to be a scientist to use it well, but you do need a few principles.

**Table 1.2** A simplified hierarchy of evidence

| Level     | Type of study                                             | What it can tell you                                             |
|-----------|-----------------------------------------------------------|------------------------------------------------------------------|
| Strongest | Systematic reviews and meta-analyses of randomised trials | whether an effect is consistent across many trials               |
|           | Randomised controlled trials (RCTs)                       | whether an intervention causes an effect compared with a control |
|           | Cohort and case-control studies                           | associations in real populations; cannot prove cause             |
|           | Case reports and case series                              | that something can happen (useful for rare harms)                |
| Weakest   | Expert opinion, mechanism, tradition                      | ideas and hypotheses to be tested                                |

Keep five cautions in mind:

- ◆ **Small studies are fragile.** Many yoga trials have a few dozen participants; their results often shrink or vanish in larger studies.
- ◆ **“Yoga” is not one thing.** A trial of twelve weeks of gentle Hatha is not evidence about hot power yoga, and vice versa.
- ◆ **Control groups matter.** Yoga compared with doing nothing will almost always look good. Yoga compared with other exercise is a fairer test.
- ◆ **Absence of evidence is not evidence of absence.** That no one has tested a traditional claim does not make it false; it makes it untested.
- ◆ **Mechanism is not outcome.** A plausible mechanism (“the pose stretches the psoas, which attaches to the lumbar spine...”) does not show that the pose relieves back pain. Only outcome studies can show that.

### 🔍 WHERE YOGA RESEARCH IS STRONGEST

The best-studied use of yoga is chronic non-specific low back pain, where systematic reviews find small-to-moderate improvements in pain and function compared with no exercise, and results broadly similar to other exercise (Wieland et al., 2022; Cramer et al., 2013); large randomised trials found yoga comparable to stretching or physical therapy (Sherman et al., 2011; Saper et al., 2017) and better than usual care (Tilbrook et al., 2011). Clinical guidelines include yoga among the recommended non-drug options for chronic low back pain (Qaseem et al., 2017). Evidence for anxiety, depression and balance in older adults is encouraging but more mixed and generally of lower quality (Cramer et al., 2018; Cramer et al., 2013; Youkhana et al., 2016). Evidence for many specific claims made about individual postures is absent.

## 1.7 Yoga in Kerala

You are studying in a region with a remarkable concentration of traditions related to yoga. The great Advaita Vedānta teacher Ādi Śaṅkara is traditionally held to have been born at Kaladi, about forty kilometres north-east of Kochi **TRADITIONAL**; Kerala is home to *kalarippayattu*, the martial art whose training shares with Haṭha Yoga an emphasis on breath, flexibility and the body's vital points; and Āyurveda has been continuously practised here for centuries. Studying here is an opportunity to see yoga as part of a living culture rather than as an exported product.

## 1.8 How this course works

The course follows an old pattern: **practise, study, teach**. Each morning you practise; you then study what you practised; and in the afternoons you begin to teach it. The *Yoga Sūtra* names three elements of “the yoga of action” (*kriyā yoga*) — *tapas*, disciplined effort; *svādhyāya*, study, including study of oneself; and *īśvara-praṇidhāna*, surrender to something greater than oneself (YS 2.1). A teacher-training course is an intense exercise in all three.

### ✎ YOUR WORKING DEFINITION

On the first evening, write a definition of yoga in no more than twenty-five words. Seal it in an envelope. On the last day of the course, write a new definition before you open the envelope, then compare the two. Most students find that their second definition is shorter.

## CHAPTER REVIEW

## 🍃 KEY TAKEAWAYS

- ◆ *Yoga* comes from the root *yuj*, meaning both “to yoke” and “to be absorbed”; it names both a set of methods and the state they aim at.
- ◆ The classical definitions — Kaṭha Upaniṣad, Bhagavad Gītā, Yoga Sūtra — describe a transformation of mind and senses, not physical skill.
- ◆ The traditional purpose of yoga is freedom from suffering; modern students usually arrive for health and calm, and both motives deserve respect.
- ◆ Distinguish historical evidence, traditional teaching, interpretation and scientific evidence — and say which you are offering.
- ◆ Judge research claims by study type, size, control group and whether the “yoga” studied resembles what you teach.

## 🍃 REFLECTION QUESTIONS

1. Which definition of yoga in Table 1.1 describes what you personally hope to gain from the course? Which one challenges you most?
2. Think of something you were told in a yoga class that you believed. Which of the four kinds of knowledge was it? Was it labelled honestly?
3. What is lost if a yoga class contains only physical exercise? What is lost if it contains no physical exercise?

## ✂ TEACHER-PRACTICE EXERCISES

1. With a partner, take turns explaining in two minutes, to an imagined beginner, what yoga is — without using the words “union” or “flexibility”.
2. Rewrite these three cues so that they are honest about the kind of knowledge they express: (a) “This twist detoxifies your organs.” (b) “Headstand sends blood to the brain.” (c) “Yoga is five thousand years old.”
3. Find one popular article that makes a health claim about yoga. Identify the study it cites (if any), its size, and its control group.

## ☑ KNOWLEDGE CHECK

1. The popular translation of *yoga* as “union” derives from which meaning of the root *yuj*? (a) to be absorbed (b) to yoke or join (c) to breathe (d) to stand firm
2. Which text contains the definition *yogaś citta-vṛtti-nirodhaḥ*? (a) Bhagavad Gītā (b) Kaṭha Upaniṣad (c) Yoga Sūtra (d) Haṭha Yoga Pradipikā
3. “Yoga is skill in action” comes from: (a) YS 2.46 (b) Gītā 2.50 (c) HYP 1.17 (d) Kaṭha Up. 2.3.11
4. A randomised controlled trial is stronger evidence than a case report because it: (a) involves more experts (b) can test whether an intervention causes an effect (c) is always larger (d) is always recent
5. The claim “yoga is five thousand years old” is best described as: (a) historical fact (b) scientific evidence (c) a traditional or popular claim not supported by historical evidence (d) an interpretation of the Yoga Sūtra

Answers 1b · 2c · 3b · 4b · 5c

 **TEACHING ASSIGNMENT**

Prepare a three-minute talk titled “What is yoga?” for a group of complete beginners at the start of their first class. It must include one classical definition with its source, one sentence on what students can expect from the practice, and nothing that you could not defend with a source. Deliver it to your practice group in week 1 and ask for one piece of feedback.

# Origins and Historical Development of Yoga

*“There is no knowledge equal to Sāṃkhya, no power equal to Yoga.” —  
Mahābhārata, Śānti Parvan (Mokṣadharma)*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ outline the main phases in the history of yoga, from the Vedic period to the present, with approximate dates;
- ♦ explain why claims about yoga in the Indus civilisation are contested;
- ♦ describe the contribution of the Upaniṣads, the śramaṇa movements, the epics, classical Yoga, tantra and Haṭha Yoga;
- ♦ explain how modern posture-based yoga developed in the nineteenth and twentieth centuries;
- ♦ tell the history of yoga to students accurately, without either inflating or dismissing its antiquity.

## 2.1 Why history matters to a teacher

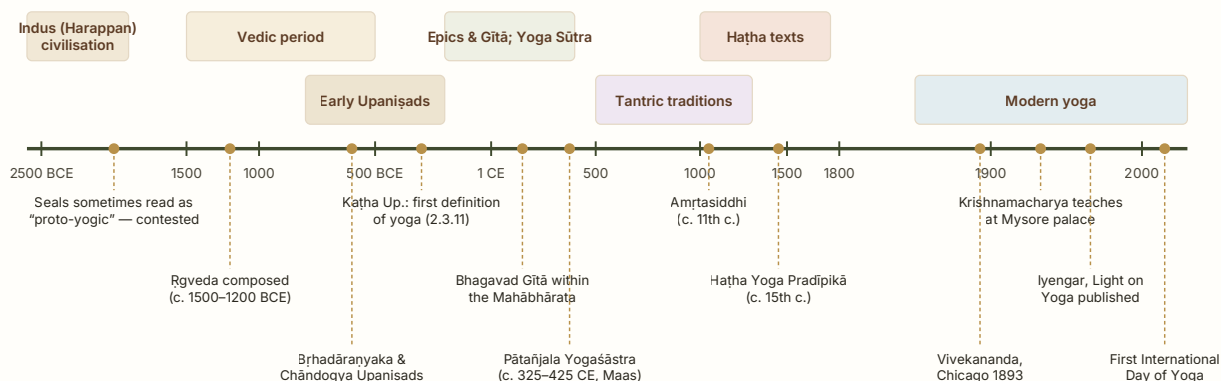
Students ask historical questions — “How old is yoga?”, “Did the ancient yogis do sun salutations?” — and many teachers answer them with myths. This matters for more than accuracy. A teacher who tells students that every posture is thousands of years old invites them to treat every instruction as sacred and unchangeable. A teacher who knows the real history can honour the ancient roots of yoga *and* explain why a posture may be adapted for a modern body. History, in other words, supports both reverence and good teaching.

Two warnings before we begin. First, **dates for early Indian texts are uncertain**: most were composed orally, transmitted for centuries before being written down, and edited over time. The dates below are the ranges most scholars currently accept, and they will be revised. Second, **the history of yoga is not one line**. Yoga developed in many streams — Brahmanical, Buddhist, Jain, Śaiva, Vaiṣṇava and others — which borrowed from and argued with one another.



## A TIMELINE OF YOGA

Dates are approximate and scholarly estimates differ; the scale is compressed before 1000 CE.



**Figure 2.1** Major phases in the history of yoga. Dates for early texts are scholarly estimates, not certainties (Samuel, 2008; Maas, 2013; Mallinson & Singleton, 2017). The Indus seals are listed because they are often cited, not because they are evidence of yoga practice.

## 2.2 The Indus civilisation: a contested beginning

The mature phase of the Indus (Harappan) civilisation flourished in the north-west of the subcontinent from about 2600 to 1900 BCE **HISTORICAL**. Among thousands of small carved seals found at its cities, a few show a figure seated with the knees spread and the heels drawn together, sometimes with a horned headdress. When these seals were published in the 1920s and 1930s, the archaeologist John Marshall suggested the best-known of them depicted a “proto-Śiva” in a yogic posture. The idea has been repeated ever since.

Most scholars now treat this interpretation with great caution (Samuel, 2008). The Indus script has not been deciphered, so we cannot read what the seals meant to the people who made them; the seated posture may represent many things; and there is a gap of more than a thousand years between the seals and the first texts that describe yoga. It is possible that practices ancestral to yoga existed in the Indus world. It is not demonstrated.

### Q HOW TO ANSWER “HOW OLD IS YOGA?”

An honest answer runs like this: “The practices of disciplining the body, breath and mind that became yoga have roots going back at least three thousand years. The first texts that describe yoga clearly date from the last few centuries BCE. Many of the postures we practise today are much more recent — some only a century or two old.”

### 2.3 The Vedic period (c. 1500–500 BCE)

The **Vedas** are the oldest texts of the Indian tradition: collections of hymns, ritual formulae and speculation composed in an early form of Sanskrit. The oldest, the *R̥gveda*, was composed roughly between 1500 and 1200 BCE **HISTORICAL** (Jamison & Brereton, 2014). The Vedic religion centred on *yajña*, the fire sacrifice, performed by priests with exact chanting.

The Vedas do not teach yoga in the later sense, but several threads that yoga would later weave together are already present:

- ◆ **Sacred sound.** Precise chanting, the power of the word (*vāc*) and the syllable *Om* (prominent in later Vedic literature).
- ◆ **Tapas.** Literally “heat”, the ascetic effort by which seers and gods gain power and insight.
- ◆ **Ascetic figures.** A late hymn of the *R̥gveda* (10.136) describes the *keśin*, a long-haired, silent sage (*muni*) who “flies with the wind” — an early image of an ecstatic ascetic.
- ◆ **The breath.** Vedic ritual speculation already treats *prāṇa*, breath, as a life-principle, and later Vedic texts begin to “internalise” the sacrifice: the fire is kindled within, and breathing itself becomes an offering.

### 2.4 The śramaṇa movements and the Upaniṣads (c. 700–200 BCE)

In the middle of the first millennium BCE the Ganges plain saw the rise of towns, trade and a new religious type: the wandering renouncer, or **śramaṇa**, who left household life to seek liberation through austerity, meditation and insight. The Buddha (fifth century BCE on most estimates) and Mahāvīra, the Jain teacher, came from this milieu, and both taught systematic methods of meditation and ethical restraint **HISTORICAL** (Samuel, 2008). Early Buddhist texts describe graded states of meditative absorption (*jhāna*) that closely parallel later yogic descriptions of concentration.

At the same time, Brahmanical thinkers composed the **Upaniṣads**, philosophical dialogues attached to the Vedas. The oldest — the *Bṛhadāraṇyaka* and *Chāndogya* — probably predate the Buddha. They introduce ideas that became central to yoga: that the true Self (*ātman*) is identical with the ground of all being (*brahman*); that actions (*karma*) bind a person to rebirth; and that liberation comes through knowledge. Slightly later Upaniṣads describe yoga explicitly:

- ◆ The **Kaṭha Upaniṣad** gives the first clear definition of yoga as “the firm holding of the senses” (2.3.11) and the image of the Self as the lord of a chariot whose horses are the senses (1.3.3–9).
- ◆ The **Śvetāśvatara Upaniṣad** instructs the practitioner to hold the body erect with chest, neck and head in line, to restrain the breath, and to practise in a clean, level place sheltered from wind (2.8–10). It lists the first signs of progress in yoga: lightness, health, steadiness, clear complexion, a pleasant voice (2.13) (Olivelle, 1996).
- ◆ The **Maitrī (Maitrāyaṇīya) Upaniṣad** lists a six-limbed yoga: breath control, withdrawal of the senses, meditation, concentration, reasoning (*tarka*) and absorption (6.18) — five of Patañjali’s later eight limbs, in a slightly different order.

Chapter 3 returns to the Upaniṣads in detail.

## 2.5 The epics and the Bhagavad Gītā (c. 400 BCE–400 CE)

The great Sanskrit epic, the **Mahābhārata**, grew over many centuries, roughly between 400 BCE and 400 CE **HISTORICAL**. Among its many teachings is a long section on liberation (*Mokṣadharmā*) in which Sāṃkhya and Yoga appear as paired paths, and the **Bhagavad Gītā**, a dialogue of 700 verses between the warrior Arjuna and Kṛṣṇa on the battlefield. The Gītā broadened the meaning of yoga dramatically: alongside meditation it taught *karma yoga*, the yoga of action performed without attachment to its fruits, and *bhakti yoga*, the yoga of devotion (Chapter 4).

## 2.6 Classical Yoga: Patañjali (c. 325–425 CE)

The *Yoga Sūtra* attributed to Patañjali is a compact collection of 195 or 196 aphorisms which systematised yoga as a philosophy of mind and a path to liberation. Traditionally the sūtras are studied together with a commentary (*bhāṣya*) attributed to Vyāsa. Philipp Maas has argued that sūtras and commentary form a single work, the *Pātañjalayogaśāstra*, composed around 325–425 CE **HISTORICAL** (Maas, 2013); older estimates range more widely, from the second century BCE onwards. Patañjali's yoga shares its metaphysics largely with Sāṃkhya and much of its meditative vocabulary with Buddhism. Chapters 5 to 8 study it in detail.

## 2.7 Tantra (from c. 500 CE)

From around the middle of the first millennium CE, a large family of traditions now called **tantra** developed within Śaivism, Vaiṣṇavism, Buddhism and Jainism. Tantric texts introduced or elaborated much of what modern students think of as “yogic”: detailed maps of a subtle body with channels (*nāḍī*) and centres (*cakra*), the coiled power *kuṇḍalinī*, extensive use of *mantra*, visualisation of deities, and ritual gestures (*mudrā*) **HISTORICAL** (Samuel, 2008; Flood, 1996). Tantra is often misunderstood in the West as primarily sexual; in the texts themselves, sexual ritual is a minor and often esoteric element within vast systems of ritual, meditation and philosophy.

## 2.8 Haṭha Yoga (c. 1000–1800 CE)

**Haṭha Yoga** emerged in the early second millennium. The earliest text to teach its characteristic methods, the *Amṛtasiddhi*, dates from around the eleventh century and comes from a Buddhist tantric milieu; many later Haṭha texts are associated with the Nāth yogis, who trace their lineage to Matsyendra and Gorakṣa **HISTORICAL** (Mallinson & Singleton, 2017). The word *haṭha* means “force”, referring to the effortful methods of the practice; the popular explanation that *ha* means “sun” and *ṭha* “moon” appears only in later texts (Birch, 2011).

Haṭha Yoga's distinctive contribution was to use the body — posture, breath retention, locks (*bandha*), seals (*mudrā*) and cleansing practices (*śaṭkarma*) — to control the vital energy and bring about the states of meditation earlier traditions pursued more directly. The classic compendium, Svātmārāma's *Haṭha Yoga Pradīpikā* (fifteenth century), gathered earlier teachings into four chapters; it describes fifteen postures. Later texts expanded the repertoire: the *Gheraṇḍa Saṃhitā* (c. 1700) describes thirty-two āsanas, and an eighteenth-century text, the *Haṭhābhyāsapaddhati*, describes over a hundred, many of them strenuous and dynamic (Mallinson & Singleton, 2017). Chapter 9 studies this tradition.

### THE 84 ĀSANAS

The *Haṭha Yoga Pradīpikā* says that Śiva taught 84 postures (HYP 1.33), and some Haṭha texts add that there are as many postures as there are species of living beings — traditionally 8,400,000. The numbers are symbolic — 84 is an auspicious number in Indian tradition — and lists of “the 84 postures” differ from text to text. The idea is best understood as “there are countless postures, and a few are essential”, which is exactly how the *Pradīpikā* goes on to name four seated postures as the most important.

## 2.9 Yoga in the modern world (from c. 1850)

Modern yoga was shaped by encounters between Indian traditions and the colonial, scientific and global world.

- ◆ **Swami Vivekananda** (1863–1902) addressed the World’s Parliament of Religions in Chicago in 1893 and published *Raja Yoga* (1896), presenting Patañjali’s yoga as a universal science of mind. He was notably dismissive of Haṭha Yoga (Vivekananda, 1896; De Michelis, 2004).
- ◆ **Scientific research** on yoga began in India in the 1920s, notably at Kaivalyadhama, founded by Swami Kuvalayananda at Lonavla in 1924.
- ◆ **Physical culture.** In the early twentieth century Indian reformers, influenced in part by European gymnastics and physical-culture movements, developed exercise systems that drew on yogic postures. The Raja of Aundh popularised a systematic *sūrya namaskār* in the 1920s and 1930s, and T. Krishnamacharya (1888–1989) taught a dynamic posture practice at the Mysore palace in the 1930s whose students included B. K. S. Iyengar, K. Pattabhi Jois and his son T. K. V. Desikachar (Singleton, 2010).
- ◆ In 2014 the United Nations General Assembly proclaimed 21 June as the **International Day of Yoga**, first observed in 2015.

## 2.10 Continuity and change

Historians such as Mark Singleton have shown that much of today’s posture practice — especially flowing sequences, standing postures and the dominance of āsana in a class — is a modern development, drawing on Haṭha Yoga but also on twentieth-century physical culture (Singleton, 2010). Some practitioners find this unsettling. It need not be. The goals of the practice — steadiness of body, regulation of breath, one-pointed attention, freedom from suffering — are ancient. The forms in which those goals are pursued have always changed, and the *Haṭha Yoga Pradīpikā* itself was a work of gathering and innovation. What matters for a teacher is to know which is which: to teach a modern posture as a modern posture, and an ancient teaching as an ancient teaching.

### OLD AIMS, CHANGING FORMS

Yoga’s aims — a steady body, a regulated breath, a focused mind and ultimately freedom — are ancient. Its forms have changed in every era. Teach both honestly.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ Early Indus seals are sometimes presented as evidence of yoga; the interpretation is speculative.
- ◆ Threads of yoga appear in the Vedas (sacred sound, tapas, breath); the first explicit teachings on yoga come in the Upaniṣads and early Buddhist and Jain texts.
- ◆ The Bhagavad Gītā broadened yoga to include action and devotion; Patañjali systematised it as a discipline of mind around 325–425 CE.
- ◆ Tantra contributed the subtle body, cakras, kuṇḍalinī and much mantra practice; Haṭha Yoga (from c. 1000 CE) made the body a primary instrument.
- ◆ Modern posture-based yoga developed in the late nineteenth and twentieth centuries, shaped by reformers, physical culture and teachers such as Krishnamacharya.

 REFLECTION QUESTIONS

1. What did you believe about the age of yoga before reading this chapter? Where did that belief come from?
2. Does learning that many postures are modern change how you feel about practising them? Why, or why not?
3. How might you explain the relationship between tradition and innovation to a student who is anxious about “doing yoga correctly”?

 TEACHER-PRACTICE EXERCISES

1. Draw the timeline in Figure 2.1 from memory, marking at least eight events with approximate dates. Check it against the figure.
2. In pairs, role-play a student asking “Is yoga a Hindu practice?” Give an answer that is accurate, respectful and brief (under one minute).
3. Choose one posture you love. Find out when it is first described in a text. Share your finding with the group.

 KNOWLEDGE CHECK

1. The mature Indus civilisation flourished approximately: (a) 5000–4000 BCE (b) 2600–1900 BCE (c) 1000–500 BCE (d) 200 BCE–200 CE
2. The first explicit definition of yoga appears in: (a) the Ṛgveda (b) the Kaṭha Upaniṣad (c) the Haṭha Yoga Pradīpikā (d) the Gheraṇḍa Saṃhitā
3. How many āsanās does the Haṭha Yoga Pradīpikā describe? (a) 12 (b) 15 (c) 32 (d) 84
4. The word *haṭha* literally means: (a) sun and moon (b) force (c) posture (d) union
5. The Haṭha Yoga Pradīpikā was compiled in approximately the: (a) 2nd century BCE (b) 5th century CE (c) 15th century CE (d) 19th century CE

Answers 1 b · 2 b · 3 b · 4 b · 5 c

 TEACHING ASSIGNMENT

Write and rehearse a five-minute “history of yoga in five moments” talk for a workshop audience, choosing five events from this chapter. For each, say what we know, how we know it, and why it matters to practice today. Include at least one myth you correct gently.

## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER 3

## The Vedic Roots and the Upaniṣads

“Lead me from the unreal to the real, from darkness to light, from death to immortality.” — *Bṛhadāraṇyaka Upaniṣad* 1.3.28

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the structure of the Vedic literature and the place of the Upaniṣads within it;
- ♦ explain the key Upaniṣadic teachings on *ātman*, *brahman*, *karma* and liberation;
- ♦ recount the chariot image of the Kaṭha Upaniṣad and the yoga instructions of the Śvetāśvatara Upaniṣad;
- ♦ chant and translate the Gāyatrī mantra and two peace invocations used in class;
- ♦ use Upaniṣadic stories and images responsibly in teaching.

## 3.1 The Veda

*Veda* means “knowledge”. The Vedic literature is regarded in the Hindu tradition as *śruti*, “that which was heard” — revealed truth received by ancient seers (*ṛṣi*) — as distinct from *smṛti*, “that which is remembered”, the later body of epics, law books and Purāṇas composed by human authors **TRADITIONAL**. There are four Vedas:

**Table 3.1** The four Vedas

| Veda        | Contents                                                | Relevance to yoga                                                                |
|-------------|---------------------------------------------------------|----------------------------------------------------------------------------------|
| Ṛgveda      | about 1,000 hymns to the deities                        | oldest layer; Gāyatrī mantra; the keśin ascetic; speculation on origins (10.129) |
| Sāmaveda    | Ṛgvedic verses set to melody for chanting               | the tradition of sacred sound and chant                                          |
| Yajurveda   | ritual formulae for the sacrifice                       | Kaṭha, Taittiriya and Śvetāśvatara Upaniṣads belong to its branches              |
| Atharvaveda | hymns and charms for healing, protection and daily life | early ideas on breath and the body; Māṇḍūkya, Muṇḍaka, Praśna Upaniṣads          |

Each Veda grew in layers. The **Samhitā** is the core collection of hymns or formulae. The **Brāhmaṇas** explain the meaning and performance of ritual. The **Āraṇyakas**, “forest books”, reflect on ritual symbolically. The **Upaniṣads**, the final layer, are philosophical. For this reason they are also called *Vedānta*, “the end of the Veda” — both its concluding portion and, as the tradition understands it, its culmination.

GĀYATRĪ MANTRA · ṚGVEDA 3.62.10

ॐ भूर्भुवः स्वः । तत्सवितुर्वरेण्यं भर्गो देवस्य धीमहि । धियो यो नः प्रचोदयात् ॥

om bhūr bhuvaḥ svaḥ · tat savitur vareṇyaṁ bhargo devasya dhīmahi · dhiyo yo naḥ  
pracodayāt

We meditate on the excellent radiance of the divine Savitṛ (the sun as impeller); may it inspire our minds.

The Gāyatrī is the most famous Vedic verse. The opening “om bhūr bhuvaḥ svaḥ” (earth, atmosphere, heaven) is a later ritual prefix, not part of the Ṛgvedic verse itself. It is chanted at dawn and dusk by many practitioners and is often sung at the opening of traditional yoga classes.

### 3.2 What the Upaniṣads are

The word *upaniṣad* is traditionally explained as *upa* (near), *ni* (down) and *śad* (sit) — “sitting down near” a teacher to receive secret teaching **INTERPRETIVE**. Śaṅkara, the great commentator, preferred to explain it as “that which destroys ignorance”. Both readings capture something true: the Upaniṣads are records of intimate dialogue between teacher and student about the deepest questions.

Over two hundred texts carry the name *Upaniṣad*, composed over more than two thousand years. A traditional canon lists 108. Scholars distinguish a group of about thirteen early or “principal” Upaniṣads, composed roughly between the seventh century BCE and the first centuries CE **HISTORICAL** (Olivelle, 1996). These, together with the Bhagavad Gītā and the Brahma Sūtras, form the threefold foundation of Vedānta philosophy (Chapter 10).

### 3.3 The central teachings

#### Ātman and brahman

The Upaniṣads ask: what is the true Self? And what is the ultimate reality behind the changing world? Their answer — or rather, the answer of their most influential passages — is that these are the same. *Ātman*, the innermost Self, is *brahman*, the ground of all being **TRADITIONAL**. Four short statements, the *mahāvākyas* or “great sayings”, became the classic formulations:

**Table 3.2** The four mahāvākyas

| Mahāvākya        | Meaning                  | Source                               |
|------------------|--------------------------|--------------------------------------|
| prajñānaṁ brahma | Consciousness is brahman | Aitareya Up. 3.3 (Ṛgveda)            |
| ahaṁ brahmāsmi   | I am brahman             | Bṛhadāraṇyaka Up. 1.4.10 (Yajurveda) |
| tat tvam asi     | You are that             | Chāndogya Up. 6.8.7 (Sāmaveda)       |



| Mahāvākya        | Meaning              | Source                       |
|------------------|----------------------|------------------------------|
| ayam ātmā brahma | This Self is brahman | Māṇḍūkya Up. 2 (Atharvaveda) |

In the *Chāndogya*, the sage Uddālaka teaches his son Śvetaketu this truth through a series of demonstrations: salt dissolved in water cannot be seen but can be tasted everywhere in it; so the subtle essence pervades all things, “and you are that” (*tat tvam asi*).

## Karma and rebirth

The *Bṛhadāraṇyaka* gives one of the earliest statements of the law of action: “As a person acts, as a person behaves, so does he become. Doing good, he becomes good; doing evil, he becomes evil” (4.4.5) (Olivelle, 1996). Actions leave impressions; impressions shape desire; desire shapes further action and future birth. Liberation (*mokṣa*) is release from this cycle through knowledge of the Self.

## The breath as life

Many Upaniṣadic passages place *prāṇa* at the centre of life. In a famous story, the bodily faculties — speech, sight, hearing, mind and breath — quarrel over which is most important. Each leaves the body in turn, and the body carries on, impaired. When breath begins to leave, all the others are torn from their places and beg it to stay (*Bṛhadāraṇyaka* 6.1; *Chāndogya* 5.1) (Olivelle, 1996). The story explains why yoga gives such weight to the breath: it is the faculty on which all others depend.

## 3.4 Yoga in the Upaniṣads

### The chariot of the Kaṭha Upaniṣad

The *Kaṭha Upaniṣad* frames its teaching as a dialogue between a boy, Naciketas, and Yama, the lord of death. Offered every pleasure the world can give, Naciketas refuses them all and asks only what happens after death. Yama, impressed, teaches him about the Self. His central image is the chariot:

#### ↓ THE CHARIOT · KAṬHA UPANIṢAD 1.3.3–9

Know the Self as the rider in a chariot, and the body as the chariot. Know the intellect (*buddhi*) as the charioteer and the mind (*manas*) as the reins. The senses are the horses, and the objects of the senses the roads they run on. When the charioteer lacks understanding and the reins are slack, the senses are like wild horses. When the charioteer is wise and the reins held firm, the senses are like good horses — and the rider reaches the end of the journey.

A little later the Upaniṣad defines yoga: “When the five perceptions are still, together with the mind, and the intellect does not stir — that, they say, is the highest state. This firm holding of the senses is what they consider yoga” (2.3.10–11) (Olivelle, 1996). Every later definition of yoga stands in the shadow of this image.

### Instructions in the Śvetāśvatara Upaniṣad

The *Śvetāśvatara Upaniṣad* gives the oldest practical instructions for yoga practice we have. It advises:

- ♦ holding the body steady with the three upper parts — chest, neck and head — erect (2.8);
- ♦ restraining the breath gently and breathing out through the nose when the breath is subdued (2.9);

- ♦ controlling the mind “as a wise man controls a wagon yoked to unruly horses” (2.9);
- ♦ practising in a level, clean place, free from pebbles, fire and gravel, pleasant to the mind and sheltered from wind (2.10).

It describes forms seen in meditation, such as mist, smoke, sun, wind and fire, and the first fruits of yoga: “lightness, health, freedom from desire, a clear complexion, a pleasant voice, a sweet smell and scanty excretions” (2.11–13) (Olivelle, 1996). These are traditional statements of what practitioners experienced **TRADITIONAL**, and it is striking how closely the practical advice — erect posture, gentle breath, quiet surroundings — matches what a meditation teacher would advise today.

### The sheaths and the states of consciousness

Two further Upaniṣadic teachings are essential for teachers:

- ♦ The *Taittirīya Upaniṣad* describes the person as a series of sheaths (*kośa*) — made of food, breath, mind, understanding and bliss — each subtler than the last (2.1–5). This model is studied in Chapter 13.
- ♦ The *Māṇḍūkya Upaniṣad*, only twelve verses long, analyses the syllable *Om* as A-U-M plus silence, corresponding to waking, dreaming, deep sleep and *turiya*, “the fourth”, pure awareness underlying all three.

#### THE FOUR QUARTERS OF OM · MĀṆḌŪKYA UPANIṢAD 3–12



**Figure 3.1** The *Māṇḍūkya Upaniṣad* maps the three sounds of AUM onto waking, dreaming and deep sleep, and the silence that follows onto *turiya* (Olivelle, 1996).

### 3.5 Peace invocations

Each Upaniṣad is traditionally chanted with a peace invocation (*śānti pāṭha*). Two are widely used in yoga classes:

#### BṚHADĀRAṆYAKA UPANIṢAD 1.3.28

असतो मा सद्गमय । तमसो मा ज्योतिर्गमय । मृत्योर्माऽमृतं गमय ॥

*asato mā sad gamaya · tamaso mā jyotir gamaya · mṛtyor mā'mṛtaṁ gamaya*

Lead me from the unreal to the real; lead me from darkness to light; lead me from death to immortality.

## ĪŚA UPANIṢAD (INVOCATION)

ॐ पूर्णमदः पूर्णमिदं पूर्णात् पूर्णमुदच्यते । पूर्णस्य पूर्णमादाय पूर्णमेवावशिष्यते ॥

om pūrṇam adaḥ pūrṇam idaṁ pūrṇāt pūrṇam udacyate · pūrṇasya pūrṇam ādāya  
pūrṇam evāvaśiṣyate

That is whole; this is whole. From the whole the whole arises. Taking the whole from the whole, the whole alone remains.

The threefold *śānti* that closes these chants is traditionally explained as peace from three sources of disturbance: from within oneself (*ādhyātmika*), from other beings (*ādhibhautika*) and from forces beyond control (*ādhidaivika*) **TRADITIONAL**.

## Q USING THE UPANIṢADS IN CLASS

A single line well chosen — “lead me from darkness to light” before relaxation, or the chariot image before *prāṇāyāma* — can give a class depth without turning it into a lecture. Always give the source, translate the Sanskrit, and never ask students to affirm a belief. Offer chanting as an invitation: “Join in if you wish, or simply listen.”

## CHAPTER REVIEW

## KEY TAKEAWAYS

- ◆ The Veda has four collections, each in layers ending in the Upaniṣads, also called Vedānta.
- ◆ The central Upaniṣadic teaching is the identity of the Self (*ātman*) with ultimate reality (*brahman*), expressed in the four *mahāvākyas*.
- ◆ The Upaniṣads introduce karma and rebirth, the primacy of *prāṇa*, the five sheaths and the analysis of Om.
- ◆ The Kaṭha Upaniṣad gives the chariot image and the first definition of yoga; the Śvetāśvatara gives the first practical instructions.
- ◆ Chants from the Upaniṣads can enrich a class when offered with translation, context and freedom to participate.

## REFLECTION QUESTIONS

1. In the chariot image, which part of you — horses, reins or charioteer — is strongest? Which is weakest?
2. Naciketas refused every pleasure in order to ask one question. What question would you ask?
3. How does the Śvetāśvatara’s advice on where to practise apply to the room in which you will teach?

## TEACHER-PRACTICE EXERCISES

1. Learn to chant the Gāyatrī mantra and the Asato mā invocation from memory, with correct pronunciation. Chant each for your practice group.

2. Explain the chariot image to a partner in under two minutes, relating it to what happens in a difficult posture.
3. Design a one-minute opening for a class using one line from an Upaniṣad, with its source and translation.

#### ☑ KNOWLEDGE CHECK

1. Which layer of the Veda is philosophical? (a) Saṃhitā (b) Brāhmaṇa (c) Āraṇyaka (d) Upaniṣad
2. *Tat tvam asi* comes from which Upaniṣad? (a) Kaṭha (b) Chāndogya (c) Māṇḍūkya (d) Taittirīya
3. In the Kaṭha chariot image, the reins represent: (a) the senses (b) the intellect (c) the mind (d) the body
4. The oldest practical yoga instructions (posture, breath, place of practice) are found in the: (a) Śvetāśvatara Upaniṣad (b) Ṛgveda (c) Īśa Upaniṣad (d) Aitareya Upaniṣad
5. The Māṇḍūkya Upaniṣad maps A-U-M onto: (a) the three guṇas (b) waking, dreaming and deep sleep (c) the three nāḍīs (d) past, present and future

Answers 1d · 2b · 3c · 4a · 5b

#### 📄 TEACHING ASSIGNMENT

Choose one Upaniṣadic passage from this chapter. Prepare a five-minute “philosophy moment” for the end of a class: read the passage, translate or paraphrase it, connect it to one experience students have just had in practice, and close with a short silence. Deliver it in your practice group and invite one question.

## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER

## 4

## The Bhagavad Gītā

*“You have a right to your action, but never to its fruits. Let the fruits of action not be your motive, nor let your attachment be to inaction.” —  
Bhagavad Gītā 2.47*

🕒 **LEARNING OBJECTIVES**

By the end of this chapter you will be able to:

- ◆ place the Bhagavad Gītā in its literary and historical context;
- ◆ summarise the structure of its eighteen chapters and the dramatic situation that frames them;
- ◆ explain the Gītā’s teachings on karma yoga, jñāna yoga, bhakti yoga and dhyāna yoga;
- ◆ describe the figure of the *sthitaprajña*, the person of steady wisdom;
- ◆ apply the Gītā’s teaching on the three guṇas to diet, practice and teaching.

### 4.1 Context

The **Bhagavad Gītā**, “the Song of the Lord”, is a poem of 700 verses in eighteen chapters, forming part of the sixth book (*Bhīṣma Parvan*) of the *Mahābhārata*. It was probably composed between the second century BCE and the second century CE **HISTORICAL** (Flood, 1996). Though technically *smṛti*, it has been treated in the Hindu tradition with the authority of an Upaniṣad, and it is one of the three foundational texts of Vedānta.

The setting is the eve of a catastrophic war between two branches of one royal family. Arjuna, the finest archer of his age, asks his charioteer Kṛṣṇa to drive between the armies. Seeing teachers, cousins and friends ranged against him, he is overcome: his bow slips from his hand and he refuses to fight. The Gītā is Kṛṣṇa’s reply. By its end Arjuna has recovered his clarity and resolve, and the poem has laid out one of the most comprehensive syntheses of Indian spiritual teaching.

## READING THE BATTLEFIELD

Commentators have read the battlefield both literally and symbolically. Mahatma Gandhi, committed to non-violence, read the war as an allegory of the struggle within every person between higher and lower tendencies **INTERPRETIVE**. Others insist on the literal setting: the Gītā is a teaching on how to act rightly when every choice has painful consequences. Both readings are ancient in spirit, and both are useful for teachers.

## 4.2 Structure

**Table 4.1** An overview of the eighteen chapters (titles vary between editions)

| Chapters | Theme                     | Key teachings                                                                                                           |
|----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1        | Arjuna's despair          | the crisis that prompts the teaching                                                                                    |
| 2        | Knowledge and the Self    | the eternal Self; duty; yoga as evenness (2.48) and skill in action (2.50); the person of steady wisdom (2.54–72)       |
| 3–5      | Action                    | karma yoga; acting without attachment; action as offering; renunciation in action                                       |
| 6        | Meditation                | instructions for sitting, moderation, the restless mind (6.10–36)                                                       |
| 7–12     | Devotion and the divine   | the nature of the Lord; the cosmic vision (ch. 11); bhakti yoga; qualities of the devotee (12.13–20)                    |
| 13–15    | Knowledge                 | the field and the knower of the field; the three guṇas (ch. 14); the cosmic tree (ch. 15)                               |
| 16–18    | Discernment and surrender | divine and demonic qualities; threefold faith, food, austerity and giving (ch. 17); renunciation and surrender (ch. 18) |

## 4.3 Karma yoga: the yoga of action

The Gītā's most distinctive teaching is that action itself can be yoga. Arjuna's instinct is to withdraw — to renounce the world and avoid the harm of acting. Kṛṣṇa answers that no one can avoid action altogether, even for a moment (3.5), and that withdrawing from one's duty is itself a choice with consequences. The solution is not to stop acting but to change the relationship to action:

### BHAGAVAD GĪTĀ 2.47

कर्मण्येवाधिकारस्ते मा फलेषु कदाचन ।

*karmany evādhikāras te mā phaleṣu kadācana*

Your right is to the action alone, never to its fruits.

This is *niṣkāma karma*, action without craving for results. It does not mean indifference to outcomes — Kṛṣṇa wants Arjuna to fight well — but freedom from the anxiety and pride that attach to success and failure. Hence the definitions of yoga in chapter 2: *samatvaṃ yoga ucyate*, “evenness of mind is called yoga” (2.48), and *yogaḥ karmasu kauśalam*, “yoga is skill in action” (2.50). The Gītā also teaches that each person should follow their own path of duty (*svadharma*): “Better one's own duty, even imperfectly performed, than the duty of another well performed” (3.35) (Easwaran, 2007).

### Q KARMA YOGA FOR TEACHERS

Teaching is karma yoga in its purest modern form. You prepare carefully, teach with full attention — and then let go of whether students praise you, return or perform the poses beautifully. Teachers who teach for applause burn out; teachers who teach as an offering tend to last.

## 4.4 Jñāna yoga: the yoga of knowledge

In chapter 2 Kṛṣṇa begins not with action but with the nature of the Self: it is not born and does not die; weapons do not cut it, fire does not burn it (2.20–23). Wisdom is to know oneself as this unchanging awareness rather than as the body and mind that change. Chapter 13 develops this as the distinction between the *field* (*kṣetra*: body, mind and all that can be known) and the *knower of the field* (*kṣetrajña*). Jñāna yoga is the path of discrimination between the two (Chapter 10).

### The person of steady wisdom

At the end of chapter 2, Arjuna asks what a person established in wisdom looks like — how they speak, sit and walk. Kṛṣṇa's answer (2.54–72) is one of the great portraits of the goal of yoga: one who is not shaken by sorrow nor elated by pleasure, free from craving, fear and anger; who can withdraw the senses from their objects “as a tortoise draws in its limbs” (2.58) — an image of *pratyāhāra* centuries before Patañjali.

## 4.5 Dhyāna yoga: meditation

Chapter 6 contains detailed instructions for meditation that remain practical today (Easwaran, 2007):

- ◆ Sit in a clean place, on a seat neither too high nor too low (6.11).
- ◆ Hold the body, head and neck erect and still, gazing as if at the tip of the nose, not looking around (6.13).
- ◆ Practise **moderation**: “Yoga is not for one who eats too much, nor for one who eats too little; not for one who sleeps too much, nor for one who stays awake too long” (6.16). It is for one moderate in food, recreation, effort, sleep and waking (6.17).
- ◆ When the mind wanders, bring it back again and again (6.26).

Arjuna objects that the mind is as restless and hard to hold as the wind (6.34). Kṛṣṇa agrees — and gives the answer Patañjali would also give: through **practice** (*abhyāsa*) and **detachment** (*vairāgya*) it can be held (6.35).

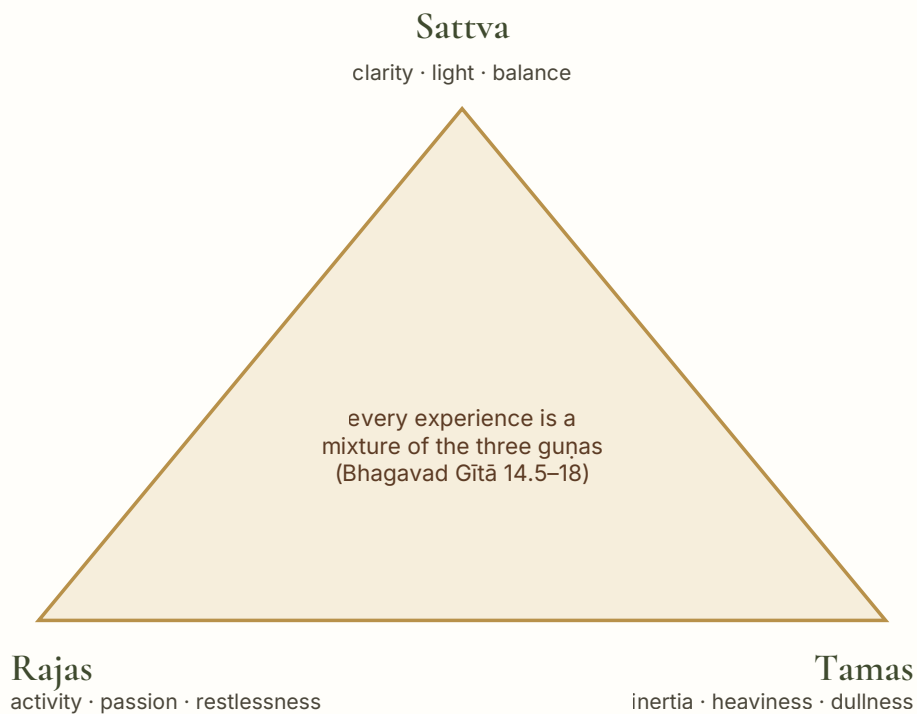
## 4.6 Bhakti yoga: the yoga of devotion

From chapter 7 onward the Gītā turns increasingly to devotion. Kṛṣṇa reveals himself as the Lord present in all beings, and in chapter 11 grants Arjuna a terrifying vision of his universal form. The path of love and surrender is presented as the most accessible: whatever is offered with devotion — “a leaf, a flower, a fruit, water” — is accepted (9.26). Chapter 12 describes the devotee dear to the Lord: without

hatred, friendly and compassionate, free from possessiveness and ego, even-minded in pain and pleasure, forgiving (12.13–20). The Gītā closes with the call to surrender: “Abandoning all duties, take refuge in me alone” (18.66).

## 4.7 The three guṇas

The Gītā adopts from Sāṃkhya the teaching that all of nature is composed of three qualities, the **guṇas**: *sattva* (clarity, harmony), *rajas* (activity, passion) and *tamas* (inertia, darkness) (ch. 14). Every experience, food and activity is a mixture of the three. Chapter 17 applies this to daily life, and it remains the basis for the yogic approach to diet and lifestyle taught in the tradition taught here.



**Figure 4.1** The three *guṇas* of Sāṃkhya and the Bhagavad Gītā. Yogic practice aims to strengthen *sattva*, while the Gītā ultimately points beyond all three (14.19–26).

**Table 4.2** The *guṇas* applied to food, practice and teaching (after Bhagavad Gītā 17)

|                                  | Sattva                                                                   | Rajas                                                          | Tamas                                       |
|----------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------|
| <b>Food</b> (17.8–10)            | fresh, wholesome, nourishing, agreeable; promotes life, strength, health | very bitter, sour, salty, hot, pungent, dry; causes discomfort | stale, tasteless, spoiled, leftover, impure |
| <b>Practice</b> (interpretation) | steady, attentive, breath-led, ending in rest                            | competitive, rushed, pushing into pain                         | sluggish, distracted, habitual              |
| <b>Teaching</b> (interpretation) | clear, calm, responsive to students                                      | performing, showing off, over-talking                          | unprepared, disengaged                      |

The first row follows the text **TRADITIONAL**; the second and third are applications, not quotations **INTERPRETIVE**. The Gītā’s ultimate teaching goes beyond cultivating *sattva*: the liberated person transcends all three *guṇas* (14.19–26).



### ◆ THE GĪTĀ'S SYNTHESIS

The Gītā does not ask you to choose one path. Action, knowledge, meditation and devotion are presented as mutually supporting — the foundation of the integrated practice of the four paths (Chapter 11).

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ The Bhagavad Gītā is a 700-verse dialogue within the Mahābhārata, composed around the turn of the Common Era.
- ◆ It teaches action without attachment to results (*niṣkāma karma*) and defines yoga as evenness of mind and skill in action.
- ◆ It presents the paths of knowledge, meditation and devotion alongside action, and portrays the goal in the *sthitaprajña*.
- ◆ Chapter 6 gives practical meditation instructions and the principle of moderation; practice and detachment steady the mind.
- ◆ The three guṇas provide a framework for diet, lifestyle and the quality of practice and teaching.

### 🌿 REFLECTION QUESTIONS

1. Where in your life do you act mainly for the fruits? What would change if you acted for the action itself?
2. Chapter 6 asks for moderation in food, sleep and effort. Which of these is hardest for you during the training?
3. Which description of the *sthitaprajña* (2.54–72) would you most like a student to say about you as a teacher?

### ✋ TEACHER-PRACTICE EXERCISES

1. For one day, keep a “fruits” journal: each time you notice anxiety about an outcome (praise, performance, comparison), note it without judgement.
2. Teach a five-minute seated meditation using only the instructions in Gītā 6.11–13 and 6.26, in your own words.
3. With a partner, classify the last three meals you ate by the guṇas, and discuss how you felt after each.

 **KNOWLEDGE CHECK**

1. The Bhagavad Gītā is part of which text? (a) Rāmāyaṇa (b) Mahābhārata (c) Ṛgveda (d) Yoga Sūtra
2. *Niṣkāma karma* means: (a) renunciation of all action (b) action without craving for its fruits (c) action for the sake of karma (d) ritual action
3. According to Gītā 6.35, the mind is controlled by: (a) prāṇāyāma and āsana (b) practice and detachment (c) knowledge and devotion (d) diet and sleep
4. Which chapter contains the vision of the universal form? (a) 2 (b) 6 (c) 11 (d) 18
5. Food that is stale, spoiled or leftover is described as: (a) sattvic (b) rajasic (c) tamasic (d) none of these

Answers 1b · 2b · 3b · 4c · 5c

 **TEACHING ASSIGNMENT**

Choose one verse from this chapter. Plan a 20-minute practice (movement, breath and a short meditation) that embodies its teaching — for example, a practice on “evenness of mind” in which students hold a challenging posture while noticing their reactions. Write a one-page plan and teach it to two classmates.

## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER 5

## Classical Yoga — Sāṃkhya and the Yoga Sūtras of Patañjali

“Then the seer abides in its own nature.” — Yoga Sūtra 1.3

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ explain the dualist metaphysics of Sāṃkhya — puruṣa, prakṛti, the guṇas and the twenty-five tattvas — on which classical Yoga is built;
- ◆ describe the structure of the Yoga Sūtra and the theme of each of its four chapters;
- ◆ explain the five *vṛttis*, *abhyāsa* and *vairāgya*, the obstacles to practice and their remedies;
- ◆ explain *kriyā yoga* and the five *kleśas*;
- ◆ read a sūtra with a commentary and explain it to a student in plain language.

## 5.1 Yoga as a darśana

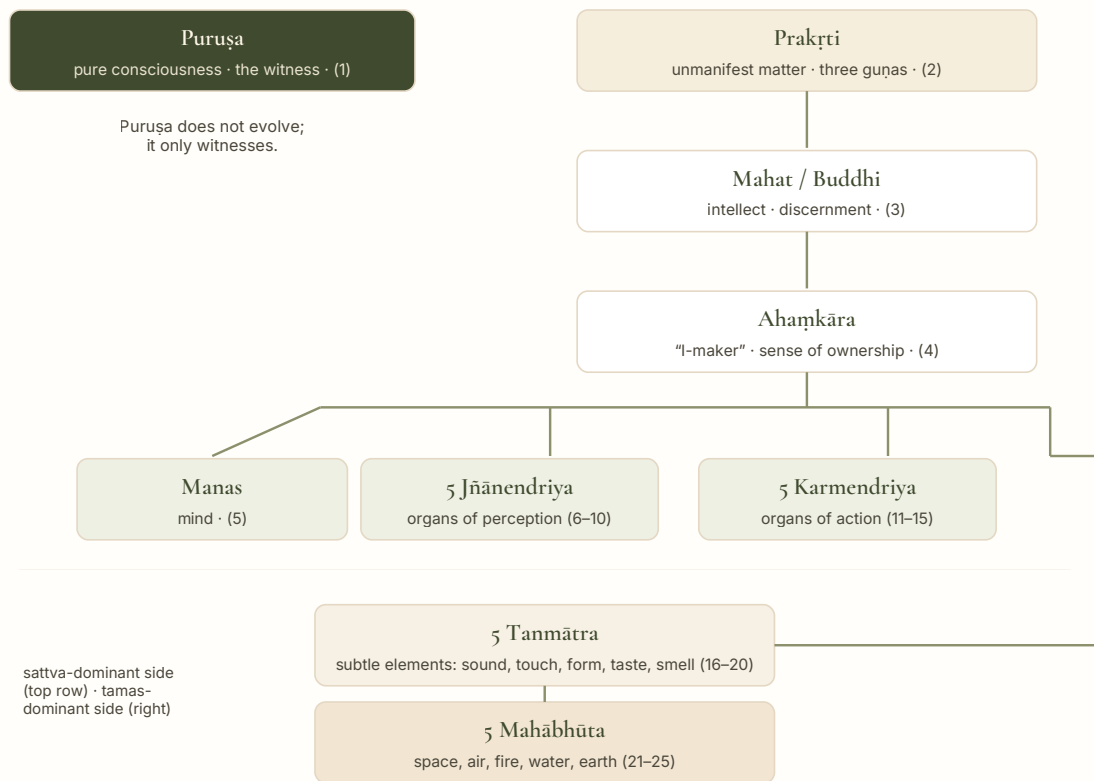
By the early centuries CE, Brahmanical thought had organised itself into schools, or *darśanas* (“ways of seeing”). Six were eventually recognised as orthodox: Nyāya and Vaiśeṣika (logic and natural philosophy), Mīmāṃsā and Vedānta (interpretation of the Veda) and **Sāṃkhya and Yoga**, which were paired as theory and practice. When Indian philosophers speak of “Yoga” as a school, they mean the system of the *Yoga Sūtra*.

## 5.2 Sāṃkhya: the map beneath the practice

Sāṃkhya (“enumeration”) is one of the oldest Indian philosophies; its classical text is Īśvarakṛṣṇa’s *Sāṃkhyakārikā*, of the fourth or fifth century CE **HISTORICAL** (Larson, 1979). It is a **dualism**: reality consists of two fundamentally different principles.

- ◆ **Puruṣa** is pure consciousness — the witness, the seer. It does not act, change or suffer. There are many puruṣas, one for each being.
- ◆ **Prakṛti** is nature or matter in the widest sense — everything that can be experienced, including the mind. Prakṛti is made of three strands, the **guṇas**: *sattva* (lucidity), *rajas* (activity) and *tamas* (inertia).

Unmanifest prakṛti evolves into twenty-three further principles (*tattvas*): the intellect (*buddhi* or *mahat*), the sense of “I” (*ahaṁkāra*), the mind (*manas*), the five organs of perception and five of action, the five subtle elements and the five gross elements.



**Figure 5.1** The twenty-five *tattvas* of classical Sāṃkhya, the philosophical background of Patañjali’s Yoga, following the *Sāṃkhyakārikā* (c. 4th–5th century CE) (Larson, 1979). The diagram is a map of categories, not of physical anatomy.

The crucial and surprising point is that **the mind is on the side of matter**. Thinking, feeling, deciding and remembering are all movements of prakṛti. Consciousness itself is puruṣa, which only witnesses them. Suffering arises because puruṣa is confused with the movements of the mind it observes — as if the audience believed it was the film. Liberation is the discriminating recognition (*viveka*) that seer and seen are different.

#### ◆ WHY THIS MATTERS IN THE PRACTICE HALL

When a teacher says “observe your breath” or “notice the thought, and let it go”, they are applying Sāṃkhya-Yoga: the practice of standing back from the contents of experience as the witness of them. You can teach this without teaching the metaphysics — but you should understand where it comes from.

## 5.3 The Yoga Sūtra

The *Yoga Sūtra* consists of 195 aphorisms (196 in some editions) in four chapters called *pādas*. A *sūtra* (“thread”) is a condensed statement meant to be memorised and explained by a teacher. Almost every *sūtra* is ambiguous without a commentary, and the text has always been studied with the commentary

(*bhāṣya*) attributed to Vyāsa — which recent scholarship suggests may be Patañjali's own (Maas, 2013). Later commentators include Vācaspati Miśra (tenth century), King Bhoja (eleventh century) and Vijñān-abhikṣu (sixteenth century) (Bryant, 2009; White, 2014).

**Table 5.1** The four chapters of the Yoga Sūtra

| Pāda              | Sūtras | Theme                                                                                                     |
|-------------------|--------|-----------------------------------------------------------------------------------------------------------|
| 1 · Samādhi pāda  | 51     | what yoga is; the mind and its fluctuations; practice and detachment; Īśvara; obstacles; kinds of samādhi |
| 2 · Sādhana pāda  | 55     | the practice: kriyā yoga, the kleśas, suffering and its cause, the eight limbs (first five)               |
| 3 · Vibhūti pāda  | 55–56  | the inner limbs; samyama; the extraordinary powers and their dangers                                      |
| 4 · Kaivalya pāda | 34     | the mind, karma and impressions; liberation (kaivalya)                                                    |

## 5.4 The mind and its fluctuations (YS 1.1–1.16)

The text begins: *atha yogānuśāsanam* — “Now, the teaching of yoga” (1.1). The word *atha*, “now”, is traditionally read as marking a beginning that follows preparation: the student is ready. The definition follows (1.2), and then its consequence: “Then the seer abides in its own nature” (1.3). Otherwise, the seer takes on the form of the fluctuations (1.4).

*Citta* is the whole mind-field; *vṛttis* are its movements or “turnings”. Patañjali lists five kinds (1.5–11), each of which may be afflicted (*kliṣṭa*) or unafflicted (*akliṣṭa*):

**Table 5.2** The five *vṛttis* (YS 1.6–1.11)

| Vṛtti     | Meaning                                                               | Example                       |
|-----------|-----------------------------------------------------------------------|-------------------------------|
| pramāṇa   | valid cognition — through perception, inference or reliable testimony | seeing that the room is dark  |
| viparyaya | error, a cognition not matching its object                            | mistaking a rope for a snake  |
| vikalpa   | conceptualisation without a real object; verbal construction          | “the horns of a hare”         |
| nidrā     | sleep — a cognition whose content is absence                          | dreamless sleep               |
| smṛti     | memory — retention of experienced objects                             | remembering yesterday's class |

The inclusion of sleep and memory is instructive: yoga does not aim at replacing “bad” thoughts with good ones, but at stilling *all* the movements — even pleasant ones — so that awareness can know itself.

How is this done? Through *abhyāsa* and *vairāgya* (1.12). Practice is steady effort to remain in stillness (1.13), and it becomes firmly grounded when cultivated **for a long time, without interruption and with devotion** (*dīrgha-kāla-nairantarya-satkāra*, 1.14). Detachment is the mastery of one who no longer thirsts for objects seen or heard of (1.15); its highest form is indifference even to the *guṇas* themselves (1.16).

## YOGA SŪTRA 1.14

स तु दीर्घकालनैरन्तर्यसत्कारासेवितो दृढभूमिः ।

*sa tu dīrgha-kāla-nairantarya-satkārāsevito dṛḍha-bhūmiḥ*

Practice becomes firmly grounded when it is cultivated for a long time, without interruption, and with devotion.

## 5.5 Īśvara (YS 1.23–1.29)

Among the means to samādhi Patañjali includes *īśvara-praṇidhāna*, devotion or surrender to Īśvara, “the Lord” (1.23). Īśvara is a special puruṣa, untouched by affliction, action or its fruits (1.24), the teacher of even the ancient teachers (1.26). Its designator is the sacred syllable *Om* (1.27), and its repetition with contemplation of its meaning is a practice (1.28) which brings inward-turned awareness and removes obstacles (1.29). Scholars debate how theistic Patañjali’s system is; in practice, the sūtras leave room both for devotional students and for those who understand Īśvara as an ideal of pure awareness INTERPRETIVE (Bryant, 2009).

## 5.6 Obstacles and their remedies (YS 1.30–1.39)

Every practitioner meets obstacles. Patañjali names nine (1.30): illness, mental dullness, doubt, carelessness, laziness, sensual indulgence, false perception, failure to reach a stage of practice, and inability to maintain a stage once reached. They are accompanied by four symptoms (1.31): distress, dejection, trembling of the body and **irregular breathing**. The remedy is one-pointed practice (1.32), supported by four attitudes of mind:

### YOGA SŪTRA 1.33 · THE FOUR “IMMEASURABLES”

Friendliness towards the happy, compassion towards the suffering, gladness towards the virtuous, and equanimity towards the non-virtuous — through these the mind becomes clear.

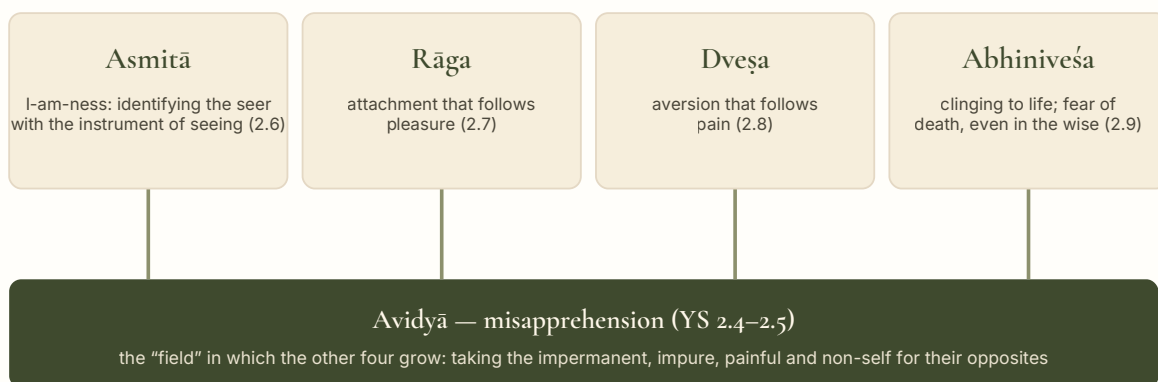
*maitrī · karuṇā · muditā · upekṣā*

These four attitudes are shared with Buddhism, where they are called the *brahmavihāras*. For teachers they are a practical ethic: friendliness towards the confident student, compassion for the struggling one, delight in another’s progress, and equanimity towards the difficult one.

## 5.7 Kriyā yoga and the kleśas (YS 2.1–2.17)

The second chapter opens with a practical yoga for those not yet able to sustain deep meditation: *kriyā yoga*, consisting of *tapas* (disciplined effort), *svādhyāya* (study and self-study, including recitation) and *īśvara-praṇidhāna* (2.1). Its purpose is to cultivate samādhi and weaken the afflictions (2.2).

## THE FIVE KLEŚAS (AFFLICTIONS) · YOGA SŪTRA 2.3–2.9



**Figure 5.2** The five *kleśas* in the Yoga Sūtra. Patañjali describes them as the root of the causes of suffering; *kriyā yoga* (2.1–2.2) is offered to weaken them (Bryant, 2009).

The five *kleśas* — ignorance, I-am-ness, attachment, aversion and clinging to life — are the roots of suffering (2.3). Ignorance (*avidyā*) is the field in which the others grow (2.4); it is defined as taking the impermanent for permanent, the impure for pure, suffering for happiness and the non-self for the Self (2.5). The *kleśas* may be dormant, weak, interrupted or fully active (2.4); in their subtle form they are removed by returning them to their source, and their active forms by meditation (2.10–11).

Patañjali then makes the remarkable statement that for the discerning, all is suffering (2.15) — because of change, anxiety and conditioning, and because the *guṇas* are always in conflict. What can be done is to prevent suffering that has not yet come (2.16). Its cause is the conjunction of the seer and the seen (2.17), and its removal is discriminating knowledge (2.26), for which the **eight limbs** are the means (2.28–29). These are the subject of the next three chapters.

## 5.8 Saṃskāra: how practice changes the mind

Throughout the text runs the idea of *saṃskāra*, a latent impression left by every experience. Impressions produce tendencies, tendencies produce further *vṛttis*, which leave further impressions. Practice works by laying down impressions of stillness, which in turn suppress the impressions of distraction (1.50; 3.9–10). This is a remarkably modern-sounding model of habit and learning, and it underlies the Sūtra’s insistence on long, uninterrupted practice **INTERPRETIVE**.

## 5.9 Powers and liberation (YS 3–4)

The third chapter describes *saṃyama* — concentration, meditation and absorption practised together on one object (3.4) — and the extraordinary powers (*siddhis*) that the tradition says arise from it, such as knowledge of past and future or of other minds. These are traditional claims **TRADITIONAL**, and Patañjali himself warns that they are obstacles to *saṃādhi* even if they appear as powers to the outward-turned mind (3.37). The fourth chapter discusses the mind, karma and time, and closes with *kaivalya*, the “aloneness” of *puruṣa*: the *guṇas*, having served their purpose, return to their source, and pure consciousness is established in its own nature (4.34).

### 📖 READING THE SŪTRAS WITH STUDENTS

Choose one sūtra per week. Write it on a board in transliteration and translation. Explain it in two sentences, connect it to the practice students have just done, and leave it. The sūtras were designed to be carried in memory and chewed slowly.

## CHAPTER REVIEW

### 📌 KEY TAKEAWAYS

- ◆ Classical Yoga rests on Sāṃkhya dualism: consciousness (puruṣa) is distinct from nature (prakṛti), which includes the mind.
- ◆ The Yoga Sūtra has four chapters (samādhi, sādhana, vibhūti, kaivalya) and around 195 aphorisms, studied with commentary.
- ◆ The five vṛttis are valid cognition, error, conceptualisation, sleep and memory; they are stilled by long, uninterrupted, devoted practice and by detachment.
- ◆ Nine obstacles, with symptoms including irregular breathing, are met by one-pointed practice and the four attitudes of friendliness, compassion, gladness and equanimity.
- ◆ Kriyā yoga (tapas, svādhyāya, īśvara-praṇidhāna) weakens the five kleśas, rooted in avidyā.

### 📖 REFLECTION QUESTIONS

1. Which of the nine obstacles (1.30) do you meet most often in your own practice? What does Patañjali suggest?
2. Think of a difficult person. How would 1.33 change how you meet them?
3. Is it helpful or unhelpful for a beginner to hear that “all is suffering for the discerning” (2.15)? How would you present it?

### ✂️ TEACHER-PRACTICE EXERCISES

1. Memorise YS 1.2, 1.12, 1.14, 2.1 and 2.46 in Sanskrit transliteration with a translation.
2. Sit for ten minutes and, each time the mind moves, silently label the vṛtti: *perception, error, imagination, sleepiness, memory*. Discuss what you noticed.
3. Explain the relationship of puruṣa and prakṛti to a partner using an analogy of your own (not the cinema).



☑ **KNOWLEDGE CHECK**

1. In Sāṃkhya, the mind belongs to: (a) puruṣa (b) prakṛti (c) both (d) neither
2. Which is *not* one of the five vṛttis? (a) memory (b) sleep (c) desire (d) error
3. According to YS 1.14, practice becomes firmly grounded when cultivated: (a) with great force (b) for a long time, without interruption, with devotion (c) under a guru only (d) in solitude
4. Kriyā yoga consists of: (a) āsana, prāṇāyāma, pratyāhāra (b) tapas, svādhyāya, īśvara-praṇidhāna (c) yama, niyama, āsana (d) dhāraṇā, dhyāna, samādhi
5. The root kleśa, in which the others grow, is: (a) rāga (b) dveṣa (c) avidyā (d) abhiniveśa

Answers 1 b · 2 c · 3 b · 4 b · 5 c

📄 **TEACHING ASSIGNMENT**

Choose one sūtra from this chapter. Write a 300-word explanation for a student who has never read the Yoga Sūtra, using no Sanskrit except the sūtra itself and one everyday example. Then design a two-minute practice that lets students *experience* the sūtra rather than only hear it.

# The Eight Limbs I — Yamas and Niyamas

*“These are the great vow, universal, not limited by birth, place, time or circumstance.” — Yoga Sūtra 2.31*

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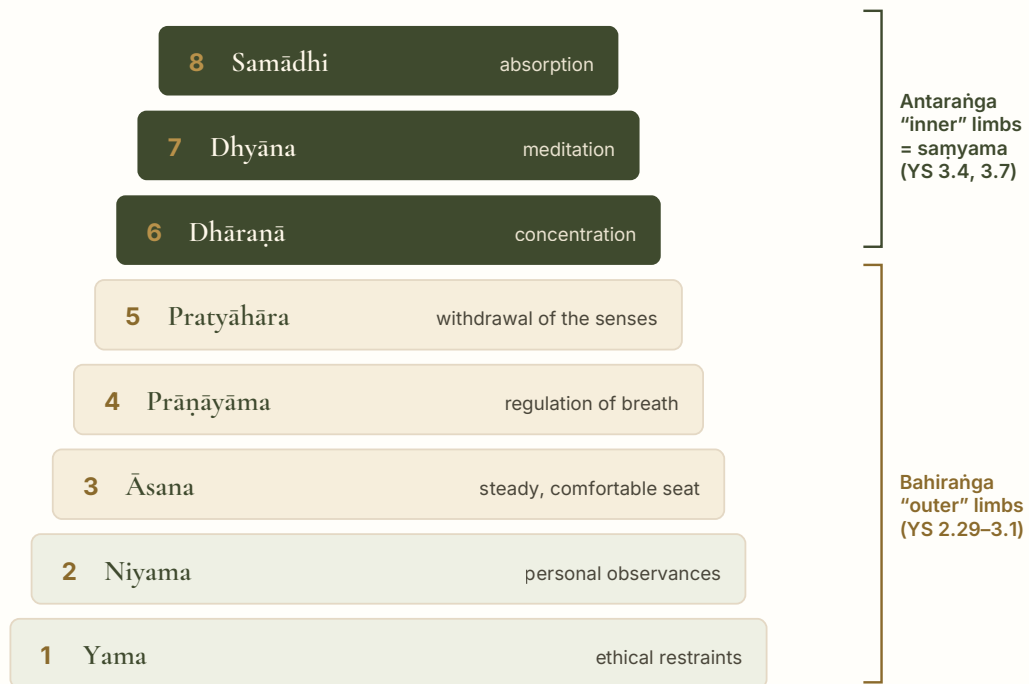
## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ list the eight limbs of yoga with their meanings and explain their relationship;
- ◆ define each of the five yamas and five niyamas, with the result Patañjali attributes to each;
- ◆ explain *pratipakṣa-bhāvanam*, the cultivation of the opposite;
- ◆ apply the yamas and niyamas to personal practice, to the conduct of a class and to professional life;
- ◆ discuss the yamas and niyamas with students without moralising.

## 6.1 The eight limbs

Patañjali presents the eight limbs (*aṣṭāṅga*) as the means to discriminating knowledge, which removes the impurities that obscure it (YS 2.28–29):



**Figure 6.1** The eight limbs (*aṣṭāṅga*) of Patañjali's yoga (YS 2.29). Patañjali calls the last three "inner" relative to the first five (YS 3.7); practised together on one object they are called *saṁyama* (YS 3.4). The ladder image is a teaching device: the text presents the limbs as mutually supporting, not as strictly sequential stages.

It is common to present the limbs as rungs on a ladder, to be climbed in order. This is partly right: Patañjali lists them in a logical sequence from outer to inner. But the word *aṅga* means "limb" — as of a body — and the limbs grow together. A beginner practises all eight in some form in the first class: behaving kindly (*yama*), arriving on time (*niyama*), sitting (*āsana*), breathing (*prāṇāyāma*), closing the eyes (*pratyāhāra*), focusing (*dhāraṇā*). The sequence describes increasing subtlety, not a waiting list

INTERPRETIVE .

## 6.2 The yamas: restraints

The five **yamas** are restraints — primarily in our dealings with others (YS 2.30). Patañjali calls them the "great vow" (*mahāvratā*) when they are practised universally, unlimited by birth, place, time or circumstance (2.31). This is a demanding standard: *ahiṁsā* is not only for the practice hall, or for people we like, or for days when we feel calm.

**Table 6.1** The five yamas (YS 2.30, 2.35–2.39)

| Yama   | Meaning                               | Result according to Patañjali                                           |
|--------|---------------------------------------|-------------------------------------------------------------------------|
| ahiṁsā | non-harming in thought, word and deed | in the presence of one established in it, hostility is abandoned (2.35) |
| satya  | truthfulness                          | actions and their fruits come to rest on one's word (2.36)              |
| asteya | non-stealing                          | all jewels present themselves (2.37)                                    |

| Yama         | Meaning                                                          | Result according to Patañjali                  |
|--------------|------------------------------------------------------------------|------------------------------------------------|
| brahma-carya | "conduct in brahman"; sexual restraint, moderation of the senses | vigour is gained (2.38)                        |
| aparigraha   | non-possessiveness, non-grasping                                 | knowledge of the "how" of births arises (2.39) |

The results are traditional teachings **TRADITIONAL**; their language is poetic, but their psychology is recognisable. People relax in the presence of someone who will not harm them. A truthful person's word becomes reliable. Someone who does not grasp becomes clear-sighted about what drives their life.

## Ahiṃsā

Non-harming is the root of the other yamas; Vyāsa's commentary says the others are practised for the sake of ahiṃsā (Bryant, 2009). Traditionally it includes vegetarianism, which many yoga schools, including this one, still recommend. In the practice hall, ahiṃsā has a very direct meaning: **do not harm the body in the name of yoga**. Pushing through pain, forcing a student into a posture or practising prāṇāyāma aggressively are failures of ahiṃsā before they are failures of technique.

## Satya

Truthfulness is to be practised in the service of ahiṃsā: the tradition holds that truth should be spoken so that it is also beneficial and kind. For teachers, satya means honesty about what you know ("I don't know – I will find out"), about your training and qualifications, and about the benefits of yoga. Exaggerated health claims in marketing are a breach of satya.

## Asteya

Non-stealing extends beyond property to time, credit and ideas. A teacher who starts late steals students' time; one who presents another teacher's sequence as their own takes credit that is not theirs.

## Brahmacarya

Literally "moving in brahman", *brahmacarya* has meant celibacy for renunciates and faithfulness and moderation for householders. In modern professional terms its most important application is unambiguous: **a teacher must never exploit the trust of the teaching relationship for sexual or romantic purposes** (Chapter 30). More widely, it means moderation – conserving energy rather than scattering it.

## Aparigraha

Non-possessiveness includes not grasping at achievements. A student fixated on "getting" a posture practises aparigraha on the mat; a teacher who needs students to be "theirs" practises it in the studio.

## 6.3 The niyamas: observances

The five **niyamas** concern one's relationship with oneself (YS 2.32). The last three are identical with kriyā yoga (2.1).

**Table 6.2** The five niyamas (YS 2.32, 2.40–2.45)

| Niyama            | Meaning                              | Result according to Patañjali                                                                                                                                             |
|-------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| śauca             | cleanliness, purity of body and mind | detachment from one's own body and from contact with others; purity of mind, cheerfulness, one-pointedness, mastery of the senses, fitness for self-realisation (2.40–41) |
| santoṣa           | contentment                          | unexcelled happiness (2.42)                                                                                                                                               |
| tapas             | disciplined effort, austerity        | perfection of body and senses through destruction of impurities (2.43)                                                                                                    |
| svādhyāya         | study, self-study, recitation        | connection with one's chosen deity (2.44)                                                                                                                                 |
| īśvara-praṇidhāna | devotion to Īśvara; surrender        | perfection of samādhi (2.45)                                                                                                                                              |

## Śauca

Cleanliness of the body, the practice space and the mind. Haṭha Yoga develops external śauca into the six cleansing practices (Chapter 9). For a teacher, śauca includes a clean, well-ventilated room, clean mats, personal hygiene, and a mind cleared of the day's irritations before entering class.

## Santoṣa

Contentment is not complacency. It is the capacity to be at peace with what is while continuing to practise. The student who is content with a modified posture is practising santoṣa; so is the teacher content with a small class.

## Tapas

The word means “heat”. It is the willingness to endure discomfort for a worthwhile purpose — rising at dawn, holding a posture, keeping silence. Tapas must be balanced by ahiṃsā: the heat that transforms is not the heat that burns.

## Svādhyāya

Study of sacred texts, recitation of mantra, and — in modern readings — study of oneself **INTERPRETIVE**. This manual, your practice journal and your reflections on your own reactions in class are all svādhyāya.

## Īśvara-praṇidhāna

Surrender to something greater than oneself. For a devotional student this is surrender to God; for others, it may be a letting go of the ego's need to control outcomes — which brings us back to the Gītā's *niṣkāma karma*.

## 6.4 Cultivating the opposite

When disturbing thoughts arise — thoughts of harming, lying or grasping — Patañjali advises *pratipakṣa-bhāvanam*, “cultivating the opposite” (2.33). He explains that such thoughts, whether acted upon, caused in others or approved of, and whether mild, moderate or intense, bear the fruit of endless suffering and

ignorance (2.34). Reflecting on this is itself the cultivation of the opposite. This is not the suppression of feeling but a deliberate turning of attention — remarkably close to modern cognitive approaches to emotion **INTERPRETIVE**.

#### ✧ THE YAMA AND NIYAMA WEEK

Choose one yama or niyama each day for ten days. In the morning, write one concrete way you will practise it. In the evening, note one moment when you did and one when you did not. Share one observation in the group at the end of the week.

## 6.5 The yamas and niyamas in the practice hall

**Table 6.3** Applying the yamas and niyamas to teaching

|                   | In your own practice                    | In the class you teach                                           |
|-------------------|-----------------------------------------|------------------------------------------------------------------|
| Ahiṃsā            | practise without pain; rest when needed | offer options; never force; create a room where no one is shamed |
| Satya             | be honest about your limits             | make no claims you cannot support; admit what you do not know    |
| Asteya            | give full effort and attention          | start and end on time; credit sources                            |
| Brahmacarya       | moderation in effort, food, sleep       | maintain clear professional boundaries                           |
| Aparigraha        | let go of attachment to poses           | let students progress at their own pace; do not compare them     |
| Śauca             | clean body, clean space, clear mind     | clean room and props; hygiene; tidy preparation                  |
| Santoṣa           | contentment with today's practice       | celebrate small progress; reduce comparison                      |
| Tapas             | daily discipline                        | keep your own practice alive as a teacher                        |
| Svādhyāya         | study and self-observation              | keep learning; invite students to observe themselves             |
| Īśvara-praṇidhāna | surrender the ego's need to achieve     | teach as an offering, not a performance                          |

#### 🗨️ SPEAKING ABOUT ETHICS WITHOUT PREACHING

Students do not come to class to be told how to live. Let the yamas and niyamas show in *how you teach* — your punctuality, honesty and kindness — and speak about them lightly: “Today, see if you can practise without harming yourself even a little.” One sentence, lived, is worth an hour of instruction.

## CHAPTER REVIEW

## KEY TAKEAWAYS

- ◆ The eight limbs are yama, niyama, āsana, prāṇāyāma, pratyāhāra, dhāraṇā, dhyāna and samādhi; they grow together like limbs of one body.
- ◆ The yamas — ahimsā, satya, asteya, brahmacarya, aparigraha — are a “great vow” when practised without exception (2.31).
- ◆ The niyamas — śauca, santoṣa, tapas, svādhyāya, īśvara-praṇidhāna — shape one’s relationship with oneself; the last three form kriyā yoga.
- ◆ Pratipakṣa-bhāvanam, cultivating the opposite, is Patañjali’s method for working with harmful thoughts.
- ◆ For a teacher the yamas and niyamas become a professional ethic: non-harm, honesty, punctuality, boundaries, humility and continued practice.

## REFLECTION QUESTIONS

1. Which yama is easiest for you? Which is hardest? Why?
2. Where does tapas become harmful in your own practice? How do you recognise the moment?
3. Is it possible to practise ahimsā towards a student and still challenge them? What does that look like?

## TEACHER-PRACTICE EXERCISES

1. Lead a partner through a posture while giving one cue for each of three yamas, embedded naturally in the instruction.
2. Write the “ethics statement” you would display on your studio website, drawing on the yamas and niyamas, in under 150 words.
3. In small groups, discuss a case: a student asks you to “push them deeper” in a forward bend. Which yamas and niyamas are in play?

## KNOWLEDGE CHECK

1. How many limbs does Patañjali list? (a) five (b) six (c) eight (d) ten
2. Which yama does Vyāsa treat as the root of the others? (a) satya (b) ahimsā (c) brahmacarya (d) aparigraha
3. According to YS 2.42, santoṣa brings: (a) vigour (b) unexcelled happiness (c) jewels (d) perfection of samādhi
4. Which three niyamas together form kriyā yoga? (a) śauca, santoṣa, tapas (b) tapas, svādhyāya, īśvara-praṇidhāna (c) santoṣa, svādhyāya, śauca (d) ahimsā, satya, tapas
5. Pratipakṣa-bhāvanam means: (a) surrender to God (b) cultivating the opposite (c) withdrawal of the senses (d) non-possessiveness

Answers 1 c · 2 b · 3 b · 4 b · 5 b

## TEACHING ASSIGNMENT

Design a 45-minute class themed on one yama or niyama. The theme must appear in the opening words, in at least three cues during the āsana practice, and in the closing relaxation — without a lecture. Submit the plan with the exact wording of the themed cues.



## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER

## 7

# The Eight Limbs II — Āsana, Prāṇāyāma and Pratyāhāra

“Posture should be steady and comfortable.” — Yoga Sūtra 2.46

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ explain Patañjali’s three sūtras on āsana and how the meaning of āsana changed in Haṭha Yoga;
- ◆ explain Patañjali’s account of prāṇāyāma, including its three movements and three measures;
- ◆ define pratyāhāra and describe practical ways to cultivate it;
- ◆ relate these three limbs to the structure of a classical Haṭha class;
- ◆ teach “steadiness and ease” as a quality within any posture.

## 7.1 Āsana

### What Patañjali says

Patañjali devotes only three sūtras to posture:

#### YOGA SŪTRA 2.46–2.48

स्थिरसुखमासनम् ॥ प्रयत्नशैथिल्यानन्तसमापत्तिभ्याम् ॥ ततो द्वन्द्वानभिघातः ॥

*sthira-sukham āsanam · prayatna-śaithilyānanta-samāpattibhyām · tato dvandvānab-  
highātaḥ*

Posture should be steady and comfortable. It is perfected by relaxation of effort and by absorption in the infinite. From that, one is no longer assailed by the pairs of opposites.

Āsana literally means “seat” — both the act of sitting and the thing one sits on. In the Yoga Sūtra it refers mainly to the seated posture for meditation **HISTORICAL**. Vyāsa’s commentary names about a dozen such postures, including the lotus (*padmāsana*), the hero (*virāsana*) and the auspicious (*bhadrāsana*) (Bryant,

2009). The purpose is stillness: a body that no longer demands attention, so that attention can turn inward. The “pairs of opposites” (*dvandva*) — heat and cold, pleasure and pain — no longer disturb the one whose seat is mastered.

The second sūtra (2.47) is often read as a teaching method: posture is perfected by **relaxing effort** — releasing unnecessary tension — and by **absorbing the mind in the infinite**, sometimes interpreted as the image of Ananta, the endless serpent who supports the world **INTERPRETIVE**.

## What Haṭha Yoga added

In Haṭha Yoga, āsana became a practice in its own right with its own benefits. The *Haṭha Yoga Pradīpikā* places it first: “Being the first limb of Haṭha Yoga, āsana is described first. It brings steadiness, health and lightness of body” (1.17) (Akers, 2002). It then describes fifteen postures, some seated, some strenuous, such as *mayūrāsana*, the peacock. Over the following centuries the repertoire grew into the hundreds (Chapter 9), and in the twentieth century āsana came to dominate yoga as it is practised worldwide.

### ◆ STHIRA AND SUKHA IN EVERY POSE

Every posture — headstand as much as seated meditation — can be judged by Patañjali’s two words. **Sthira**: is it stable, alert, well-founded? **Sukha**: is there ease, space and a steady breath? A pose that is strong but strained lacks sukha; a pose that is comfortable but slack lacks sthira. Much of the art of teaching is helping each student find both.

### Q CUEING STHIRA AND SUKHA

- ◆ *Sthira*: “Press the feet down evenly... feel the ground under you... let the legs be strong.”
- ◆ *Sukha*: “Soften the jaw... let the eyes be quiet... can the breath move a little more freely?”
- ◆ *Relaxation of effort* (2.47): “Notice any effort you don’t need — in the face, shoulders, hands — and let it go.”

## 7.2 Prāṇāyāma

### Patañjali’s account

The word *prāṇāyāma* combines *prāṇa* (breath, vital force) with *āyāma* (extension, restraint) — or, by another division, *prāṇa* and *yama* (control) **INTERPRETIVE**. Patañjali defines it as the regulation or cessation (*viccheda*) of the flow of inhalation and exhalation, practised once posture is established (2.49). He analyses it precisely:

- ◆ It has three movements: **external** (after exhalation), **internal** (after inhalation) and **suspended** (stopping) (2.50).
- ◆ It is observed in terms of **place** (*deśa* — where the breath is felt, how far it reaches), **time** (*kāla* — duration) and **number** (*saṅkhyā* — count of breaths), becoming long and subtle (*dirgha-sūkṣma*) (2.50).
- ◆ A fourth kind goes beyond the external and internal spheres (2.51).
- ◆ Through it the covering over the inner light is diminished (2.52), and the mind becomes fit for concentration (2.53).

Notice that Patañjali describes prāṇāyāma as becoming **long and subtle**, not forceful. The dramatic breathing techniques of Haṭha Yoga — breath retention with locks, rapid abdominal breathing — belong to a later tradition, which also warns emphatically against forcing: “Just as lions, elephants and tigers are tamed gradually, so the breath is brought under control gradually; otherwise it kills the practitioner” (HYP 2.15) (Akers, 2002).

## The link between breath and mind

The central yogic insight about breathing is that breath and mind move together. The *Haṭha Yoga Pradīpikā* puts it directly: “When the breath is unsteady, the mind is unsteady; when the breath is steady, the mind is steady” (2.2) **TRADITIONAL**. Patañjali lists irregular breathing among the symptoms of a distracted mind (1.31) and recommends the exhalation and retention of breath as one of the means of steadying it (1.34).

### 🔍 BREATH AND THE NERVOUS SYSTEM

Modern physiology supports a two-way link between breathing and mental state **SCIENTIFIC**. Breathing is the only vital rhythm that is both automatic and easily controlled at will, and slow, regular breathing — around six breaths a minute — increases heart rate variability and baroreflex sensitivity and is associated with reduced subjective anxiety in many studies (Zaccaro et al., 2018; Russo et al., 2017). The mechanisms and the size of effects are still being studied, and most trials are small; but the traditional claim that steadying the breath steadies the mind is broadly consistent with current evidence. The anatomy and physiology are covered in Chapter 21 and the practices in Chapter 27.

## 7.3 Pratyāhāra

**Pratyāhāra** is the fifth limb, a bridge between the outer and inner practices. Patañjali defines it as the senses imitating the nature of the mind, as it were, by disengaging from their objects (2.54); from it comes supreme mastery of the senses (2.55). The Bhagavad Gītā’s image is the tortoise drawing in its limbs (2.58).

Pratyāhāra is not sensory deprivation or suppression. The senses keep working; attention simply stops being dragged outward by them. Think of reading an absorbing book in a noisy café: the sound is there, but it does not take you. Pratyāhāra is that skill, developed deliberately.

## Pratyāhāra in practice

**Table 7.1** Practical doorways into pratyāhāra

| Practice                                                             | How it works                                                       |
|----------------------------------------------------------------------|--------------------------------------------------------------------|
| Closing the eyes in relaxation and between postures                  | removes the most dominant sense from the outer world               |
| Internal focus in āsana (breath, sensation)                          | turns attention from appearance to experience                      |
| Ṣaṇmukhī mudrā (closing ears, eyes, nose and mouth with the fingers) | reduces sensory input and turns attention to inner sound           |
| Bhṛāmarī (humming breath)                                            | draws attention to an internal sound and vibration                 |
| Yoga nidrā and systematic relaxation                                 | rotates awareness through the body while outer attention withdraws |
| Trāṭaka (candle gazing) followed by the inner image                  | shifts attention from an outer object to an inner one              |

## ✧ SENSING WITHOUT BEING PULLED

Sit for five minutes with the eyes closed. For one minute each, notice sounds, then bodily sensations, then smells, then the taste in the mouth, then the play of light behind the eyelids — simply registering each without following it. For the last minute, let all the senses be present at once while attention rests on the breath. What changed between the first and last minute?

## 7.4 The three limbs in a classical class

The traditional structure of the class taught in this course follows the logic of these three limbs:

1. It begins with **relaxation** and **prāṇāyāma** — *kapālabhāti* and alternate-nostril breathing — to steady breath and mind before moving.
2. **Āsana** follows, with short relaxations between postures so that each is absorbed.
3. The class ends with a long **final relaxation**, a guided practice of *pratyāhāra* in which awareness withdraws from the body part by part.

In this sense a well-taught classical class is not an exercise class with breathing added; it is a practice of the outer limbs in the service of the inner ones.

### CHAPTER REVIEW

#### 🍃 KEY TAKEAWAYS

- ◆ In the *Yoga Sūtra*, *āsana* is primarily the steady, comfortable seat for meditation (2.46); *Haṭha Yoga* developed it into a large practice with benefits of its own.
- ◆ Posture is perfected by relaxing effort and absorbing the mind (2.47); *sthira* and *sukha* are criteria for any pose.
- ◆ *Prāṇāyāma*, for Patañjali, regulates inhalation, exhalation and suspension, observed by place, time and number, becoming long and subtle.
- ◆ Tradition and modern physiology agree that breath and mental state are closely linked; slow breathing has measurable autonomic effects.
- ◆ *Pratyāhāra* is freedom of attention from the pull of the senses, not their suppression, and can be taught through concrete practices.

#### 🍃 REFLECTION QUESTIONS

1. In which posture do you find *sthira* most easily? *Sukha*? Which posture lacks both for you?
2. When did you last notice your breath change before you noticed your emotion change?
3. What pulls your senses outward most strongly during practice — sights, sounds, comparison? What helps?

#### ✧ TEACHER-PRACTICE EXERCISES

1. Teach one standing posture to a partner twice: once cueing only *sthira*, once cueing only *sukha*. Ask which felt better and why.

2. Practise and then teach “counting the breath” using Patañjali’s place, time and number: where do you feel it, how long is it, how many?
3. Lead a five-minute pratyāhāra practice using two of the doorways in Table 7.1.

#### ☑ KNOWLEDGE CHECK

1. *Sthira-sukham āsanam* is Yoga Sūtra: (a) 1.2 (b) 2.46 (c) 2.49 (d) 3.1
2. In the Yoga Sūtra, āsana mainly refers to: (a) inversions (b) the seated posture for meditation (c) dynamic sequences (d) standing postures
3. Patañjali says prāṇāyāma becomes: (a) forceful and fast (b) long and subtle (c) loud and deep (d) held with locks
4. Which text says “When the breath is unsteady, the mind is unsteady”? (a) Yoga Sūtra (b) Bhagavad Gītā (c) Haṭha Yoga Pradīpikā (d) Kaṭha Upaniṣad
5. Pratyāhāra is best described as: (a) sensory deprivation (b) freedom of attention from the pull of the senses (c) suppression of thought (d) breath retention

Answers 1b · 2b · 3b · 4c · 5b

#### 📄 TEACHING ASSIGNMENT

Write a script for the final five minutes of a class — the transition from the last posture into relaxation — that deliberately guides students from āsana through breath awareness into pratyāhāra. Use no more than 200 words. Record yourself reading it and listen back for pace and silence.

## CHAPTER

## 8

# The Eight Limbs III — Dhāraṇā, Dhyāna and Samādhi

*“Meditation is the continuous flow of the same awareness towards that object.” — Yoga Sūtra 3.2*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ define dhāraṇā, dhyāna and samādhi according to Yoga Sūtra 3.1–3.3 and explain how they differ;
- ♦ explain *saṃyama* and why Patañjali calls these three the “inner limbs”;
- ♦ distinguish the main kinds of samādhi described in the Yoga Sūtra;
- ♦ describe what a beginner can realistically expect from concentration practice;
- ♦ speak about samādhi to students honestly and without exaggeration.

## 8.1 From outer to inner

The first five limbs prepare the ground. The last three — concentration, meditation and absorption — are the practice of yoga in the strict sense defined in YS 1.2: the stilling of the mind. Patañjali calls them “inner” (*antaraṅga*) relative to the first five (3.7), and yet “outer” relative to the seedless samādhi that is the final goal (3.8). Everything is relative to the next step.

## 8.2 Dhāraṇā: concentration

*deśa-bandhaś cittasya dhāraṇā* — “Concentration is the binding of the mind to a place” (YS 3.1).

The “place” may be a point in the body (the navel, the heart, the space between the eyebrows, the tip of the nose), an external object (a candle flame, an image), a sound (a mantra) or an idea. Vyāsa gives examples of bodily places: the navel, the lotus of the heart, the light in the head, the tip of the nose, the tongue (Bryant, 2009).

In dhāraṇā the mind is placed on the object and keeps wandering away; it is brought back again and again. This is the stage at which almost all practitioners spend most of their time — and it is not a failure. Each return is the practice (Figure 28.1).

### 8.3 Dhyāna: meditation

*tatra pratyayaikatānatā dhyānam* — “Meditation is the continuous flow of the same awareness towards that object” (YS 3.2).

When concentration becomes steady, the effort of returning falls away. Attention flows towards the object without interruption, like oil poured from one vessel to another — a traditional image. The meditator is still aware of meditating: there is a subject, an object and the act of attending.

### 8.4 Samādhi: absorption

*tad evārtha-mātra-nirbhāsaṃ svarūpa-śūnyam iva samādhiḥ* — “Absorption is when the object alone shines forth, as if empty of its own form” (YS 3.3).

In samādhi the sense of a separate meditator falls away. Only the object remains in awareness. The phrase “as if empty of its own form” refers to the mind: it becomes so transparent to its object that it is as if it were not there.

### 8.5 Saṁyama

Practised together on one object, the three are called *saṁyama* (3.4). From its mastery comes the light of insight (*prajñā*) (3.5), which is applied in stages (3.6). The third chapter of the Sūtra describes many applications of saṁyama and the powers traditionally said to result **TRADITIONAL**; as we saw, Patañjali also warns that they are obstacles to the final goal (3.37).

**Table 8.1** Dhāraṇā, dhyāna and samādhi compared

|                    | Dhāraṇā                                     | Dhyāna                                   | Samādhi                                         |
|--------------------|---------------------------------------------|------------------------------------------|-------------------------------------------------|
| Definition         | binding the mind to one place (3.1)         | continuous flow of awareness to it (3.2) | the object alone shines, mind as if empty (3.3) |
| Effort             | frequent, deliberate return                 | effortless continuity                    | effort transcended                              |
| Sense of meditator | strong                                      | present                                  | absent                                          |
| Everyday analogy   | repeatedly bringing a puppy back to its bed | the puppy lying quietly                  | — (no ordinary analogy is adequate)             |

## 8.6 The kinds of samādhi

Patañjali distinguishes two broad kinds (Bryant, 2009):

- ◆ **Samprajñāta** (“with cognition”), also called *sabīja*, “with seed”: absorption in an object. It is accompanied, in progressively subtler stages, by gross thought (*vitarka*), subtle reflection (*vicāra*), bliss (*ānanda*) and the pure sense of “I-am” (*asmitā*) (1.17).
- ◆ **Asamprajñāta** (“beyond cognition”), also *nirbīja*, “seedless”: the stilling of even the last impressions, in which only latent impressions remain and then dissolve (1.18, 1.51). This leads to *kaivalya*.

Later Vedānta speaks of *savikalpa* samādhi (with distinctions) and *nirvikalpa* samādhi (without distinctions); these terms are often used interchangeably with Patañjali’s, though the two systems do not map onto each other exactly **INTERPRETIVE**.

### ◆ SAMĀDHI IS NOT A SPECIAL FEELING

It is easy to confuse samādhi with pleasant or unusual experiences in meditation — lights, bliss, a sense of floating. Patañjali’s own list of obstacles includes *bhrānti-darśana*, false perception (1.30). Traditional teachers consistently advise students not to seek or cling to experiences, but to continue practising. As a teacher, receive students’ reports of experiences calmly and without inflating them.

## 8.7 What a beginner can expect

Honesty is the first duty of a meditation teacher. In the first months, a student can expect:

- ◆ a great deal of mind-wandering, which is normal and is the material of practice;
- ◆ moments — seconds, then longer — of steady attention;
- ◆ growing awareness of how busy the mind is (which may feel like the practice is making things worse, but is usually the opposite);
- ◆ improved ability to notice and step back from thoughts and emotions in daily life;
- ◆ occasional drowsiness, restlessness, boredom or strong emotion.

### 🔍 WHAT RESEARCH SAYS ABOUT BEGINNERS

Meta-analyses of structured meditation programmes find moderate benefits for anxiety, depression and pain compared with no treatment, and effects generally similar to other active treatments such as exercise **SCIENTIFIC** (Goyal et al., 2014). Researchers caution that the field has often overstated its findings, and that terms such as “mindfulness” and “meditation” cover very different practices (Van Dam et al., 2018). Meditation can also occasionally bring difficult experiences — anxiety, re-experiencing of trauma, disturbing perceptual changes — which teachers should know how to meet (Chapter 28) (Britton et al., 2021).

### 🧘 TEACHING THE INNER LIMBS IN A POSTURE CLASS

You cannot teach samādhi; you can teach dhāraṇā, and create the conditions in which the rest may happen. In a posture class, a single point of focus held through a pose (the breath at the nostrils, the soles of the feet, a *drishti*) is dhāraṇā. Long holds with a quiet room invite dhyāna. Final relaxation is where many students first glimpse deep absorption. Guard the silence.



## CHAPTER REVIEW

## KEY TAKEAWAYS

- ◆ Dhāraṇā binds the mind to one place; dhyāna is its uninterrupted flow; in samādhi the object alone shines and the sense of a separate meditator falls away.
- ◆ Together, practised on one object, they are called samyama.
- ◆ Samādhi with seed (samprajñāta) has stages of vitarka, vicāra, ānanda and asmitā; seedless samādhi (asamprajñāta) leads to liberation.
- ◆ Unusual experiences in meditation are not samādhi and should not be sought or inflated.
- ◆ Beginners spend most of their time in dhāraṇā; teachers should set honest expectations and know that meditation can occasionally be difficult.

## REFLECTION QUESTIONS

1. Recall a moment in your life — not necessarily in meditation — of complete absorption. What conditions made it possible?
2. What is your mind's favourite place to wander to? What does that tell you?
3. How would you respond to a student who says, "I can't meditate — my mind won't stop"?

## TEACHER-PRACTICE EXERCISES

1. Practise ten minutes of dhāraṇā on the breath at the nostrils each morning for a week, noting in your journal roughly how many times you returned.
2. Teach a partner a five-minute concentration practice on a candle flame (trāṭaka), including setting up, timing, closing and the after-image (see Chapter 28 for safety).
3. Role-play: a student reports seeing "blue light" in meditation and asks if it means they are awakening. Respond in under one minute.

## KNOWLEDGE CHECK

1. *Deśa-bandhaś cittasya* defines: (a) samādhi (b) dhyāna (c) dhāraṇā (d) pratyāhāra
2. Samyama refers to: (a) the yamas and niyamas together (b) dhāraṇā, dhyāna and samādhi on one object (c) the five kleśas (d) breath retention
3. Which kind of samādhi is called "seedless"? (a) savikalpa (b) samprajñāta (c) asamprajñāta (d) vitarka
4. Patañjali calls the siddhis, relative to samādhi: (a) signs of success (b) obstacles (c) its goal (d) irrelevant
5. What should a teacher do when a student reports an unusual meditation experience? (a) declare it a sign of enlightenment (b) tell them to stop meditating (c) receive it calmly, neither dismissing nor inflating it (d) ignore it

Answers 1c · 2b · 3c · 4b · 5c

## TEACHING ASSIGNMENT

Write and teach a ten-minute guided concentration practice that moves from pratyāhāra (settling the senses) to dhāraṇā (one chosen point) and then invites effortless resting (towards dhyāna). Mark the silences in your script with their durations. Afterwards, ask your listeners where they found it easiest and hardest to follow.

# The Haṭha Yoga Tradition and the Haṭha Yoga Pradīpikā

*“Haṭha Yoga is a stairway for those who wish to climb to the lofty Rāja Yoga.” — Haṭha Yoga Pradīpikā 1.1*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the origins of Haṭha Yoga and the main texts of the tradition;
- ♦ explain the meaning of the word *haṭha* and the traditional rationale of Haṭha practice;
- ♦ summarise the contents of the four chapters of the Haṭha Yoga Pradīpikā;
- ♦ list the six cleansing practices, eight kumbhakas and ten mudrās of the Pradīpikā;
- ♦ identify which traditional Haṭha practices are appropriate for a general class and which are not.

## 9.1 Origins

Haṭha Yoga — the yoga that uses the body, breath and vital energy as its primary means — took shape between the eleventh and fifteenth centuries CE **HISTORICAL** (Mallinson & Singleton, 2017). Its earliest surviving text, the *Amṛtasiddhi* (eleventh century), teaches techniques for controlling the vital essence (*bindu*) and breath and comes from a Buddhist tantric environment. The *Dattātreyayogasāstra* (thirteenth century) is the first text to call its methods *haṭha yoga*; the Nāth yogis, a Śaiva order tracing its lineage to Matsyendra and Gorakṣa, became the tradition most closely associated with it.

The word *haṭha* means “force” or “obstinacy” — the forceful, effortful nature of the methods (Birch, 2011). Later texts gave it an esoteric etymology: *ha* stands for the sun and *ṭha* for the moon, and Haṭha Yoga is the union of these two currents — the right and left channels, *piṅgalā* and *idā* — in the central channel, *suṣumṇā* **TRADITIONAL**. The etymology is later than the name, but it expresses the tradition’s own understanding of its aim.

## The logic of Haṭha practice

The traditional rationale runs roughly as follows **TRADITIONAL** : mind and breath are bound together; vital energy (*prāṇa*) ordinarily flows through the left and right channels; when *prāṇa* is made to enter the central channel through posture, breath retention, locks and seals, the mind becomes still, the dormant power (*kuṇḍalinī*) rises, and the practitioner attains Rāja Yoga — the state of *śamādhi*. The *Pradīpikā* insists that Haṭha and Rāja Yoga need each other: “Without Haṭha, Rāja Yoga cannot be attained; without Rāja Yoga, Haṭha is fruitless” (2.76) (Akers, 2002).

## 9.2 The Haṭha Yoga Pradīpikā

The *Haṭha Yoga Pradīpikā* (“Light on Haṭha Yoga”) was compiled by **Svātmārāma** in the fifteenth century, drawing on at least twenty earlier texts **HISTORICAL** (Mallinson & Singleton, 2017). It became the best-known Haṭha text and is the foundation of the Haṭha practice taught in this course. It has four chapters (*upadeśas*). Verse numbering differs slightly between editions; this manual follows Akers (2002); the Bihar School edition with extensive commentary is Muktibodhananda (1998).

### Chapter 1: preparation and āsana

The first chapter describes the ideal practitioner and place of practice — a small hut in a peaceful, well-governed land, free from stones, fire and water, with a small door (1.12–13) — and the conditions for success and failure:

#### 📌 SIX THINGS THAT DESTROY YOGA, AND SIX THAT BRING SUCCESS · HYP 1.15–16

**Yoga is destroyed by:** overeating, over-exertion, too much talking, adhering to rules rigidly, too much company, and unsteadiness (fickleness).

**Yoga succeeds through:** enthusiasm, courage, patience, knowledge of the truth, determination, and giving up excessive company.

Āsana is presented as the first limb, bringing “steadiness, health and lightness” (1.17). Fifteen postures are described: *svastikāsana*, *gomukhāsana*, *virāsana*, *kūrmāsana*, *kukkuṭāsana*, *uttāna-kūrmāsana*, *dhanurāsana*, *matsyendrāsana*, *paścimottānāsana*, *mayūrāsana*, *śavāsana*, and the four principal seated postures *siddhāsana*, *padmāsana*, *siṃhāsana* and *bhadrāsana* (Akers, 2002). *Siddhāsana* is praised as the foremost (1.35–43). Several of the twelve basic postures appear here in early form: *dhanurāsana* (bow), *matsyendrāsana* (the ancestor of the half spinal twist), *paścimottānāsana* (forward bend) and *śavāsana*. The chapter closes with guidance on moderate diet (*mitāhāra*), recommending pleasant, nourishing food eaten while leaving a quarter of the stomach empty **TRADITIONAL** .

### Chapter 2: prāṇāyāma and cleansing

The second chapter begins with the link between breath and mind (2.2) and the importance of purifying the channels through alternate-nostril breathing. It warns that correctly practised *prāṇāyāma* removes disease while incorrect practice creates it (2.16–17), and that the breath must be tamed gradually, as a lion or elephant is tamed (2.15).

For those with excess fat or phlegm, six cleansing practices are recommended (2.21–22) **TRADITIONAL** :

**Table 9.1** The six cleansing practices (ṣaṭkarma), HYP 2.22

| Kriyā       | Traditional description                                                      | Taught in a general class?                                                 |
|-------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Dhauti      | swallowing and withdrawing a long strip of cloth to clean the stomach        | <b>No</b> — specialist practice under direct supervision                   |
| Basti       | drawing water into the colon by abdominal control                            | <b>No</b>                                                                  |
| Neti        | cleansing the nasal passages with water (jala neti) or a thread (sūtra neti) | jala neti may be taught with hygiene precautions (sterile or boiled water) |
| Trāṭaka     | gazing steadily at a small object until the eyes water                       | yes, with eye-health precautions                                           |
| Nauli       | isolating and rotating the rectus abdominis muscles                          | only to experienced students on an empty stomach; many contraindications   |
| Kapālabhāti | rapid, forceful exhalations like the bellows of a blacksmith                 | yes, with screening and precautions ( <b>Chapter 27</b> )                  |

The chapter then describes eight **kumbhakas** — breath retentions, each with its own breathing pattern (2.44): *sūryabhedana*, *ujjāyī*, *sītkārī*, *śītalī*, *bhastrikā*, *bhrāmarī*, *mūrcchā* and *plāvinī*. Several of these are now practised as breathing techniques without long retention; **Chapter 27** teaches those suitable for general classes.

#### ⚠ NOT EVERYTHING IN A CLASSICAL TEXT BELONGS IN A MODERN CLASS

The Haṭha texts were written for renunciate practitioners working one-to-one with a teacher over years. Some practices they describe (swallowing cloth, internal washes, very long breath retention, the more esoteric mudrās) carry real risks and are **not** taught in general classes or on this course except as theory. Respecting a tradition includes respecting its insistence on direct, individual guidance for advanced practices.

## Chapter 3: mudrās and bandhas

The third chapter teaches ten **mudrās** — “seals” — which are advanced techniques combining posture, breath retention and locks to direct prāṇa into the central channel and awaken kuṇḍalinī (3.6–7): *mahāmudrā*, *mahābandha*, *mahāvedha*, *khecārī*, *uḍḍiyāna bandha*, *mūla bandha*, *jālandhara bandha*, *viparītakaraṇī*, *vajrolī* and *śakticālana* (Akers, 2002). The three locks and *viparītakaraṇī* — the inverted posture that is the textual ancestor of shoulderstand — are taught in this course; *khecārī* (turning the tongue back into the nasal cavity after gradually cutting its frenulum) and *vajrolī* (a urethral practice) are described historically only **HISTORICAL**. **Chapter 14** covers the practices taught.

## Chapter 4: samādhi

The final chapter describes the goal: samādhi, in which prāṇa dissolves in the central channel and the mind in its source. It gives many synonyms for this state — Rāja Yoga, *unmanī*, *amanaska*, *laya* and others — and teaches meditation on the inner sound (*nāda*), heard first as loud sounds like the ocean, drums and thunder, then as subtle sounds like bells, flutes and bees (4.84–86) **TRADITIONAL**. It ends with the image of the yogi freed from all states, “like a dead man”, unaffected by the world — in the sense of utterly still, not lifeless.

### 9.3 Other Haṭha texts

**Table 9.2** Principal Haṭha texts

| Text                           | Approximate date | Notes                                                                  |
|--------------------------------|------------------|------------------------------------------------------------------------|
| Amṛtasiddhi                    | 11th century     | earliest text teaching Haṭha methods; Buddhist tantric context         |
| Dattātreyayogaśāstra           | 13th century     | first to name haṭha yoga; teaches it alongside Patañjali's eight limbs |
| Vivekamārtaṇḍa / Gorakṣaśataka | c. 13th century  | Nāth texts on āsana, prāṇāyāma, bandha and cakras                      |
| Haṭha Yoga Pradīpikā           | 15th century     | Svātmārāma's classic compilation                                       |
| Śiva Saṃhitā                   | c. 15th century  | combines Haṭha practice with Vedāntic philosophy                       |
| Gheraṇḍa Saṃhitā               | c. 1700          | "sevenfold" yoga; 32 āsanās, 25 mudrās                                 |
| Haṭhābhyāsapaddhati            | 18th century     | over 100 āsanās, many dynamic and strenuous                            |

### 9.4 Haṭha Yoga today

The Haṭha tradition reaches a modern yoga class mostly through three routes: its postures, its breathing practices, and its attitude – that the body is not an obstacle to the spiritual life but its instrument. The classical class taught in this course preserves the Haṭha order of practice (āsana, prāṇāyāma, relaxation, meditation) and its emphasis on the breath, while teaching a set of postures suited to householders. It is worth remembering the *Pradīpikā*'s own balance: success comes not from wearing the robes or talking about yoga, but from practice (1.66) **TRADITIONAL**.

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ♦ Haṭha Yoga developed between the eleventh and fifteenth centuries; *haṭha* means “force”; the sun-moon etymology is later and symbolic.
- ♦ Its rationale is to lead prāṇa into the central channel through posture, breath, locks and seals, as a stairway to Rāja Yoga.
- ♦ The Haṭha Yoga Pradīpikā (15th century) has four chapters: āsana; prāṇāyāma and cleansing; mudrās and bandhas; samādhi and nāda.
- ♦ It lists six things that destroy and six that support yoga, fifteen āsanās, six śaṭkarmas, eight kumbhakas and ten mudrās.
- ♦ Many classical Haṭha practices are unsuitable for general classes; respecting the tradition means respecting its insistence on individual guidance.

### REFLECTION QUESTIONS

1. Which of the six things that “destroy yoga” (HYP 1.15) do you recognise in your own life?
2. Why might a tradition that aims at stillness of mind begin with the body?
3. What would be lost if yoga classes taught only āsana from the Haṭha tradition?

## ✧ TEACHER-PRACTICE EXERCISES

1. Recite the six obstacles and six supports of HYP 1.15–16 from memory and give a modern example of each.
2. Practise jala neti under supervision, then teach it to a partner, including hygiene and safety instructions.
3. Compare the fifteen āsanās of the Pradīpikā with the twelve basic postures of this course. Which appear in both? Which are new?

## ☑ KNOWLEDGE CHECK

1. The Haṭha Yoga Pradīpikā was compiled by: (a) Patañjali (b) Svātmārāma (c) Gheraṇḍa (d) Gorakṣa
2. How many śaṭkarmas are listed? (a) four (b) six (c) eight (d) ten
3. Which is *not* one of the eight kumbhakas? (a) bhrāmārī (b) ujjāyī (c) kapālabhāti (d) śītalī
4. According to HYP 1.15, yoga is destroyed by: (a) enthusiasm (b) overeating (c) patience (d) determination
5. In HYP chapter 4, the inner sound meditated upon is called: (a) mantra (b) nāda (c) bindu (d) japa

Answers 1 b · 2 b · 3 c · 4 b · 5 b

## 📄 TEACHING ASSIGNMENT

Prepare a ten-minute presentation on the Haṭha Yoga Pradīpikā for fellow trainees, covering one chapter of the text in depth. Include at least two verse references, one practice from the chapter that is safe to teach and how you would teach it, and one practice you would not teach and why.

## CHAPTER 10

## Vedānta — Yoga and Self-Knowledge

*“Brahman is real; the world is appearance; the individual self is not other than Brahman.” — a traditional summary of Advaita Vedānta*

### © LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ identify the three foundational texts of Vedānta and its three main schools;
- ♦ explain the core concepts of Advaita Vedānta: brahman, ātman, māyā, superimposition and levels of reality;
- ♦ describe the fourfold qualification of the student and the threefold method of hearing, reflection and meditation;
- ♦ compare the views of Yoga and Vedānta on the nature of liberation;
- ♦ explain how Yoga and Vedānta are combined in practice as “Yoga Vedānta”.

## 10.1 What Vedānta is

*Vedānta* means “the end of the Veda”: the Upaniṣads, and the philosophy built on them. Its three foundational texts, the *prasthāna-traya* (“three points of departure”), are the **Upaniṣads**, the **Bhagavad Gītā** and the **Brahma Sūtras** (attributed to Bādarāyaṇa), a terse systematisation of Upaniṣadic teaching. Every Vedāntic school had to show that its reading of these three was coherent.

**Table 10.1** Three schools of Vedānta

| School                                | Founder (traditional dates)  | Relation of self, world and brahman                                                 |
|---------------------------------------|------------------------------|-------------------------------------------------------------------------------------|
| Advaita (non-dualism)                 | Śaṅkara (c. 8th century)     | the self is identical with brahman; the world of multiplicity is appearance         |
| Viśiṣṭādvaita (qualified non-dualism) | Rāmānuja (11th–12th century) | souls and world are real, forming the “body” of brahman (God), on which they depend |
| Dvaita (dualism)                      | Madhva (13th century)        | God, souls and world are eternally distinct; liberation is loving service of God    |



Most modern teachers of “Vedānta” in yoga follow **Advaita**, while honouring devotion as a path. Śaṅkara, its greatest teacher, is traditionally said to have been born at Kaladi, on the Periyar river in central Kerala, and to have travelled the length of India before his early death **TRADITIONAL**. Many works are attributed to him; scholars accept his commentaries on the Upaniṣads, Gītā and Brahma Sūtras as genuine and debate the authorship of others, such as the popular *Vivekacūḍāmaṇi* (“Crest-jewel of discrimination”) **HISTORICAL**.

## 10.2 Core ideas of Advaita

### Brahman and ātman

Brahman is described as *sat-cit-ānanda*: being, consciousness and bliss. It is not an object that can be known, because it is the knower in every act of knowing. The Upaniṣadic method for pointing to it is negation — *neti, neti*, “not this, not this” (Bṛhadāraṇyaka 2.3.6): whatever can be observed is not the observer. When everything observed has been set aside, what remains is the Self, *ātman*, which is not different from brahman.

### Superimposition and māyā

How, then, do we experience ourselves as limited individuals in a world of separate things? Śaṅkara’s answer is *adhyāsa*, superimposition: the properties of one thing are mistakenly projected onto another. The classic example is the rope mistaken for a snake in dim light. The snake is not real — but the fear it causes is real, and it vanishes not when the snake is fought but when the rope is seen **TRADITIONAL**. In the same way, the qualities of body and mind (“I am tired”, “I am anxious”) are superimposed on the Self, and the qualities of the Self (awareness, being) are lent to body and mind. *Māyā* is the power by which the one appears as many.

### Levels of reality

Advaita distinguishes three orders of reality **INTERPRETIVE**:

- ♦ **Absolute** (*pāramārthika*): brahman alone.
- ♦ **Empirical** (*vyāvahārika*): the everyday world, which is fully real for all practical purposes — the rope-snake does not apply to paying your rent.
- ♦ **Illusory** (*prātibhāsika*): dreams, mirages, the snake in the rope.

This distinction is often forgotten by enthusiastic students (“nothing matters, it is all māyā”). Advaita insists that ethics, duty and care for others remain binding at the empirical level, where we all live.

## 10.3 The path of knowledge

Advaita teaches that liberation is not produced by action — it is the recognition of what already is. But the mind must be prepared for that recognition.

### The fourfold qualification

The student needs four qualifications (*sādhana-catustaya*), described in the *Vivekacūḍāmaṇi* and other works (Madhavananda, 1921):

1. **Viveka** — discrimination between the eternal and the transient.

2. **Vairāgya** — dispassion towards the fruits of action, here and hereafter.
3. **Ṣaṭ-sampatti**, the “six treasures”: *śama* (calm of mind), *dama* (restraint of the senses), *uparati* (withdrawal from distraction), *titikṣā* (forbearance), *śraddhā* (faith in teacher and teaching) and *samādhāna* (concentration).
4. **Mumukṣutva** — intense longing for liberation.

### Hearing, reflection, meditation

The method itself has three stages, from Bṛhadāraṇyaka 2.4.5 (“the Self should be heard about, reflected upon and meditated upon”):

- ♦ **Śravaṇa** — hearing the teaching from a qualified teacher and the texts;
- ♦ **Manana** — reflecting on it until every doubt is resolved;
- ♦ **Nididhyāsana** — sustained contemplation until the understanding becomes direct and stable.

## 10.4 Yoga and Vedānta compared

**Table 10.2** Classical Yoga and Advaita Vedānta

|                  | Classical Yoga (Patañjali)               | Advaita Vedānta (Śaṅkara)                                 |
|------------------|------------------------------------------|-----------------------------------------------------------|
| Metaphysics      | dualist: many puruṣas and one prakṛti    | non-dualist: brahman alone is ultimately real             |
| Cause of bondage | conjunction of seer and seen; avidyā     | superimposition; avidyā                                   |
| Means            | eight limbs; stilling the mind (nirodha) | knowledge (jñāna) through hearing, reflection, meditation |
| Goal             | kaivalya: aloneness of puruṣa            | mokṣa: recognition that ātman is brahman                  |
| Role of samādhi  | central                                  | useful preparation; knowledge liberates                   |

Despite these differences, the two traditions have long been practised together. Vedāntins adopt the methods of Yoga to prepare the mind; many Yoga teachers adopt the non-dual vision of Vedānta. The phrase **Yoga Vedānta**, used by several modern teachers, expresses exactly this: the practices of yoga in the service of Vedāntic self-knowledge. In this framework, Haṭha and Rāja Yoga purify and steady the mind; Karma and Bhakti Yoga soften the ego; and Jñāna Yoga brings the final recognition.

#### ♦ THE WITNESS

Both Yoga and Vedānta point to the **witness** (*sākṣī*): the awareness that observes sensations, thoughts and emotions without being any of them. The practical instruction “observe the breath” or “notice the thought” is the doorway to this teaching. You can offer the doorway to every student; what they find there is their own.

#### Q VEDĀNTA IN A POSTURE CLASS

In final relaxation: “Notice that sensations come and go... thoughts come and go... and something in you notices them all. Rest as that.” Four sentences carry the heart of Vedānta without a single Sanskrit term. Offer it; do not insist.

## CHAPTER REVIEW

## KEY TAKEAWAYS

- ◆ Vedānta rests on the Upaniṣads, the Bhagavad Gītā and the Brahma Sūtras; its main schools are Advaita, Viśiṣṭādvaita and Dvaita.
- ◆ Advaita teaches that the Self is brahman — being, consciousness, bliss — and that multiplicity arises through superimposition and māyā.
- ◆ Empirical reality remains fully binding in everyday life; the teaching is not an excuse to ignore ethics.
- ◆ The student is prepared by the fourfold qualification and liberated through hearing, reflection and meditation.
- ◆ “Yoga Vedānta” uses the practices of yoga to prepare the mind for Vedāntic self-knowledge.

## REFLECTION QUESTIONS

1. Apply *neti, neti* for five minutes: “I am not this sensation... not this thought...” What remains?
2. Where in your life have you been afraid of a “snake” that turned out to be a “rope”?
3. Which of the six treasures (śama, dama, uparati, titikṣā, śraddhā, samādhāna) do you most need during this training?

## TEACHER-PRACTICE EXERCISES

1. Explain the rope-and-snake analogy to a partner and apply it to a fear commonly met in yoga practice (e.g. fear of inversions).
2. Write a four-line relaxation script that points to the witness without using any Sanskrit word.
3. In a group, debate: “If the world is māyā, why should a yogi care about the environment?” Use the three levels of reality.

## KNOWLEDGE CHECK

1. The three foundational texts of Vedānta are: (a) Vedas, Upaniṣads, Purāṇas (b) Upaniṣads, Bhagavad Gītā, Brahma Sūtras (c) Yoga Sūtra, Gītā, HYP (d) Ṛgveda, Gītā, Upaniṣads
2. Who is the principal teacher of Advaita Vedānta? (a) Rāmānuja (b) Madhva (c) Śaṅkara (d) Patañjali
3. *Adhyāsa* means: (a) practice (b) superimposition (c) liberation (d) devotion
4. Which is *not* one of the “six treasures”? (a) śama (b) titikṣā (c) mumukṣutva (d) śraddhā
5. *Śravaṇa, manana, nididhyāsana* are: (a) three kinds of samādhi (b) hearing, reflection and meditation (c) three bandhas (d) three guṇas

Answers 1 b · 2 c · 3 b · 4 c · 5 b

## TEACHING ASSIGNMENT

Prepare and lead a 15-minute “Vedānta in practice” session for classmates: five minutes of teaching on the witness using one analogy from this chapter, five minutes of guided self-enquiry in meditation, and five minutes of questions. Record the questions you were asked and how you answered them.

CHAPTER 11

# The Four Paths — Karma, Bhakti, Jñāna and Rāja Yoga

*“Whatever you do, whatever you eat, whatever you offer or give, whatever austerity you practise — do it as an offering to me.” — Bhagavad Gītā 9.27*

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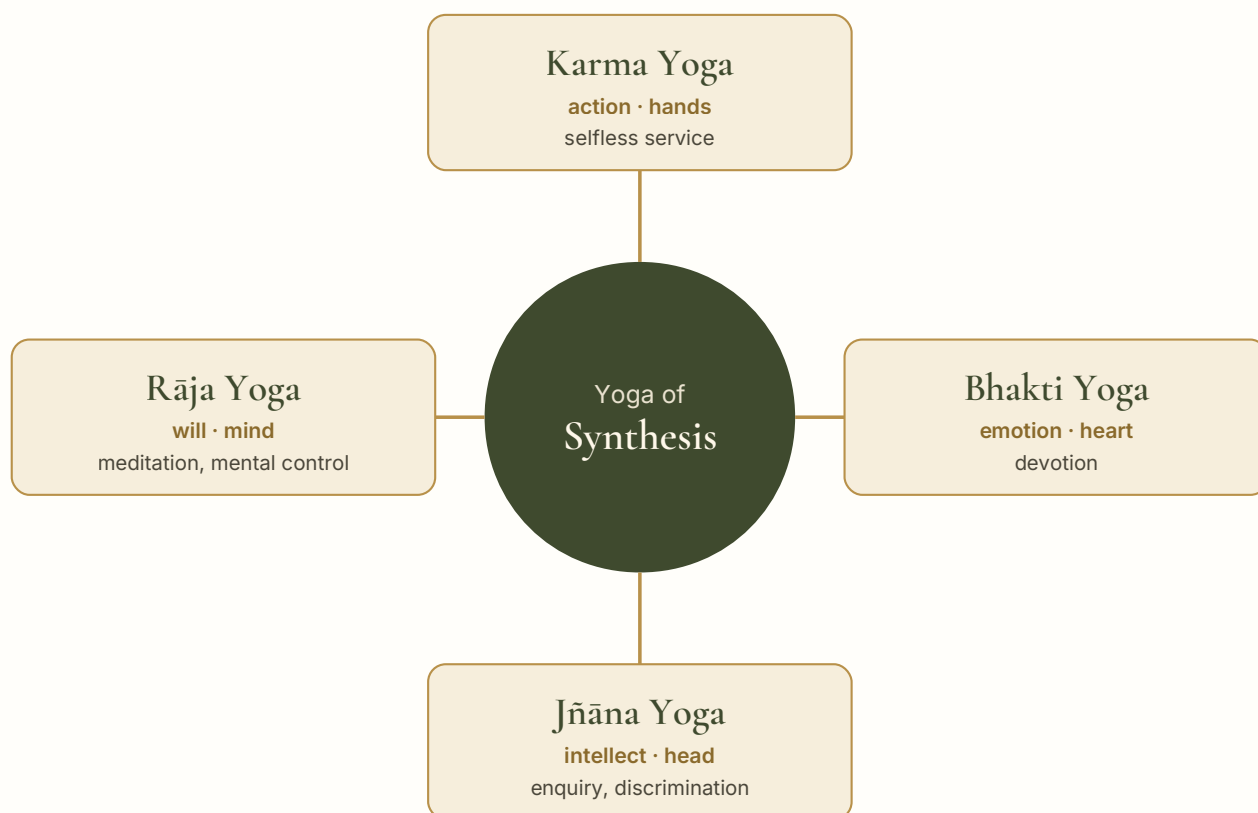
🕒 **LEARNING OBJECTIVES**

By the end of this chapter you will be able to:

- ◆ describe the four classical paths of yoga, their methods and the temperament each suits;
- ◆ explain why the four paths are practised together;
- ◆ list the nine forms of devotion and describe practices of karma, bhakti, jñāna and rāja yoga suitable for daily life;
- ◆ recognise the characteristic imbalances of each path practised alone;
- ◆ bring each path into a yoga class in a way that respects students of all backgrounds.

## 11.1 Four paths, one goal

By the time of the Bhagavad Gītā, Indian traditions recognised several paths to liberation, suited to different temperaments. Modern teaching, following Swami Vivekananda, commonly presents four (Vivekananda, 1896):



**Figure 11.1** Integrating the four classical paths: they are practised together so that head, heart, hands and will develop in balance. The associations with faculties are a traditional teaching aid.

**Table 11.1** The four paths compared

| Path        | Faculty            | Method                                      | Key texts                                            | Danger when practised alone             |
|-------------|--------------------|---------------------------------------------|------------------------------------------------------|-----------------------------------------|
| Karma yoga  | action ("hands")   | selfless service; action without attachment | Bhagavad Gītā ch. 3–5                                | busyness; activism without inner change |
| Bhakti yoga | emotion ("heart")  | love and devotion to the divine             | Gītā ch. 9–12; Nārada Bhakti Sūtra; Bhāgavata Purāṇa | sentimentality; fanaticism              |
| Jñāna yoga  | intellect ("head") | discrimination and self-enquiry             | Upaniṣads; Vivekacūḍāmaṇi                            | dry intellectualism; spiritual pride    |
| Rāja yoga   | will and mind      | meditation; the eight limbs                 | Yoga Sūtra                                           | withdrawal; coldness                    |

## 11.2 Karma yoga

Karma yoga is the yoga of action performed as an offering, without attachment to its results (Chapter 4). Its practice is not a special activity but a way of doing ordinary things. The tests are simple: *Why am I doing this? Would I do it equally well if no one noticed?*

In traditional yoga schools, karma yoga is a daily practice — cleaning, cooking, gardening, serving — and it is part of this course. Its effect is to **purify the heart**, weakening the ego's constant question, "What do I get?"

## ✧ ANONYMOUS SERVICE

During the second week of the course, perform one act of useful service each day that no one will know was yours. Note in your journal how it feels to receive no credit.

### 11.3 Bhakti yoga

Bhakti yoga is the path of love for the divine, conceived in a personal form. It transforms emotion rather than suppressing it: the energy of love, longing and even grief is turned towards God. The *Bhāgavata Purāṇa* (7.5.23) lists nine forms of devotion **TRADITIONAL**:

**Table 11.2** The nine forms of devotion (navadhā bhakti)

|   | Form          | Practice                              |
|---|---------------|---------------------------------------|
| 1 | śravaṇa       | listening to sacred stories and names |
| 2 | kīrtana       | singing the divine names              |
| 3 | smaraṇa       | remembrance                           |
| 4 | pāda-sevana   | service at the feet of the Lord       |
| 5 | arcana        | ritual worship                        |
| 6 | vandana       | prostration, reverence                |
| 7 | dāśya         | the attitude of a servant             |
| 8 | sakhya        | the attitude of a friend              |
| 9 | ātma-nivedana | complete self-offering                |

In a traditional yoga setting, bhakti appears most visibly in **kīrtan** (call-and-response chanting), in the opening and closing prayers of class, and in the *ārati* (waving of lights) at satsaṅg. For many students raised in other faiths or none, devotional practice is the most unfamiliar part of the tradition. A teacher can offer it generously and without pressure: devotion to God, to a teacher, to one's own highest ideal, or simply wholehearted love expressed through practice.

### 11.4 Jñāna yoga

Jñāna yoga is the path of knowledge, whose method is described in @ch:vedanta: discrimination between the real and the unreal, the four qualifications, and hearing, reflection and meditation. Its most direct modern form is **self-enquiry** (*ātma-vicāra*), famously taught by Ramana Maharshi (1879–1950) of Tiruvannamalai: holding attention on the sense of “I” and asking, “Who am I?”, until the question dissolves into its source **TRADITIONAL**.

Jñāna yoga suits those with a strong intellect — and the tradition warns that it is the hardest path, requiring great purity of mind. Without the other paths, it easily becomes clever talk about non-duality.

## 11.5 Rāja yoga

*Rāja yoga*, “royal yoga”, meant in the Haṭha texts the state of samādhi itself — the goal to which Haṭha is the stairway. Since Vivekananda’s *Raja Yoga* (1896), the term has commonly referred to the eight-limbed path of Patañjali, the yoga of mental discipline and meditation (Vivekananda, 1896; De Michelis, 2004). It suits those with a strong will and a scientific temperament, who want to experiment with their own minds.

## 11.6 Integrating the paths

The paths are not rival sects but complementary disciplines, as the Gītā itself suggests (Chapter 4). A human being has a head, a heart and hands; developing one alone produces distortion. The traditional prescription is to practise all four, adjusting the balance to one’s temperament — with Haṭha Yoga providing a healthy body and steady breath as the foundation for all **TRADITIONAL**.

### ◆ INTEGRATION, NOT ECLECTICISM

Integrating the paths does not mean doing a little of everything at random. It means recognising which faculty is overdeveloped and which neglected, and practising accordingly. An intellectual student may need kīrtan more than philosophy; a devotional one may need study; a busy activist may need to sit still.

## 11.7 The four paths in the practice hall

**Table 11.3** Bringing the four paths into a general class

| Path   | In class                                                                  | In teaching                                                                           |
|--------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Karma  | leaving the room tidier than you found it; helping a neighbour with props | teaching as service; preparing well; free or low-cost places for those who cannot pay |
| Bhakti | an optional opening or closing chant; dedicating the practice             | care and warmth towards every student                                                 |
| Jñāna  | a short reflection or reading; witnessing sensations                      | continued study; honesty about what you know                                          |
| Rāja   | concentration within postures; final relaxation; meditation               | clarity, discipline and punctuality                                                   |

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ The four classical paths — karma, bhakti, jñāna and rāja yoga — develop action, emotion, intellect and will respectively.
- ◆ Karma yoga is action as offering; bhakti yoga turns emotion towards the divine through nine forms of devotion; jñāna yoga is discrimination and self-enquiry; rāja yoga is the discipline of the mind.
- ◆ Each path has a characteristic imbalance when practised alone.
- ◆ An integrated practice uses all four, adjusted to temperament, on the foundation of Haṭha Yoga.
- ◆ A teacher can bring each path into a class without imposing belief.

 REFLECTION QUESTIONS

1. Which path attracts you most naturally? Which do you resist? What might the resisted path offer you?
2. Have you seen the “danger” of a single path in yourself or others?
3. How can devotional practice be offered respectfully to a mixed international group?

 TEACHER-PRACTICE EXERCISES

1. Lead the group in a simple call-and-response chant (e.g. *Oṃ Namaḥ Śivāya*), with a sentence of introduction that invites but does not require participation.
2. Practise ten minutes of self-enquiry (“Who am I?”) and ten minutes of mantra repetition on different days; compare them in your journal.
3. Plan one karma yoga project the whole group could do for the school or local community.

 KNOWLEDGE CHECK

1. Which path corresponds to the “heart”? (a) karma (b) bhakti (c) jñāna (d) rāja
2. The nine forms of devotion are listed in the: (a) Yoga Sūtra (b) Bhāgavata Purāṇa (c) Haṭha Yoga Pradīpikā (d) Kaṭha Upaniṣad
3. Self-enquiry (“Who am I?”) is most associated with: (a) Ramana Maharshi (b) Svātmārāma (c) Madhva (d) Vivekananda
4. Before Vivekananda, in the Haṭha texts *rāja yoga* referred mainly to: (a) royal ritual (b) the state of samādhi (c) the yamas (d) bhakti
5. The Yoga of Synthesis teaches that the four paths should be: (a) chosen between (b) practised in strict sequence (c) practised together, balanced to temperament (d) practised only by monks

Answers 1 b · 2 b · 3 a · 4 b · 5 c

 TEACHING ASSIGNMENT

Design a one-day “Yoga of Synthesis” retreat schedule for twelve beginners, including at least one practice from each path and a Haṭha foundation. Explain in a short paragraph why you placed each practice where you did.



## CHAPTER 12

## Meditation, Mindfulness and the Yogic Model of the Mind

*“As a lamp in a windless place does not flicker — that is the image of the yogi whose mind is controlled.” — Bhagavad Gītā 6.19*

### 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the yogic model of the mind — manas, buddhi, ahaṁkāra and citta — and the role of saṁskāras and vāsanās;
- ♦ distinguish concentration (focused attention) from mindfulness (open monitoring);
- ♦ explain the origins of the word “mindfulness” in Buddhist practice and its modern secular forms;
- ♦ describe the traditional foundations of a regular meditation practice;
- ♦ explain the relationship between the traditional and modern understandings of meditation.

## 12.1 The inner instrument

Yoga and Vedānta analyse the mind as an “inner instrument” (*antaḥkaraṇa*) with four functions

**TRADITIONAL :**

**Table 12.1** The fourfold inner instrument

| Function     | Sanskrit | Role                                                       | Everyday example                               |
|--------------|----------|------------------------------------------------------------|------------------------------------------------|
| Mind         | manas    | receives sense impressions; doubts, wavers, weighs options | “Shall I get up for morning practice or not?”  |
| Intellect    | buddhi   | decides, discriminates, judges                             | “I will get up — it matters to me.”            |
| Ego          | ahaṁkāra | the sense of “I” and “mine” that claims experiences        | “I am the one who gets up early.”              |
| Memory-store | citta    | holds impressions and memories                             | remembering how good yesterday’s practice felt |

(In the Yoga Sūtra *citta* is used more broadly for the whole mind-field.) The model is not neurological; it describes how experience *feels* from the inside, and it gives teachers a vocabulary. When a student “can’t decide whether to come out of the pose”, that is *manas*; when they choose wisely, *buddhi*; when they cannot bear to be seen modifying, *ahaṃkāra*.

### Samskāras and vāsanās

Every experience leaves an impression (*saṃskāra*) in the mind. Impressions cluster into tendencies (*vāsanā*) that shape future perception and action. This is why the Yoga Sūtra insists on long, uninterrupted practice (1.14): meditation lays down impressions of stillness strong enough to counter the impressions of distraction **TRADITIONAL**. Modern students recognise the idea immediately as a theory of habit.

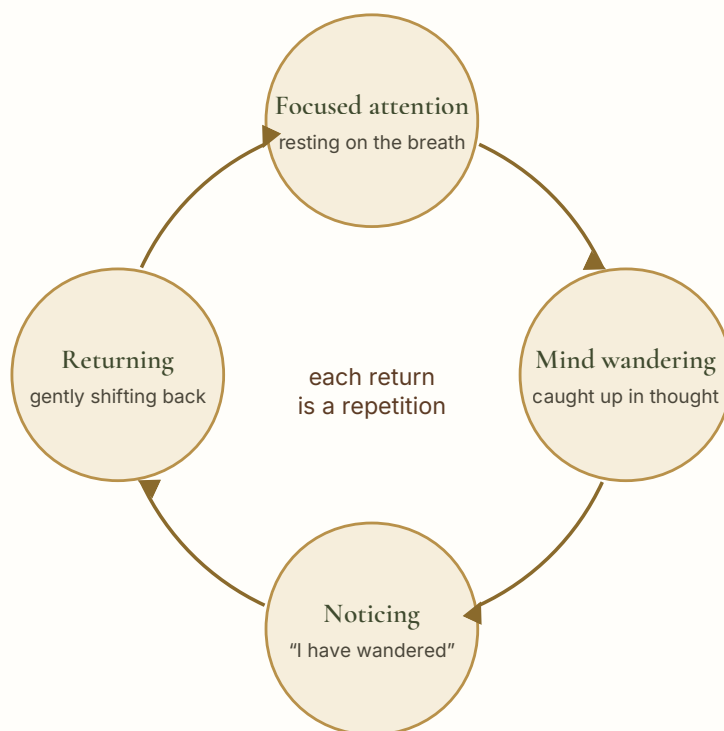
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## 12.2 Two families of meditation

Across traditions, meditation methods fall broadly into two families, a distinction also used by researchers (Lutz et al., 2008):

- ♦ **Focused attention** (*concentration*): attention is placed on one object — the breath, a mantra, a candle flame, a point in the body — and returned whenever it wanders. This is Patañjali’s *dhāraṇā* developing into *dhyāna*.
- ♦ **Open monitoring** (*mindfulness* or *choiceless awareness*): attention rests in awareness itself, noticing whatever arises — sensations, sounds, thoughts, emotions — without selecting or pursuing any of it. This resembles the witness practice of Vedānta.

The two support each other. Concentration steadies the mind enough to observe without being swept away; open awareness prevents concentration from becoming rigid.



**Figure 12.1** The cycle of attention in focused meditation, after Hasenkamp et al. (2012). Noticing that the mind has wandered is not failure; it is the moment in which the skill is trained.

### 12.3 Mindfulness: origins and modern forms

The English word *mindfulness* was chosen by nineteenth-century translators for the Pāli *sati* (Sanskrit *smṛti*), which means “memory” or “remembering” — in practice, remembering to be present **HISTORICAL**. In the Buddhist *Satipaṭṭhāna Sutta*, mindfulness is established on four foundations: the body, feelings (pleasant, unpleasant, neutral), states of mind, and mental objects. Classic introductions for modern readers include Thich Nhat Hanh (1975) and Gunaratana (2011).

In 1979 Jon Kabat-Zinn developed **Mindfulness-Based Stress Reduction (MBSR)** at the University of Massachusetts Medical Center, an eight-week secular programme combining sitting meditation, body scan and gentle yoga (Kabat-Zinn, 1990). MBSR and its offshoot, Mindfulness-Based Cognitive Therapy, brought mindfulness into medicine and psychology and generated the large research literature discussed in Chapter 28.

#### MINDFULNESS AND THE YOGA TRADITION

Some teachers present mindfulness as a Buddhist import foreign to yoga; others present all meditation as “mindfulness”. Both are simplifications. Yoga and Buddhism developed in close dialogue, share much vocabulary (the Yoga Sūtra’s four attitudes of 1.33 are the Buddhist *brahmavihāras*) and differ in philosophy. It is accurate and respectful to name each practice’s origin.

## 12.4 Traditional guidance for meditation

Traditional teachers present meditation as the culmination of practice, with a strong emphasis on regularity and on **mantra**. Their practical advice, which echoes the Gītā's instructions on the seat and posture of meditation (BG 6.10–14), can be summarised as follows **TRADITIONAL** :

- ◆ **Place.** Keep a special, clean place for meditation; the atmosphere builds with use.
- ◆ **Time.** Practise at the same time daily; the hours around dawn (*brahma muhūrta*) and dusk are recommended.
- ◆ **Direction.** Face north or east (a traditional recommendation).
- ◆ **Posture.** Sit in a steady, comfortable cross-legged posture with the back, neck and head in a straight line.
- ◆ **Mind and breath.** Instruct the mind to be quiet for the period of practice; begin with deep breathing, then let the breath become slow and rhythmic.
- ◆ **Focal point.** Fix the mind on one point — the space between the eyebrows (*ājñā*) for intellectual temperaments, or the heart (*anāhata*) for emotional ones — and keep it there.
- ◆ **Object.** Use one object consistently: a personal mantra (*japa*), repeated mentally and coordinated with the breath, or an image.
- ◆ **Duration.** Start with twenty minutes and increase gradually; consistency matters more than length.
- ◆ **Letting go.** When the mind wanders, bring it back gently; do not fight thoughts.

This structure — same time, same place, same object, gradually increasing duration — is the backbone of a stable practice in any tradition.

### ◆ "LIKE A LAMP IN A WINDLESS PLACE"

The Gītā's image (6.19) is precise: the flame does not stop burning — the mind does not go dark — but it no longer flickers. Meditation is not blankness but steady, bright awareness.

## 12.5 Traditional and modern understandings

The traditional purpose of meditation is liberation — the stilling of the mind so that consciousness knows itself. Modern secular programmes aim at stress reduction, emotional regulation and wellbeing. The practices overlap heavily; the frameworks differ. A yoga teacher should be able to teach both honestly: offering a student with anxiety a simple breath-focused practice without spiritual claims, and offering a spiritually motivated student the full traditional framework.

### 🔍 WHAT MEDITATION RESEARCH DOES AND DOES NOT SHOW

Research suggests that regular meditation practice can produce modest improvements in anxiety, depression and perceived stress, and changes in attention and emotional regulation **SCIENTIFIC** (Goyal et al., 2014; Tang et al., 2015). Many popular claims — about permanent brain changes, dramatic health effects or superiority over other treatments — go well beyond the evidence (Van Dam et al., 2018). A small but real proportion of meditators experience difficult effects such as anxiety or dissociation, particularly on intensive retreats (Lindahl et al., 2017; Britton et al., 2021).

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ The yogic model divides the mind into *manas*, *buddhi*, *ahaṃkāra* and *citta*; *saṃskāras* and *vāsanās* explain how habits form and how practice changes them.
- ◆ Meditation methods fall into focused attention (concentration) and open monitoring (mindfulness), which support each other.
- ◆ Mindfulness translates the Buddhist *sati*; modern secular programmes such as MBSR date from 1979.
- ◆ Traditional guidance emphasises a regular time and place, a steady posture, rhythmic breath, one focal point and mantra.
- ◆ Research supports modest benefits; exaggerated claims should be avoided and possible difficulties recognised.

 REFLECTION QUESTIONS

1. Watch your mind for one day using the fourfold model. Which function dominates?
2. Do you naturally prefer concentration or open awareness? Why?
3. How would you describe meditation to a sceptical scientist? To a devout Christian? To a Hindu grandmother?

 TEACHER-PRACTICE EXERCISES

1. Practise twenty minutes of meditation daily for the week, following every point in the list above. Record time, place and quality in your journal.
2. Teach a partner a five-minute focused-attention practice, then a five-minute open-monitoring practice. Ask which they preferred and why.
3. Explain *saṃskāra* to a partner using an example from learning a posture.

 KNOWLEDGE CHECK

1. In the fourfold inner instrument, the faculty that decides and discriminates is: (a) *manas* (b) *buddhi* (c) *ahaṃkāra* (d) *citta*
2. “Mindfulness” translates the Pāli word: (a) *jhāna* (b) *sati* (c) *dukkha* (d) *prajñā*
3. Open-monitoring meditation involves: (a) fixing attention on one object (b) noticing whatever arises without pursuing it (c) chanting aloud (d) visualising a deity
4. MBSR was developed by: (a) Swami Vivekananda (b) Thich Nhat Hanh (c) Jon Kabat-Zinn (d) Patañjali
5. In traditional guidance, the space between the eyebrows is recommended as a focal point for: (a) emotional temperaments (b) intellectual temperaments (c) beginners only (d) no one

Answers 1 b · 2 b · 3 b · 4 c · 5 b

 TEACHING ASSIGNMENT

Write a one-page guide, “Your first month of meditation”, for your future students. Include when and where to sit, posture, a simple method, what to expect, what to do when the mind wanders, and when to seek help. Use plain language; test it on a non-yogi friend.

## CHAPTER 13

## The Subtle Body — Prāṇa, Vāyus, Nāḍīs, Cakras and Kośas

*“As the spokes of a wheel are fixed in the hub, so everything is fixed in prāṇa.” — Prasna Upaniṣad 2.6*

## © LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ explain the traditional concept of the subtle body and its relation to the three bodies and five sheaths;
- ◆ describe prāṇa and the five vāyus;
- ◆ describe the nāḍīs, especially iḍā, piṅgalā and suṣumṇā, and the idea of kuṇḍalinī;
- ◆ describe the six cakras and sahasrāra as commonly taught, and explain how the cakra system developed historically;
- ◆ speak about the subtle body to students accurately, distinguishing traditional teaching from anatomy.

## 13.1 A map of experience

Alongside the physical body, the yoga traditions describe a **subtle body** (*sūkṣma śarīra*) made of vital energy and mind. It is not an object that can be dissected; it is a map of experience used to guide practice **TRADITIONAL**. Practitioners feel the breath “move” through the body, sense warmth or tingling in certain regions, notice that attention can be gathered at a point. The subtle-body maps give these experiences names and a structure, so that a teacher can say “take your awareness to the space between the eyebrows” and the student knows where to go.

## ◆ TWO LANGUAGES, BOTH VALUABLE

Anatomy describes the body from outside — what can be measured and dissected. The subtle-body teachings describe it from inside — what can be felt and directed with attention. Neither replaces the other. A cakra is not the solar plexus; the solar plexus is not a cakra. The teacher’s task is to use each language accurately and never to present one as the other.

### WHAT SCIENCE SAYS ABOUT SUBTLE ANATOMY

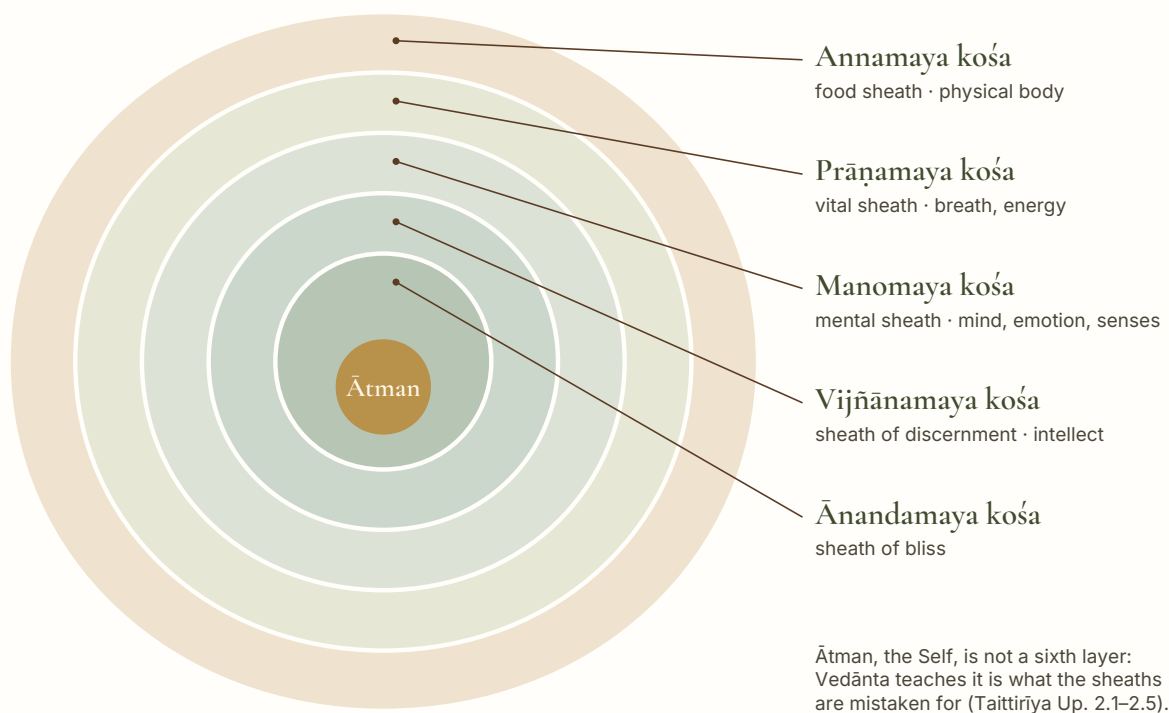
No anatomical structures corresponding to nāḍīs or cakras have been identified **SCIENTIFIC**. Attempts to equate them with nerves, nerve plexuses or endocrine glands are modern interpretations, not findings, and the correspondences proposed differ from author to author **INTERPRETIVE**. This does not make the traditional maps useless: as maps of attention and felt experience, they have guided practitioners for centuries. It does mean that they should not be taught as physiology.

## 13.2 Three bodies and five sheaths

Vedānta describes three “bodies” (*śarīra*) **TRADITIONAL**:

- ♦ The **gross body** (*sthūla śarīra*) — the physical body, made of food.
- ♦ The **subtle body** (*sūkṣma śarīra*) — vital energy, mind and intellect.
- ♦ The **causal body** (*kāraṇa śarīra*) — the seed-state of deep sleep and latent impressions.

The five sheaths (*pañca-kośa*) of the Taittirīya Upaniṣad map onto these: the food sheath is the gross body; the vital, mental and intellectual sheaths form the subtle body; and the bliss sheath is the causal body.



**Figure 13.1** The five sheaths (*pañca-kośa*) of the Taittirīya Upaniṣad (Olivelle, 1996). The model is a map for self-enquiry and for teaching; it describes levels of experience, not anatomical structures.

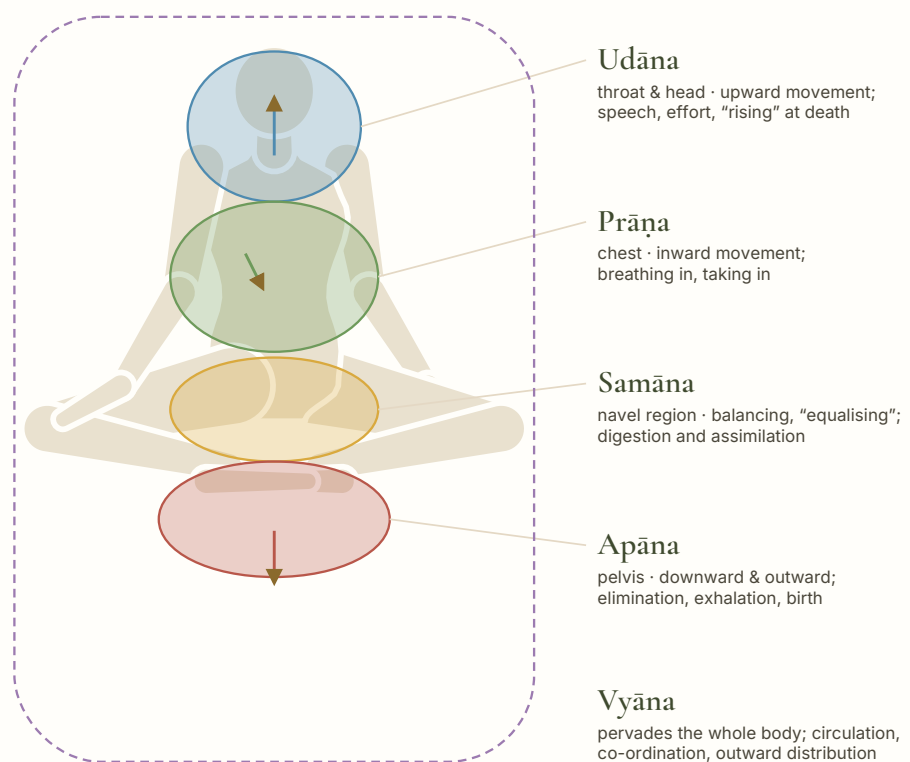
**Table 13.1** The five sheaths and practices associated with them

| Kośa      | Made of              | Body   | Practices that work with it (traditional view) |
|-----------|----------------------|--------|------------------------------------------------|
| Annamaya  | food                 | gross  | āsana, diet, cleansing                         |
| Prāṇamaya | vital energy, breath | subtle | prāṇāyāma, bandhas, mudrās                     |

| Kośa        | Made of                | Body   | Practices that work with it (traditional view) |
|-------------|------------------------|--------|------------------------------------------------|
| Manomaya    | mind, emotion, senses  | subtle | pratyāhāra, mantra, relaxation, devotion       |
| Vijñānamaya | intellect, discernment | subtle | study, self-enquiry, meditation                |
| Ānandamaya  | bliss                  | causal | deep meditation; glimpsed in deep sleep        |

### 13.3 Prāṇa and the five vāyus

*Prāṇa* means both “breath” and the vital force of which breath is the most obvious expression. The Upaniṣads already treat it as the chief of the bodily faculties (Chapter 3). Later texts divide it into five functions, the *vāyus* (“winds”), each with a direction and a region **TRADITIONAL** :



**Figure 13.2** The five *vāyus* (“winds”) of *prāṇa* as usually taught. Texts disagree on their exact seats and functions; the dashed outline stands for *vyāna*, which pervades the whole body. These are traditional categories of experience, not physiological systems.

**Table 13.2** The five *vāyus* (placements vary between texts)

| Vāyu   | Region                | Direction           | Functions traditionally attributed                               |
|--------|-----------------------|---------------------|------------------------------------------------------------------|
| Prāṇa  | chest, heart          | inward, upward      | inhalation, taking in (food, impressions)                        |
| Apāna  | pelvis, lower abdomen | downward, outward   | elimination, exhalation, childbirth, menstruation                |
| Samāna | navel region          | centring, balancing | digestion, assimilation                                          |
| Udāna  | throat, head          | upward              | speech, effort, growth; the departure of the life-force at death |
| Vyāna  | whole body            | outward, pervading  | circulation, co-ordination                                       |



Haṭha texts describe the union of prāṇa and apāna in the navel region, drawn together by the bandhas, as a key step in awakening kuṇḍalinī (Chapter 14). Five minor vāyus (*upaprāṇas*) are also named in some texts, associated with belching, blinking, sneezing, yawning and the persistence of the body after death.

### 13.4 The nāḍīs

Nāḍī means “channel” or “tube”. The texts say the body contains many thousands — the figure 72,000 is most often given, though some texts give other numbers — through which prāṇa flows **TRADITIONAL**. Three are primary:

- ♦ *Suṣumṇā*, the central channel, running from the base of the spine to the crown of the head.
- ♦ *Iḍā*, on the left, associated with the moon, coolness, the left nostril and a calm, receptive quality.
- ♦ *Piṅgalā*, on the right, associated with the sun, heat, the right nostril and an active quality.

Iḍā and piṅgalā are said to begin at the base of the spine and end at the left and right nostrils. In the traditional view, prāṇa usually flows through one of them at a time, which is experienced as the alternation of nostril dominance; when it flows equally, or enters suṣumṇā, the mind becomes still and meditation deepens. Alternate-nostril breathing (*anuloma viloma*, *nāḍī śodhana* — “purification of the channels”) is the principal practice for balancing them.

#### 🔍 THE NASAL CYCLE

It is well established that in most people the nostrils alternately congest and decongest over a period of hours — the **nasal cycle** — so that one nostril is usually more open than the other; in 24-hour recordings the dominant side switched roughly every two hours while awake (Kahana-Zweig et al., 2016)

**SCIENTIFIC** . Traditional yogic observation of this alternation is accurate. Whether breathing through one nostril selectively affects the two halves of the brain or the branches of the autonomic nervous system is less clear: small studies report such effects, but results are inconsistent and have not been firmly established (Raghuraj & Telles, 2008; Saoji et al., 2019).

### 13.5 Kuṇḍalinī

In tantric and Haṭha teaching, a dormant power, *kuṇḍalinī* (“the coiled one”), lies at the base of the spine, coiled like a serpent and blocking the entrance to suṣumṇā. Through prāṇāyāma, bandhas and mudrās, and through devotion and meditation, she is awakened, pierces the cakras and unites with Śiva at the crown — the symbolic image of liberation **TRADITIONAL**. The *Haṭha Yoga Pradīpikā* devotes much of its third chapter to this process.

#### ⚠️ SPEAKING ABOUT KUṆḌALINĪ

Traditional teachers treat kuṇḍalinī practices as advanced and potentially destabilising, to be undertaken only under experienced guidance. Some modern practitioners report intense physical and emotional experiences they attribute to kuṇḍalinī. In a general class: do not promise awakenings, do not teach forceful practices aimed at them, and if a student reports distressing experiences — sleeplessness, intense anxiety, unusual perceptions — reduce or stop intensive breath work and meditation and encourage them to seek appropriate support, including medical and mental-health care where needed (Chapter 28).

### 13.6 The cakras

*Cakra* means “wheel” or “circle”. In the subtle-body maps, cakras are centres along the central channel where nāḍīs converge, visualised as lotuses with a given number of petals, each associated with an element, a seed syllable (*bīja mantra*), a deity and qualities.

#### A history of the cakra systems

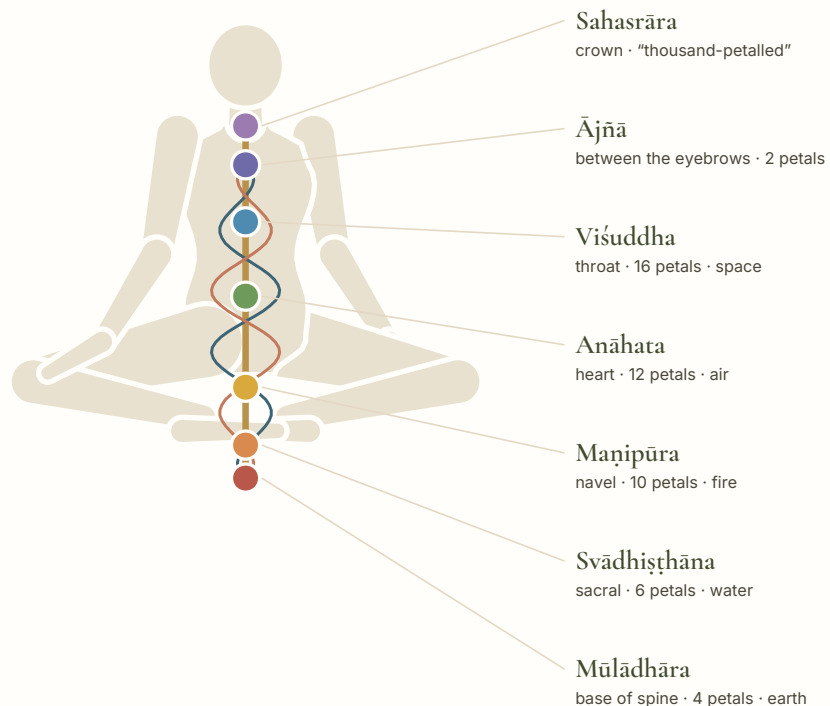
The cakra system most students know — six cakras plus sahasrāra — is one of many **HISTORICAL**. Earlier tantric texts describe systems with four, five, six, nine, twelve or more centres, differently placed (Mallinson & Singleton, 2017; Samuel, 2008). The six-cakra system became dominant largely through the *Ṣaṭcakraṇirūpaṇa* (“Description of the six cakras”), written by Pūrṇānanda in 1577, which reached the English-speaking world in Arthur Avalon’s (Sir John Woodroffe’s) translation *The Serpent Power* (1919) (Avalon, 1919).

Several popular associations are much more recent. The rainbow sequence of colours (red at the base to violet at the crown) and the correspondence of each cakra with an endocrine gland are twentieth-century Western innovations, not found in the traditional texts **HISTORICAL**. So are many of the psychological meanings now attributed to each cakra (“the heart cakra is about love”, “the throat cakra is about self-expression”) — useful teaching metaphors, perhaps, but modern ones **INTERPRETIVE**.

#### Nāḍīs

- Suṣumṇā · central channel
- Iḍā · “lunar”, left nostril
- Piṅgalā · “solar”, right nostril

The crossing, spiral form is a common modern rendering; many texts simply place iḍā and piṅgalā to the left and right of the central channel.



**Figure 13.3** The six *cakras* and *sahasrāra* as commonly taught in modern yoga, following the arrangement of the 16th-century *Ṣaṭcakraṇirūpaṇa* (Avalon, 1919). Earlier texts describe different numbers and placements of cakras (Mallinson & Singleton, 2017). This is a map of traditional meditative experience; no anatomical structures corresponding to cakras or nāḍīs have been identified.

**Table 13.3** The cakras as commonly taught, following the *Ṣaṭcakraṇirūpaṇa*

| Cakra        | Meaning of name   | Location                 | Petals | Element           | Bīja |
|--------------|-------------------|--------------------------|--------|-------------------|------|
| Mūlādhāra    | root support      | base of spine / perineum | 4      | earth             | laṃ  |
| Svādhiṣṭhāna | one's own seat    | genital region / sacrum  | 6      | water             | vaṃ  |
| Maṇipūra     | city of jewels    | navel                    | 10     | fire              | raṃ  |
| Anāhata      | unstruck (sound)  | heart                    | 12     | air               | yaṃ  |
| Viśuddha     | purified          | throat                   | 16     | space (ether)     | haṃ  |
| Ājñā         | command           | between the eyebrows     | 2      | (mind)            | oṃ   |
| Sahasrāra    | thousand-petalled | crown of the head        | 1000   | (beyond elements) | —    |

## Using cakras in teaching

The most traditional and least problematic use of cakras is as **points of concentration**: “Bring your awareness to the space between the eyebrows” is a clear, safe instruction that does not depend on any belief. Chanting the bīja mantras while focusing on each region is a traditional practice many students enjoy. Claims that a posture “opens” or “balances” a particular cakra, or that cakras cause physical or psychological conditions, should be presented — if at all — as traditional or interpretive, never as fact.

### Q A SAFE, HONEST CAKRA CUE

“In the tradition, this region is associated with *maṇipūra*, the navel centre, linked to fire. Let your attention rest there and notice any warmth or movement of the breath.” This teaches the tradition, invites experience, and claims nothing.

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ◆ The subtle body is a traditional map of felt experience used to guide practice; it is not physical anatomy.
- ◆ Vedānta describes three bodies (gross, subtle, causal) and five sheaths (food, vital, mental, intellectual, bliss).
- ◆ Prāṇa is divided into five vāyus: prāṇa, apāna, samāna, udāna and vyāna, whose placements vary between texts.
- ◆ The three main nāḍīs are suṣumṇā, idā and piṅgalā; alternation of nostril dominance (the nasal cycle) is real, but lateralised brain or nervous-system effects are unproven.
- ◆ The popular six-cakra system derives mainly from a 1577 text; rainbow colours and gland correspondences are modern additions.

### REFLECTION QUESTIONS

1. Before this chapter, what did you believe about cakras? Where did those beliefs come from?
2. Can a map be useful without being literally true? Give an example from outside yoga.
3. How will you answer a student who asks, “Is my heart cakra blocked?”

## ✂ TEACHER-PRACTICE EXERCISES

1. Notice which nostril is more open at five times during one day; record the results.
2. Teach a partner a five-minute concentration practice moving attention through the seven cakra locations with their bija mantras, using only traditional attributions.
3. Rewrite three common cakra claims you have heard so that they are honestly labelled.

## ☑ KNOWLEDGE CHECK

1. How many petals does anāhata have in the traditional system? (a) 4 (b) 10 (c) 12 (d) 16
2. Which nāḍī is associated with the left nostril and the moon? (a) piṅgalā (b) iḍā (c) suṣumṇā (d) vajra
3. The vāyu associated with elimination and downward movement is: (a) prāṇa (b) udāna (c) apāna (d) vyāna
4. The rainbow colour scheme for the cakras is: (a) from the Haṭha Yoga Pradīpikā (b) from the Ṣaṭ-cakranirūpaṇa (c) a twentieth-century innovation (d) from the Upaniṣads
5. Which kośa corresponds to the causal body? (a) annamaya (b) prāṇamaya (c) vijñānamaya (d) ānandamaya

Answers 1 c · 2 b · 3 c · 4 c · 5 d

## 📄 TEACHING ASSIGNMENT

Write a 400-word handout, “The subtle body: a guide for students”, that explains prāṇa, nāḍīs and cakras accurately and respectfully, clearly labelling traditional teachings and stating what science has and has not found. Include one practice students can try at home.

# Bandhas, Mudrās and Mantra

*“Through the practice of the bandhas, the prāṇa is made to enter the suṣumṇā.” — after Haṭha Yoga Pradīpikā 3.1–3*

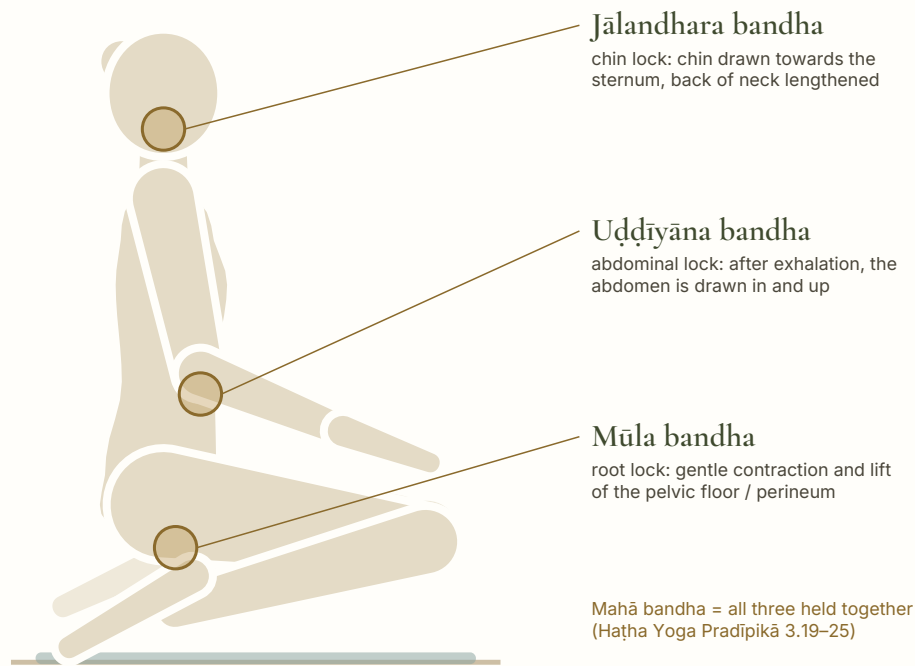
## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ describe the technique, traditional purpose, physical action and contraindications of mūla, uḍḍiyāna and jālāndhara bandha;
- ◆ teach the hand mudrās used in a classical class and describe the principal Haṭha mudrās;
- ◆ define mantra, explain the three modes of japa and the use of the mālā;
- ◆ chant Om̐ and several traditional mantras correctly and explain their meaning;
- ◆ introduce mantra and mudrā in class in a way that is accurate, safe and inclusive.

## 14.1 Bandhas: the locks

A *bandha* (“lock”, “bond”) is a sustained muscular contraction that, in the traditional understanding, seals or directs the flow of prāṇa **TRADITIONAL**. The Haṭha texts teach three principal bandhas, usually combined with breath retention (*kumbhaka*), and a fourth, *mahā bandha*, in which all three are held together.



**Figure 14.1** The three principal *bandhas* shown on a seated practitioner (schematic). Uḍḍīyāna bandha is practised only on an empty stomach and after exhalation; see the safety notes that follow.

### Jālandhara bandha — the chin lock

**Technique.** Sitting in a steady cross-legged posture with hands on knees, inhale; lift the sternum slightly, lengthen the back of the neck and lower the chin towards the notch between the collarbones, without forcing. Hold for the duration of the retention. To release, lift the head *before* exhaling.

**Physical action.** Flexion of the cervical spine with elevation of the chest, compressing the front of the throat. **Traditional purpose:** to prevent prāṇa escaping upward during retention and to prevent pressure rising into the head (HYP 3.70–73) **TRADITIONAL**.

### Uḍḍīyāna bandha — the abdominal lock

**Technique.** Standing with knees bent and hands on thighs, or sitting: exhale completely. Holding the breath out, close the throat and make a “mock inhalation” — expand the ribs as if breathing in, but let no air enter. The abdomen is drawn in and up under the rib cage. Hold for a few seconds only, release the abdomen, and then inhale gently.

**Physical action.** With the glottis closed after full exhalation, the expansion of the rib cage lowers the pressure inside the chest; the diaphragm, relaxed, is drawn upward and the abdominal wall is sucked inward. **Traditional purpose:** *uḍḍīyāna* means “flying up”; the text says it causes prāṇa to “fly up” through suṣūmṇā (HYP 3.55) **TRADITIONAL**.

### △ UDDĪYĀNA BANDHA

Practise only on an **empty stomach**. Commonly advised contraindications **CLINICAL CAUTION** : pregnancy; high blood pressure or heart disease; hernia; recent abdominal or chest surgery; peptic ulcer or other active abdominal conditions; glaucoma and conditions aggravated by straining. Many traditional teachers also advise against it during menstruation. Never practise it after an inhalation, and never combine it with vigorous breathing before the course has taught you both.

## Mūla bandha — the root lock

**Technique.** Sitting steadily, draw up the muscles of the pelvic floor — the area between the anus and genitals — as if gently stopping the flow of urine and wind at the same time, and hold. The contraction should be moderate and should not tense the buttocks, abdomen or face.

**Physical action.** Voluntary contraction of the pelvic-floor muscles (principally levator ani). **Traditional purpose:** to draw apāna upward to unite with prāṇa, awakening kuṇḍalinī (HYP 3.61–64) **TRADITIONAL**.

### 🔍 PELVIC FLOOR AND BANDHAS

The pelvic floor muscles support the pelvic organs and work together with the diaphragm and deep abdominal muscles in regulating intra-abdominal pressure **SCIENTIFIC** (Hodges & Gandevia, 2000). Pelvic-floor muscle training is an established physiotherapy approach for some continence problems, but that is a specific clinical protocol; mūla bandha as taught in yoga has not itself been tested as a treatment. Some people have pelvic floors that are *overactive* (too tense) rather than weak, and for them repeated strong contraction may not help. Teach mūla bandha gently, and refer students with pelvic pain or continence problems to a pelvic-health physiotherapist.

## Mahā bandha

With the breath held out after exhalation, all three locks are applied together, held briefly and released in reverse order (jālandhara last) before inhaling (HYP 3.19–25). This is an advanced practice, introduced only after each lock is secure.

## 14.2 Mudrās: seals and gestures

*Mudrā* means “seal” or “gesture”. The word covers two quite different things:

- ♦ **Haṭha mudrās** — whole-body techniques combining posture, breath and locks, such as the ten in HYP chapter 3 (Chapter 9);
- ♦ **Hasta mudrās** — hand gestures used in meditation, prāṇāyāma, ritual and Indian classical dance.

**Table 14.1** Mudrās used in a classical-style class

| Mudrā       | How it is formed                                                                      | When it is used       | Traditional meaning                                                                         |
|-------------|---------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------|
| Cin mudrā   | tip of index finger touches tip of thumb; other fingers extended; palms down on knees | meditation, prāṇāyāma | the individual self (index finger) united with the universal (thumb); palms down: grounding |
| Jñāna mudrā | same finger position, palms facing up                                                 | meditation            | receptivity; knowledge                                                                      |

| Mudrā           | How it is formed                                                                                                                               | When it is used                      | Traditional meaning                                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------|
| Viṣṇu mudrā     | right hand: index and middle fingers folded into the palm; thumb and ring (and little) fingers extended                                        | alternate - nostril breathing        | a practical hand position for closing each nostril    |
| Añjali mudrā    | palms pressed together at the heart                                                                                                            | greeting, prayer, namaste            | reverence; the meeting of opposites                   |
| Dhyāna mudrā    | hands resting in the lap, right on left, palms up, thumb tips touching                                                                         | seated meditation                    | stillness, receptivity                                |
| Ṣaṇmukhī mudrā  | thumbs close the ears, index fingers rest on the eyelids, middle fingers beside the nostrils, ring and little fingers above and below the lips | pratyāhāra; listening to inner sound | "closing the six gates" of the senses                 |
| Viparīta karaṇī | inverted posture with hips supported (Chapter 24)                                                                                              | as a Haṭha mudrā                     | "reversing" — traditionally said to preserve vitality |

### ② "THIS MUDRĀ TREATS YOUR THYROID."

Books and websites sometimes claim that particular hand mudrās cure specific diseases. There is no scientific evidence for such claims **SCIENTIFIC**. Hand mudrās are traditional aids to attention, feeling and devotion; they can be valuable exactly as that. Teach their traditional meaning — and do not make health promises.

## 14.3 Mantra

### What mantra is

A **mantra** is a sacred sound, syllable, word or verse, repeated as a meditative or devotional practice. A traditional etymology divides the word into *man* ("to think") and *tra* ("instrument", "protection"): a mantra is an instrument of the mind, or that which protects the one who reflects on it **INTERPRETIVE**. The tradition teaches that the sound itself has power, which is why Sanskrit mantras are preserved in their original form rather than translated **TRADITIONAL**.

### Om

Om (also written *AUM*) is the most important mantra, the *praṇava*. The Māṇḍūkya Upaniṣad analyses it into three sounds and silence (Figure 3.1); Patañjali calls it the expression of Īśvara and recommends its repetition with reflection on its meaning (YS 1.27–28). Chanting Om at the start and end of class is a tradition in almost every school of yoga.

#### 🕉 TEACHING OM

Divide it into three parts: *A* (open mouth, sound felt in the abdomen), *U* (lips rounding, sound in the chest), *M* (lips closed, a hum felt in the head), followed by silence. A common proportion is roughly 1 : 1 : 2, with the humming *M* longest. Let students listen first, and invite rather than require them to join.

### Kinds of mantra

- ◆ **Bīja (seed) mantras** — single syllables such as *om*, *hrīṃ*, or the elemental bījas of the cakras (*laṃ*, *vaṃ*, *raṃ*, *yaṃ*, *haṃ*).



- ♦ **Saṅga mantras** — invoking a personal form of the divine, e.g. *Oṃ Namaḥ Śivāya*, *Oṃ Namo Nārāyaṇāya*, *Oṃ Namo Bhagavate Vāsudevāya*.
- ♦ **Nirguṇa mantras** — pointing to the formless absolute, e.g. *Soham* (“I am That”) or *Oṃ* alone.
- ♦ **Vedic verses** — such as the Gāyatrī (Chapter 3) and the *Mahāmṛtyuñjaya* mantra (R̥gveda 7.59.12), traditionally chanted for healing and protection.
- ♦ **Peace chants** (*śānti pāṭha*) — such as *Oṃ saha nāvavatu* and *Asato mā sad gamaya*.

## Japa

*Japa* is the repetition of a mantra. The tradition describes three modes **TRADITIONAL** :

1. **Vaikhari** — aloud; easiest for a restless mind.
2. **Upāṃśu** — whispered or with moving lips.
3. **Mānasika** — mental; the subtlest and traditionally the most powerful.

A **mālā** of 108 beads, with a larger *meru* bead that is not crossed, is used to count repetitions: the beads are moved with the thumb and middle finger, and at the *meru* the **mālā** is turned back. Traditionally a personal mantra is given at initiation (*mantra dīkṣā*) by a teacher; students without initiation can practise with *Oṃ* or a universal mantra **TRADITIONAL** .

### Q CHANTING AND BREATHING

In a small experimental study, reciting the Ave Maria in Latin and a yoga mantra (*Om mani padme om*) both slowed breathing to around six breaths per minute and increased cardiovascular rhythms associated with baroreflex sensitivity **SCIENTIFIC** (Bernardi et al., 2001). This suggests one physiological pathway by which rhythmic chanting may be calming: it structures and slows the breath. Humming, as in *M* of *Oṃ* and in *bhṛāmari*, also greatly increases nasal nitric oxide levels (Weitzberg & Lundberg, 2002), though the practical significance of this is not established.

## 14.4 Mantra, mudrā and inclusion

Chanting and devotional gestures are among the most culturally rich parts of yoga — and among the most sensitive. Some students love them; others feel they are being asked to participate in a religion not their own. Good practice:

- ♦ **Translate** what you chant, briefly, so no one repeats words they do not understand.
- ♦ **Invite** rather than require: “Join in if you wish, or simply listen.”
- ♦ **Pronounce** Sanskrit carefully; learning correct pronunciation is a form of respect.
- ♦ **Give context**: “This is a traditional Hindu chant for peace.”
- ♦ **Never** use sacred symbols or chants as decoration or as a sales device.

## CHAPTER REVIEW

## KEY TAKEAWAYS

- ♦ The three bandhas are the chin lock (jālandhara), the abdominal lock (uḍḍiyāna, only after exhalation on an empty stomach) and the root lock (mūla); together they form mahā bandha.
- ♦ Uḍḍiyāna bandha has important contraindications, including pregnancy, hypertension, heart disease and hernia.
- ♦ Hand mudrās are aids to attention and devotion; claims that they treat disease are unsupported.
- ♦ Mantra is a sacred sound repeated aloud, in a whisper or mentally (japa), often with a 108-bead mālā.
- ♦ Chanting slows and structures breathing, one plausible reason for its calming effect; offer it with translation and freedom to participate.

## REFLECTION QUESTIONS

1. How do you feel about chanting in Sanskrit? Where do those feelings come from?
2. Which bandha is easiest for you to feel? Which is hardest? What does that tell you?
3. How would you introduce Om̐ to a class of corporate employees in their lunch hour?

## TEACHER-PRACTICE EXERCISES

1. Practise each bandha individually, three times, under supervision; then teach jālandhara bandha to a partner with full safety instructions.
2. Chant one mālā (108 repetitions) of a mantra of your choice aloud, then one mentally. Compare.
3. Teach the formation and meaning of cin, jñāna and Viṣṇu mudrā in under two minutes.

## KNOWLEDGE CHECK

1. Uḍḍiyāna bandha is practised: (a) after inhalation (b) after full exhalation (c) during rapid breathing (d) at any time
2. Which bandha involves contraction of the pelvic floor? (a) jālandhara (b) uḍḍiyāna (c) mūla (d) mahā
3. In Viṣṇu mudrā, which fingers are folded into the palm? (a) thumb and little finger (b) index and middle fingers (c) ring and little fingers (d) none
4. Mental repetition of a mantra is called: (a) vaikhari (b) upāṃśu (c) mānasika (d) kīrtan
5. How many beads does a traditional mālā have (excluding the meru)? (a) 27 (b) 54 (c) 108 (d) 1,000

Answers 1b · 2c · 3b · 4c · 5c

## TEACHING ASSIGNMENT

Prepare a five-minute opening for a mixed international class that includes chanting Om̐ three times. Script your introduction (context, meaning, invitation) in under 60 words. Teach it three times to different partners and refine it after each.

## SECTION I · YOGA PHILOSOPHY, TRADITION &amp; PRACTICE

## CHAPTER 15

## The Role and Responsibilities of a Yoga Teacher

*“Om — may it protect us both. May it nourish us both. May we work together with energy.” — Taittirīya Upaniṣad, invocation*

### 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ describe the traditional guru-śiṣya relationship and how the modern role of yoga teacher differs from it;
- ◆ state the core responsibilities of a yoga teacher to students, to the tradition and to oneself;
- ◆ explain the idea of *adhikāra* (fitness to receive teaching) and apply it to modern classes;
- ◆ describe the limits of a yoga teacher’s scope of practice;
- ◆ write a personal statement of your intentions as a teacher.

## 15.1 From guru to teacher

In the Indian traditions, spiritual knowledge was transmitted through the relationship between **guru** and **śiṣya** (disciple), often living together over years. A traditional (and popular) etymology explains *guru* as *gu*, darkness, and *ru*, that which removes it — the dispeller of darkness **TRADITIONAL**. The chain of teachers through which a practice is handed down is called *paramparā*, and much of the authority of a tradition rests on its unbroken lineage.

A modern yoga teacher, even one teaching in a traditional lineage, is usually not a guru in this sense. You will teach students for an hour, a week, perhaps a few years. They will pay you, give feedback, and leave if they are not served. This is not a loss; it is a different and honest relationship. The modern teacher’s authority rests on **competence, integrity and care** — and on transparency about what they know and do not know.

**A NECESSARY CAUTION**

The history of modern yoga includes well-documented cases in which teachers used the authority of the guru role to exploit students sexually, financially and psychologically (Remski, 2019). These cases have taught the yoga community that devotion to a teaching must never become unconditional obedience to a person. A good teacher welcomes questions, respects students' autonomy and submits to ethical standards and accountability like any other professional (Chapter 30).

## 15.2 What a teacher owes

### To students

1. **Safety.** Teach practices appropriate to the students in front of you; screen for relevant health conditions; offer modifications; know when to refer (Chapter 26).
2. **Honesty.** Represent your training and experience truthfully; distinguish tradition from evidence; make no health promises.
3. **Respect.** Treat every student with dignity regardless of body, ability, age, gender, ethnicity, religion or background; respect their right to decline any practice or touch.
4. **Competence.** Know what you teach, and teach within what you know.
5. **Boundaries.** Keep the relationship professional; never exploit the trust or power the role carries.
6. **Confidentiality.** Keep students' health information and personal disclosures private.
7. **Presence.** Give your full attention during class — not to your phone, your own practice or your image.

### To the tradition

- ◆ Transmit teachings accurately, with sources, and name the lineage you learned from.
- ◆ Pronounce Sanskrit carefully, and use sacred sounds and symbols with respect.
- ◆ Neither exaggerate the tradition's claims nor dismiss them.
- ◆ Acknowledge your teachers.

### To yourself

- ◆ Keep your own practice alive. A teacher who has stopped practising teaches from memory, not experience.
- ◆ Keep learning (Chapter 39).
- ◆ Look after your body, voice and energy; teaching many classes a week is physically demanding.
- ◆ Seek supervision or mentoring when facing difficult situations.

## 15.3 Adhikāra: meeting students where they are

The traditions teach that a practice should be given according to the student's *adhikāra* — their fitness, readiness or eligibility to receive it **TRADITIONAL**. The *Haṭha Yoga Pradīpikā* insists that advanced techniques be learned from a teacher; the Upaniṣads show teachers testing students before instructing them. In modern terms, *adhikāra* is the principle behind **progression and individualisation**: a beginner is not taught advanced *prāṇāyāma*; a student with glaucoma is not taught headstand; a student in acute grief may need relaxation rather than a vigorous practice.

## 15.4 Scope of practice

A 200-hour yoga teacher is trained to teach yoga to generally healthy people in group and private settings. A yoga teacher is **not** a physician, physiotherapist, psychotherapist or dietitian, and should not diagnose, treat or advise on medical or psychological conditions.

**Table 15.1** Within and outside a yoga teacher's scope

| Within scope                                                                     | Outside scope                                                                                                              |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| teaching yoga practices and their traditional context                            | diagnosing any condition ("that sounds like a disc problem")                                                               |
| screening for health conditions relevant to practice                             | prescribing exercises to treat a diagnosed condition, unless specifically trained and working with the student's clinician |
| offering general modifications for common limitations                            | advising a student to reduce or stop medication                                                                            |
| sharing general, evidence-based information about the benefits and risks of yoga | offering psychotherapy or trauma processing                                                                                |
| recognising warning signs and referring to a healthcare professional             | giving dietary prescriptions for medical conditions                                                                        |

### ◆ "REFER, DON'T DIAGNOSE"

When a student describes symptoms, your job is not to name them but to recognise whether they call for a professional opinion, adapt the practice meanwhile, and encourage the student to seek that opinion. "I can't tell you what's causing that, but it would be worth having it checked by a doctor or physiotherapist. Until then, let's leave out deep forward bends."

## 15.5 The teacher as student

The tradition taught here speaks of the teacher as an instrument through which the teaching flows. Practically, this means humility: every class teaches the teacher something. The invocation that opens this manual asks for protection and nourishment for teacher *and* student together, and that they work together with energy. That is the relationship this course hopes to form in you.

### ✧ YOUR TEACHING INTENTION

Write a statement of up to 200 words beginning "As a yoga teacher, I intend to...". Include at least one commitment to students, one to the tradition and one to yourself. Keep it in your journal; re-read it at the end of the course and after your first year of teaching.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ The traditional guru–śiṣya relationship differs from the modern teacher’s role; the modern teacher’s authority rests on competence, integrity and care.
- ◆ History shows that unquestioning obedience to teachers enables abuse; accountability and students’ autonomy are essential.
- ◆ Teachers owe students safety, honesty, respect, competence, boundaries, confidentiality and presence.
- ◆ *Adhikāra* – teaching according to readiness – is the traditional basis of progression and individualisation.
- ◆ A yoga teacher does not diagnose or treat; recognise, adapt and refer.

 REFLECTION QUESTIONS

1. Think of the best teacher you have had in any subject. What exactly did they do?
2. Where is the line, for you, between respect for a teacher and obedience?
3. Which of the responsibilities listed will be hardest for you to keep? Why?

 TEACHER-PRACTICE EXERCISES

1. Role-play: a student asks you to tell them whether their knee pain is “arthritis or a torn meniscus”. Respond within scope.
2. Draft a simple health questionnaire for new students (no more than ten questions).
3. In a group, list five ways a teacher could unintentionally misuse their authority in a class, and how each could be prevented.

 KNOWLEDGE CHECK

1. *Paramparā* refers to: (a) a posture sequence (b) a lineage of teachers (c) a breathing technique (d) a mantra
2. *Adhikāra* means: (a) authority of the teacher (b) the student’s fitness or readiness for a practice (c) non-harming (d) a sacred text
3. Which is within a 200-hour yoga teacher’s scope? (a) diagnosing a hernia (b) recommending stopping medication (c) recognising warning signs and referring (d) offering trauma therapy
4. The traditional etymology of *guru* is: (a) heavy one (b) remover of darkness (c) old teacher (d) holy man
5. The best response when a student describes new, unexplained symptoms is to: (a) name the likely condition (b) ignore them (c) adapt the practice and encourage them to seek a professional opinion (d) advise rest for a month

Answers 1b • 2b • 3c • 4b • 5c

 TEACHING ASSIGNMENT

Write your “teacher’s charter”: a one-page document you could give to students describing how you teach, what you promise them, what you ask of them, how you handle touch and consent, and how they can give you feedback. You will revise it in **Chapter 30**.





## SECTION II

# Asana, Pranayama, Meditation & Anatomy

The body is the first instrument of Haṭha Yoga, and a teacher must know it both from the inside, through practice, and from the outside, through anatomy. This section builds a working knowledge of the skeleton, muscles, breath and nervous system — always connected to what happens on the mat — and then applies it to the twelve basic postures, the safety of students, and the practices of prāṇāyāma and meditation.

- 16 Foundations — The Language of Anatomy and Movement
- 17 The Skeletal System in Practice
- 18 How Muscles Work — Contraction, Flexibility and Strength
- 19 A Yoga Teacher's Muscle Atlas
- 20 Yoga Biomechanics — How Postures Load the Body
- 21 The Anatomy and Physiology of Breathing

- 22 The Nervous System, Stress and Regulation
- 23 Preparing the Body — Relaxation, Sūrya Namaskār and Leg Raises
- 24 The Twelve Basic Postures
- 25 Asana–Anatomy Mapping Tables
- 26 Injury Prevention and Teacher Safety
- 27 Prāṇāyāma — Theory and Practice
- 28 Meditation — Practice and Teaching



## CHAPTER 16

# Foundations — The Language of Anatomy and Movement

*“Āsana brings steadiness, health and lightness of body.” — Haṭha Yoga Pradīpikā 1.17*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

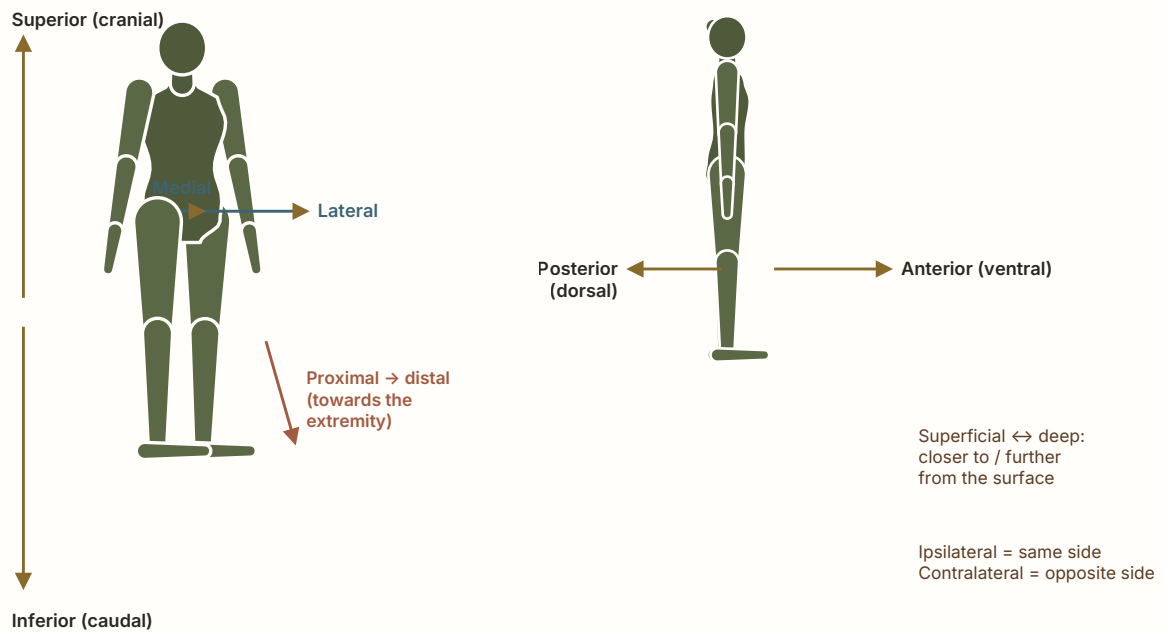
- ♦ describe anatomical position and use directional terms correctly, including in inverted postures;
- ♦ name the three cardinal planes and their axes and identify the plane of common yoga movements;
- ♦ classify joints by structure and describe the movements available at each type of synovial joint;
- ♦ name and locate the skeletal landmarks most useful for teaching;
- ♦ describe the main connective tissues and how each responds to load.

## 16.1 Why a shared language matters

“Lift your hips” means different things in cobra, shoulderstand and downward dog. Anatomical language exists to remove that ambiguity. A teacher who uses it well — mostly in training, observation and conversation with healthcare professionals rather than in class cues — can describe precisely what a student is doing, communicate with physiotherapists and doctors, and read the anatomy literature. In class you will translate it back into plain words.

## 16.2 Anatomical position

All anatomical description refers to one reference posture, **anatomical position**: standing upright, facing forward, feet together and pointing forward, arms by the sides with the palms facing forward (Moore et al., 2018). Every directional term keeps its meaning relative to this position, whatever position the body is actually in. In headstand, the head is still **superior** to the feet; in forward bend, the chest is still **anterior** to the spine.



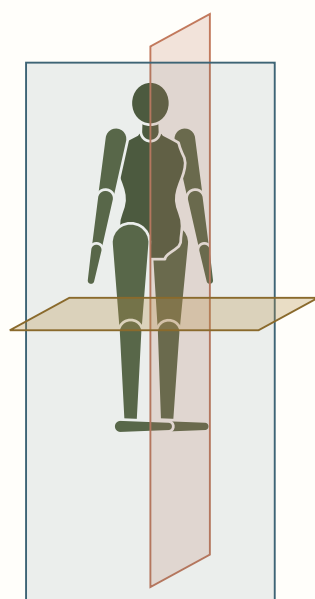
**Figure 16.1** Directional terms are always relative to anatomical position, even when the body is upside down: in headstand the head is still *superior* to the feet.

**Table 16.1** Directional terms

| Term                                            | Meaning                                        | Example in yoga                                                               |
|-------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------|
| Superior / inferior                             | towards the head / towards the feet            | "draw the shoulder blades inferiorly" = down the back                         |
| Anterior (ventral) / posterior (dorsal)         | towards the front / back                       | the hamstrings are posterior thigh muscles                                    |
| Medial / lateral                                | towards / away from the midline                | the big toe is on the medial side of the foot                                 |
| Proximal / distal                               | closer to / further from the trunk (on a limb) | the knee is proximal to the ankle                                             |
| Superficial / deep                              | nearer to / further from the surface           | transversus abdominis is deep to the obliques                                 |
| Ipsilateral / contralateral                     | same side / opposite side                      | in triangle to the right, the right hand reaches down on the ipsilateral side |
| Cranial / caudal                                | towards head / tail (used for the trunk)       | the coccyx is the most caudal part of the spine                               |
| Palmar / dorsal (hand); plantar / dorsal (foot) | palm and sole / back of hand and top of foot   | in cobra the dorsal surface of the feet rests on the floor                    |

### 16.3 Planes and axes

Movement is described in three cardinal **planes**, each perpendicular to the others, with an **axis** of rotation perpendicular to each plane (Neumann, 2017):



### Sagittal plane

divides left from right; flexion and extension (e.g. forward bends, backbends)

### Frontal (coronal) plane

divides front from back; abduction, adduction, lateral flexion (e.g. triangle, side bends)

### Transverse (horizontal) plane

divides upper from lower; rotation (e.g. twists, hip rotation)

Axes: a movement in a plane turns about the axis perpendicular to it —  
sagittal plane ↔ mediolateral axis · frontal plane ↔ anteroposterior axis ·  
transverse plane ↔ longitudinal (vertical) axis.

**Figure 16.2** Anatomical position (standing, facing forward, palms forward) with the three cardinal planes. Real yoga movements usually combine planes; the planes are a vocabulary for describing them (Neumann, 2017).

**Table 16.2** Planes, axes and yoga movements

| Plane                                                    | Axis                            | Typical movements                     | Yoga examples                                    |
|----------------------------------------------------------|---------------------------------|---------------------------------------|--------------------------------------------------|
| <b>Sagittal</b> (divides left and right)                 | mediolateral (side to side)     | flexion, extension                    | forward bends, backbends, lunges, sun salutation |
| <b>Frontal / coronal</b> (divides front and back)        | anteroposterior (front to back) | abduction, adduction, lateral flexion | triangle, side bends, star, arms rising sideways |
| <b>Transverse / horizontal</b> (divides upper and lower) | longitudinal (vertical)         | rotation                              | twists, hip rotation in lotus and pigeon         |

Real postures combine planes. Triangle is mainly a frontal-plane posture, yet the front hip also flexes (sagittal) and the trunk often rotates (transverse). Naming the planes helps you see what each part of the body is doing.

## 16.4 Joints

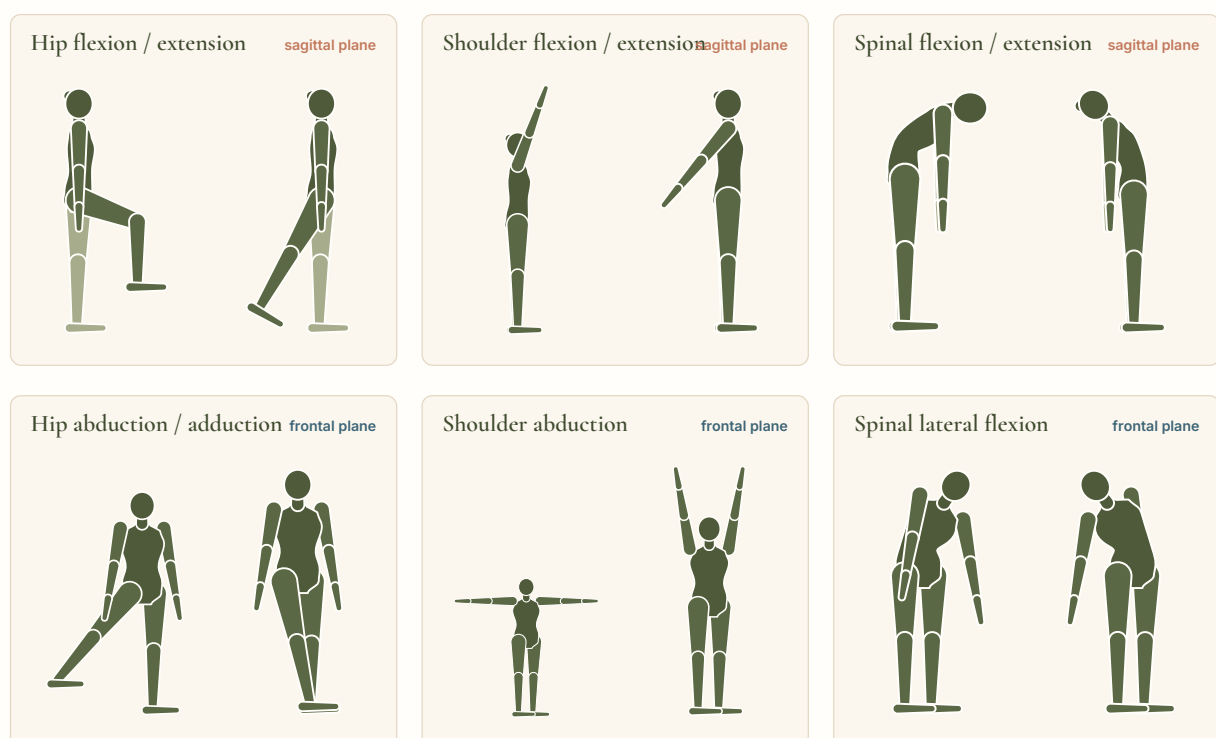
A joint (articulation) is where two or more bones meet. Joints are classified by structure (Moore et al., 2018):

- ♦ **Fibrous joints** — bones joined by dense connective tissue with little or no movement (e.g. the sutures of the skull).
- ♦ **Cartilaginous joints** — bones joined by cartilage, allowing limited movement (e.g. the intervertebral discs, the pubic symphysis).
- ♦ **Synovial joints** — bones separated by a joint cavity with lubricating synovial fluid, enclosed in a capsule and reinforced by ligaments. These are the freely movable joints of yoga practice.

**Table 16.3** Types of synovial joint

| Type                  | Movement                                            | Examples                                                                 |
|-----------------------|-----------------------------------------------------|--------------------------------------------------------------------------|
| Ball-and-socket       | movement in all three planes                        | hip, shoulder (glenohumeral)                                             |
| Hinge                 | mainly one plane (flexion/extension)                | elbow (humeroulnar), knee (with slight rotation), ankle (talocrural)     |
| Pivot                 | rotation about a long axis                          | atlanto-axial joint (turning the head), proximal radioulnar joint        |
| Condyloid (ellipsoid) | two planes (flexion/extension, abduction/adduction) | wrist (radiocarpal), knuckles (metacarpophalangeal)                      |
| Saddle                | two planes plus some rotation                       | base of the thumb (carpometacarpal)                                      |
| Plane (gliding)       | small sliding movements                             | facet joints of the spine, between carpal bones, acromioclavicular joint |

## 16.5 Joint movements



Rotation (internal/external, spinal rotation) occurs in the transverse plane and is shown in the twist and hip-rotation figures later in this section.

**Figure 16.3** Basic joint movements illustrated with the manual's figure. In each panel the left figure shows the first-named movement.

**Table 16.4** Movement terms used in this manual

| Movement | Definition                                                    | Example                                 |
|----------|---------------------------------------------------------------|-----------------------------------------|
| Flexion  | decreasing the angle between segments (in the sagittal plane) | folding at the hips in pascimottānāsana |

| Movement                                        | Definition                                                          | Example                                                         |
|-------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| Extension                                       | increasing the angle; returning from flexion                        | straightening the knees; backbending the spine                  |
| Hyperextension                                  | extension beyond the neutral or anatomical position                 | the lumbar spine in bow; “locked” knees or elbows               |
| Abduction / adduction                           | moving away from / towards the midline                              | legs apart in triangle / legs together in tādāsana              |
| Internal (medial) / external (lateral) rotation | rotating a limb inward / outward about its long axis                | the front thigh externally rotates in warrior II                |
| Circumduction                                   | circular movement combining the above                               | arm circles                                                     |
| Lateral flexion                                 | side-bending of the spine                                           | the trunk in triangle                                           |
| Rotation (spine)                                | turning of the trunk about its long axis                            | half spinal twist                                               |
| Pronation / supination                          | turning the palm down / up (forearm rotation)                       | palms down in plank / palms up in savāsana                      |
| Dorsiflexion / plantarflexion                   | lifting the top of the foot / pointing the toes                     | heels towards floor in downward dog / pointed feet in headstand |
| Inversion / eversion                            | turning the sole inward / outward                                   | balancing on the outer foot                                     |
| Elevation / depression (scapula)                | lifting / lowering the shoulder blade                               | shrugging / drawing shoulders down                              |
| Protraction / retraction (scapula)              | sliding forward around the ribs / drawing back                      | pushing the floor away in plank / squeezing the shoulder blades |
| Upward / downward rotation (scapula)            | turning the glenoid cavity up / down                                | arms overhead / hands behind the back                           |
| Nutation / counternutation (sacrum)             | small nodding of the sacrum forward / backward relative to the ilia | occurs in forward and backward bending                          |

#### ♀ TRANSLATING ANATOMY INTO CUES

“Flex your hips” becomes “fold from the hip creases”. “Depress and retract the scapulae” becomes “draw the shoulder blades gently down and back”. Use anatomical language to *think* and plain language to *teach*. Many students also respond better to external images (“press the floor away”) than to internal instructions (“contract your triceps”) (Chapter 22).

## 16.6 Landmarks you can feel

These bony landmarks can be felt through the skin and are the teacher’s reference points for observing alignment. Learn to find each on yourself and, with consent, on a partner.

**Table 16.5** Palpable landmarks for teachers

| Landmark                          | Where                                                | Why it matters in teaching                                                          |
|-----------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| C7 spinous process                | most prominent bump at the base of the neck          | reference point for neck position in shoulderstand; “lengthen the back of the neck” |
| Spine and acromion of the scapula | ridge across the shoulder blade; tip of the shoulder | observing shoulder position in plank, headstand                                     |

| Landmark                              | Where                                    | Why it matters in teaching                            |
|---------------------------------------|------------------------------------------|-------------------------------------------------------|
| Inferior angle of the scapula         | lower point of the shoulder blade        | scapular upward rotation in arm raises                |
| Sternum and xiphoid process           | breastbone and its lower tip             | "lift the sternum" in backbends; hand position in CPR |
| Iliac crest                           | top rim of the pelvis at the waist       | pelvic level in standing poses                        |
| ASIS (anterior superior iliac spine)  | front "hip points"                       | pelvic orientation in lunges, triangle, forward bends |
| PSIS (posterior superior iliac spine) | "dimples" at the low back                | sacroiliac region; pelvic tilt                        |
| Ischial tuberosities                  | "sitting bones"                          | seated postures; origin of the hamstrings             |
| Greater trochanter                    | outer bony prominence of the upper thigh | hip rotation; lying on the side                       |
| Patella and tibial tuberosity         | kneecap and bump below it                | knee tracking over the foot                           |
| Medial and lateral malleoli           | ankle bones                              | foot alignment; lotus safety                          |
| Olecranon                             | point of the elbow                       | elbow position in crow, chaturaṅga, headstand         |

## 16.7 The body's connective tissues

Yoga loads several kinds of tissue, which adapt differently (Neumann, 2017):

- ◆ **Bone** — a living, remodelling tissue that strengthens in response to regular loading and weakens with disuse. Relevant to older students and osteoporosis (**Chapter 26**).
- ◆ **Articular (hyaline) cartilage** — the smooth covering of joint surfaces; has no blood supply and is nourished by movement and loading of the joint.
- ◆ **Fibrocartilage** — tough cartilage of the menisci, intervertebral discs and pubic symphysis.
- ◆ **Ligaments** — dense connective tissue joining bone to bone; they limit movement and contribute to joint position sense. They are not designed to be stretched: a chronically overstretched ligament does not reliably return to its original length.
- ◆ **Tendons** — join muscle to bone; they adapt to load over weeks to months — more slowly than muscle.
- ◆ **Muscle** — contractile tissue that adapts rapidly in strength and in tolerance to stretch.
- ◆ **Fascia** — sheets and wrappings of connective tissue around and within muscles and organs. Fascia is an active research area; many popular claims about "fascial release" and "storing emotions in fascia" go well beyond the evidence **SCIENTIFIC**.

### ◆ STRETCH MUSCLES, RESPECT LIGAMENTS

In a well-taught posture the sensation of stretch is felt in the *belly* of a muscle, not at a joint. Sensation at the back of the knee, in the front of the shoulder joint, at the sacroiliac joints or at the sitting bones is a signal to back off and modify, not to go deeper.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ All anatomical terms refer to anatomical position, even when the body is inverted.
- ◆ The three planes — sagittal, frontal, transverse — and their axes describe flexion/extension, abduction/adduction/lateral flexion and rotation.
- ◆ Synovial joints are classified as ball-and-socket, hinge, pivot, condyloid, saddle and plane; each type allows characteristic movements.
- ◆ Palpable landmarks (C7, ASIS, PSIS, ischial tuberosities, greater trochanter, malleoli and others) are the teacher's reference points.
- ◆ Bone, cartilage, ligament, tendon, muscle and fascia each respond differently to load; ligaments should not be deliberately stretched.

 REFLECTION QUESTIONS

1. Which movement terms did you already use correctly? Which surprised you?
2. Why might anatomical language in class cues help some students and alienate others?
3. Where in your own body do you feel stretch “in the joint” rather than “in the muscle”?

 TEACHER-PRACTICE EXERCISES

1. In pairs, one partner performs a pose; the other describes it aloud using only anatomical terms (joints, movements, planes). Swap.
2. Find each landmark in Table 16.5 on yourself. With consent, find three of them on a partner.
3. Take five postures from the classical sequence and name the dominant plane of movement for each.

 KNOWLEDGE CHECK

1. In headstand, the feet are \_\_\_ to the head: (a) superior (b) inferior (c) anterior (d) proximal
2. Twists occur mainly in which plane? (a) sagittal (b) frontal (c) transverse (d) all equally
3. The knee is best described as a: (a) ball-and-socket joint (b) hinge joint with slight rotation (c) plane joint (d) saddle joint
4. Pointing the toes is: (a) dorsiflexion (b) plantarflexion (c) eversion (d) supination
5. Which tissue should *not* be deliberately stretched? (a) muscle (b) ligament (c) fascia around muscle (d) none

Answers 1 b · 2 c · 3 b · 4 b · 5 b

 TEACHING ASSIGNMENT

Choose three postures from different planes of movement. For each, write a two-column description: on the left the anatomical description of what each major joint is doing; on the right the plain-language cue you would give a beginner for the same action. Teach one of the postures to a partner using only your right-hand column.

## CHAPTER 17

# The Skeletal System in Practice

*“Alignment serves the student’s body; the student’s body does not serve an ideal image.” — a working principle of this manual*

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## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

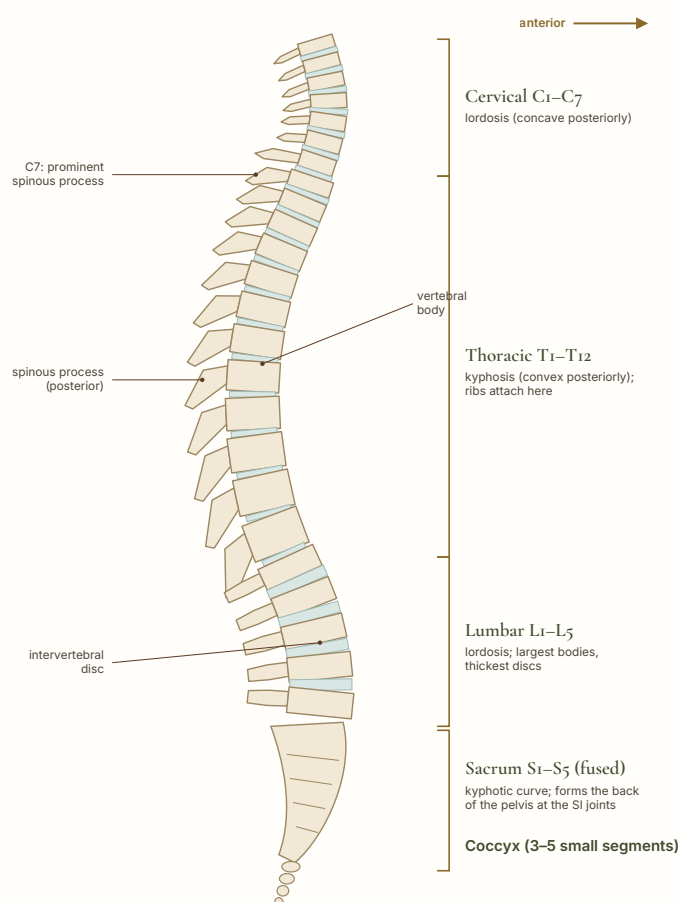
- ♦ describe the regions, curves and structure of the spine and how movement differs between regions;
- ♦ explain the structure and function of the pelvis and sacroiliac joints, and the meaning of pelvic tilt;
- ♦ describe the rib cage, shoulder girdle, hip, knee, ankle, foot, elbow, wrist and hand in terms relevant to yoga;
- ♦ explain how differences in bone shape between individuals affect what postures look like;
- ♦ relate skeletal anatomy to specific postures in the classical sequence.

## 17.1 The skeleton at a glance

The adult skeleton typically has 206 bones: 80 in the **axial skeleton** (skull, vertebral column, ribs and sternum) and 126 in the **appendicular skeleton** (the limbs and the girdles that attach them) (Moore et al., 2018). In yoga the axial skeleton is the axis around which postures are organised; the appendicular skeleton supplies the base of support and the levers.



## 17.2 The spine



**Figure 17.1** The vertebral column in lateral view (schematic; anterior to the right). Typical counts are 7 cervical, 12 thoracic and 5 lumbar vertebrae, a sacrum of 5 fused segments and a small coccyx; variation exists between individuals (Moore et al., 2018).

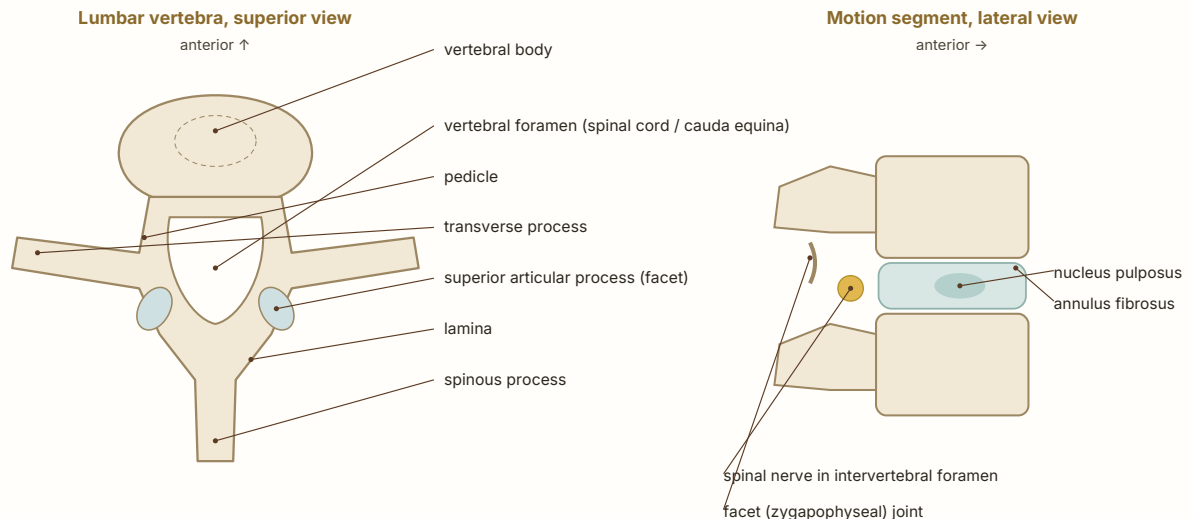
### Regions and curves

The vertebral column usually consists of 7 cervical, 12 thoracic and 5 lumbar vertebrae, the sacrum (5 fused vertebrae) and the coccyx (3–5 small fused segments) (Moore et al., 2018). In a newborn the whole spine forms a single forward-concave curve (the **primary** curve). As the child lifts its head and later stands, **secondary** curves develop in the neck and lower back. The adult spine therefore has four curves: cervical **lordosis**, thoracic **kyphosis**, lumbar **lordosis** and sacral kyphosis.

The curves are not defects to be flattened; they allow the spine to absorb load like a spring and to balance the head over the pelvis. Their depth varies considerably between healthy people. The cue “flatten your back” is anatomically misleading; “lengthen the spine” or “let the spine find its natural curves” is closer to the truth.

### The vertebra and the motion segment

A typical vertebra has a weight-bearing **body** in front and a **vertebral arch** behind, which encloses the vertebral foramen through which the spinal cord passes. Processes project from the arch: a **spinous process** backward (the bumps you feel along the back), two **transverse processes** sideways, and four **articular processes** forming the **facet (zygapophyseal) joints** with the vertebrae above and below (Bogduk, 2005).



**Figure 17.2** A lumbar vertebra from above and a spinal motion segment from the side (schematic). Each segment moves at three joints: the intervertebral disc in front and the paired facet joints behind (Bogduk, 2005).

Two adjacent vertebrae and the tissues between them form a **motion segment**: the intervertebral disc in front, two facet joints behind, and the ligaments and muscles around them. Spinal nerves leave the canal through the **intervertebral foramina** between each pair of vertebrae.

### The intervertebral discs

Each disc has a tough outer ring of fibrocartilage, the **annulus fibrosus**, and a gel-like centre, the **nucleus pulposus**. Discs make up roughly a quarter of the length of the column above the sacrum and allow movement between vertebral bodies while transmitting load (Moore et al., 2018). They have almost no blood supply in adults and are nourished by diffusion, assisted by the changes in pressure that come with movement and rest.

#### 🔍 SPINAL LOADING AND POSTURE

In a frequently cited study, pressure within a lumbar disc was measured with a sensor in a single healthy volunteer during daily activities **SCIENTIFIC** (Wilke et al., 1999). Pressure was lowest lying down, similar in relaxed standing and relaxed sitting, roughly doubled when bending forward while standing, and much higher when lifting a weight with the back rounded. Such measurements help explain why loaded spinal flexion deserves care — but they come from very few people, and the human spine adapts to the loads it regularly meets. Imaging studies also show that disc bulges and “degenerative” changes are extremely common in people with no back pain at all, rising with age (Brinjikji et al., 2015); findings on a scan do not by themselves predict pain.

### How the regions move

The shape and orientation of the facet joints, the thickness of the discs and the presence of the ribs give each region its own pattern of movement (Neumann, 2017; Bogduk, 2005):

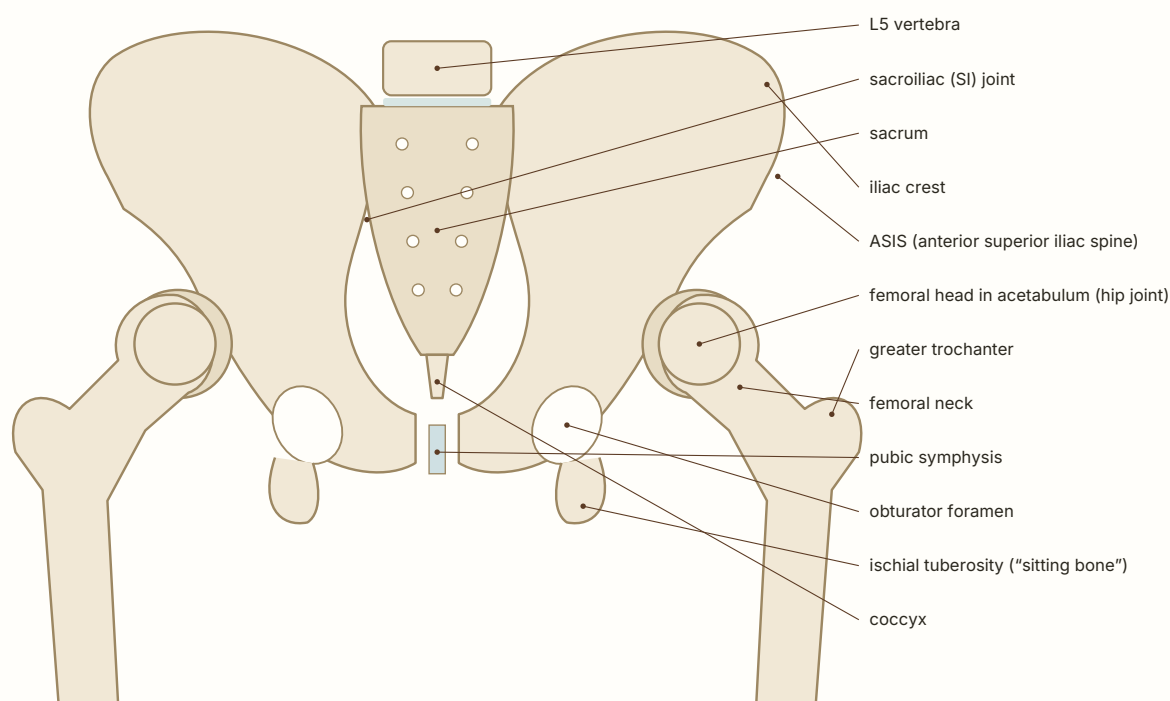
**Table 17.1** Movement of the spinal regions

| Region          | Freest movements                                                                                     | Most limited                                                  | Yoga relevance                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Cervical</b> | flexion, extension, rotation (about half of neck rotation occurs between C1 and C2), lateral flexion | — (most mobile region)                                        | vulnerable in inversions and deep backbends; shoulderstand and plough flex it strongly   |
| <b>Thoracic</b> | rotation, lateral flexion                                                                            | extension and flexion (limited by ribs and facet orientation) | where most twisting should happen; often stiff in extension — the true work of backbends |
| <b>Lumbar</b>   | flexion and extension                                                                                | rotation (only a few degrees in total)                        | backbends tend to concentrate here; do not “twist from the lower back”                   |
| <b>Sacrum</b>   | small nutation / counternutation at the SI joints                                                    | —                                                             | moves with the pelvis; see below                                                         |

#### ◆ SHARE THE MOVEMENT

The most common pattern of strain in yoga is movement concentrating where the spine is already most mobile — the lower neck and lower lumbar spine — while the stiffer thoracic spine does little. Good teaching distributes the movement: lengthen before bending, lift the chest in backbends, rotate the rib cage in twists, and keep the neck in line with the rest of the spine.

## 17.3 The pelvis and sacroiliac joints



**Figure 17.3** The pelvis and hip joints, anterior view (schematic). The pelvis is a ring of two hip bones and the sacrum; the femoral head sits deep in the acetabulum. Depth and orientation of the socket, and the angle of the femoral neck, vary considerably between people (Clark, 2016; Neumann, 2017).

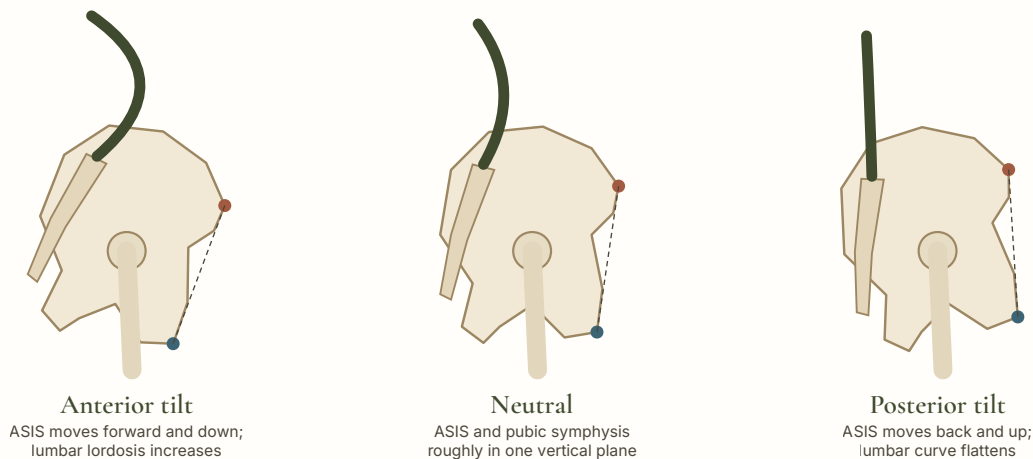
The pelvis is a ring formed by the two **hip bones** (each fused from ilium, ischium and pubis) and the **sacrum**. In front, the hip bones meet at the cartilaginous **pubic symphysis**; behind, each meets the sacrum at a **sacroiliac (SI) joint**. The SI joints are strong, irregular joints bound by some of the body's strongest ligaments; they move only a little — a few degrees at most — but that small movement matters for shock absorption **SCIENTIFIC** (Vleeming et al., 2012).

## Pelvic tilt

The pelvis sits on the heads of the femurs and can rotate forward and backward on them. This rotation, **pelvic tilt**, directly changes the lumbar curve, because the sacrum is the base of the spine.

PELVIC TILT · lateral view, facing right · rotation occurs at the hip joints

● ASIS ● pubic symphysis



**Figure 17.4** Pelvic tilt and its effect on the lumbar curve (schematic). Much “alignment” teaching in forward bends and backbends is really about where the pelvis is allowed to move.

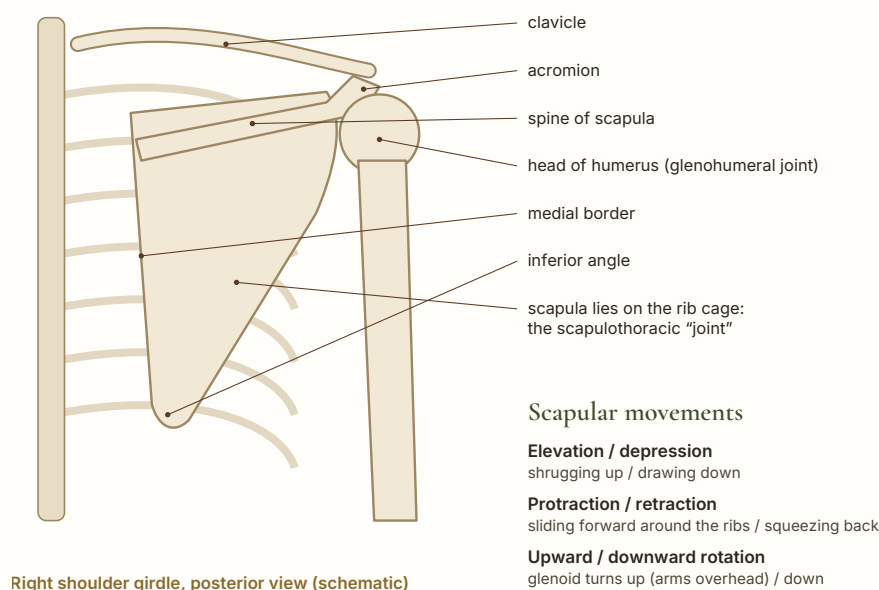
- ♦ **Anterior tilt** (ASIS forward and down) increases the lumbar lordosis. It happens in the lift of the chest in cobra and at the start of a forward bend when the pelvis “leads”.
- ♦ **Posterior tilt** (ASIS back and up; tailbone under) flattens the lumbar curve. It happens when tight hamstrings pull the sitting bones down in a seated forward bend, forcing the lower back to round.

Understanding pelvic tilt explains why a student with tight hamstrings is taught to sit on a folded blanket in *pāścimottānāsana*: raising the pelvis lets it tilt forward, so the fold can come from the hips rather than from a rounded lower back.

## 17.4 The rib cage

Twelve pairs of ribs articulate with the thoracic vertebrae. Ribs 1–7 attach directly to the sternum by their own costal cartilage (“true” ribs); ribs 8–10 attach indirectly through the cartilage above (“false” ribs); ribs 11–12 end free in the abdominal wall (“floating” ribs) (Moore et al., 2018). The rib cage protects the heart and lungs, stiffens the thoracic spine, and moves with every breath (Chapter 21). Because the ribs connect to the thoracic vertebrae, a stiff rib cage limits thoracic extension and rotation — one reason breathing practice and backbends support each other.

## 17.5 The shoulder girdle



**Figure 17.5** The shoulder girdle is a chain: sternum → clavicle → scapula → humerus. Raising the arms overhead needs both glenohumeral movement and upward rotation of the scapula (scapulohumeral rhythm) (Neumann, 2017).

The arm attaches to the trunk only through the **clavicle** at the sternoclavicular joint. The shoulder complex has four parts: the sternoclavicular joint, the acromioclavicular joint, the **glenohumeral joint** (the ball-and-socket joint between the humerus and the shallow glenoid of the scapula) and the **scapulothoracic** “joint” — the gliding of the scapula over the rib cage (Neumann, 2017).

The glenohumeral joint is the most mobile joint in the body: its socket is small relative to the head of the humerus and deepened only by a rim of cartilage, the labrum. That mobility is bought at the cost of stability, which depends largely on muscles, especially the rotator cuff (Chapter 19). When the arm rises overhead, roughly two degrees of glenohumeral movement accompany every one degree of upward rotation of the scapula — the classic **scapulohumeral rhythm** (Neumann, 2017).

### 🔑 THE SHOULDERS IN WEIGHT-BEARING

In plank, downward dog, crow and headstand the shoulder girdle carries body weight. Two cues protect it: keep the shoulder blades **broad and gently protracted** (not collapsing between them) and keep the shoulders **away from the ears** (not shrugged or sagging). Students who have trouble raising the arms fully overhead often compensate by arching the lower back — watch for it in hasta uttānāsana and headstand.

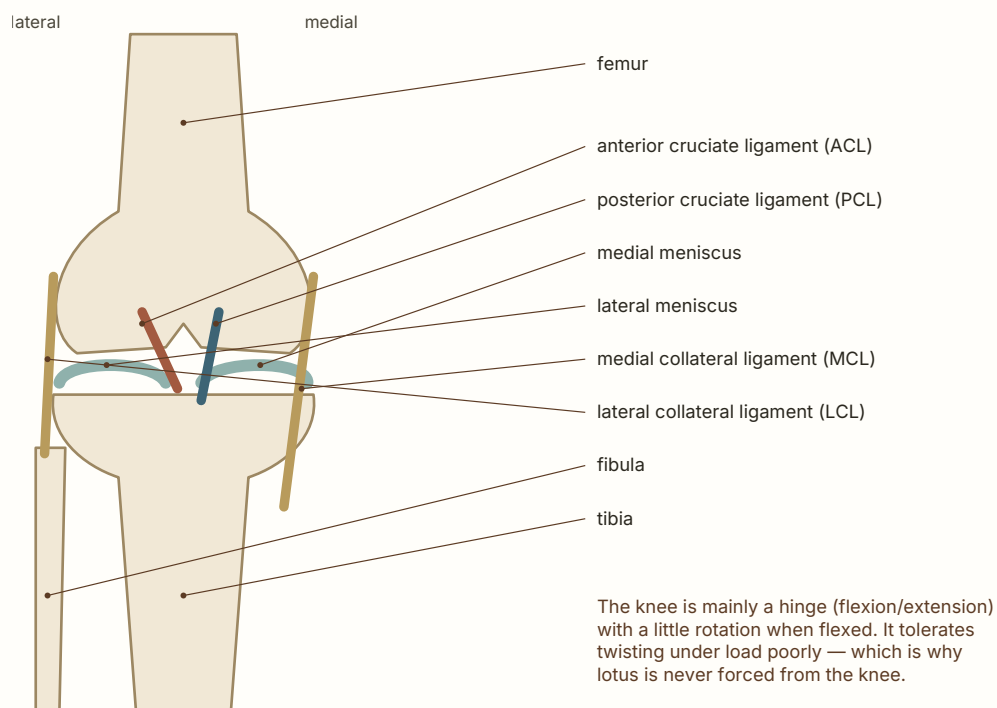
## 17.6 The hip joint

The hip is a deep ball-and-socket joint: the femoral head sits well inside the acetabulum, held by a thick capsule and strong ligaments, making it far more stable — and less mobile — than the shoulder. Its movements are flexion, extension, abduction, adduction and internal and external rotation.

The shape of the hip varies strikingly between people (Clark, 2016; Neumann, 2017). The angle between the femoral neck and shaft averages about 125° but ranges widely; the femoral neck is usually angled slightly forward (anteversion, averaging roughly 15°), again with wide variation; and the depth and direction of the socket differ. These differences — present from birth, not the result of practice — mean that for some people lotus or wide splits are impossible without injury, however long they stretch, because **bone meets bone**.

## 17.7 The knee

The knee is essentially a hinge between femur and tibia, with a little rotation available when it is bent. Two C-shaped **menisci** deepen the flat tibial plateau; the **cruciate ligaments** (anterior and posterior) prevent forward and backward sliding of the tibia; the **collateral ligaments** limit sideways bending (Moore et al., 2018). The kneecap (patella) glides in a groove on the femur within the quadriceps tendon.

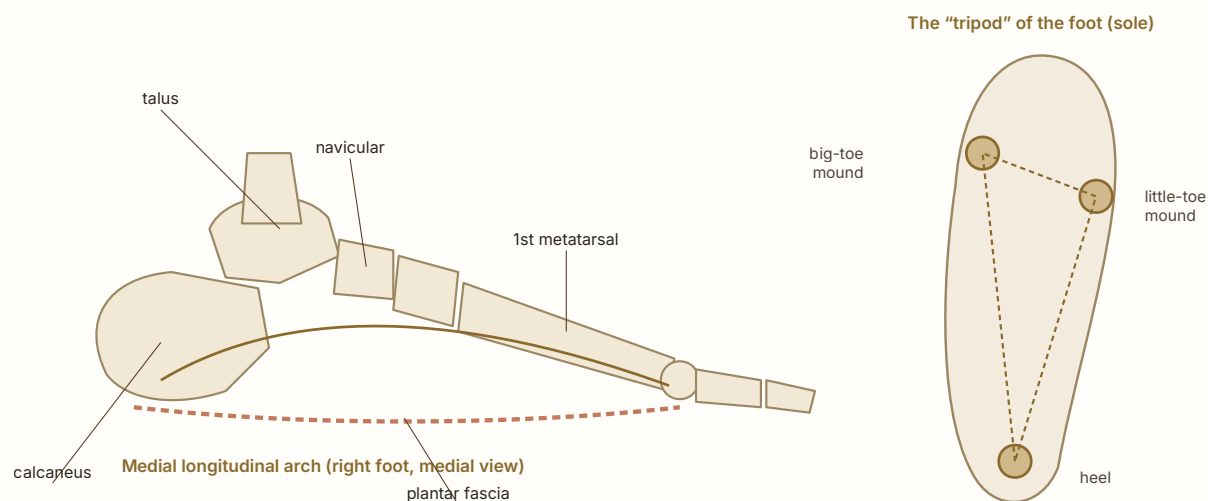


**Figure 17.6** Right knee, anterior view with the patella removed (schematic). The menisci deepen the flat tibial plateau; the cruciate and collateral ligaments limit forward/backward and side-to-side movement of the tibia on the femur (Moore et al., 2018).

The knee is the joint most at risk when a student forces the hip. In lotus, if the hip does not rotate outward sufficiently, the twisting force goes to the knee — and the knee, a hinge, cannot rotate safely under load. **The rule is: rotation comes from the hip; the knee is protected.**

## 17.8 The ankle and foot

Each foot has 26 bones (Moore et al., 2018). The **talocrural (ankle) joint** allows dorsiflexion and plantarflexion; the **subtalar joint** below it allows inversion and eversion. The foot's arches — medial longitudinal, lateral longitudinal and transverse — are shaped by bone and ligament and supported dynamically by muscle.



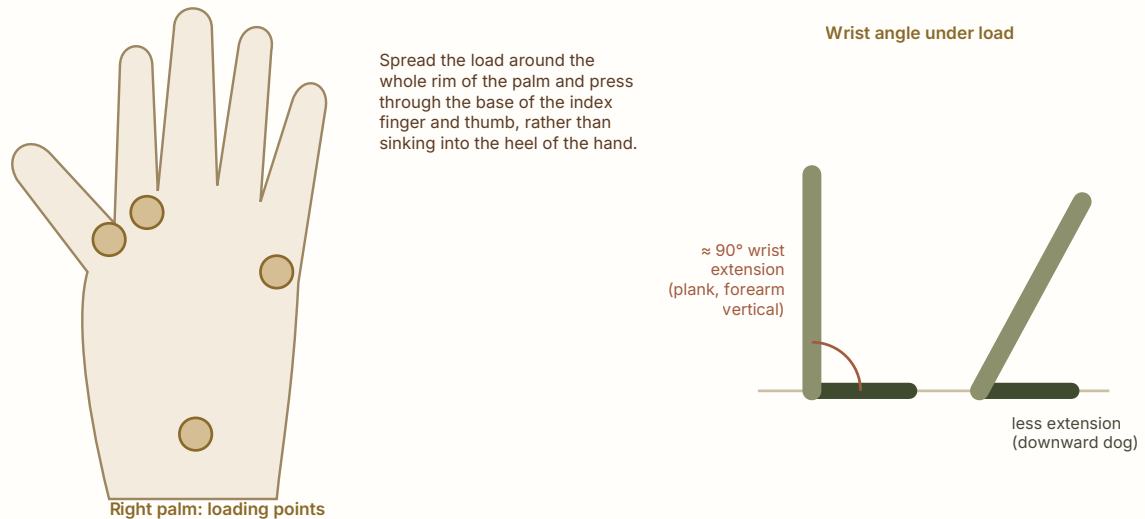
**Figure 17.7** The foot as a foundation: the medial arch (left) and the three weight-bearing points often cued in standing poses (right). Arch height varies widely and a low arch is not in itself a problem.

In standing postures the foot is the foundation. Cueing weight evenly through three points — heel, big-toe mound and little-toe mound — gives a stable base. Arch height varies greatly between healthy people; a low arch is not in itself a problem to be corrected.

## 17.9 The elbow, wrist and hand

The **elbow** combines a hinge (between humerus and ulna) with a pivot (between radius and ulna) that allows pronation and supination. Many people, especially the hypermobile, can hyperextend the elbow; in weight-bearing postures they should keep a micro-bend so that load is carried by muscle rather than by the joint's end range.

The **wrist and hand** contain 27 bones: 8 carpals, 5 metacarpals and 14 phalanges (Moore et al., 2018). The wrist was not designed as a weight-bearing joint in the way the ankle was. Plank, crow and other arm balances place it in strong extension under load (Figure 17.8).



**Figure 17.8** Weight-bearing through the hands. Plank and crow place the wrist near the end of its extension range under load; commonly reported active wrist extension is around 70°, though individuals differ (Neumann, 2017). Spreading load and building tolerance gradually are the practical responses.

## 17.10 Bones, variation and teaching

### ◆ TENSION, COMPRESSION AND THE SHAPE OF BONES

When a movement stops, the limit is either **tension** — tissues on the opposite side being pulled — or **compression** — tissues or bones meeting on the same side (Clark, 2016). Tension can often change with practice; compression by bone cannot. A teacher who assumes every limit is “tightness” will push some students into bone-on-bone compression. Ask: *Where do you feel it?* Sensation on the side being stretched suggests tension; a hard, blocked feeling on the side the body is folding towards suggests compression.

This is the foundation of the approach to alignment taken in this manual: **alignment serves the student's body; the student's body does not serve an ideal image.** The illustrations in this book show one possible shape of each posture, not the correct one.



## CHAPTER REVIEW

## 🍃 KEY TAKEAWAYS

- ◆ The spine's four curves are normal and variable; the cervical and lumbar regions are most mobile, the thoracic region most limited in flexion–extension but freest in rotation.
- ◆ Discs and facet joints make up each motion segment; loaded flexion raises disc pressure, but imaging changes are common without pain.
- ◆ The pelvis tilts on the femoral heads, changing the lumbar curve; tight hamstrings cause posterior tilt in forward bends.
- ◆ The shoulder trades stability for mobility and relies on muscle; the hip is deep and stable, and its bone shape varies widely.
- ◆ The knee is a hinge that must be protected from twisting; the foot and hand are foundations whose loading can be taught.
- ◆ Movement limits come from tension or compression; bone shape sets a limit no amount of stretching can change.

## 🕯 REFLECTION QUESTIONS

1. Which region of your spine moves most easily? Least? How does this show in your backbends?
2. Have you ever felt a “hard stop” in a posture? What did your teacher say about it?
3. How does the idea that bone shape varies change the way you feel about poses you cannot do?

## ✋ TEACHER-PRACTICE EXERCISES

1. Sit on the floor and then on a folded blanket; in each case, fold forward and note what happens to your lower back. Explain to a partner using “pelvic tilt”.
2. With a partner, observe arm-raising from behind: watch the scapulae rotate upward. Then observe someone who lifts the arms by arching the lower back.
3. Test the “where do you feel it?” question in three hip postures (e.g. butterfly, pigeon preparation, lotus preparation). Classify each limit as tension or compression.

## ☑ KNOWLEDGE CHECK

1. How many lumbar vertebrae are typical? (a) 4 (b) 5 (c) 7 (d) 12
2. Which region of the spine permits the least rotation? (a) cervical (b) thoracic (c) lumbar (d) they are equal
3. Tight hamstrings tend to pull the pelvis into: (a) anterior tilt (b) posterior tilt (c) lateral tilt (d) rotation
4. Floating ribs are ribs: (a) 1–7 (b) 8–10 (c) 11–12 (d) 1–2
5. In lotus, if the hip lacks external rotation, the joint most at risk is the: (a) ankle (b) knee (c) sacroiliac joint (d) lumbar spine

Answers 1 b · 2 c · 3 b · 4 c · 5 b

## 📄 TEACHING ASSIGNMENT

Choose one of the twelve basic postures. Prepare a one-page “skeletal briefing” for fellow trainees: the position of each major joint, which spinal regions move, where tension and compression are likely, and one way that differences in bone shape might change what the posture looks like in two different students. Present it in three minutes.

## CHAPTER 18

# How Muscles Work — Contraction, Flexibility and Strength

*“It is perfected by relaxation of effort.” — Yoga Sūtra 2.47*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the structure of skeletal muscle and how it produces force;
- ♦ distinguish concentric, eccentric and isometric contraction and identify each in yoga postures;
- ♦ explain the roles of agonist, antagonist, synergist and stabiliser, and the idea of open and closed chains;
- ♦ explain the stretch reflex, reciprocal inhibition and the current understanding of how flexibility changes;
- ♦ apply evidence-based principles of stretching and strengthening to yoga practice.

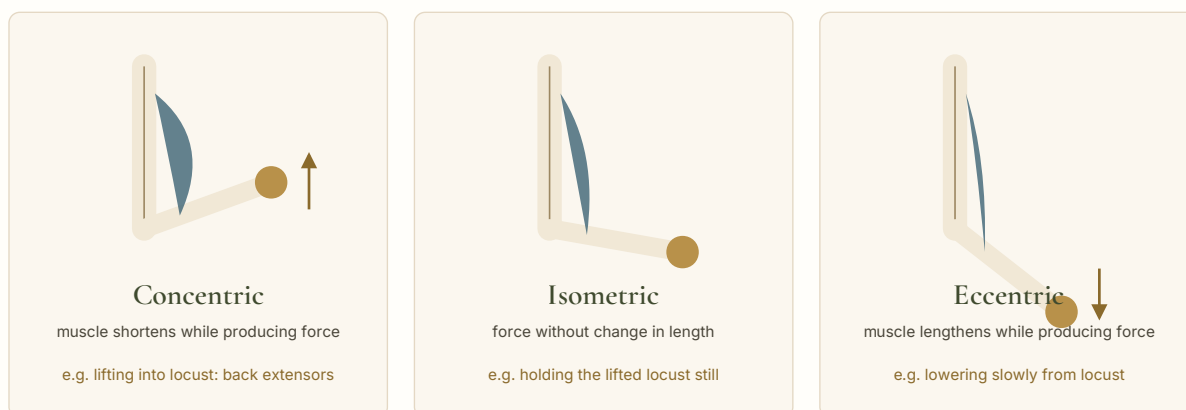
## 18.1 What a muscle is

A skeletal muscle is an organ made of thousands of long cells, the **muscle fibres**, bundled into fascicles and wrapped at every level in connective tissue that merges at each end into the tendon. Inside each fibre, repeating units called **sarcomeres** contain overlapping filaments of the proteins actin and myosin. When the nervous system signals a fibre to contract, myosin heads bind to actin and pull the filaments past each other — the **sliding-filament** mechanism — shortening the sarcomere and generating force (Hall & Hall, 2021).

A single motor neuron and all the muscle fibres it controls form a **motor unit**. The nervous system grades force by recruiting more motor units and by firing them faster. A muscle does not “switch on” all at once; fine control in balances and slow transitions depends on this grading.

Muscles can only **pull**. They produce movement by pulling on bones across joints, and they are arranged in opposing groups so that every movement can be reversed.

## 18.2 Types of contraction



**Figure 18.1** Three kinds of muscle contraction, shown for the elbow flexors for clarity; the yoga examples beneath each apply the same idea to the back extensors in locust.

- ◆ **Concentric** contraction: the muscle shortens while producing force — lifting into locust (back extensors), rising from a squat (quadriceps and gluteus maximus).
- ◆ **Isometric** contraction: the muscle produces force without changing length — holding the locust, holding plank, holding warrior II. Much of a classical class is isometric work.
- ◆ **Eccentric** contraction: the muscle lengthens while producing force, controlling a movement against gravity — lowering slowly from locust, lowering the legs in double leg raises, descending from plank to the floor.

Muscles can produce more force eccentrically than concentrically, and eccentric work is particularly effective for building strength and tendon capacity — but it also produces more post-exercise soreness in the unaccustomed (Hall & Hall, 2021). Slow, controlled lowering is one of the most valuable and least used tools in a posture class.

### ♀ GRAVITY DECIDES WHICH MUSCLE WORKS

To know which muscle is working, ask which way gravity is pulling. In lowering the legs from double leg raise, the hip flexors work *eccentrically* against gravity — the hip extensors are not what lowers the legs. In a standing forward bend, the hamstrings and back extensors control the descent eccentrically; once the torso hangs, they may largely relax and gravity does the work.

## 18.3 Roles muscles play

**Table 18.1** Muscle roles

| Role                  | Definition                                            | Example: lifting the arms overhead                |
|-----------------------|-------------------------------------------------------|---------------------------------------------------|
| Agonist (prime mover) | the main muscle producing a movement                  | anterior deltoid (shoulder flexion)               |
| Antagonist            | the muscle opposing the movement, which must lengthen | latissimus dorsi, pectoralis major (lower fibres) |

| Role                 | Definition                                                          | Example: lifting the arms overhead                          |
|----------------------|---------------------------------------------------------------------|-------------------------------------------------------------|
| Synergist            | assists the agonist or neutralises unwanted actions                 | serratus anterior and trapezius rotating the scapula upward |
| Stabiliser (fixator) | holds a neighbouring segment steady so that the movement can happen | trunk muscles preventing the lower back from arching        |

**Open and closed chains.** When the distal end of a limb is free (lifting the leg in leg raises) the chain is *open*; when it is fixed against the floor (standing, plank) the chain is *closed* and the same muscles may move the trunk on the limb rather than the limb on the trunk. In warrior I the hip flexors of the back leg act in a closed chain on the pelvis — which is why tight hip flexors tip the pelvis forward.

## 18.4 Length and tension

A muscle produces its greatest force at intermediate lengths, where actin and myosin overlap optimally; at very short or very long lengths it is weaker (the **length–tension relationship**) (Hall & Hall, 2021). This has two practical consequences:

- ♦ **Active insufficiency.** A muscle that crosses two joints cannot shorten fully over both at once. Try bending the knee strongly while the hip is extended: the hamstrings cramp or weaken because they are already short at the hip.
- ♦ **Passive insufficiency.** A two-joint muscle cannot be fully lengthened over both joints at once. The hamstrings limit a forward bend with straight legs far more than with bent knees. This is why bending the knees in pascimottānāsana or pāda hastāsana immediately allows the hips to flex further — and why it is a legitimate modification, not a cheat.

## 18.5 Reflexes and the nervous control of stretch

Muscles contain **muscle spindles**, sensory receptors that detect changes in muscle length and its rate of change. When a muscle is stretched quickly, the spindles trigger a reflex contraction of the same muscle — the **stretch reflex**. This is one reason why bouncing into a stretch is counter-productive: the muscle resists. **Golgi tendon organs**, in the tendons, sense tension. The nervous system also coordinates opposing muscles through **reciprocal inhibition**: when an agonist contracts, its antagonist is reflexly inhibited, so that, for instance, actively contracting the quadriceps can help the hamstrings to yield (Kandel et al., 2021).

These reflexes are real and useful, but popular accounts exaggerate their role in long-term flexibility. Contract–relax (“PNF”) stretching, often explained by tendon-organ inhibition, appears to work largely through increased tolerance to stretch (Weppeler & Magnusson, 2010).

## 18.6 How flexibility actually changes

It is widely assumed that regular stretching makes muscles physically longer. The evidence suggests otherwise, at least in the short and medium term. Several weeks of stretching increase range of motion mainly by increasing **tolerance to the sensation of stretch** — the nervous system allows the muscle to go

further before signalling “stop” — with little measurable change in the muscle’s mechanical properties **SCIENTIFIC** (Magnusson et al., 1996; Wepppler & Magnusson, 2010). Structural changes may occur over longer periods, and the question is not fully settled.

#### ◆ FLEXIBILITY IS PARTLY A SKILL OF THE NERVOUS SYSTEM

Much of what students experience as “getting more flexible” is the nervous system learning that a range is safe. This is good news: it responds to calm, regular, unforced practice — to breath, relaxation and trust — exactly the qualities a classical class cultivates. It also means that forcing, which alarms the nervous system, tends to be counter-productive.

## Stretching principles

**Table 18.2** Evidence-informed principles for stretching

| Principle                 | Guidance                                                                                                                                             | Source                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Frequency                 | regular practice (at least 2–3 days a week; daily is more effective)                                                                                 | Garber et al. (2011)    |
| Duration                  | holding 10–30 seconds is effective for most adults; older adults may benefit from 30–60 seconds; about 60 seconds total per muscle group per session | Garber et al. (2011)    |
| Intensity                 | to the point of mild tension or discomfort, not pain                                                                                                 | Garber et al. (2011)    |
| Before explosive activity | long static holds (over about 60 seconds) immediately before power or speed tasks can slightly reduce performance; shorter holds have little effect  | Behm & Chaouachi (2011) |
| Warm tissue               | stretching is more comfortable after a warm-up                                                                                                       | general consensus       |

Yoga postures are usually held longer than the minimum and combine stretching with isometric strength, balance and breathing, so they do not map neatly onto these guidelines — but the principles of regularity, moderate intensity and patience apply fully.

#### ② “MORE FLEXIBILITY IS ALWAYS BETTER.”

Flexibility beyond what one’s activities require has no demonstrated health benefit, and some researchers argue that flexibility should not be treated as a major component of fitness at all (Nuzzo, 2020). For hypermobile students, *more* range can mean *less* stability (Chapter 26). The yogic aim is freedom of movement and ease in the body — not maximum range.

## 18.7 Strength in yoga

Muscle strength develops through **progressive overload** — gradually increasing the demand — and is **specific**: muscles become stronger in the ranges and types of contraction in which they are trained (Garber et al., 2011). Yoga develops strength mainly through isometric holds and slow concentric and eccentric movement against body weight: chaturaṅga, plank, crow, locust and the leg raises of the classical class all build meaningful strength. Yoga is less effective for building maximal strength or bone density in the way that heavy resistance training does; for most students it is best seen as one part of a physically active life that also includes walking and strength training (Bull et al., 2020).

### 🔗 STRENGTH AND FLEXIBILITY TOGETHER

Build strength at the ends of the range you are gaining. In practice: after passively stretching the hamstrings in a forward bend, practise lifting the leg actively in leg raises. The nervous system is more willing to allow a range it can control.

## 18.8 Soreness

Unaccustomed exercise, especially eccentric work, commonly produces **delayed-onset muscle soreness** (DOMS), peaking one to three days later and resolving within about a week. It is normal after the first days of an intensive training course. Soreness that is sharp, localised to a joint, accompanied by swelling, or that worsens rather than eases should be reported (Chapter 26).

### CHAPTER REVIEW

#### 📌 KEY TAKEAWAYS

- ◆ Muscle fibres shorten by the sliding of actin and myosin; force is graded by recruiting motor units; muscles can only pull.
- ◆ Contractions are concentric (shortening), isometric (holding) or eccentric (lengthening under control); gravity decides which muscles work.
- ◆ Muscles act as agonists, antagonists, synergists and stabilisers; two-joint muscles show active and passive insufficiency.
- ◆ The stretch reflex resists fast stretch; flexibility gains in the medium term come mainly from increased stretch tolerance.
- ◆ Regular, moderate, unforced stretching and progressive strength work are the evidence-based foundations; more flexibility is not always better.

#### 🍷 REFLECTION QUESTIONS

1. When you lower from a posture, do you control it or drop? What would change if you lowered eccentrically?
2. Has your flexibility changed quickly in a class and then returned to normal next day? What does this suggest?
3. Where in your practice would more strength serve you better than more flexibility?

#### 🙏 TEACHER-PRACTICE EXERCISES

1. Demonstrate the difference between a forward bend with straight legs and with bent knees; explain passive insufficiency to a partner.
2. In pairs, identify the type of contraction in the main working muscles in each phase of double leg raise.
3. Teach a partner a hamstring stretch using contract-relax and explain honestly why it seems to work.

 **KNOWLEDGE CHECK**

1. In the lowering phase of double leg raise, the hip flexors contract: (a) concentrically (b) eccentrically (c) isometrically (d) not at all
2. The stretch reflex is triggered by: (a) Golgi tendon organs (b) muscle spindles (c) ligaments (d) the vagus nerve
3. Bending the knees in a forward bend increases hip flexion mainly because of: (a) active insufficiency of the quadriceps (b) passive insufficiency of the hamstrings (c) reciprocal inhibition (d) compression
4. Short-term flexibility gains are thought to come mainly from: (a) muscle lengthening (b) increased stretch tolerance (c) ligament stretching (d) fascial release
5. According to ACSM guidance, a stretch should be held for about: (a) 2–5 seconds (b) 10–30 seconds (c) 5 minutes (d) as long as possible

Answers 1b · 2b · 3b · 4b · 5b

 **TEACHING ASSIGNMENT**

Design a 20-minute sequence for a student who wants to improve their seated forward bend. Include at least one strengthening element, one element that works with stretch tolerance (breath, relaxation, time), and one use of passive insufficiency. Explain the reasoning in a paragraph and teach it to a partner.



## CHAPTER 19

## A Yoga Teacher's Muscle Atlas

*“Know the muscle, then forget it, and teach the person.” — a teaching maxim*

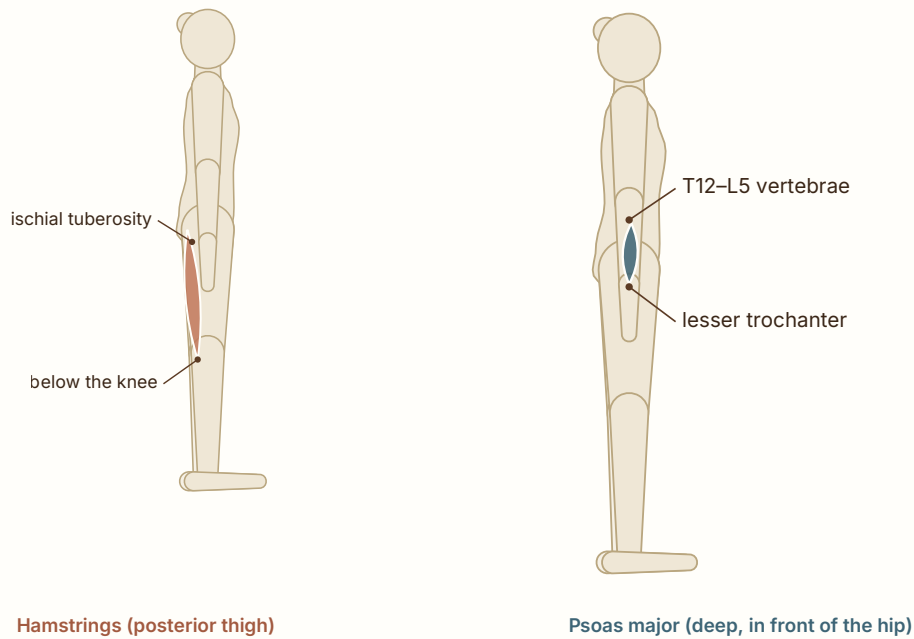
## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ state the origin, insertion and primary actions of the major muscles used in yoga;
- ◆ explain the role of each muscle in specific asanas, and whether it is typically lengthened or working;
- ◆ decide whether a student is more likely to need strengthening or lengthening of a muscle group, and how to offer each;
- ◆ correct common misconceptions about muscles taught in yoga classes;
- ◆ use the atlas as a reference when planning sequences.

## 19.1 How to use this atlas

Each card gives a muscle's attachments and actions, followed by the information a yoga teacher actually needs: its role in practice, whether practice typically lengthens or strengthens it, the key asanas that involve it and a common misconception. The small figure beside each card shows the approximate location of the muscle on the manual's figure, coloured **blue** where it is typically *working* and **terracotta** where it is typically *lengthening* in the postures named. These are location sketches, not dissections; consult Standring (2021) or Moore et al. (2018) for detailed anatomical drawings, and Long (2009) and Kaminoff & Matthews (2021) for yoga-oriented anatomical illustration.



**Figure 19.1** Location sketches of two muscles that dominate yoga conversations. The hamstrings cross the back of hip and knee; the psoas runs from the lumbar spine, in front of the hip joint, to the femur. Sketches are schematic; see the muscle atlas for attachments.

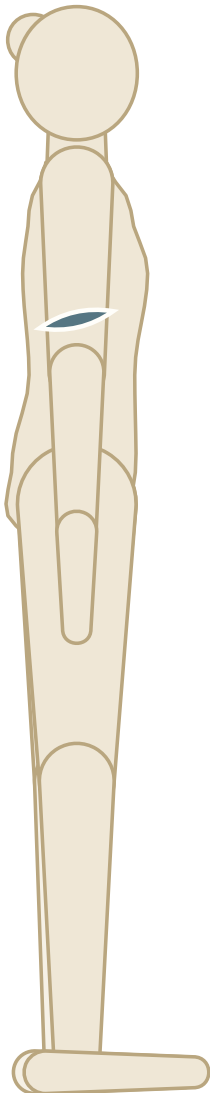
Remember two principles from **Chapter 18** as you read. First, **no muscle acts alone**: every posture is produced by patterns of coordinated activity, and naming one muscle is always a simplification. Second, **“tight” is a sensation, not a diagnosis**: the feeling of tightness can reflect muscle length, stretch tolerance, protective tension, bone shape or fatigue.

## 19.2 Breathing and trunk

The diaphragm and abdominal wall work together in breathing, in regulating intra-abdominal pressure and in moving and stabilising the trunk.

## Diaphragm

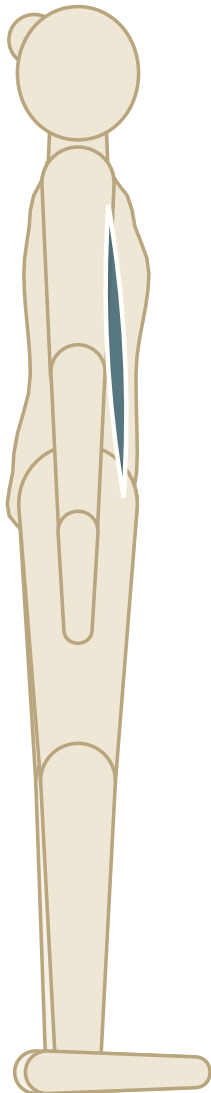
### DIAPHRAGMA



|                               |                                                                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Xiphoid process; inner surfaces of the lower six costal cartilages and ribs; lumbar vertebrae L1–L3 via the right crus and L1–L2 via the left crus, and the arcuate ligaments.                                                                        |
| <b>Insertion</b>              | Its own central tendon.                                                                                                                                                                                                                               |
| <b>Primary actions</b>        | Principal muscle of inspiration: contracts and descends, enlarging the thorax. Helps raise intra-abdominal pressure (with the abdominal wall and pelvic floor) for posture, lifting, coughing and elimination. Supplied by the phrenic nerve (C3–C5). |
| <b>Role in yoga</b>           | Every breath; the central muscle of prāṇāyāma; works with the abdominal wall in kapālabhāti, uḍḍiyāna bandha and full yogic breathing.                                                                                                                |
| <b>Stretch or strengthen?</b> | Neither “stretched” nor strengthened in the usual sense; its coordination and endurance respond to breathing practice.                                                                                                                                |
| <b>Key asanas</b>             | All; especially śavāsana breathing, full yogic breath, prāṇāyāma, and inversions (where it works against the weight of the abdominal organs).                                                                                                         |
| <b>Common misconception</b>   | “Belly breathing uses the diaphragm; chest breathing doesn’t.” The diaphragm contracts in every inhalation; belly and chest describe where the volume change is most visible ( <b>Chapter 21</b> ).                                                   |

## Rectus abdominis

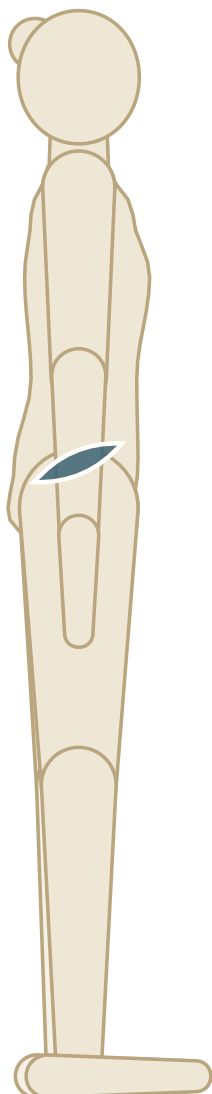
## M. RECTUS ABDOMINIS



|                               |                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Pubic crest and pubic symphysis.                                                                                                                      |
| <b>Insertion</b>              | Xiphoid process and costal cartilages of ribs 5–7.                                                                                                    |
| <b>Primary actions</b>        | Flexes the trunk; tilts the pelvis posteriorly; compresses the abdomen; assists forced exhalation.                                                    |
| <b>Role in yoga</b>           | Stabilises the pelvis in leg raises and boat; lengthened in cobra, bow, fish, camel and wheel.                                                        |
| <b>Stretch or strengthen?</b> | Strengthened by leg raises, boat and plank; lengthened eccentrically in backbends.                                                                    |
| <b>Key asanas</b>             | Leg raises, navāsana, plank (isometric); bhujāṅgāsana, dhanurāsana, matsyāsana (lengthening).                                                         |
| <b>Common misconception</b>   | “The six-pack is the core.” Trunk stability depends on coordinated action of many muscles, and on timing more than maximal strength (Lederman, 2010). |

## Transversus abdominis

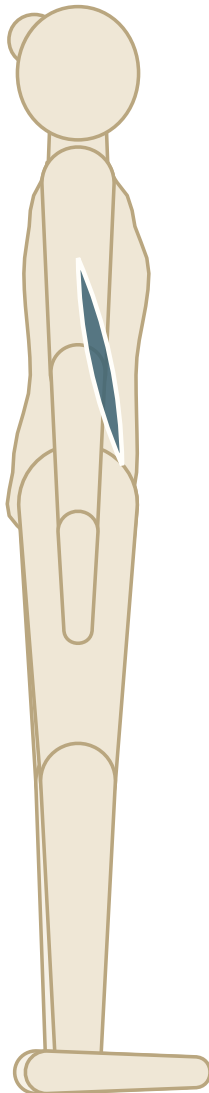
### M. TRANSVERSUS ABDOMINIS



|                               |                                                                                                                                                                                                                                                                                          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Inner surfaces of the lower six costal cartilages; thoracolumbar fascia; anterior iliac crest; lateral inguinal ligament.                                                                                                                                                                |
| <b>Insertion</b>              | Linea alba (via its aponeurosis) and pubic crest.                                                                                                                                                                                                                                        |
| <b>Primary actions</b>        | Compresses the abdomen and raises intra-abdominal pressure; assists forced exhalation. Produces little movement of the trunk.                                                                                                                                                            |
| <b>Role in yoga</b>           | The “drawing in” of the lower abdomen; active in kapālabhāti exhalations, in uḍḍiyāna (with the other abdominal muscles) and in maintaining trunk stiffness in arm balances.                                                                                                             |
| <b>Stretch or strengthen?</b> | Trained through breathing practices and through any posture requiring trunk stability.                                                                                                                                                                                                   |
| <b>Key asanas</b>             | Kapālabhāti, plank, crow, leg raises.                                                                                                                                                                                                                                                    |
| <b>Common misconception</b>   | “You must pre-tense the TrA before every movement to protect your back.” Delayed TrA timing was found in people with back pain (Hodges & Richardson, 1996), but constant bracing is neither natural nor shown to prevent pain; flexible, well-timed control is the aim (Lederman, 2010). |

## External and internal obliques

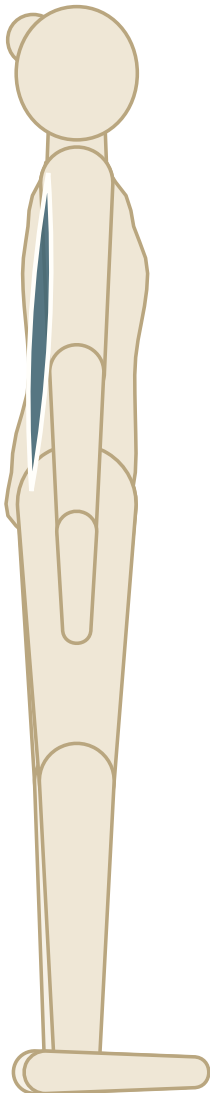
MM. OBLIQUUS EXTERNUS ET INTERNUS ABDOMINIS



|                               |                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | External: outer surfaces of ribs 5–12. Internal: thoracolumbar fascia, anterior iliac crest, lateral inguinal ligament.                                                                                                                   |
| <b>Insertion</b>              | External: linea alba, pubic tubercle, anterior iliac crest. Internal: lower borders of ribs 10–12, linea alba, pubis.                                                                                                                     |
| <b>Primary actions</b>        | Both sides together: flex the trunk and compress the abdomen. One side: lateral flexion to the same side; the <b>external</b> oblique rotates the trunk to the <b>opposite</b> side, the <b>internal</b> oblique to the <b>same</b> side. |
| <b>Role in yoga</b>           | Prime movers of twisting: turning to the left uses the right external and left internal obliques. Work in side bends and side plank; lengthened on the upper side in triangle.                                                            |
| <b>Stretch or strengthen?</b> | Strengthened in twists performed actively, side plank and leg raises with rotation.                                                                                                                                                       |
| <b>Key asanas</b>             | Ardha matsyendrāsana, trikoṇāsana, side plank, revolved postures.                                                                                                                                                                         |
| <b>Common misconception</b>   | "Twists wring toxins out of the organs." There is no evidence for this <b>SCIENTIFIC</b> ; twists move the spine and ribs and work the obliques.                                                                                          |

## Erector spinae

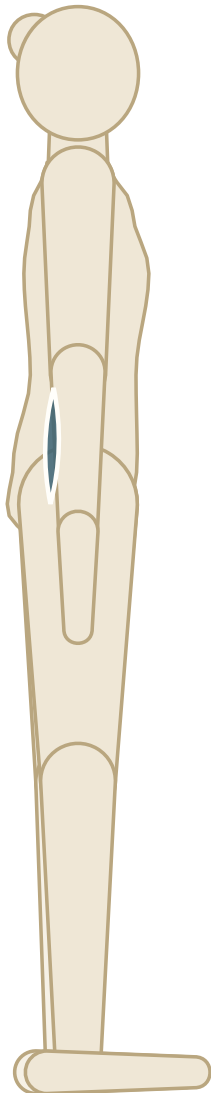
### MM. ERECTOR SPINAE (ILIOCOSTALIS, LONGISSIMUS, SPINALIS)



|                               |                                                                                                                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | A broad common tendon from the posterior iliac crest, the back of the sacrum and the lumbar spinous processes.                                                                                                 |
| <b>Insertion</b>              | Ribs; transverse and spinous processes of the vertebrae; mastoid process (longissimus capitis).                                                                                                                |
| <b>Primary actions</b>        | Both sides: extend the spine and resist flexion (eccentrically controlling forward bending). One side: lateral flexion.                                                                                        |
| <b>Role in yoga</b>           | Lift the chest in cobra and locust; control the descent into forward bends. At the end of a full relaxed forward bend they fall quiet and passive tissues bear the load (the “flexion–relaxation” phenomenon). |
| <b>Stretch or strengthen?</b> | Strengthened in locust, cobra without the hands, bow; lengthened in forward bends and child’s pose.                                                                                                            |
| <b>Key asanas</b>             | Śalabhāsana, bhujaṅgāsana, dhanurāsana (working); pāścimottānāsana, pāda hastāsana (lengthening).                                                                                                              |
| <b>Common misconception</b>   | “Backbends are done by the back muscles alone.” Hip extensors, the upper-back and shoulder muscles and the breath all contribute; the extensors of the lower back should not do all the work.                  |

## Multifidus

MM. MULTIFIDI

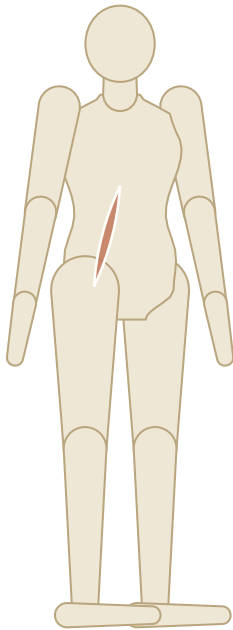


|                               |                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Sacrum, posterior superior iliac spine, mammillary processes of the lumbar vertebrae, transverse processes of the thoracic vertebrae and articular processes of C4–C7.                                                                        |
| <b>Insertion</b>              | Spinous processes two to four segments higher.                                                                                                                                                                                                |
| <b>Primary actions</b>        | Stabilises and extends individual spinal segments; minor rotation. Thickest in the lumbar region.                                                                                                                                             |
| <b>Role in yoga</b>           | Segmental control in all postures; active in locust and in “bird-dog” (opposite arm and leg lifted from tabletop).                                                                                                                            |
| <b>Stretch or strengthen?</b> | Low-load holding positions (tabletop limb lifts, half locust) and graded extension.                                                                                                                                                           |
| <b>Key asanas</b>             | Ardha śalabhāsana, cat-cow, tabletop variations.                                                                                                                                                                                              |
| <b>Common misconception</b>   | “You need special exercises to isolate the multifidus.” Research links multifidus changes to back pain, but general graded exercise, including yoga, is as effective as specific “stabilisation” programmes for most people (Lederman, 2010). |



## Quadratus lumborum

M. QUADRATUS LUMBORUM



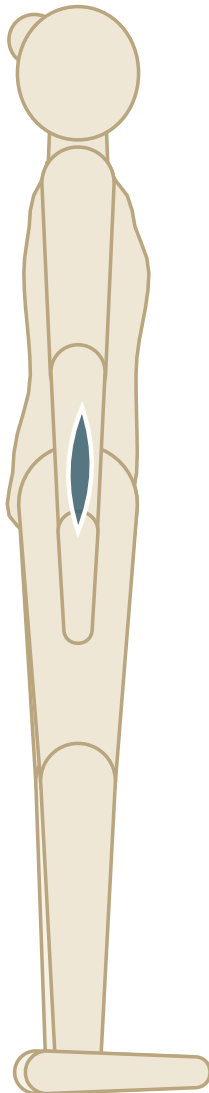
|                               |                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Posterior iliac crest and iliolumbar ligament.                                                                                         |
| <b>Insertion</b>              | 12th rib and transverse processes of L1–L4.                                                                                            |
| <b>Primary actions</b>        | One side: lateral flexion of the trunk; “hikes” the hip. Both sides: extend the lumbar spine; fix the 12th rib during inspiration.     |
| <b>Role in yoga</b>           | Lengthened on the upper side in triangle and side bends; works in balancing on one leg to keep the pelvis level (with gluteus medius). |
| <b>Stretch or strengthen?</b> | Side bends and side plank; lengthened in trikoṇāsana and seated side bends.                                                            |
| <b>Key asanas</b>             | Trikoṇāsana, pārśva bends, gate pose.                                                                                                  |
| <b>Common misconception</b>   | “QL is always the cause of low back pain.” Back pain is rarely attributable to a single muscle (NICE, 2016).                           |

### 19.3 Hip and pelvis

The muscles around the hip are among the strongest in the body; they position the pelvis and therefore the lumbar spine.

## Psoas major

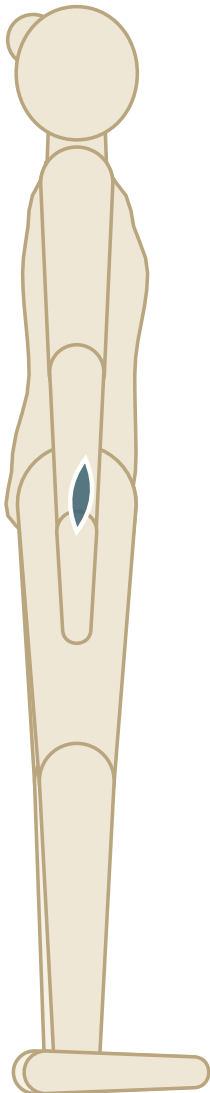
M. PSOAS MAJOR



|                               |                                                                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Sides of the bodies of T12–L5 and the intervening discs; transverse processes of L1–L5.                                                                                                                                |
| <b>Insertion</b>              | Lesser trochanter of the femur (together with iliacus, as the iliopsoas).                                                                                                                                              |
| <b>Primary actions</b>        | Flexes the hip; with the thigh fixed, flexes the trunk on the thigh (as in sitting up from lying). Its effect on the lumbar spine is small and position-dependent; it contributes to lumbar stability (Neumann, 2017). |
| <b>Role in yoga</b>           | Works in leg raises, boat and lifting the knee; lengthened in the back leg of lunges and warrior I, in bridge, bow, camel and wheel.                                                                                   |
| <b>Stretch or strengthen?</b> | Strengthen with leg raises and navāsana; lengthen in low lunge with the pelvis tucked slightly under.                                                                                                                  |
| <b>Key asanas</b>             | Leg raises, navāsana (working); aśva sañcalanāsana, setu bandha, dhanurāsana (lengthening).                                                                                                                            |
| <b>Common misconception</b>   | "The psoas stores trauma and emotions." This popular claim has no scientific basis <b>SCIENTIFIC</b> .                                                                                                                 |

## Iliacus

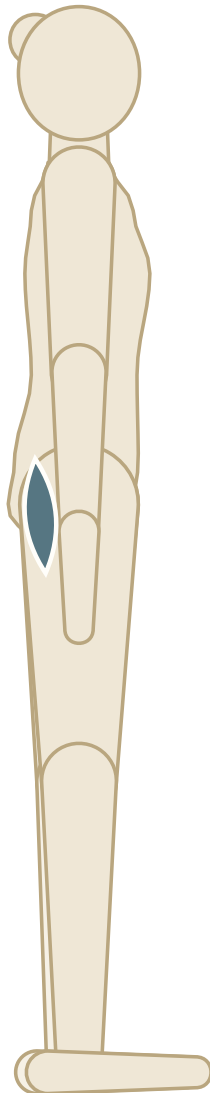
M. ILIACUS



|                               |                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Iliac fossa, inner lip of the iliac crest and the ala of the sacrum.                                                                                                   |
| <b>Insertion</b>              | Lesser trochanter, via the psoas tendon.                                                                                                                               |
| <b>Primary actions</b>        | Flexes the hip; with the femur fixed, tilts the pelvis anteriorly.                                                                                                     |
| <b>Role in yoga</b>           | Partner of psoas in all hip flexion; tight hip flexors tilt the pelvis forward in warrior I and arch the lower back in backbends.                                      |
| <b>Stretch or strengthen?</b> | As psoas.                                                                                                                                                              |
| <b>Key asanas</b>             | As psoas.                                                                                                                                                              |
| <b>Common misconception</b>   | "Hip flexors are always tight in desk workers." Sitting shortens the hip flexors while you sit; whether this produces lasting tightness varies greatly between people. |

## Gluteus maximus

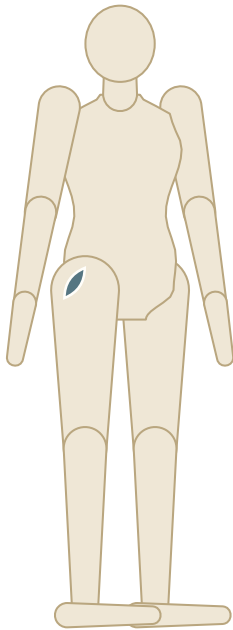
## M. GLUTEUS MAXIMUS



|                               |                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Posterior ilium, dorsal surface of the sacrum and coccyx, sacrotuberous ligament, thoracolumbar fascia.                                                                                                             |
| <b>Insertion</b>              | Iliotibial tract (most fibres) and gluteal tuberosity of the femur.                                                                                                                                                 |
| <b>Primary actions</b>        | Extends the hip, powerfully from a flexed position; externally rotates the thigh; upper fibres abduct. Stabilises the pelvis and, through the iliotibial tract, the knee.                                           |
| <b>Role in yoga</b>           | Works in bridge, locust, warrior and rising from a squat; lengthened in forward bends, child's pose and pigeon.                                                                                                     |
| <b>Stretch or strengthen?</b> | Bridge, locust, chair, warrior III; lengthened in pāścimottānāsana and knee-to-chest.                                                                                                                               |
| <b>Key asanas</b>             | Setu bandha, śalabhāsana (working); pāścimottānāsana, bālāsana (lengthening).                                                                                                                                       |
| <b>Common misconception</b>   | "Always squeeze — or never squeeze — the buttocks in backbends." Moderate activation assists hip extension; hard gripping externally rotates the thighs and may compress the lower back. Teach balance, not a rule. |

## Gluteus medius and minimus

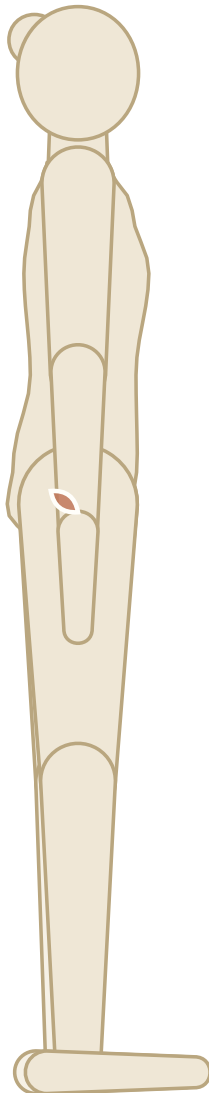
MM. GLUTEUS MEDIUS ET MINIMUS



|                               |                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Outer surface of the ilium, between the gluteal lines.                                                                                                              |
| <b>Insertion</b>              | Greater trochanter of the femur (medius: lateral surface; minimus: anterior surface).                                                                               |
| <b>Primary actions</b>        | Abduct the hip; anterior fibres rotate the thigh internally, posterior fibres of medius externally. Crucially, they hold the pelvis level when standing on one leg. |
| <b>Role in yoga</b>           | The key stabilisers in every one-legged balance; work in triangle and warrior II; weakness shows as the pelvis dropping on the lifted-leg side.                     |
| <b>Stretch or strengthen?</b> | Tree, warrior III, half-moon, side-lying leg lifts.                                                                                                                 |
| <b>Key asanas</b>             | Vṛkṣāsana, vīrabhadrāsana III, ardha candrāsana.                                                                                                                    |
| <b>Common misconception</b>   | "You can stretch the IT band." The iliotibial tract is extremely stiff connective tissue; stretching and rolling change sensation rather than its length.           |

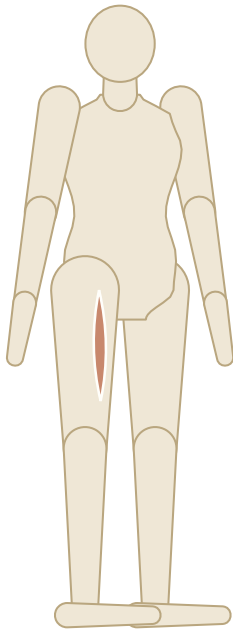
## Deep hip rotators

PIRIFORMIS, GEMELLI, OBTURATOR INTERNUS AND EXTERNUS, QUADRATUS FEMORIS



|                               |                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Piriformis: anterior surface of the sacrum. Others: ischium, obturator membrane and the rim of the obturator foramen.                                                                                               |
| <b>Insertion</b>              | Around the greater trochanter and the trochanteric fossa.                                                                                                                                                           |
| <b>Primary actions</b>        | Externally rotate the hip and stabilise the femoral head in the socket. Piriformis also abducts the flexed hip; in deep hip flexion its rotational action changes (Neumann, 2017).                                  |
| <b>Role in yoga</b>           | Allow the external rotation needed for lotus, butterfly and warrior II; lengthened in pigeon, cow-face and thread-the-needle.                                                                                       |
| <b>Stretch or strengthen?</b> | Lengthened in hip-opening postures; strengthened in standing balance and warrior II.                                                                                                                                |
| <b>Key asanas</b>             | Gomukhāsana, pigeon preparation, ardha padmāsana.                                                                                                                                                                   |
| <b>Common misconception</b>   | "Sciatica means a tight piriformis." The sciatic nerve usually passes below piriformis, and piriformis compression is only one uncommon cause of buttock and leg pain; persistent symptoms need medical assessment. |

## Adductors

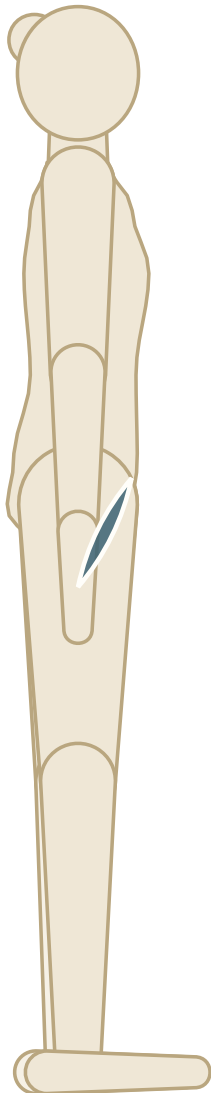


### ADDUCTOR LONGUS, BREVIS AND MAGNUS; PECTINEUS; GRACILIS

|                               |                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Body and inferior ramus of the pubis and the ischial ramus; adductor magnus also from the ischial tuberosity.                               |
| <b>Insertion</b>              | Linea aspera of the femur and adductor tubercle; gracilis on the medial tibia (pes anserinus).                                              |
| <b>Primary actions</b>        | Adduct the hip; anterior adductors assist flexion, the ischial part of adductor magnus assists extension; stabilise the pelvis in standing. |
| <b>Role in yoga</b>           | Lengthened in wide-legged postures, triangle and butterfly; work to hold the legs together in headstand and shoulderstand.                  |
| <b>Stretch or strengthen?</b> | Squeeze a block between the thighs in bridge or boat; lengthen in prasārita pādottānāsana and baddha koṇāsana.                              |
| <b>Key asanas</b>             | Baddha koṇāsana, upaviṣṭha koṇāsana (lengthening); śīrṣāsana, sarvāṅgāsana (legs together).                                                 |
| <b>Common misconception</b>   | "Wide splits are a matter of time." Bone shape at the hip limits abduction in many people (Chapter 17).                                     |

## Tensor fasciae latae

## M. TENSOR FASCIAE LATAE



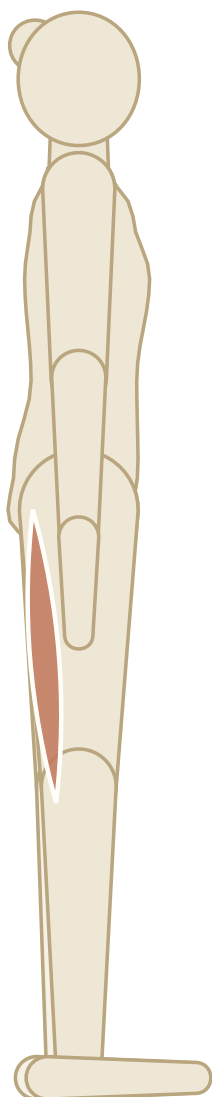
|                               |                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Anterior superior iliac spine and anterior iliac crest.                                                          |
| <b>Insertion</b>              | Iliotibial tract, which attaches to the lateral condyle of the tibia.                                            |
| <b>Primary actions</b>        | Flexes, abducts and internally rotates the hip; tensions the iliotibial tract, stabilising the knee.             |
| <b>Role in yoga</b>           | Assists hip flexion in leg raises; lengthened (with other structures) in the back leg of lunges and in cow-face. |
| <b>Stretch or strengthen?</b> | Standing balances; side-lying leg lifts.                                                                         |
| <b>Key asanas</b>             | Vṛkṣāsana, leg raises.                                                                                           |
| <b>Common misconception</b>   | See gluteus medius: the iliotibial tract itself is not meaningfully stretched.                                   |



## 19.4 Thigh

### Hamstrings

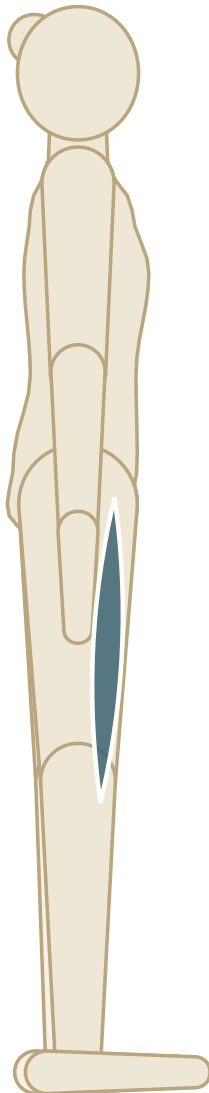
#### BICEPS FEMORIS, SEMITENDINOSUS, SEMIMEMBRANOSUS



|                               |                                                                                                                                                                                                                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Ischial tuberosity (all three, except the short head of biceps femoris, which arises from the linea aspera of the femur).                                                                                                                                                            |
| <b>Insertion</b>              | Biceps femoris: head of the fibula. Semitendinosus: upper medial tibia (pes anserinus). Semimembranosus: posterior medial condyle of the tibia.                                                                                                                                      |
| <b>Primary actions</b>        | Extend the hip and flex the knee. With the knee flexed, biceps femoris rotates the leg externally and the "semis" internally.                                                                                                                                                        |
| <b>Role in yoga</b>           | The main limit to forward bends with straight legs; work in locust, bridge and warrior III. Their upper tendon, at the sitting bone, is vulnerable to overload in aggressive forward bending (Goom et al., 2016).                                                                    |
| <b>Stretch or strengthen?</b> | Lengthen patiently in forward bends (knees bent if needed); strengthen in bridge, locust and slow lowering from standing forward bends.                                                                                                                                              |
| <b>Key asanas</b>             | Paścimottānāsana, pāda hastāsana, adho mukha śvānāsana, trikoṇāsana (lengthening); setu bandha, śalabhāsana (working).                                                                                                                                                               |
| <b>Common misconception</b>   | "Tight hamstrings cause back pain." Reduced hamstring flexibility is one of several modest predictors of developing back pain in prospective studies (Sadler et al., 2017), but back pain has many causes, and many people with tight hamstrings have no back pain (and vice versa). |

## Quadriceps femoris

### RECTUS FEMORIS, VASTUS LATERALIS, MEDIALIS AND INTERMEDIUS

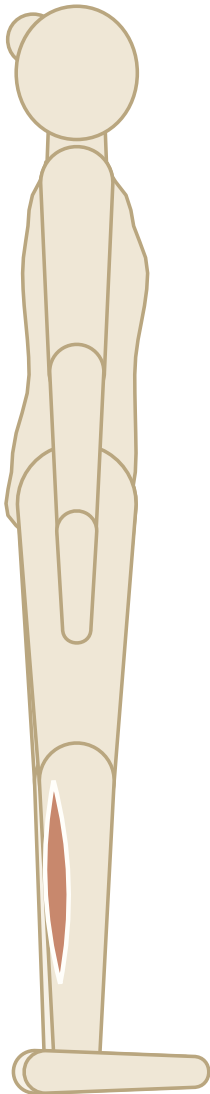


|                               |                                                                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Rectus femoris: anterior inferior iliac spine and the rim of the acetabulum.<br>Vasti: the shaft of the femur.                                                            |
| <b>Insertion</b>              | Patella and, via the patellar ligament, the tibial tuberosity.                                                                                                            |
| <b>Primary actions</b>        | Extend the knee; rectus femoris also flexes the hip.                                                                                                                      |
| <b>Role in yoga</b>           | Engaged to straighten the knees in forward bends (which also relaxes the hamstrings reciprocally); work strongly in chair and warrior; lengthened in bow, camel and hero. |
| <b>Stretch or strengthen?</b> | Chair, warrior, lunges (working); vīrāsana, dhanurāsana, knee-bent lunge (lengthening).                                                                                   |
| <b>Key asanas</b>             | Utkaṭāsana, vīrabhadrāsana (working); dhanurāsana, uṣṭrāsana (lengthening).                                                                                               |
| <b>Common misconception</b>   | "Lock the knees to straighten the legs." Straightening is done by the quadriceps; "locking" means pushing into hyperextension, which loads the back of the knee.          |

## 19.5 Lower leg

### Gastrocnemius

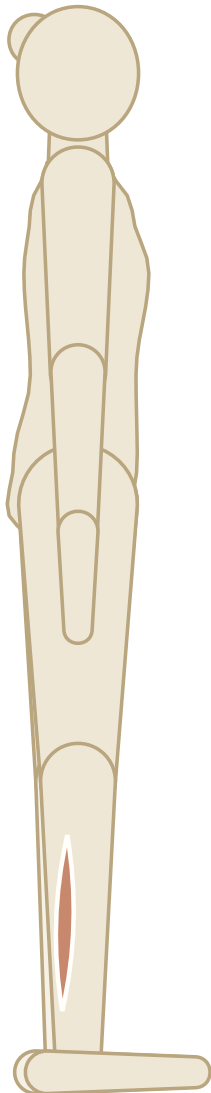
M. GASTROCNEMIUS



|                               |                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Medial and lateral condyles of the femur.                                                                                                          |
| <b>Insertion</b>              | Calcaneus, via the Achilles (calcaneal) tendon.                                                                                                    |
| <b>Primary actions</b>        | Plantarflexes the ankle; flexes the knee.                                                                                                          |
| <b>Role in yoga</b>           | Limits dorsiflexion when the knee is straight (heels in downward dog); works in rising onto the toes.                                              |
| <b>Stretch or strengthen?</b> | Lengthen in downward dog with straight knees; strengthen with heel raises.                                                                         |
| <b>Key asanas</b>             | Adho mukha śvānāsana, pārśvottānāsana (lengthening).                                                                                               |
| <b>Common misconception</b>   | "If your heels don't touch in downward dog, you're doing it wrong." Calf length, ankle shape and proportions vary; heels need not reach the floor. |

## Soleus

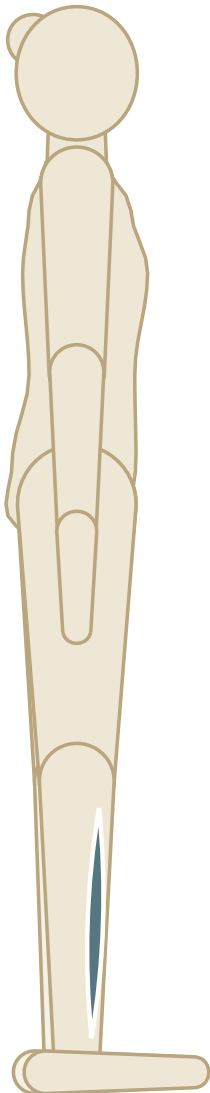
## M. SOLEUS



|                               |                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Head and upper shaft of the fibula; soleal line of the tibia.                                                                         |
| <b>Insertion</b>              | Calcaneus, via the Achilles tendon.                                                                                                   |
| <b>Primary actions</b>        | Plantarflexes the ankle (does not cross the knee). A key postural muscle and part of the “muscle pump” returning blood from the legs. |
| <b>Role in yoga</b>           | Limits dorsiflexion when the knee is bent — as in a deep squat or chair pose.                                                         |
| <b>Stretch or strengthen?</b> | Lengthen in malāsana and in downward dog with bent knees.                                                                             |
| <b>Key asanas</b>             | Malāsana, utkaṭāsana.                                                                                                                 |
| <b>Common misconception</b>   | “Calf stretches are all the same.” Straight-knee stretches bias gastrocnemius; bent-knee stretches bias soleus.                       |

## Tibialis anterior

## M. TIBIALIS ANTERIOR



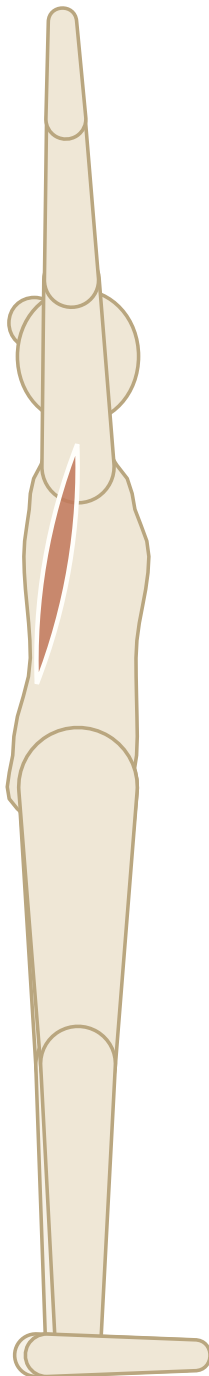
|                               |                                                                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Lateral condyle and upper lateral surface of the tibia; interosseous membrane.                                                                         |
| <b>Insertion</b>              | Medial cuneiform and base of the first metatarsal.                                                                                                     |
| <b>Primary actions</b>        | Dorsiflexes and inverts the foot; supports the medial arch.                                                                                            |
| <b>Role in yoga</b>           | Draws the toes towards the shins ("flex the feet") in <i>daṇḍāsana</i> and <i>paścimottānāsana</i> ; lengthened when kneeling on the tops of the feet. |
| <b>Stretch or strengthen?</b> | Toe lifts; lengthen in <i>vajrāsana</i> .                                                                                                              |
| <b>Key asanas</b>             | <i>Vajrāsana</i> , <i>virāsana</i> (lengthening); <i>daṇḍāsana</i> (working).                                                                          |
| <b>Common misconception</b>   | "Kneeling postures are bad for the feet." Most people adapt well; those who cannot tolerate them can place a rolled blanket under the ankles.          |

## 19.6 Shoulder girdle and arm

In yoga the arms are often weight-bearing; the shoulder's stability then depends on the coordinated work of the muscles below.

## Latissimus dorsi

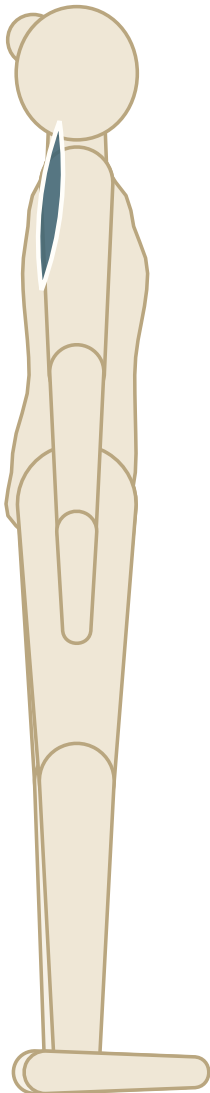
### M. LATISSIMUS DORSI



|                               |                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Spinous processes of T7–T12, thoracolumbar fascia, iliac crest and lower three or four ribs.                                                                             |
| <b>Insertion</b>              | Floor of the intertubercular (bicipital) groove of the humerus.                                                                                                          |
| <b>Primary actions</b>        | Extends, adducts and internally rotates the humerus; depresses the shoulder girdle.                                                                                      |
| <b>Role in yoga</b>           | When short it limits arms overhead — in hasta uttānāsana, downward dog and headstand — and students compensate by arching the lower back; works in chaturāṅga and plank. |
| <b>Stretch or strengthen?</b> | Lengthen in child's pose with arms extended and in side bends; strengthen in plank and in pressing up from the floor.                                                    |
| <b>Key asanas</b>             | Adho mukha śvānāsana, bālāsana with arms forward (lengthening); chaturāṅga (working).                                                                                    |
| <b>Common misconception</b>   | "Limited overhead reach is always a shoulder-joint problem." Latissimus, thoracic stiffness and rib position often limit it.                                             |

## Trapezius

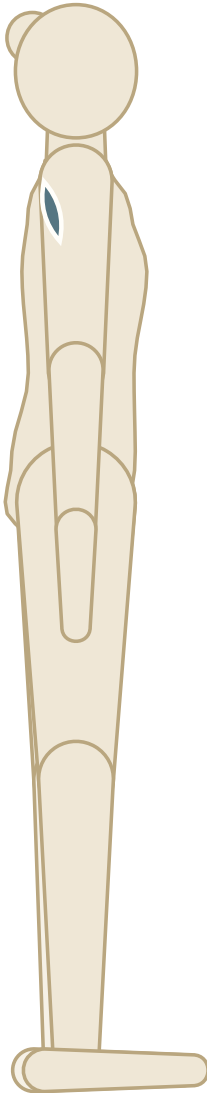
### M. TRAPEZIUS



|                               |                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | External occipital protuberance and superior nuchal line; ligamentum nuchae; spinous processes of C7–T12.                                                                                                                                                                     |
| <b>Insertion</b>              | Lateral third of the clavicle; acromion and spine of the scapula.                                                                                                                                                                                                             |
| <b>Primary actions</b>        | Upper fibres: elevate and upwardly rotate the scapula. Middle fibres: retract it. Lower fibres: depress and upwardly rotate it.                                                                                                                                               |
| <b>Role in yoga</b>           | Upper and lower fibres together rotate the scapula upward to lift the arms overhead. The cue “shoulders away from the ears” is helpful in plank, but when the arms are fully overhead the upper trapezius must work — forcing the shoulders down then blocks upward rotation. |
| <b>Stretch or strengthen?</b> | Arms-overhead postures; headstand; locust with arms forward.                                                                                                                                                                                                                  |
| <b>Key asanas</b>             | Śīrṣāsana, hasta uttānāsana, vīrabhadrāsana I.                                                                                                                                                                                                                                |
| <b>Common misconception</b>   | “The upper trapezius should always be relaxed.” It has an essential job in overhead movement.                                                                                                                                                                                 |

## Rhomboids

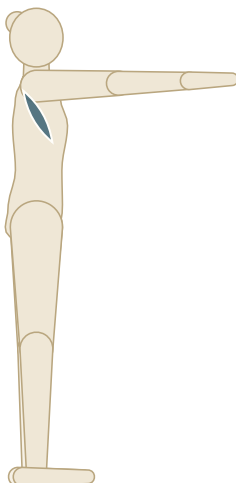
MM. RHOMBOIDEI MAJOR ET MINOR



|                               |                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Spinous processes of C7–T5 (minor: C7–T1; major: T2–T5).                                                                                                   |
| <b>Insertion</b>              | Medial border of the scapula.                                                                                                                              |
| <b>Primary actions</b>        | Retract, elevate and downwardly rotate the scapula.                                                                                                        |
| <b>Role in yoga</b>           | Draw the shoulder blades together when the hands are clasped behind the back, in bow and locust; lengthened in eagle arms and in the protraction of plank. |
| <b>Stretch or strengthen?</b> | Locust with arms back; lengthen in garuḍāsana arms.                                                                                                        |
| <b>Key asanas</b>             | Dhanurāsana, śalabhāsana (working); garuḍāsana (lengthening).                                                                                              |
| <b>Common misconception</b>   | "Always squeeze the shoulder blades together for good posture." Healthy shoulders move freely between protraction and retraction.                          |

## Serratus anterior

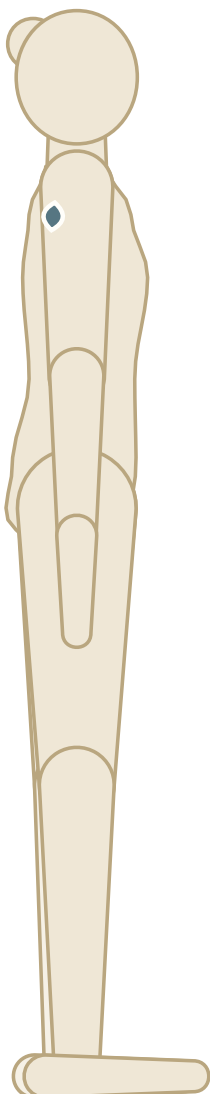
M. SERRATUS ANTERIOR



|                               |                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Outer surfaces of ribs 1–8 or 9.                                                                                             |
| <b>Insertion</b>              | Costal (front) surface of the medial border of the scapula.                                                                  |
| <b>Primary actions</b>        | Protracts the scapula and rotates it upward; holds it against the rib cage. Weakness causes "winging".                       |
| <b>Role in yoga</b>           | The muscle behind "push the floor away" in plank, downward dog, crow and headstand; essential for lifting the arms overhead. |
| <b>Stretch or strengthen?</b> | Plank and push-back to downward dog; tabletop with protraction; crow.                                                        |
| <b>Key asanas</b>             | Adho mukha śvānāsana, kākāsana, plank.                                                                                       |
| <b>Common misconception</b>   | "Collapsing between the shoulder blades in plank is a flexibility issue." It usually reflects serratus weakness or fatigue.  |



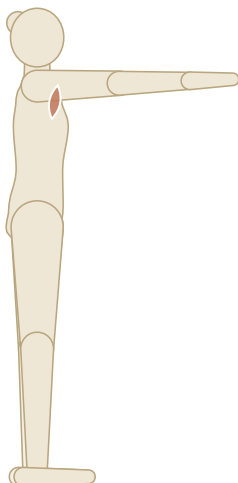
## Rotator cuff



### SUPRASPINATUS, INFRASPINATUS, TERES MINOR, SUBSCAPULARIS

|                               |                                                                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Supraspinous, infraspinous and subscapular fossae and the lateral border of the scapula.                                                                                                                                            |
| <b>Insertion</b>              | Greater tubercle of the humerus (supraspinatus, infraspinatus, teres minor) and lesser tubercle (subscapularis).                                                                                                                    |
| <b>Primary actions</b>        | Supraspinatus assists abduction; infraspinatus and teres minor rotate the humerus externally; subscapularis rotates it internally. Together they compress the humeral head into its shallow socket — the key to shoulder stability. |
| <b>Role in yoga</b>           | Stabilise the shoulder in all weight-bearing postures; overloaded by repeated, poorly controlled chaturaṅga.                                                                                                                        |
| <b>Stretch or strengthen?</b> | Plank and forearm plank holds; slow, controlled chaturaṅga with knees down; external-rotation work.                                                                                                                                 |
| <b>Key asanas</b>             | Plank, dolphin, headstand preparation.                                                                                                                                                                                              |
| <b>Common misconception</b>   | "If you can do it, your shoulders are fine." Rotator-cuff irritation often builds silently through repetition; volume and quality matter.                                                                                           |

## Pectoralis major and minor

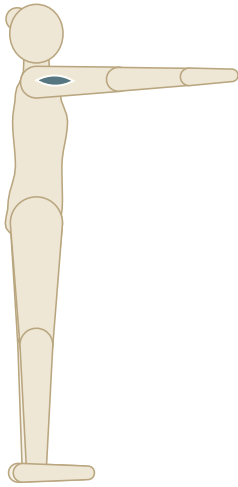


### MM. PECTORALIS MAJOR ET MINOR

|                               |                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Major: medial clavicle, sternum, costal cartilages 1–6. Minor: ribs 3–5.                                                                             |
| <b>Insertion</b>              | Major: lateral lip of the intertubercular groove of the humerus. Minor: coracoid process of the scapula.                                             |
| <b>Primary actions</b>        | Major: adducts and internally rotates the humerus; clavicular fibres flex it. Minor: draws the scapula forward and down; assists forced inspiration. |
| <b>Role in yoga</b>           | Work in chaturaṅga and crow; lengthened in fish, in hands clasped behind the back and in wheel.                                                      |
| <b>Stretch or strengthen?</b> | Lengthen in matsyāsana and in doorway or wall chest openers.                                                                                         |
| <b>Key asanas</b>             | Matsyāsana, clasped-hands forward bend (lengthening); chaturaṅga (working).                                                                          |
| <b>Common misconception</b>   | "Round shoulders are caused by tight pecs." Posture reflects habit, comfort and many tissues; stretching alone rarely changes it.                    |

## Deltoid

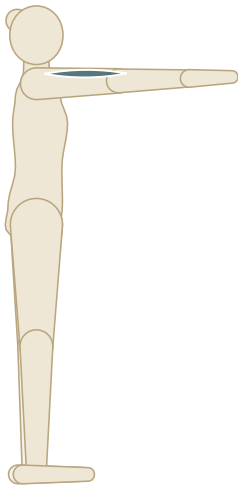
### M. DELTOIDEUS



|                               |                                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Lateral third of the clavicle; acromion; spine of the scapula.                                                                    |
| <b>Insertion</b>              | Deltoid tuberosity of the humerus.                                                                                                |
| <b>Primary actions</b>        | Anterior fibres: flex and internally rotate the arm. Middle fibres: abduct it. Posterior fibres: extend and externally rotate it. |
| <b>Role in yoga</b>           | Lifts and holds the arms in warrior II, triangle and all overhead postures.                                                       |
| <b>Stretch or strengthen?</b> | Warrior II holds, arm raises.                                                                                                     |
| <b>Key asanas</b>             | Vīrabhadrāsana II, trikoṇāsana.                                                                                                   |
| <b>Common misconception</b>   | —                                                                                                                                 |

## Biceps brachii

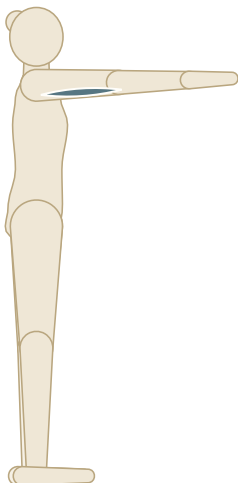
### M. BICEPS BRACHII



|                               |                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Short head: coracoid process. Long head: supraglenoid tubercle of the scapula.                   |
| <b>Insertion</b>              | Radial tuberosity and, via the bicipital aponeurosis, the forearm fascia.                        |
| <b>Primary actions</b>        | Flexes the elbow; supinates the forearm; weakly flexes the shoulder.                             |
| <b>Role in yoga</b>           | Lengthened in binds and when the arms are extended behind; works to hold the elbow bent in crow. |
| <b>Stretch or strengthen?</b> | —                                                                                                |
| <b>Key asanas</b>             | Clasped hands behind the back (lengthening).                                                     |
| <b>Common misconception</b>   | —                                                                                                |

## Triceps brachii

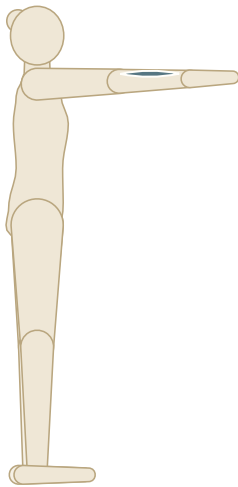
### M. TRICEPS BRACHII



|                               |                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Long head: infraglenoid tubercle of the scapula. Lateral and medial heads: posterior humerus.                           |
| <b>Insertion</b>              | Olecranon of the ulna.                                                                                                  |
| <b>Primary actions</b>        | Extends the elbow; the long head also extends and adducts the shoulder.                                                 |
| <b>Role in yoga</b>           | Controls the lowering in chaturaṅga (eccentric) and holds the bent arms in crow; lengthened in the top arm of cow-face. |
| <b>Stretch or strengthen?</b> | Chaturaṅga with knees down; dolphin push-ups.                                                                           |
| <b>Key asanas</b>             | Chaturaṅga, kākāsana (working); gomukhāsana (lengthening).                                                              |
| <b>Common misconception</b>   | —                                                                                                                       |

## Forearm flexors and extensors

### FLEXOR AND EXTENSOR GROUPS OF THE FOREARM

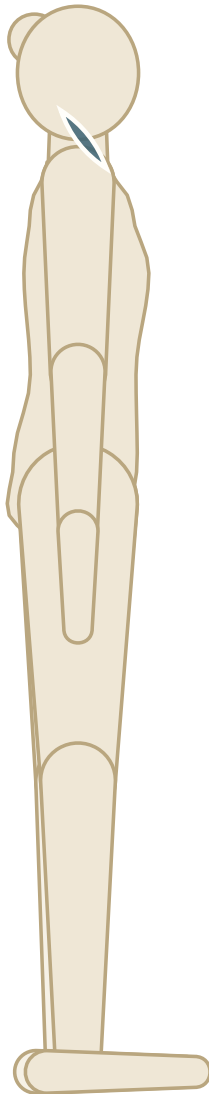


|                               |                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Flexors: common flexor origin on the medial epicondyle of the humerus.<br>Extensors: common extensor origin on the lateral epicondyle.                                                                   |
| <b>Insertion</b>              | Carpal bones, metacarpals and phalanges.                                                                                                                                                                 |
| <b>Primary actions</b>        | Flex or extend the wrist and fingers.                                                                                                                                                                    |
| <b>Role in yoga</b>           | In weight-bearing on the hands, gripping the floor with the fingertips and finger-pads recruits the flexors and takes load off the extended wrist; the flexors are lengthened in plank and downward dog. |
| <b>Stretch or strengthen?</b> | Gradually increasing time on the hands; fist and fingertip variations; wrist mobility.                                                                                                                   |
| <b>Key asanas</b>             | Plank, adho mukha śvānāsana, kākāsana.                                                                                                                                                                   |
| <b>Common misconception</b>   | "Wrist pain in plank means weak wrists." Load distribution, wrist extension range and gradual adaptation all matter (Chapter 26).                                                                        |

## 19.7 Neck

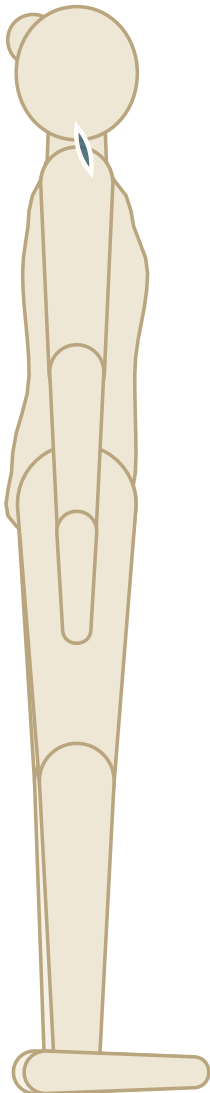
### Sternocleidomastoid

**M. STERNOCLEIDOMASTOIDEUS**



|                               |                                                                                                                                          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Manubrium of the sternum and medial clavicle.                                                                                            |
| <b>Insertion</b>              | Mastoid process and superior nuchal line of the skull.                                                                                   |
| <b>Primary actions</b>        | One side: tilts the head to the same side and turns the face to the opposite side. Both sides: flex the neck; assist forced inspiration. |
| <b>Role in yoga</b>           | Lifts the head from the floor in supine postures; lengthened in fish and when the head turns in twists.                                  |
| <b>Stretch or strengthen?</b> | Supine head lifts; lengthen with gentle neck rotations.                                                                                  |
| <b>Key asanas</b>             | Matsyāsana, jālandhara bandha (lengthening).                                                                                             |
| <b>Common misconception</b>   | "Neck tension means the neck is weak." Tension often comes from breathing with the accessory muscles or from habitual posture.           |

## Scalenes

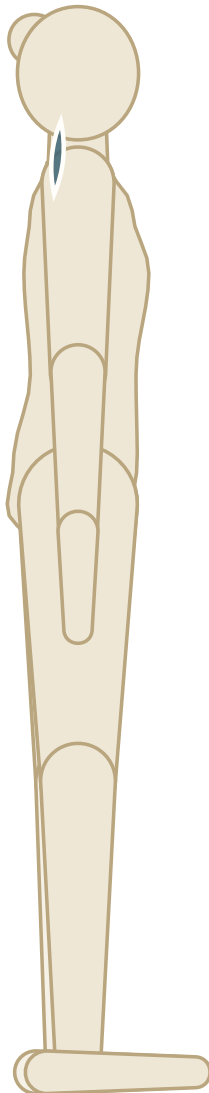


### MM. SCALENI ANTERIOR, MEDIUS ET POSTERIOR

|                               |                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Transverse processes of the cervical vertebrae (C2–C7).                                                                                     |
| <b>Insertion</b>              | First rib (anterior and middle scalene) and second rib (posterior scalene).                                                                 |
| <b>Primary actions</b>        | Raise the first two ribs (accessory inspiration); flex the neck laterally.                                                                  |
| <b>Role in yoga</b>           | Overactive in upper-chest or “clavicular” breathing; the nerves to the arm (brachial plexus) pass between the anterior and middle scalenes. |
| <b>Stretch or strengthen?</b> | Slow abdominal and full yogic breathing reduce reliance on them.                                                                            |
| <b>Key asanas</b>             | Prāṇāyāma; side neck stretches.                                                                                                             |
| <b>Common misconception</b>   | —                                                                                                                                           |

## Levator scapulae

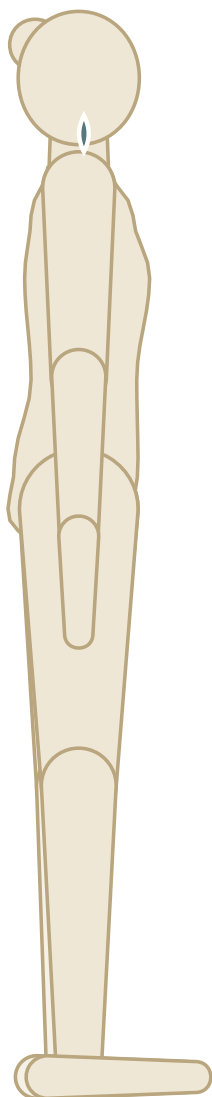
### M. LEVATOR SCAPULAE



|                               |                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Transverse processes of C1–C4.                                                                               |
| <b>Insertion</b>              | Superior angle and upper medial border of the scapula.                                                       |
| <b>Primary actions</b>        | Elevates the scapula and rotates it downward; laterally flexes the neck.                                     |
| <b>Role in yoga</b>           | Often tense in “shoulders up to the ears”; lengthened in neck side-bends with the head turned slightly down. |
| <b>Stretch or strengthen?</b> | Lengthen gently; learn to release shoulder elevation in postures.                                            |
| <b>Key asanas</b>             | Neck release sequence; relaxation.                                                                           |
| <b>Common misconception</b>   | —                                                                                                            |

## Suboccipitals and deep neck flexors

RECTUS CAPITIS AND OBLIQUUS CAPITIS; LONGUS COLLI AND LONGUS CAPITIS



|                               |                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origin</b>                 | Suboccipitals: atlas (C1) and axis (C2). Deep flexors: front of the cervical and upper thoracic vertebrae.                                                                          |
| <b>Insertion</b>              | Suboccipitals: occipital bone and atlas. Deep flexors: occipital bone and upper cervical vertebrae.                                                                                 |
| <b>Primary actions</b>        | Suboccipitals: fine extension and rotation of the head; rich in proprioceptors. Deep neck flexors: flex and stabilise the cervical spine.                                           |
| <b>Role in yoga</b>           | “Lengthen the back of the neck” and the chin tuck of jālandhara bandha use the deep flexors and release the suboccipitals; they help protect the neck in shoulderstand preparation. |
| <b>Stretch or strengthen?</b> | Gentle chin nods lying supine; jālandhara bandha.                                                                                                                                   |
| <b>Key asanas</b>             | Jālandhara bandha, sarvāṅgāsana preparation.                                                                                                                                        |
| <b>Common misconception</b>   | —                                                                                                                                                                                   |

## 19.8 From muscles to movement

**Table 19.1** Which muscle groups work and which lengthen in the twelve basic postures (simplified)

| Posture       | Mainly working                                                                                                           | Mainly lengthening                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Headstand     | deltoid, serratus anterior, trapezius, rotator cuff, triceps (forearm press); trunk and hip muscles (balance); adductors | — (neck and shoulders loaded, not stretched)             |
| Shoulderstand | back extensors and gluteals (to keep the body vertical); hip extensors; adductors                                        | neck extensors, upper back, (pectorals with arms behind) |
| Plough        | trunk muscles controlling the descent                                                                                    | hamstrings, back extensors, neck extensors               |
| Fish          | back extensors, rhomboids, triceps (elbow press)                                                                         | pectorals, rectus abdominis, neck flexors, intercostals  |

| Posture               | Mainly working                                                                  | Mainly lengthening                                                           |
|-----------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Forward bend          | quadriceps (knees straight), hip flexors (to fold actively)                     | hamstrings, gluteals, back extensors, calves                                 |
| Cobra                 | back extensors, rhomboids, triceps (lightly)                                    | rectus abdominis, hip flexors (mildly), pectorals                            |
| Locust                | back extensors, gluteus maximus, hamstrings                                     | hip flexors (mildly)                                                         |
| Bow                   | back extensors, gluteals, hamstrings (pulling feet into hands), rhomboids       | quadriceps, hip flexors, rectus abdominis, pectorals, anterior deltoid       |
| Half spinal twist     | obliques (rotation), back extensors (upright spine)                             | obliques on the opposite side, deep hip rotators and gluteals of the top leg |
| Crow                  | serratus anterior, pectorals, triceps, forearm flexors, hip flexors, abdominals | wrist flexors (in extension)                                                 |
| Standing forward bend | quadriceps; hamstrings and back extensors eccentrically during descent          | hamstrings, calves, back extensors                                           |
| Triangle              | obliques and quadratus lumborum of the lower side, gluteus medius, quadriceps   | adductors, hamstrings of the front leg, lateral trunk of the upper side      |

Chapter 25 gives the full asana–anatomy mapping tables for these and other postures.

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ◆ The major muscles of yoga can be learned in six regions: trunk, hip, thigh, lower leg, shoulder and arm, and neck.
- ◆ For each, know the attachments and actions — and translate them into the postures in which the muscle works or lengthens.
- ◆ Many popular claims (the psoas storing emotions, twists detoxifying organs, stretching the IT band) have no scientific basis.
- ◆ The trapezius, serratus anterior and rotator cuff are central to safe weight-bearing and overhead postures.
- ◆ “Tight” describes a sensation with many possible causes; teach people, not muscles.

### REFLECTION QUESTIONS

1. Which muscles do you feel most strongly in your own practice? Are they working or lengthening?
2. Which misconception in this chapter had you believed? How did you come to believe it?
3. How much anatomy belongs in a beginners’ class? What is the risk of too much?

### TEACHER-PRACTICE EXERCISES

1. Choose six muscles from the atlas. For each, demonstrate one posture in which it works and one in which it lengthens.
2. In pairs, palpate (with consent) the belly of the quadriceps, hamstrings, gluteus maximus and triceps while your partner contracts and relaxes each.



3. Rewrite three cues from your own classes that name a muscle into cues that name an action or image instead, and compare how students respond.

#### ☑ KNOWLEDGE CHECK

1. The psoas major inserts on the: (a) greater trochanter (b) lesser trochanter (c) ischial tuberosity (d) ASIS
2. Turning the trunk to the left uses the: (a) left external and right internal obliques (b) right external and left internal obliques (c) both rectus abdominis (d) right quadratus lumborum
3. Which muscle holds the pelvis level when standing on one leg? (a) gluteus maximus (b) gluteus medius (c) psoas (d) hamstrings
4. The hamstrings originate mainly from the: (a) ASIS (b) ischial tuberosity (c) iliac crest (d) lesser trochanter
5. “Push the floor away” in plank mainly recruits the: (a) rhomboids (b) serratus anterior (c) upper trapezius alone (d) biceps

Answers 1b · 2b · 3b · 4b · 5b

#### 📄 TEACHING ASSIGNMENT

Choose one posture from the twelve. Using the atlas, prepare a “muscle map” of it: list the muscles working and lengthening, the type of contraction of each, and one cue that helps each key muscle do its job. Present it in three minutes, then teach the posture using only your three best cues.

## CHAPTER 20

# Yoga Biomechanics — How Postures Load the Body

*“Sthira: steady. Sukha: easy. Between the two lies the whole art of posture.” — a reflection on Yoga Sūtra 2.46*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ define mobility, stability, flexibility, strength, active and passive range of motion, end range and load tolerance;
- ♦ distinguish tensile loading from compression and explain how each limits movement;
- ♦ describe the mechanics of balance and the roles of proprioception and motor control;
- ♦ explain how the body moves and is loaded in forward bends, backbends, twists, lateral bends, inversions, arm balances, hip openers, standing and seated postures;
- ♦ use these principles to observe students and choose appropriate modifications.

## 20.1 Key concepts

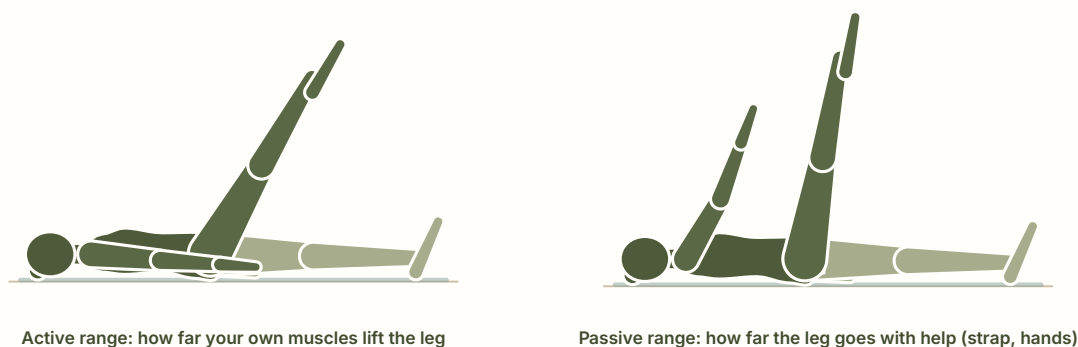
### Mobility and stability

**Mobility** is the ability to move a joint or region through its available range. **Stability** is the ability to control position and movement — to resist unwanted motion. They are not opposites but partners: a joint needs enough mobility for the task and enough control to use that mobility safely. The shoulder and the lumbar spine illustrate the two sides of this balance (Chapter 17).

### Flexibility and range of motion

**Flexibility** usually means the range through which a joint can be moved, often passively. Two measures matter:

- ♦ **Passive range of motion (PROM):** the range reached with external help — gravity, a strap, the hands, a teacher.
- ♦ **Active range of motion (AROM):** the range reached and held by the person’s own muscles.



**Figure 20.1** Active versus passive range of motion. The gap between them is the range you can reach but cannot yet control — a useful thing for a teacher to notice.

AROM is almost always smaller than PROM. The gap is range the student can reach but not control — and it is where many injuries in yoga happen, because load arrives at a position the muscles cannot support. A useful principle is to **build active control in the range you already have** before seeking more passive range.

## Strength

**Strength** is the capacity of muscles to produce force. In yoga it is expressed mostly as holding (isometric) and controlled movement (concentric and eccentric) against body weight (**Chapter 18**). Strength at the end of a range turns passive flexibility into usable mobility.

## Tension and compression

When a posture stops, something has limited it (Clark, 2016):

- ♦ **Tension (tensile loading)** — tissues on the side being lengthened are pulled taut: muscles, fascia, ligaments, joint capsules, even nerves. Tension limits are felt as pulling or stretching, usually on the *opposite* side from the direction of movement.
- ♦ **Compression** — tissues on the side the body is moving towards are pressed together: soft tissue against soft tissue (belly against thighs), or bone against bone. Compression limits are felt as blocking, pinching or a hard stop, on the *same* side as the movement.



### Tension (tensile load)

tissues on the convex side are pulled:  
hamstrings, calf, back of the trunk,  
posterior hip and spinal ligaments

### Compression

tissues on the concave side are pressed  
together: front of the hip, abdomen,  
anterior discs when the spine rounds

When a fold stops, the limit may be  
tension (felt behind) or compression  
(felt in front, e.g. belly meets thighs).

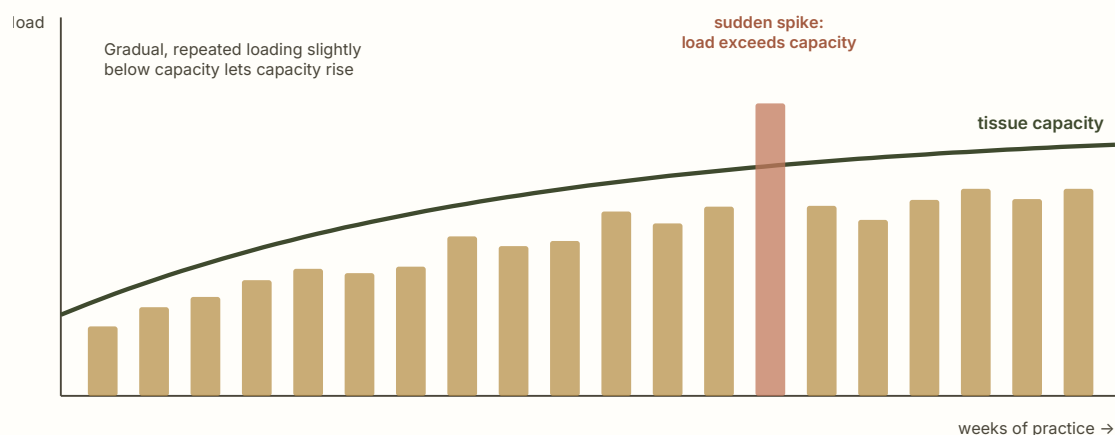
**Figure 20.2** Tension and compression in a seated forward bend. Coloured spindles show tissues that are lengthening under load (terracotta). Recognising which kind of limit a student has reached changes what you teach next (Clark, 2016).

|                        | Tension limit                 | Compression limit                                               |
|------------------------|-------------------------------|-----------------------------------------------------------------|
| Where felt             | on the side being lengthened  | on the side moved towards                                       |
| Quality                | stretching, pulling           | blocking, pinching, hard stop                                   |
| Changes with practice? | often, gradually              | soft-tissue compression sometimes; bone-on-bone compression, no |
| Teaching response      | patience, breath, time, props | change the angle, use props, accept the limit                   |

## End range and load tolerance

**End range** is the last part of a joint's range, where passive tissues become taut and the joint surfaces are least congruent. Loading a joint at end range — especially repeatedly, quickly, or with added force from a teacher or momentum — places the greatest demand on ligaments, capsules and discs.

**Load tolerance** (or capacity) is the amount of load a tissue can accept without irritation. It is not fixed: tissues adapt to loads they meet regularly and progressively, and lose capacity with disuse. Injury risk rises when load increases faster than capacity can adapt (Soligard et al., 2016) — the principle behind gradual progression.



**Figure 20.3** A conceptual model of load and capacity. Tissues adapt to loads they meet regularly; injury risk rises when load jumps faster than capacity can adapt. The curve is illustrative, not data (Soligard et al., 2016).

#### ◆ THE TWO QUESTIONS OF LOAD

For any posture and any student, ask: **how much load**, and **where is it going**? Load that is shared across many joints and muscles, at moderate ranges, with steady breath, is well tolerated. Load concentrated in one joint, at end range, held with strain, is not — however beautiful the posture looks.

## Balance, proprioception and motor control

A body is balanced when its **centre of mass** lies over its **base of support**. The wider and lower the base, the more stable the posture: triangle is more stable than tree, and tree more stable than standing on the toes. The nervous system maintains balance using three sources of information — vision, the vestibular system of the inner ear, and **proprioception**, the sense of position and movement from receptors in muscles, tendons, joints and skin (Proske & Gandevia, 2012) — and corrects small errors through the ankles, the hips and, if necessary, by stepping.

**Motor control** is the nervous system's organisation of movement: which muscles fire, when, and how strongly. Much of what looks like “weakness” or “stiffness” in a beginner is really unfamiliar motor control, which improves rapidly with practice (Chapter 22).

#### ✧ SENSES OF BALANCE

Stand in tree pose for 30 seconds with eyes open; then with the eyes closed; then with eyes open on a folded blanket. Notice how much you depend on each sense. Balance postures with the eyes closed train proprioception and the vestibular system — and are best taught near a wall.

## 20.2 How the body moves in each family of postures

### Forward bends

**Movement.** Flexion at the hips combined with flexion of the spine. In a well-organised forward bend, hip flexion dominates: the pelvis tilts forward over the thighs, and the spine follows.

**What limits it.** Chiefly tension in the hamstrings (and, with straight knees, the calves and the nerves at the back of the leg); compression of the belly against the thighs at the end.

**Loads.** Tension on the hamstrings and their attachment at the sitting bones; with the pelvis held back, flexion moves into the lumbar spine, increasing load on the discs and posterior ligaments. Seated forward bends load the spine less than standing ones held against gravity, but pulling with the arms can add considerable force.

**Teaching principles.** Sit on a blanket to allow the pelvis to tilt; bend the knees freely; fold from the hips with a long spine first, then allow the spine to round in a relaxed way; never pull or bounce. Tingling or electric sensations down the leg suggest nerve tension: bend the knees and reduce the stretch.

## Backbends

**Movement.** Extension of the spine, usually with extension at the hips (in bow, camel, wheel) and often flexion at the shoulders (in wheel).

**What limits it.** Tension at the front — hip flexors, abdominal wall, chest and shoulder muscles; compression at the back — the facet joints and spinous processes of the spine, especially in the lower lumbar region, which is the most mobile in extension.

**Loads.** Compression of the posterior elements of the lumbar spine; the neck when the head is dropped back under load.

**Teaching principles.** Lengthen before bending; distribute the extension — lift the sternum and upper back rather than pressing into the lower back; engage the legs and buttocks moderately so that the hips extend; keep the neck in line with the curve of the spine; approach gradually through cobra and locust before bow and wheel; counterpose with gentle flexion.

## Twists

**Movement.** Rotation of the spine in the transverse plane, often combined with some flexion or lateral flexion. The thoracic spine provides most rotation; the lumbar spine very little; the neck a great deal.

**What limits it.** Tension in the obliques, back muscles and rib-cage tissues; the facet joints of the lumbar spine.

**Loads.** Combined rotation and flexion under load is the combination most often associated with disc and spinal injury in biomechanical studies of the back (McGill, 2016); seated twists with a lifted, lengthened spine keep flexion low.

**Teaching principles.** Sit tall (on a blanket if the lower back rounds); inhale to lengthen, exhale to turn; turn from the rib cage, not by leveraging the arms; let the pelvis move naturally in beginners rather than locking it; keep the neck's rotation moderate; avoid deep, forced twists in students with osteoporosis (Sinaki, 2013).

## Lateral bends

**Movement.** Lateral flexion of the spine in the frontal plane, as in triangle and seated side bends.

**What limits it.** Tension on the lengthened side — quadratus lumborum, obliques, latissimus dorsi, intercostals; compression on the shortened side of the ribs and waist.

**Teaching principles.** Lengthen the spine first; keep the trunk in the frontal plane (not collapsing forward); emphasise length of the upper side rather than depth of the bend.

## Inversions

**Movement.** The body is upside down; the joints may be in neutral (headstand) or in strong flexion (shoulderstand and plough, in which the neck flexes towards its end range).

**Loads.** In headstand, part of the body weight passes through the head and cervical spine; in shoulderstand and plough, body weight rests on the shoulders and upper back while the neck is strongly flexed. Inversion also raises pressure in the veins of the head and the pressure inside the eyes (Baskaran et al., 2006; Jasien et al., 2015).

**Teaching principles.** Build strength and shoulder mobility first; in headstand, bear most weight through the forearms and shoulders and enter slowly and with control (Hector & Jensen, 2015); in shoulderstand, use folded blankets under the shoulders so that the neck flexes less and the cervical spine is not pressed into the floor; never turn the head while inverted; screen for contraindications (Chapter 26).

### 🔍 WHAT WE KNOW ABOUT INVERSIONS AND THE NECK

Case reports describe neck injuries and rare strokes associated with inversions and extreme neck positions in yoga (Cramer et al., 2013). A laboratory study of experienced practitioners found that the technique used to enter and hold headstand altered the load on the head and neck (Hector & Jensen, 2015). Neither shows that inversions are dangerous for healthy, well-prepared students; both support the traditional insistence on preparation, props and control.

## Arm balances

**Movement.** Body weight is carried on the hands with the elbows bent (crow) or straight, the wrists extended, and the centre of mass brought over the hands by leaning forward.

**Loads.** Wrist extension under load; the shoulder girdle and elbows; the rotator cuff.

**Teaching principles.** Spread the fingers and press through the whole hand, especially the knuckles of the index finger and thumb (Figure 17.8); keep the shoulder blades protracted (“push the floor away”); build time on the hands gradually; teach a controlled way out; place a cushion in front of the head for beginners.

## Hip openers

**Movement.** Mainly external rotation, abduction and flexion of the hip (lotus, butterfly, pigeon, cow-face).

**What limits it.** Soft-tissue tension (adductors, deep rotators, gluteals, capsule) and, very often, the shape of the bones of the hip (Clark, 2016).

**Loads.** When the hip reaches its limit, the rotational force travels to the next joint — the knee — which tolerates twisting badly.

**Teaching principles.** Rotation comes from the hip, not the knee; keep the foot of a bent leg flexed (dorsiflexed) to stabilise the knee; lift the pelvis on a cushion; never force a leg into lotus or press on a student’s knee. Pain in the groin or a pinching at the front of the hip may reflect bony impingement, for which more stretching is not the answer (Griffin et al., 2016).

## Standing postures

**Movement.** The legs form the foundation; the trunk is positioned over them. Standing postures load the joints of the lower limb through their functional range and demand balance.

**Teaching principles.** Build from the ground up: feet, then knees, then pelvis, then spine, arms, head. Track the knee in line with the second and third toes; in bent-knee postures keep the knee roughly over the ankle; adjust stance width and length to the student's proportions; keep the pelvis and spine organised rather than forcing an ideal shape.

## Seated postures

**Movement.** The pelvis rests on the floor with the hips flexed and often rotated.

**What limits it.** Hamstrings and hip rotators pull the pelvis into posterior tilt, rounding the lower back.

**Teaching principles.** Raise the pelvis on a folded blanket or cushion until the knees are level with or below the hips; this allows the pelvis to tilt forward and the spine to lengthen without effort — the foundation of comfortable meditation.

## 20.3 A practical checklist for observing a posture

**Table 20.1** The teacher's observation checklist

| Question                            | What to look for                                                   |
|-------------------------------------|--------------------------------------------------------------------|
| 1. Is the breath steady?            | held breath, gasping, strain in the face = too much                |
| 2. Where is the foundation?         | even weight on feet or hands; knees tracking; stable base          |
| 3. Where is the movement happening? | is it shared (hips and spine) or concentrated (lower back, neck)?  |
| 4. Where is the load going?         | end-range joints; hyperextended knees or elbows; sagging shoulders |
| 5. Tension or compression?          | ask the student where they feel it                                 |
| 6. Active or passive?               | can they hold the range with their own muscles?                    |
| 7. Sthira and sukha?                | steady and easy — or one without the other?                        |

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ♦ Mobility needs stability; active range needs building within passive range.
- ♦ Movement is limited by tension (felt on the lengthened side) or compression (felt on the side moved towards); bone compression does not change with practice.
- ♦ Load concentrated at end range is poorly tolerated; tissues adapt to gradually progressive load.
- ♦ Balance depends on keeping the centre of mass over the base of support, using vision, the vestibular system and proprioception.
- ♦ Each family of postures has characteristic movements, limits, loads and teaching principles; the observation checklist applies to all.



### REFLECTION QUESTIONS

1. Where in your own practice do you rely on passive range you cannot control?
2. Which family of postures do you find easiest? Is it because of mobility, strength or bone shape?
3. What would change in your teaching if you asked students “where do you feel it?” in every class?

### TEACHER-PRACTICE EXERCISES

1. Measure a partner’s active and passive straight-leg raise (with a strap) and discuss the gap.
2. Take a backbend and explore distributing the movement: first let it happen in the lower back, then consciously lift the chest. Compare sensations.
3. Use the observation checklist (Table 20.1) on three classmates in the same posture; compare your notes with a partner.

### KNOWLEDGE CHECK

1. The range a student can reach and hold with their own muscles is: (a) passive range (b) active range (c) end range (d) hypermobility
2. A pinching sensation at the front of the hip in a deep forward bend suggests: (a) tension (b) compression (c) weakness (d) nerve tension
3. Most spinal rotation in a twist comes from the: (a) lumbar spine (b) thoracic spine (c) sacrum (d) pelvis
4. In shoulderstand, folded blankets under the shoulders mainly: (a) increase neck flexion (b) reduce neck flexion (c) increase intraocular pressure (d) make the pose harder
5. In hip openers, if the hip reaches its limit, the joint most at risk is the: (a) ankle (b) knee (c) wrist (d) sacroiliac joint

Answers 1b · 2b · 3b · 4b · 5b

### TEACHING ASSIGNMENT

Choose one posture from each of three families (e.g. a forward bend, a backbend and a twist). For each, write: the main joint movements; the likely tension and compression limits; where load concentrates; two teaching cues; and one modification. Teach the three postures in a 15-minute mini-class and ask your students to report where they felt each posture.

## CHAPTER 21

# The Anatomy and Physiology of Breathing

*“When the breath wanders, the mind is unsteady; when the breath is still, so is the mind.” — Haṭha Yoga Pradīpikā 2.2*

## © LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the structures of the respiratory system and the muscles of quiet and forced breathing;
- ♦ explain how pressure changes move air, and how the diaphragm, ribs and abdomen move in inhalation and exhalation;
- ♦ describe lung volumes and gas exchange, and explain why carbon dioxide is the main driver of breathing;
- ♦ explain the physiology of hyperventilation, breath-holding, nasal breathing and slow breathing;
- ♦ connect breathing mechanics to prāṇāyāma, āsana, meditation and the regulation of the nervous system.

## 21.1 Why breathing is special

Breathing is the only vital rhythm that runs automatically all our lives and can also be taken over at will. That dual control — automatic and voluntary — is why yoga has always treated the breath as the bridge between body and mind. This chapter describes the machinery; **Chapter 27** teaches the practices.

## 21.2 The respiratory system

Air enters through the **nose** (or mouth), passes the **pharynx** and **larynx** — where the vocal folds form the glottis, whose partial closure produces the sound of ujjāyī — and travels down the **trachea**, which divides into the two main **bronchi** and then into ever smaller airways ending in the **alveoli**, tiny air sacs surrounded by capillaries where gas exchange occurs (West & Luks, 2020).

The lungs themselves contain no muscle that could expand them. Each lung is covered by a thin membrane, the **pleura**, which lines the inside of the chest wall too; the thin film of fluid between the two layers links the lungs to the chest wall, so that when the chest expands, the lungs are drawn open with it.

#### ◆ THE NOSE FIRST

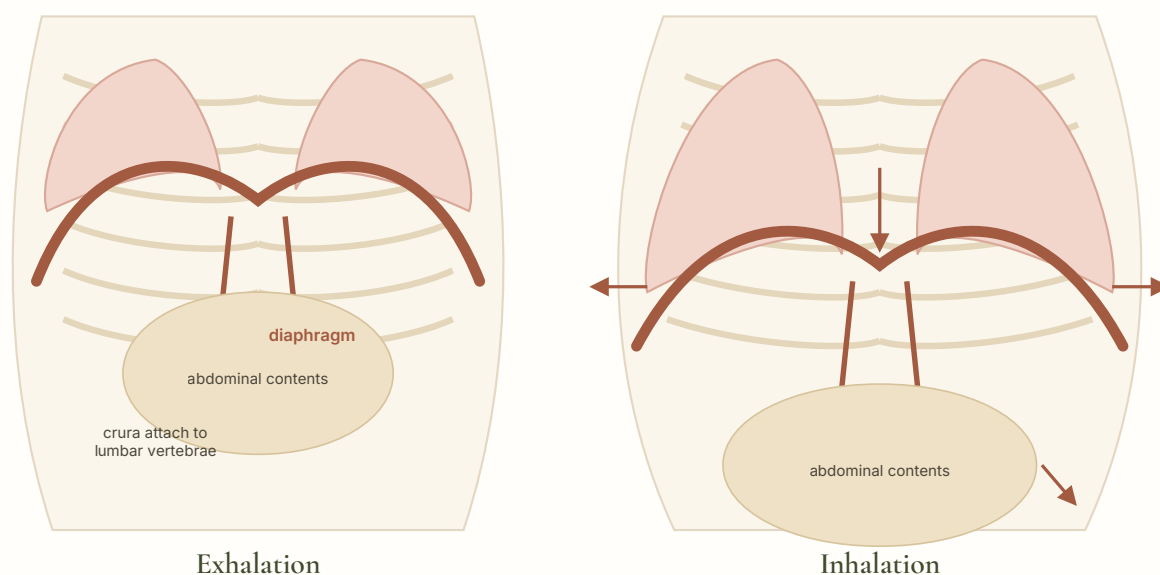
Breathing through the nose warms, humidifies and filters the air, and slows the breath by increasing resistance (West & Luks, 2020). Almost all yogic breathing is done through the nose; the exceptions (*śītalī*, *sītkārī*, and some exhalations) are deliberate.

## 21.3 The muscles of breathing

### The diaphragm

The **diaphragm** is a dome-shaped sheet of muscle separating the chest from the abdomen. It arises from the lower six ribs and their cartilages, the xiphoid process and — through two muscular pillars, the **crura** — from the upper lumbar vertebrae, and inserts into its own central tendon (Moore et al., 2018). It is supplied by the phrenic nerves (C3–C5).

When the diaphragm contracts, its dome moves downward and flattens, enlarging the chest from top to bottom. Because the abdominal organs below it cannot be compressed, they are pushed down and forward — the belly rises — and, as the abdomen resists, the diaphragm's continued contraction lifts and widens the lower ribs (De Troyer & Boriek, 2011). In quiet breathing the dome moves perhaps one to two centimetres; in a maximal breath it moves several centimetres (West & Luks, 2020).

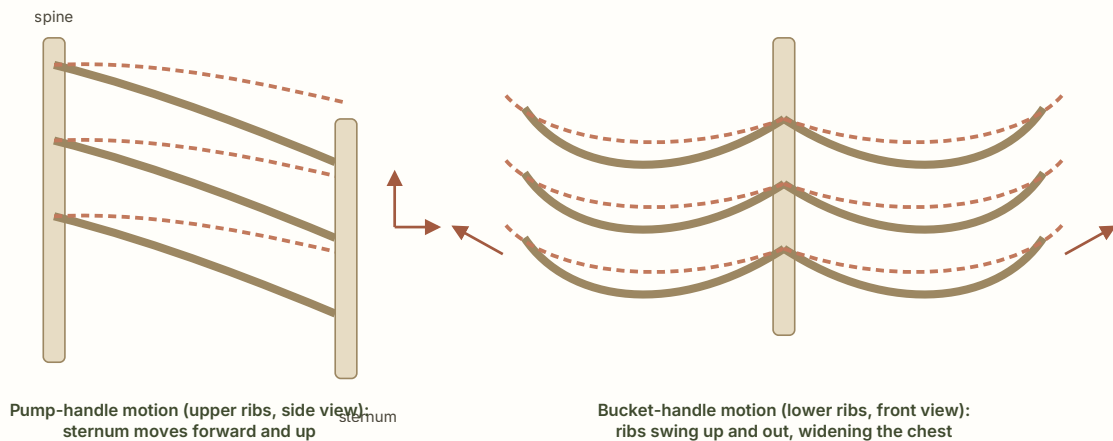


**Figure 21.1** The diaphragm in exhalation and inhalation (front view, schematic). Contracting, the dome descends and flattens, lower ribs widen, and abdominal contents are displaced downward and forward; in quiet breathing, exhalation is largely passive recoil (West & Luks, 2020; Hall & Hall, 2021).

## The intercostal muscles and the ribs

Between the ribs lie the **intercostal muscles**. The **external intercostals** are active in inspiration, helping to raise the ribs; the **internal intercostals** (their interosseous part) assist forced expiration (Moore et al., 2018). The ribs move in two ways:

- ♦ The **upper ribs** move mainly in a “**pump-handle**” fashion: as they rise, the sternum moves forward and up, increasing the front-to-back depth of the chest.
- ♦ The **lower ribs** move mainly in a “**bucket-handle**” fashion: they swing up and out, widening the chest from side to side.



**Figure 21.2** Rib movement during inhalation (dashed = inhaled position, schematic). Upper ribs mainly increase the front-to-back depth of the chest; lower ribs mainly increase its width (Calais-Germain, 2006; West & Luks, 2020).

## Accessory muscles

When more air is needed — or when breathing becomes habitually tense — **accessory inspiratory muscles** join in: the scalenes and sternocleidomastoid lift the upper ribs and sternum; the pectoralis minor and others assist when the arms are fixed (West & Luks, 2020). Habitual reliance on these muscles at rest (“clavicular” or upper-chest breathing) is associated with tension in the neck and shoulders and is one reason students are taught full, diaphragmatic breathing first.

## Muscles of exhalation

At rest, **exhalation is passive**: the diaphragm relaxes, and the elastic recoil of the lungs and chest wall returns them to their resting position. In forced or active exhalation — speaking loudly, coughing, kapālabhāti — the **abdominal muscles** (transversus abdominis, obliques, rectus abdominis) contract to push the abdominal contents up against the diaphragm, and the internal intercostals draw the ribs down (West & Luks, 2020).

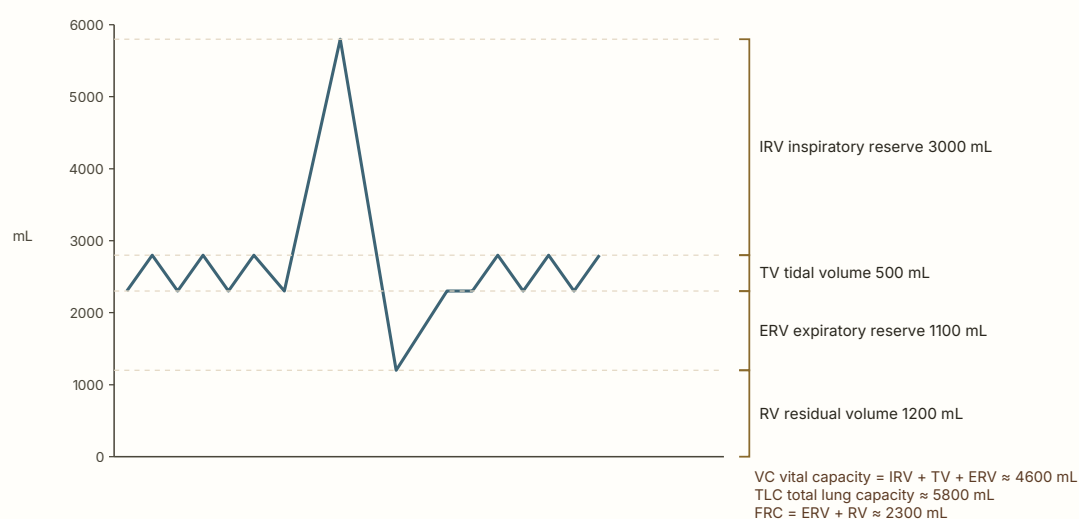
## 21.4 How air moves: pressure

Air moves from higher to lower pressure. When the chest expands, pressure in the lungs falls slightly below atmospheric pressure and air flows in; when it recoils or is compressed, pressure rises above atmospheric pressure and air flows out (Hall & Hall, 2021). The pressure differences in quiet breathing are only a few centimetres of water — which is why small changes in posture and tension can make breathing noticeably easier or harder.

## 21.5 Abdominal pressure and posture

The diaphragm is not only a breathing muscle. Together with the abdominal wall and the pelvic floor it forms a pressurised cylinder around the abdominal organs, and it contracts to raise **intra-abdominal pressure** when we lift, push or stabilise the trunk — even while it continues to breathe (Hodges & Gandevia, 2000). This is the physiological basis of the yogic locks (Chapter 14) and of the connection between steady breathing and a stable trunk in postures such as plank and crow.

## 21.6 Lung volumes



**Figure 21.3** Lung volumes and capacities, using the textbook values for an average young adult man (Hall & Hall, 2021). Values vary with body size, sex, age and training; they are shown to explain what “full breathing” changes (mainly the use of the reserve volumes), not as targets.

In quiet breathing we move only about half a litre of air in each breath (the **tidal volume**). A great deal more can be moved: the **inspiratory reserve volume** above it and the **expiratory reserve volume** below it. Some air — the **residual volume** — always remains in the lungs (Hall & Hall, 2021). “Full yogic breathing” uses these reserve volumes deliberately, which is why it feels so different from ordinary breathing, and why it should never be forced.

## 21.7 Gas exchange and the control of breathing

In the alveoli, oxygen diffuses into the blood and carbon dioxide diffuses out. The rate and depth of breathing are controlled by groups of neurons in the **brainstem** (medulla and pons), which generate the basic rhythm automatically (Hall & Hall, 2021).

Crucially, the strongest chemical drive to breathe is not a lack of oxygen but a rise in **carbon dioxide** (and the accompanying acidity), sensed mainly by **central chemoreceptors** in the brainstem. Peripheral chemoreceptors in the carotid and aortic bodies respond to falling oxygen and rising carbon dioxide (Hall & Hall, 2021). This has two major practical consequences:

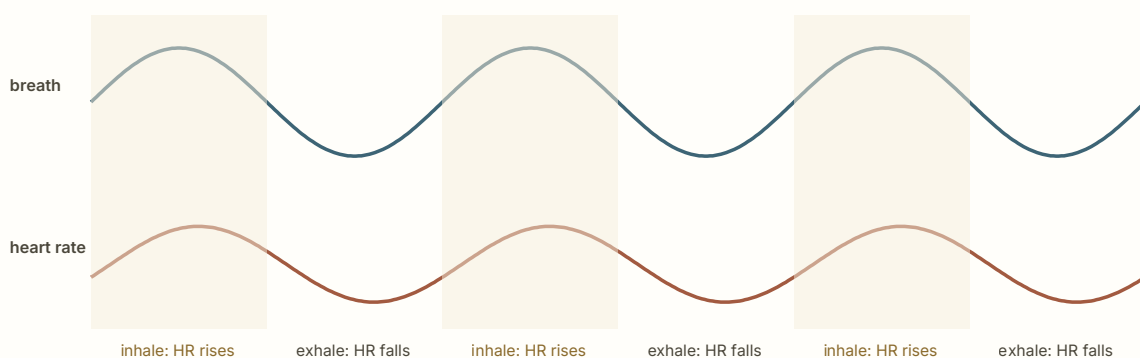
1. **Breath-holding.** The urge to breathe during retention comes chiefly from rising CO<sub>2</sub>, not falling oxygen.
2. **Over-breathing.** Breathing faster or more deeply than the body needs — as can happen in rapid breathing practices — washes out CO<sub>2</sub> (**hypocapnia**). This makes the blood more alkaline and constricts blood vessels in the brain, producing light-headedness, tingling in the fingers and around the mouth, and sometimes cramps or anxiety (West & Luks, 2020).

### ⚠ NEVER OVER-BREATHE BEFORE HOLDING THE BREATH — ESPECIALLY IN WATER

Rapid breathing before a breath-hold removes CO<sub>2</sub> and so delays the urge to breathe, while oxygen continues to fall. Consciousness can be lost without warning. This mechanism, described in underwater swimmers decades ago (Craig, 1961), has caused many drownings. **Never combine vigorous breathing and long retention, never practise breath-holding in or near water, and never practise intense prāṇāyāma while driving or standing.**

## 21.8 Breathing and the heart

Heart rate rises slightly with each inhalation and falls with each exhalation — **respiratory sinus arrhythmia** (RSA), a normal sign of a healthy, flexible autonomic nervous system (Lehrer & Gevirtz, 2014).



**Figure 21.4** Respiratory sinus arrhythmia (schematic): heart rate rises a little with each inhalation and falls with each exhalation. Slow, even breathing near six breaths a minute tends to enlarge these oscillations, which is one physiological basis of slow-breathing practices (Lehrer & Gevirtz, 2014; Russo et al., 2017).

When breathing is slowed to around six breaths a minute (for example, about five seconds in and five seconds out), these oscillations of heart rate become larger and more regular, and they interact with the rhythms of blood pressure control (the **baroreflex**). Slow breathing at this rate has been shown in many

small studies to increase heart rate variability and baroreflex sensitivity, and is associated with reports of calm and reduced anxiety **SCIENTIFIC** (Russo et al., 2017; Zaccaro et al., 2018; Lehrer & Gevirtz, 2014). In one study of yoga beginners, slow breathing with equal inhalation and exhalation increased baroreflex sensitivity most; adding ujjāyī did not increase the effect further (Mason et al., 2013).

#### Q EXHALATION AND CALM

Many teachers say that a longer exhalation “activates the parasympathetic system”. The idea has a physiological basis — heart rate slows during exhalation, and vagal influence on the heart is greater then — and a small randomised study found that daily cyclic breathing with long exhalations improved mood and reduced resting respiratory rate more than an equivalent period of mindfulness meditation (Balban et al., 2023). But the evidence is still limited, effects are modest and individual responses vary. Present it as “tends to be calming” rather than as a switch.

## 21.9 Breathing and movement

In most postures the breath and the movement support each other mechanically:

- ♦ **Inhalation** accompanies **opening and extending** movements — lifting the arms, arching the spine, raising the chest — because inhalation itself extends the thoracic spine slightly and lifts the ribs.
- ♦ **Exhalation** accompanies **folding, twisting and lowering** — forward bends, twists, coming down — because exhalation draws the ribs down and the abdomen in, making space for flexion and rotation.

This is why the classical sun salutation pairs each position with an inhalation or exhalation, and why cues such as “inhale to lengthen, exhale to fold” feel natural.

#### Q THE BREATH AS THE TEACHER’S INSTRUMENT

Watch the breath before you watch the posture. A student holding their breath, breathing through the mouth, or lifting the shoulders to breathe is working beyond their present capacity. “Can you breathe easily here?” is one of the most useful questions in teaching.

## 21.10 Breathing in prāṇāyāma, meditation and relaxation

**Table 21.1** Connecting breathing physiology to yoga practice

| Practice                              | Main mechanical feature                                              | Physiological effect (current understanding)                                   |
|---------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Diaphragmatic breathing in relaxation | slow, abdominal, nasal                                               | reduced accessory - muscle activity; tends to calm                             |
| Full yogic breath                     | uses reserve volumes; abdomen, lower ribs, upper chest               | greater tidal volume; must be slow to avoid over-breathing                     |
| Ujjāyī                                | partial closure of the glottis                                       | increases airway resistance, slows breathing, provides an audible focus        |
| Kapālabhāti                           | rapid active exhalation by the abdominal muscles, passive inhalation | increases ventilation; can cause light-headedness through CO <sub>2</sub> loss |
| Nāḍī śodhana                          | slow, alternate - nostril breathing                                  | slows breathing; lateralised effects unproven (Saoji et al., 2019)             |

| Practice                    | Main mechanical feature           | Physiological effect (current understanding)                                       |
|-----------------------------|-----------------------------------|------------------------------------------------------------------------------------|
| Bhrāmarī                    | prolonged exhalation with humming | slows breathing; greatly increases nasal nitric oxide (Weitzberg & Lundberg, 2002) |
| Breath retention (kumbhaka) | apnoea                            | rising CO <sub>2</sub> ; blood-pressure changes; needs care                        |

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ◆ The lungs are expanded by the diaphragm, intercostals and (when needed) accessory muscles; quiet exhalation is passive, forced exhalation uses the abdominal muscles.
- ◆ The diaphragm descends and flattens in inhalation, displacing the abdomen and widening the lower ribs; upper ribs move like a pump handle, lower ribs like a bucket handle.
- ◆ The diaphragm also contributes to intra-abdominal pressure and trunk stability.
- ◆ Carbon dioxide is the main chemical driver of breathing; over-breathing causes hypocapnia, and over-breathing before breath-holding is dangerous.
- ◆ Slow breathing near six breaths a minute enlarges respiratory sinus arrhythmia and increases baroreflex sensitivity; calming effects are plausible but modest.
- ◆ Inhalation supports extension and opening; exhalation supports folding, twisting and lowering.

### REFLECTION QUESTIONS

1. Lie down and observe your natural breath for five minutes. Where does it move? Where doesn't it?
2. When do you hold your breath in daily life or in practice?
3. How will you explain CO<sub>2</sub> and breath-holding safety to students without frightening them?

### TEACHER-PRACTICE EXERCISES

1. With a partner, place hands on the lower ribs at the sides, then on the upper chest, while they breathe; feel bucket-handle and pump-handle movement.
2. Breathe at six breaths a minute for five minutes (count 5 in, 5 out). Note heart rate and mood before and after.
3. Teach a partner diaphragmatic breathing in śavāsana in under three minutes, using hands on the abdomen and lower ribs.



☑ **KNOWLEDGE CHECK**

1. The main muscle of quiet inspiration is: (a) the external intercostals (b) the diaphragm (c) the scalenes (d) rectus abdominis
2. Quiet exhalation is mainly: (a) active, using the abdominal muscles (b) passive elastic recoil (c) produced by the diaphragm contracting (d) produced by the scalenes
3. The strongest chemical stimulus to breathe is: (a) low oxygen (b) high carbon dioxide (c) low nitrogen (d) high blood sugar
4. Tingling in the fingers during rapid breathing is most likely due to: (a) too much CO<sub>2</sub> (b) too little CO<sub>2</sub> (c) lack of oxygen (d) kuṇḍalinī
5. Respiratory sinus arrhythmia means that heart rate: (a) rises with exhalation (b) rises with inhalation and falls with exhalation (c) is constant (d) stops during retention

Answers 1b · 2b · 3b · 4b · 5b

📄 **TEACHING ASSIGNMENT**

Prepare a ten-minute “breath anatomy for students” talk with a short practice. Explain the diaphragm and ribs using your hands and one simple model (e.g. a balloon, an umbrella), then guide students through three-part breathing, finishing with one sentence on why the breath is never forced. Deliver it to a group of classmates who will note one thing they learned.

## CHAPTER 22

# The Nervous System, Stress and Regulation

*“Yoga is the stilling of the fluctuations of the mind.” — Yoga Sūtra 1.2*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the organisation of the central, peripheral, somatic and autonomic nervous systems;
- ♦ explain the sympathetic and parasympathetic branches, the vagus nerve and the stress response;
- ♦ explain proprioception, interoception and balance, and their relevance to āsana and meditation;
- ♦ describe how motor skills are learned and what this means for cueing;
- ♦ summarise what research suggests about yoga and the nervous system — and what remains uncertain.

## 22.1 Organisation

The **central nervous system** (CNS) is the brain and spinal cord. The **peripheral nervous system** (PNS) consists of the nerves that connect the CNS with the rest of the body: 12 pairs of cranial nerves and 31 pairs of spinal nerves (Kandel et al., 2021). Functionally the PNS is divided into:

- ♦ the **somatic** nervous system, which carries sensation from the skin, muscles and joints and controls voluntary movement of skeletal muscle; and
- ♦ the **autonomic** nervous system (ANS), which regulates the internal organs, blood vessels and glands largely outside conscious control.

Nerves carry information in both directions: **afferent** (sensory) fibres towards the CNS and **efferent** (motor) fibres away from it. A great deal of what yoga trains is afferent — noticing, sensing, feeling.

## 22.2 The autonomic nervous system

The ANS has two main branches, which generally have opposing effects on the organs they both supply (Hall & Hall, 2021):

- ◆ The **sympathetic** branch, whose nerves leave the spinal cord in the thoracic and upper lumbar regions (T1–L2), prepares the body for action: heart rate and blood pressure rise, airways dilate, digestion slows, glucose is released.
- ◆ The **parasympathetic** branch, whose fibres leave the brainstem in four cranial nerves (III, VII, IX and X) and the sacral spinal cord (S2–S4), supports rest, digestion and repair: heart rate slows, digestion and secretion increase.

| Organ             | Sympathetic<br>thoracolumbar outflow (T1–L2) | Parasympathetic<br>craniosacral (CN III, VII, IX, X; S2–S4) |
|-------------------|----------------------------------------------|-------------------------------------------------------------|
| Pupils            | dilate                                       | constrict                                                   |
| Heart rate        | increases                                    | decreases                                                   |
| Airways (bronchi) | dilate                                       | constrict                                                   |
| Digestion         | inhibited                                    | stimulated                                                  |
| Salivation        | reduced, thick                               | increased, watery                                           |
| Liver             | releases glucose                             | stores glycogen                                             |
| Bladder           | relaxes (retains)                            | contracts (empties)                                         |
| Adrenal medulla   | releases adrenaline                          | —                                                           |

Both branches are active all the time; health depends on flexible shifting between them, not on one “winning”. The vagus nerve (CN X) carries roughly three-quarters of all parasympathetic fibres.

**Figure 22.1** Typical effects of the two branches of the autonomic nervous system (Hall & Hall, 2021). The table simplifies: several organs receive only one branch, and responses depend on context.

These are sometimes called “fight or flight” and “rest and digest”. The labels are useful but oversimplified: both branches are active all the time, their balance shifts continuously from moment to moment, and health depends on the **flexibility** of that balance rather than on one branch dominating. A third division, the **enteric** nervous system, is a network of neurons in the wall of the gut.

### The vagus nerve

The **vagus nerve** (cranial nerve X, “the wanderer”) runs from the brainstem through the neck to the heart, lungs and most of the abdominal organs. It carries about three-quarters of all parasympathetic fibres (Hall & Hall, 2021) — and most of its fibres are **sensory**, carrying information from the organs to the brain (Gerritsen & Band, 2018). The vagus slows the heart; this is why heart rate falls during exhalation, when vagal influence is greatest (Figure 21.4).

## Stress and allostasis

When we meet a challenge, the sympathetic system and the hormonal **hypothalamic–pituitary–adrenal (HPA) axis** mobilise the body; cortisol and adrenaline rise. This response is healthy and necessary. Problems arise when it is triggered too often, too strongly or for too long, without recovery – a burden sometimes called *allostatic load* (Hall & Hall, 2021). Much of what yoga offers a stressed modern student is a structured practice of **recovery**: slowing the breath, releasing muscular tension, resting in relaxation.

### Q POLYVAGAL THEORY: POPULAR, CONTESTED

Many yoga trainings teach the **polyvagal theory**, which proposes that a specifically mammalian branch of the vagus supports social engagement and safety, and that the nervous system moves in a hierarchy of states (Porges, 2011). The theory has been influential in trauma-informed practice. However, several of its core anatomical and evolutionary premises have been strongly challenged by researchers in autonomic physiology (Grossman, 2023). It is reasonable to use its everyday language – feeling safe, feeling threatened, feeling shut down – as a description of experience; it is not reasonable to present its neuroanatomy as settled fact.

## 22.3 Sensing the body

### Proprioception

**Proprioception** is the sense of the position and movement of the body and its parts. It depends on muscle spindles, tendon organs, receptors in joints and skin, and the brain's processing of their signals together with vision and the vestibular system (Proske & Gandevia, 2012). Every alignment cue – “feel the weight in the heels”, “notice where your arms are” – is a proprioceptive instruction. Proprioception can be trained: balance practice improves it, and fatigue and injury impair it.

### Interoception

**Interoception** is the sense of the physiological condition of the body – heartbeat, breath, hunger, temperature, visceral sensations, the bodily feelings of emotion (Craig, 2002). Contemplative practices such as body scanning, breath awareness and yoga nidrā cultivate attention to interoceptive signals, and researchers have proposed that improved interoceptive awareness is one of the ways such practices influence emotional regulation and health (Farb et al., 2015). This is an active field; measures of interoception are still debated.

### ◆ PROPRIOCEPTION, INTEROCEPTION AND PRATYĀHĀRA

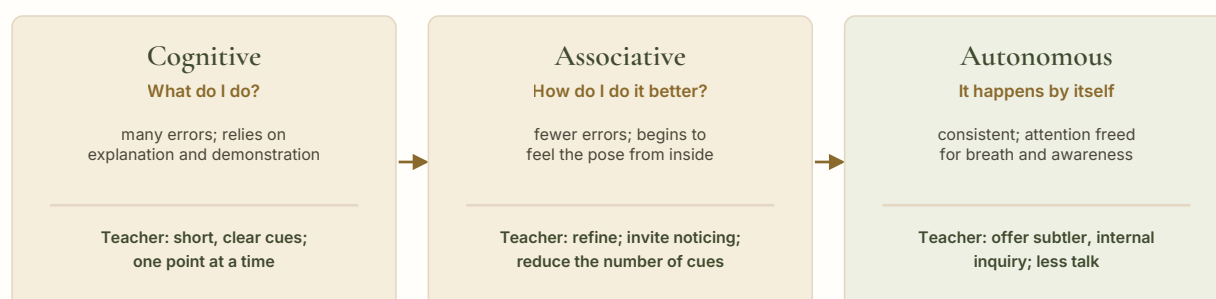
Yoga turns attention from the external world to the body's position (proprioception) and inner condition (interoception). In the language of the tradition, this is the beginning of pratyāhāra (Chapter 7). The two descriptions – traditional and physiological – illuminate each other without being the same thing.

## Balance

Balance integrates three inputs — vision, the vestibular organs of the inner ear, and proprioception — with motor responses from the ankles, hips and stepping (Chapter 20). Yoga-based exercise has been found to improve balance and mobility in adults over 60 (Youkhana et al., 2016), and exercise programmes that challenge balance reduce falls in older people (Sherrington et al., 2019) — one of the clearest practical benefits a teacher can offer older students.

### 22.4 How movement is learned

Skilled movement is learned in recognisable stages (Fitts & Posner, 1967; Schmidt et al., 2019):



**Figure 22.2** Three stages of motor learning after Fitts and Posner, with what each asks of the teacher (Fitts & Posner, 1967). Students move between stages pose by pose, not all at once.

Two findings from motor-learning research are directly useful to yoga teachers:

1. **An external focus of attention often helps.** Instructions that direct attention to the effect of a movement or to the environment (“press the floor away”, “reach towards the wall”) often produce better performance and learning than instructions that direct attention to the body’s own mechanics (“contract your triceps”) (Wulf, 2013). This does not mean internal cues are wrong — in yoga, internal awareness is often the point — but it explains why simple images frequently work better than anatomical instructions for learning a shape.
2. **Autonomy and confidence enhance learning.** Giving learners some choice and supporting their sense of competence improves learning (Wulf & Lewthwaite, 2016). Offering options (“choose the variation that suits you today”) is good pedagogy, not merely kindness.

The nervous system is also **plastic**: it changes with practice. This is the physiological counterpart of the yogic teaching on *samskāras* — repeated experience lays down patterns that shape future experience (Chapter 12).

### 22.5 Pain

Pain is defined by the International Association for the Study of Pain as “an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage” (Raja et al., 2020). Pain is produced by the brain as a protective response; it usually reflects threat to the tissues, but its intensity is also shaped by context, attention, expectation, mood and past experience

(Butler & Moseley, 2013). This has two practical implications: pain always deserves to be taken seriously; and pain does not map neatly onto damage, so a teacher should neither dismiss it nor catastrophise about it (Chapter 26).

## 22.6 Yoga and the nervous system: the evidence

**Table 22.1** A summary of research on yoga and the nervous system

| Claim                                                                        | What research suggests                                                                                    | Strength                                                   |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Slow breathing increases heart rate variability and baroreflex sensitivity   | consistent acute effects in many small studies                                                            | moderate (Zaccaro et al., 2018; Russo et al., 2017)        |
| Yoga reduces physiological markers of stress (e.g. cortisol, blood pressure) | reductions in some markers compared with controls in meta-analysis                                        | low-moderate (Pascoe et al., 2017)                         |
| Yoga improves symptoms of anxiety and depression                             | benefits over no treatment; evidence quality mixed                                                        | low-moderate (Cramer et al., 2018; Cramer et al., 2013)    |
| Yoga increases brain GABA                                                    | one small trial found increases after yoga compared with walking; mechanism hypotheses remain unconfirmed | preliminary (Streeter et al., 2010; Streeter et al., 2012) |
| Yoga “switches on” the parasympathetic system                                | an oversimplification; effects depend on practice type and person                                         | —                                                          |
| Yoga improves balance in older adults                                        | consistent small-moderate benefits                                                                        | moderate (Youkhana et al., 2016)                           |

### Q SPEAKING ABOUT THE NERVOUS SYSTEM IN CLASS

Accurate and helpful: “Slowing the breath and lengthening the exhalation tends to calm the body.” Inaccurate and unhelpful: “This pose stimulates your vagus nerve and resets your nervous system.” Choose language that invites students to notice their own experience rather than language that promises a mechanism.

## CHAPTER REVIEW

### KEY TAKEAWAYS

- ◆ The nervous system comprises the CNS and PNS; the PNS includes somatic and autonomic divisions.
- ◆ The sympathetic and parasympathetic branches both operate continuously; health reflects flexible balance. The vagus nerve carries most parasympathetic fibres, most of them sensory.
- ◆ Proprioception and interoception are trainable senses at the heart of posture and meditation practice.
- ◆ Motor learning progresses through stages; external-focus cues and learner autonomy aid learning.
- ◆ Pain is a protective output of the brain shaped by context, not a direct measure of damage.
- ◆ Evidence for yoga’s effects on the nervous system is promising in places but often preliminary; polyvagal theory is contested.

 REFLECTION QUESTIONS

1. How do you know, from the inside, that you are stressed? Which signals come first?
2. Recall learning a difficult posture. Can you identify the cognitive, associative and autonomous stages?
3. Which claims about yoga and the nervous system have you heard in classes? How would you now rephrase them?

 TEACHER-PRACTICE EXERCISES

1. Teach the same posture twice to a partner: once with only internal cues, once with only external cues. Compare.
2. Lead a three-minute interoceptive awareness practice (heartbeat, breath, belly, face).
3. Rewrite five nervous-system claims from popular yoga media into accurate, modest language.

 KNOWLEDGE CHECK

1. The parasympathetic outflow is: (a) thoracolumbar (b) craniosacral (c) only cervical (d) only lumbar
2. Most fibres in the vagus nerve are: (a) motor (b) sensory (c) sympathetic (d) somatic
3. Interoception refers to the sense of: (a) body position (b) the internal physiological condition of the body (c) balance (d) touch on the skin
4. Which cue uses an external focus of attention? (a) “contract your quadriceps” (b) “press the floor away” (c) “rotate your femur” (d) “engage your core”
5. According to the IASP definition, pain is: (a) always a sign of tissue damage (b) a sensory and emotional experience associated with actual or potential tissue damage (c) purely psychological (d) a reflex

Answers 1b · 2b · 3b · 4b · 5b

 TEACHING ASSIGNMENT

Design a 15-minute “down-regulation” practice for a stressed office worker, using only breath, gentle movement and relaxation. For each element, write one sentence of honest explanation you could give the student about why it might help, citing the kind of evidence (strong, moderate, preliminary, traditional) behind it.

## CHAPTER 23

# Preparing the Body — Relaxation, Sūrya Namaskār and Leg Raises

*“Lying flat on the ground like a corpse is śavāsana. It removes fatigue and brings rest to the mind.” — Haṭha Yoga Pradīpikā 1.32*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ teach śavāsana and a complete systematic relaxation, at the beginning, between postures and at the end of class;
- ♦ teach the twelve positions of the classical Sūrya Namaskār with correct breath, alignment and mantras;
- ♦ analyse the sun salutation anatomically and offer modifications for common limitations;
- ♦ teach single and double leg raises safely;
- ♦ explain why these practices precede the twelve basic postures.

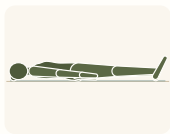
## 23.1 The shape of the opening

A traditional class does not begin with effort. It begins lying down, with a few minutes of relaxation, followed by prāṇāyāma (Chapter 27), and only then moves into sun salutation and leg raises. This order is deliberate: the student arrives, the breath settles, the body warms gradually, and the abdominal and back muscles are prepared for the inversions that follow.

## 23.2 Śavāsana: the relaxation posture

Śavāsana (“corpse posture”) is the most important posture in the class and, many teachers say, the hardest to master.





1  
Śavāsana

lying relaxed on the  
back

**Technique.** Lie on the back with the feet a little more than hip-width apart and falling outward; the arms about 45 degrees from the body with the palms facing up; the head in line with the spine (a thin folded blanket under the head if the chin tilts up). Close the eyes. Let the breath become slow and natural, moving mostly in the abdomen.

**Modifications.** A bolster or rolled blanket under the knees relieves the lower back; a blanket over the body keeps students warm; students in later pregnancy, and anyone uncomfortable lying flat, rest on the side (usually the left) with a cushion between the knees; those who become anxious lying with the eyes closed may keep them open with a soft downward gaze.

## Relaxation between postures

After each posture the classical method returns to rest — śavāsana after supine and seated postures, and **makarāsana** (the crocodile: lying on the front, forehead resting on the stacked hands or with the cheek to one side) after prone backbends. A few breaths of rest let the heart rate and breath return towards resting levels and allow students to notice the effects of the posture — an essential part of the practice, not an interruption of it.

## Final relaxation

The final relaxation lasts ten to fifteen minutes in a 90-minute class. In the classical class taught here it has three stages:

1. **Tensing and relaxing** the body part by part: lift the right leg a little off the floor, tense it, hold, and let it drop; then the left leg; each arm; the buttocks; the chest (by filling it with air); the shoulders (drawn up and released); the face (squeezed and then opened wide). Gently roll the legs, arms and head from side to side.
2. **Autosuggestion:** the teacher guides attention slowly through the body from the feet upward, suggesting relaxation of each part in turn — “relax the toes... relax the soles of the feet...” — including the internal organs and the face.
3. **Relaxation of the mind:** attention rests on the breath, then on the awareness that observes body and mind. The student remains still in silence for several minutes.

To come out, deepen the breath, move the fingers and toes, stretch, roll to one side and come up slowly.

### ||| A SHORT RELAXATION SCRIPT (OPENING OF CLASS)

*Lie down and let the body become heavy. Let the floor take your weight... Feel the back of the head, the shoulder blades, the backs of the hands, the backs of the heels... Let the breath come and go by itself...*

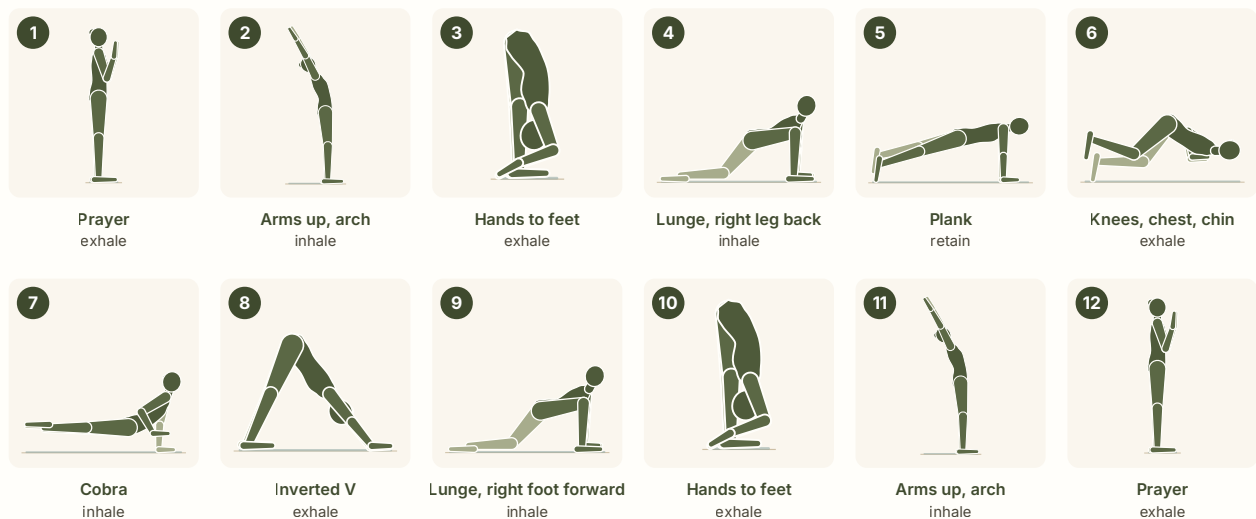
*With each exhalation, let a little more tension go... Notice the belly rising as you breathe in, falling as you breathe out... There is nothing to do. Just rest, and breathe.*

### ⚠ RELAXATION IS NOT ALWAYS RELAXING

For some students — particularly those with trauma histories, anxiety or panic — lying still with the eyes closed can be uncomfortable or distressing. Always offer choices: eyes open or closed, lying on the back, side or with knees bent, or sitting. Keep your voice steady and your guidance concrete (Chapter 31). If a student becomes distressed, speak calmly, invite them to open the eyes, feel their feet or hands, and sit up if they wish.

## 23.3 Sūrya Namaskār: the sun salutation

The sun salutation is a flowing sequence of twelve positions, each coordinated with one breath. The form taught in the tradition taught here was widely popularised in the 1920s and 1930s, though salutations of the sun are of course far older as a devotional practice (Singleton, 2010) **HISTORICAL**. It warms the whole body, links breath and movement, flexes and extends the spine in alternation, and prepares the shoulders, hips and legs for the postures to come.



**Figure 23.1** Sūrya Namaskār as taught in this course: twelve positions, each linked to one breath. The right leg steps back in position 4 and forward in position 9; the next round begins with the left leg. The lunge figures show the stepping leg in the lighter tone.

**Table 23.1** The twelve positions of Sūrya Namaskār as taught in this course

| # | Position                                       | Breath | Mantra (optional) | Key actions                                                               | Common modifications                                           |
|---|------------------------------------------------|--------|-------------------|---------------------------------------------------------------------------|----------------------------------------------------------------|
| 1 | Prayer (standing, palms together at the chest) | exhale | Oṃ Mitrāya Namaḥ  | stand evenly; feet together or hip-width                                  | feet apart for balance                                         |
| 2 | Arms raised, gentle backbend                   | inhale | Oṃ Ravaye Namaḥ   | reach up; lift the chest; hips slightly forward; arch from the upper back | arms shoulder-width; minimal backbend; hands on the lower back |
| 3 | Hands to feet (forward bend)                   | exhale | Oṃ Sūryāya Namaḥ  | fold from the hips; hands beside the feet                                 | bend the knees freely; hands on shins or blocks                |

| #  | Position                             | Breath                               | Mantra (optional)        | Key actions                                                                                   | Common modifications                                        |
|----|--------------------------------------|--------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 4  | Lunge: right leg back, knee down     | inhale                               | Oṃ Bhānave Namaḥ         | front knee over the ankle; hands beside the front foot; lift the chest and look forward or up | hands on blocks; back knee cushioned                        |
| 5  | Plank (both legs back)               | retain the breath in, or breathe out | Oṃ Khagāya Namaḥ         | one straight line from heels to head; hands under shoulders                                   | knees on the floor                                          |
| 6  | Knees, chest and chin to the floor   | exhale                               | Oṃ Pūṣṇe Namaḥ           | lower knees, then chest and chin; hips raised; elbows close to the body                       | lower the whole body to the floor                           |
| 7  | Cobra                                | inhale                               | Oṃ Hiraṇyagarbhāya Namaḥ | slide forward and up; elbows bent and close; shoulders down                                   | low cobra; sphinx                                           |
| 8  | Inverted V (downward dog)            | exhale                               | Oṃ Marīcaye Namaḥ        | press hips up and back; lengthen the spine; heels towards the floor                           | knees bent; heels lifted                                    |
| 9  | Lunge: right foot forward, knee down | inhale                               | Oṃ Ādityāya Namaḥ        | step the foot between the hands; chest lifted                                                 | help the foot forward with the hand                         |
| 10 | Hands to feet                        | exhale                               | Oṃ Savitre Namaḥ         | bring the back foot forward; fold                                                             | knees bent                                                  |
| 11 | Arms raised, gentle backbend         | inhale                               | Oṃ Arkāya Namaḥ          | come up with a long spine, arms alongside the ears                                            | bend the knees and come up with the back rounded, head last |
| 12 | Prayer                               | exhale                               | Oṃ Bhāskarāya Namaḥ      | return to stillness                                                                           | —                                                           |

The second half-round repeats the sequence with the **left** leg stepping back in position 4 and forward in position 9. Beginners start with two to four rounds at a slow pace and gradually increase; the classical class allows up to twelve rounds. The mantras are twelve names of the sun, one for each position; they may be chanted aloud, mentally, or omitted.

#### ☯ PACE AND BREATH

The breath sets the pace, not the teacher's count. If students are gasping, slow down or add a breath in positions 5 and 8. For beginners, “step” rather than “jump”, and move through positions 4–9 with an extra breath where needed.

### What the sun salutation does, anatomically

The sequence moves the spine alternately into extension (positions 2, 4, 7, 9, 11) and flexion (3, 6, 8, 10); flexes and extends the hips (the back leg of the lunge extends the hip and lengthens the hip flexors; the forward bends flex it); loads the arms and shoulders in weight-bearing (5–8); and strengthens the trunk in plank. The upward flow of positions 1–2 and 11–12 involves shoulder flexion and scapular upward rotation. Repeated rounds raise heart rate moderately. It is therefore a whole-body warm-up — and, practised briskly for many rounds, a meaningful cardiovascular exercise.

**Table 23.2** Sūrya Namaskār: positions of the sun salutation

| Asana                                   |                                                                                     | Joints                                     | Movement                                                | Key muscles                                                                  | Muscle action | Loading                                                    | Common limitations                     | Modifications                            |
|-----------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------|---------------|------------------------------------------------------------|----------------------------------------|------------------------------------------|
| <b>1 Prayer</b><br>(Prajñāmasana)       |    | Ankles, knees, hips, spine                 | Standing neutral; elbow flexion                         | Postural muscles of legs and trunk                                           | I (low level) | Standing load                                              | —                                      | Feet hip-width                           |
| <b>2 Arms raised</b> (Hasta Uttānāsana) |   | Shoulders, thoracic and lumbar spine, hips | Shoulder flexion; spinal extension; hip extension       | Deltoids, serratus anterior, trapezius; back extensors; rectus abdominis (L) | C then I      | Lumbar extension if thoracic spine and shoulders are stiff | Shoulder flexion; low-back sensitivity | Minimal backbend; arms wide              |
| <b>3 Hands to feet</b>                  |  | Hips, spine, knees                         | Hip and spinal flexion                                  | Hamstrings, back extensors (E/L)                                             | E then L      | Loaded flexion                                             | Hamstring length                       | Bend knees                               |
| <b>4 Lunge</b> (Āśva Sañcalanāsana)     |  | Hips, knees, ankles, spine                 | Front hip flexion; back hip extension; spinal extension | Hip flexors of back leg (L); quadriceps and gluteals of front leg (I)        | I / L         | Back knee on floor; front knee flexion                     | Hip-flexor length; knee comfort        | Blanket under back knee; hands on blocks |

| Asana                               |                                                                                   | Joints                                 | Movement                                              | Key muscles                                                                  | Muscle action                   | Loading                                  | Common limitations                                  | Modifications                             |
|-------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------|------------------------------------------|-----------------------------------------------------|-------------------------------------------|
| 5 Plank                             |  | Wrists, shoulders, trunk, hips         | Wrist extension; shoulder flexion ~90°; neutral trunk | Serratus anterior, pectorals, triceps, abdominals, quadriceps                | I                               | Body weight on hands and toes            | Upper-body and trunk strength; wrists               | Knees down                                |
| 6 Knees, chest, chin                |  | Shoulders, elbows, spine, hips         | Elbow flexion; spinal extension; hip flexion          | Triceps, pectorals (E)                                                       | E lowering                      | Controlled lowering onto chest and chin  | Upper-body strength                                 | Lower whole body to the floor             |
| 7 Cobra                             |  | Spine, shoulders                       | Spinal extension                                      | Back extensors (C/I)                                                         | C then I                        | Lumbar compression                       | Upper-back stiffness                                | Sphinx; low cobra                         |
| 8 Inverted V (Adho Mukha Śvānāsana) |  | Shoulders, wrists, hips, knees, ankles | Shoulder flexion; hip flexion; ankle dorsiflexion     | Serratus anterior, deltoids, triceps (I); hamstrings, calves, latissimus (L) | I upper body; L posterior chain | Weight on hands and feet; mild inversion | Hamstring and calf length; shoulder flexion; wrists | Knees bent; heels lifted; hands on blocks |

Positions 9–12 repeat positions 4, 3, 2 and 1.

## 23.4 Leg raises

After the sun salutation, the classical class includes **leg raises** lying on the back, to strengthen the abdominal and hip-flexor muscles in preparation for headstand, shoulderstand and plough.



**Single leg raises.** Lie on the back with the legs together and the palms flat on the floor beside the body (or under the buttocks for support). Inhale and raise one leg as high as comfortable, keeping it straight and the other leg on the floor; exhale and lower it slowly. Alternate legs, several times each.

**Double leg raises.** Keeping the lower back in contact with the floor, inhale and raise both legs together to vertical; exhale and lower them slowly and with control. Repeat several times.

### ⚠️ LEG RAISES AND THE LOWER BACK

When both legs are lowered, the hip flexors, including psoas, pull strongly on the pelvis and lumbar spine; if the abdominal muscles cannot hold the pelvis, the lower back arches off the floor. Beginners and students with low-back pain should raise one leg at a time, bend the knees, keep the hands under the buttocks, or lower the legs only as far as the back stays grounded. Leg raises are best avoided or modified in pregnancy and after recent abdominal surgery.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ The classical class opens with relaxation and prāṇāyāma before movement; śavāsana is taught with care at the beginning, between postures and at the end.
- ◆ Final relaxation has three stages: tensing and relaxing, autosuggestion through the body, and relaxation of the mind.
- ◆ The twelve positions of Sūrya Namaskār are each linked to a breath and, optionally, to one of twelve names of the sun; the stepping leg alternates between half-rounds.
- ◆ The sun salutation alternates spinal extension and flexion and warms the whole body; its modifications should be offered freely.
- ◆ Leg raises strengthen the abdomen and hip flexors; the lower back must remain supported.

### 🍷 REFLECTION QUESTIONS

1. Can you relax in śavāsana? What happens in your mind in the first two minutes?
2. Which position of the sun salutation is weakest in your own practice?
3. Why might the tradition insist on relaxation between postures when many modern classes omit it?

## ✧ TEACHER-PRACTICE EXERCISES

1. Memorise the twelve positions with their breaths and mantras; teach one full round (both sides) to a partner by voice alone, without demonstrating.
2. Lead a ten-minute final relaxation following the three stages. Time each stage.
3. Teach leg raises to a partner, including two modifications, and observe the lower back throughout.

## ☑ KNOWLEDGE CHECK

1. Position 5 of the classical Sūrya Namaskār is: (a) cobra (b) plank (c) inverted V (d) lunge
2. The breath in position 7 (cobra) is: (a) exhale (b) inhale (c) retain (d) any
3. In the second half-round, which leg steps back in position 4? (a) right (b) left (c) both (d) either
4. The first stage of the classical final relaxation is: (a) autosuggestion (b) tensing and relaxing body parts (c) breath counting (d) chanting
5. During double leg raises, the lower back tends to arch because: (a) the hamstrings are tight (b) the hip flexors pull on the pelvis and spine when the abdominals cannot hold it (c) the diaphragm contracts (d) of weak calves

Answers 1 b · 2 b · 3 b · 4 b · 5 b

## 📄 TEACHING ASSIGNMENT

Teach the first 25 minutes of a classical-style class to three classmates: opening relaxation (3 min), a short breathing practice (5 min), four rounds of sun salutation with breath and optional mantras, and leg raises with modifications. Record the class. Watching it back, note every place your instructions and the students' breath were out of step.

## CHAPTER 24

## The Twelve Basic Postures

*“Posture should be steady and comfortable.” — Yoga Sūtra 2.46*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ practise and teach each of the twelve basic postures of the classical sequence, with preparation, technique, breathing and alignment;
- ◆ describe the joints, muscles, spinal actions, planes and loads involved in each posture;
- ◆ recognise common mistakes and limitations and offer appropriate modifications, beginner and advanced variations and props;
- ◆ apply the contraindications and precautions for each posture, and know when hands-on help is and is not appropriate;
- ◆ distinguish, for each posture, the effects traditionally attributed to it from what modern evidence supports.

## 24.1 How to use the posture monographs

The twelve basic postures are the heart of the classical class (Chapter 9, Chapter 34). Each is presented in a monograph with the same structure, so that you can compare them and later use the same framework for any posture you teach:

1. **Names, meaning and facts** — Sanskrit (IAST and Devanāgarī), English, an approximate pronunciation guide, the family of postures, level, hold and counterpose.
2. **Purpose and preparation.**
3. **Technique** step by step, **breathing** and **alignment**.
4. **Anatomy** — joints, muscles and their actions, spinal action and planes, and the **biomechanics** of how the posture loads the body.
5. **Common mistakes** and **common student limitations**.
6. **Contraindications** and **precautions**.
7. **Modifications, beginner and advanced variations** and **props**.
8. **Teaching** — safe cues and hands-on guidance.
9. **Transitions** and **counterpose**.



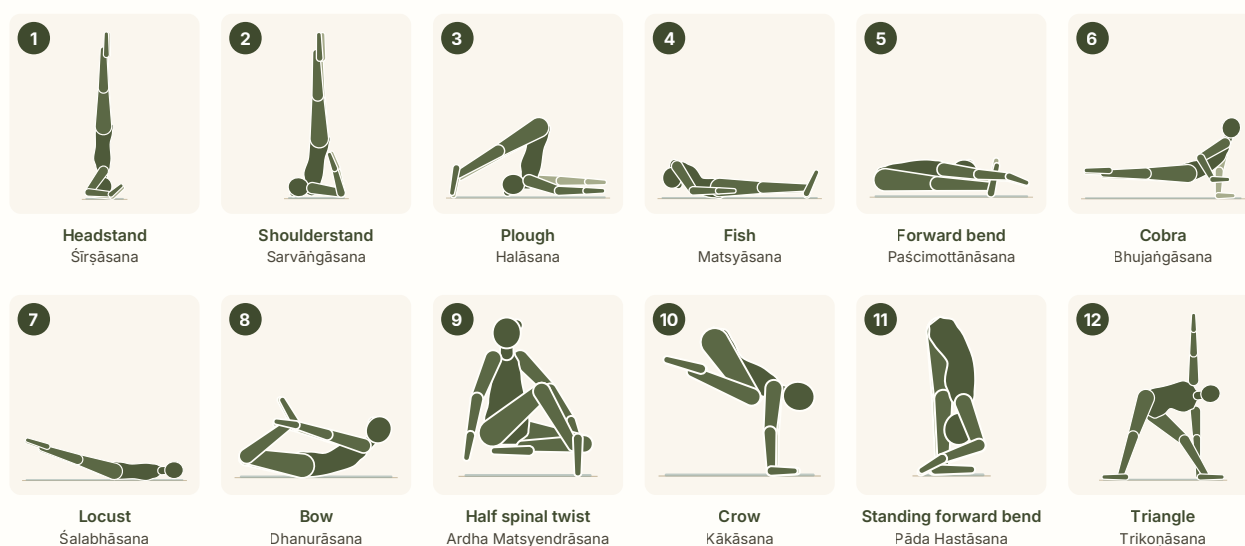
### 10. Effects traditionally attributed and what modern evidence does and does not support.

The illustrations show one way the posture may look; they are not a template every body must match (Chapter 17). Where a figure carries coloured overlays, **terracotta** marks muscles that are lengthening under load and **blue** marks muscles that are working; these are schematic locations.

#### ⚠ BEFORE TEACHING ANY INVERSION

Inversions come first in the classical sequence and carry the highest demands. Before teaching headstand, shoulderstand or plough to any group, ask about neck problems, high blood pressure, heart disease, glaucoma and other eye conditions, pregnancy, osteoporosis, recent head injury and dizziness. Offer the supported alternatives (dolphin, half shoulderstand on blankets, legs up the wall) without making anyone feel singled out.

## 24.2 The sequence at a glance



**Figure 24.1** The twelve basic postures of the classical sequence in their traditional order. Savāsana is taken between postures.

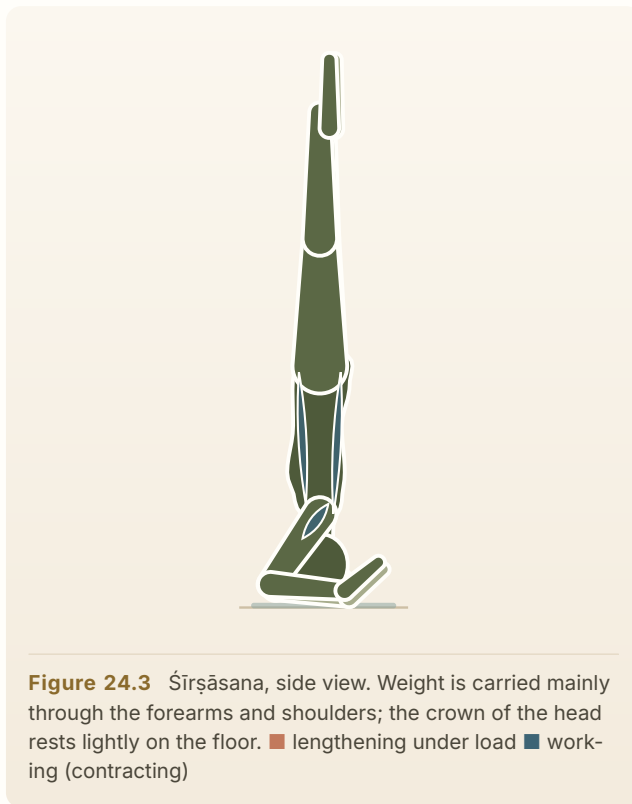
A brief relaxation in śavāsana (or makarāsana after the prone postures) follows each posture. In a 90-minute class the twelve take roughly 50–60 minutes, including relaxations; in a shorter class some are shortened or replaced by their preparations, but the order is kept.

# 1 Śīrṣāsana

## HEADSTAND

शीर्षासन

sheer-SHAH-sa-na (approximate English guide)

**FAMILY**

Inversion; balance

**LEVEL**

Intermediate (preparations suit beginners)

**TRADITIONAL HOLD**

Build from 10 seconds to 3 minutes or more

**MAIN PLANES**

Vertical alignment; balance in the sagittal and frontal planes

**COUNTERPOSE**

Bālāsana (child's pose), then śavāsana

Śīrṣa means “head”. Headstand is traditionally called the “king of āsanās” (Iyengar, 1966), and the classical sequence places it first. Although inverted practice is described in the Haṭha texts as the mudrā *viparītakaraṇī*, the headstand as taught today is largely a modern development of that idea (Mallinson & Singleton, 2017).

### 01 Purpose

To build strength, balance and concentration in an inverted position; to reverse the habitual relationship of the body to gravity; and, in the traditional view, to calm and focus the mind. It is the most demanding of the twelve and is taught progressively.

### 02 Preparation

- ◆ Regular practice of dolphin, downward dog and plank to build shoulder and trunk strength.
- ◆ Sufficient shoulder flexion to bring the arms beside the head without arching the back.
- ◆ Child's pose (bālāsana) before and after to accustom the head to lower positions.
- ◆ A folded blanket or firm mat under the head and forearms; a clear space or a wall behind for beginners.
- ◆ Practise on an empty stomach, after the sun salutation and leg raises, while fresh.

### 03 Technique, step by step

- 1 Kneel and sit back on the heels. Hold the elbows with the opposite hands and place them on the floor under the shoulders; this sets the distance between the elbows.
- 2 Keeping the elbows where they are, release the hands and interlock the fingers, forming a triangle with the forearms. Tuck the lower little finger in so that the wrists stand upright.
- 3 Place the crown of the head on the floor with the back of the head cupped in the hands.
- 4 Straighten the knees and lift the hips, forming an inverted V. Press the forearms down and lift the shoulders away from the floor.
- 5 Walk the feet slowly towards the face until the hips are above the shoulders and the back is nearly vertical.
- 6 Bend the knees and draw them towards the chest, lifting the feet off the floor with control, not a jump. Pause here.
- 7 Raise the bent knees towards the ceiling, then straighten the legs. The body forms one vertical line; the feet are relaxed or gently flexed.
- 8 Hold, breathing slowly through the nose. To come down, reverse the steps: bend the knees, lower them to the chest, lower the feet to the floor, and rest in child's pose for several breaths before sitting up.

### 04 Breathing

Breathe slowly and evenly through the nose throughout. Never hold the breath to get up. A steady breath is the best sign that the student is ready; a held or strained breath means come down.

### 05 Alignment

- ◆ Forearms and shoulders carry most of the weight; the head rests lightly. A student should be able to lift the head slightly off the floor at any time.
- ◆ Elbows no wider than the shoulders; wrists vertical; forearms pressing firmly.
- ◆ Shoulders lifted away from the ears (active serratus anterior and trapezius); the neck long, neither flexed nor arched.
- ◆ Hips stacked over shoulders; ribs drawn in so that the lower back does not arch (a “banana” shape).
- ◆ Legs together, active; weight evenly on both forearms.

### 06 Major joints and their positions

| Joint                    | Position / movement                                                   |
|--------------------------|-----------------------------------------------------------------------|
| Cervical spine           | Near neutral, under axial load; must not flex, extend or turn         |
| Shoulders (glenohumeral) | Flexed (arms beside the head); scapulae upwardly rotated and elevated |
| Elbows                   | Flexed about 90°; forearms on the floor                               |
| Spine                    | Neutral, vertical                                                     |
| Hips and knees           | Extended (neutral); flexed during entry and exit                      |
| Ankles                   | Neutral or gently plantarflexed                                       |

## 07 Major muscles and what they are doing

| Muscle(s)                                      | Action in this pose               | Role                                                                                       |
|------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------|
| Serratus anterior, trapezius (upper and lower) | Isometric                         | Rotate the scapulae upward and lift the shoulders away from the floor, protecting the neck |
| Deltoids, rotator cuff, triceps                | Isometric                         | Stabilise the shoulders; press the forearms down                                           |
| Abdominals and back extensors                  | Isometric, continuously adjusting | Keep the trunk vertical; prevent the lower back from arching                               |
| Hip flexors                                    | Concentric then eccentric         | Lift and lower the legs during entry and exit                                              |
| Gluteals and adductors                         | Isometric                         | Keep the legs together and extended                                                        |

## 08 Spinal action and planes of movement

The spine is held in neutral, vertical and inverted. Load travels down through the forearms and shoulder girdle and — partly — through the head into the cervical spine. The critical skill is keeping the neck in neutral while it is loaded.

Movement into and out of the pose occurs in the sagittal plane (hip flexion and extension). Holding is a balancing task in the sagittal and frontal planes, controlled from the forearms and trunk.

## 09 Biomechanics: how the pose loads the body

The base of support is the triangle of forearms and head; the centre of mass must stay over it. A wider elbow position reduces the triangle and increases load on the head. A laboratory study of experienced practitioners found that entry and holding technique altered head and neck loading (Hector & Jensen, 2015), consistent with the traditional emphasis on slow, controlled entry. Kicking or jumping into the pose applies sudden loads to the neck.

### COMMON MISTAKES

- ◆ Elbows sliding wider than the shoulders.
- ◆ Most of the weight on the head, with shoulders collapsing towards the floor.
- ◆ Kicking or jumping up; coming down suddenly.
- ◆ Arching the lower back (“banana” shape).
- ◆ Turning the head or looking around while inverted.
- ◆ Staying up with a held breath or a red, strained face.

### COMMON STUDENT LIMITATIONS

- ◆ **Weak shoulders or trunk:** remain in the preparation (dolphin, walking in) until it is steady for a minute.
- ◆ **Limited shoulder flexion:** the back arches to compensate; work on shoulder mobility and practise at a wall.
- ◆ **Fear:** normal; build confidence at a wall and in the tuck position before straightening the legs.
- ◆ **Tight hamstrings:** make walking in difficult; bend the knees.

### ⚠️ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Neck injury, cervical disc problems or neck pain; cervical instability
- ◆ Uncontrolled high blood pressure, heart disease, history of stroke
- ◆ Glaucoma, retinal detachment or other serious eye conditions; recent eye surgery (Baskaran et al., 2006)
- ◆ Osteoporosis affecting the spine
- ◆ Pregnancy, unless the student has an established practice and medical approval; never learn it while pregnant
- ◆ Recent head injury; migraine or headache at the time; vertigo or inner-ear problems
- ◆ Acute sinus infection or ear infection

#### **Precautions:**

- ◆ Learn with a teacher, near a wall, on a soft but firm surface.
- ◆ Build time gradually; come down at the first sign of strain, pressure in the head or eyes, or dizziness.
- ◆ Many traditional teachers advise avoiding inversions during menstruation; there is no clear medical evidence of harm, and the choice can be left to the student.
- ◆ Rest in child's pose afterwards, and sit up slowly.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Preparation: dolphin — forearms down, hips lifted



Walking in: crown down, feet stepping towards the face



Tuck: knees to chest, balanced before straightening

### **MODIFICATIONS**

- ◆ Practise with the back a few centimetres from a wall; use the wall only for security, not to lean on.
- ◆ Stay at the walking-in stage or in the tuck, with the feet close to the floor.
- ◆ Substitute dolphin or supported half shoulderstand for students who should not load the head.

### **BEGINNER VARIATION**

Dolphin and walking in (shown above), held for 5–10 breaths, then child's pose. When these are steady for a minute, try lifting one foot, then both knees into the tuck, at a wall.

### **ADVANCED VARIATION**

Variations with the legs (splits, lotus in headstand, legs lowered to horizontal with control) and the tripod headstand (hands flat, śīrṣāsana II), taught only when the classic pose is held steadily for several minutes.

## USE OF PROPS

- ◆ Folded blanket or firm mat under head and forearms.
- ◆ A wall for security.
- ◆ A strap around the upper arms just above the elbows to keep them shoulder-width.

## 11 Teaching the pose

## SAFE TEACHING CUES

- ◆ “Press the forearms down and lift the shoulders away from the floor.”
- ◆ “The head rests; the arms carry you.”
- ◆ “Walk in until the hips are over the shoulders — no jumping.”
- ◆ “Draw the ribs in and lengthen up through the legs.”
- ◆ “If the breath becomes strained, come down slowly.”

## HANDS-ON GUIDANCE

Hands-on help, when used, is for **safety and confidence**, not for lifting a student into the pose. With consent, stand beside the student and place a hand lightly at the hips or thighs as they lift, or steady the legs once they are up. Never pull or lift a student’s legs, never stand in front of them, and never adjust the neck or head. Teach the exit before the entry.

## 12 Transitions and counterpose

|                 |                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------|
| Into the pose   | From kneeling (after leg raises and a brief rest) or from child’s pose.                           |
| Out of the pose | Reverse the entry with control; rest in child’s pose for at least 5–10 breaths before sitting up. |
| Counterpose     | Child’s pose, then śavāsana. In the classical sequence, shoulderstand follows after rest.         |

### ⚠ EFFECTS TRADITIONALLY ATTRIBUTED

In the tradition taught here headstand is said to reverse the effects of gravity on the body, increase circulation to the brain, improve memory and concentration, and bring the mind to calmness and balance; it is called the king of postures (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

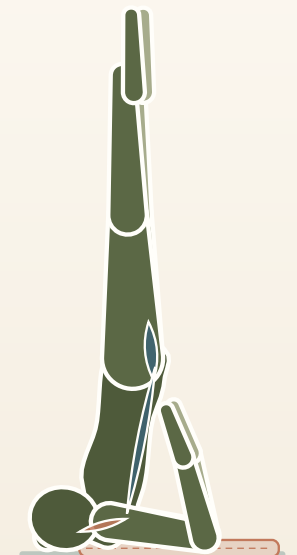
There are no controlled trials of headstand’s long-term effects. Physiology does not support the idea that headstand substantially increases blood flow to the brain: cerebral blood flow is closely autoregulated and kept relatively constant across a wide range of blood pressures (Hall & Hall, 2021). Headstand roughly doubles intraocular pressure while held (Baskaran et al., 2006), which is why it is avoided in glaucoma. Case reports describe neck injuries and rare vascular events associated with headstand (Cramer et al., 2013). The pose is an effective way to train strength, balance and focus; claims about memory or “rejuvenation” are traditional, not demonstrated.

# 2 Sarvāṅgāsana

## SHOULDERSTAND

सर्वाङ्गसन

sar-vahn-GAH-sa-na



**Figure 24.4** Sarvāṅgāsana. In the version shown the trunk is close to vertical and supported by the hands; the neck is strongly flexed, which is why blankets under the shoulders are recommended. ■ lengthening under load ■ working (contracting)

**FAMILY**

Inversion

**LEVEL**

Beginner (supported half shoulderstand) to intermediate

**TRADITIONAL HOLD**

Build from 30 seconds to 3–5 minutes

**MAIN PLANES**

Sagittal (neck flexion, hip extension)

**COUNTERPOSE**

Matsyāsana (fish), after halāsana

*Sarva* means “all” and *aṅga* “limb” — the posture of the whole body. The tradition calls it the “queen of āsanās”. Its textual ancestor is *viparītakaraṇī*, “the reversing action”, described in the Haṭha Yoga Pradīpikā as a mudrā (HYP 3.78–82) (Akers, 2002).

### 01 Purpose

To invert the body with the weight on the shoulders rather than the head; to strengthen the back and legs; and, traditionally, to calm the mind and prepare for the plough and fish that follow.

### 02 Preparation

- ◆ Two or three firmly folded blankets, stacked neatly, under the shoulders with the head on the floor beyond them. The shoulders rest near the edge of the blankets.
- ◆ Bridge pose and legs-up-the-wall to prepare the back, legs and shoulders.
- ◆ Leg raises to strengthen the abdomen for the lift.
- ◆ Empty stomach; no tight clothing around the neck.

### 03 Technique, step by step

- 1 Lie on the back on the blankets, shoulders near the edge, head on the floor, arms by the sides, palms down.

- 2 Inhale and raise the legs to vertical (as in double leg raise).
- 3 Exhale, press the hands down and lift the hips, bringing the legs over the head.
- 4 Bend the elbows and place the hands on the back, fingers pointing towards the spine and up towards the hips; bring the elbows as close together as comfortable.
- 5 Walk the hands higher up the back towards the shoulder blades and straighten the body, bringing the legs vertical. The chest moves towards the chin — the chin does not press into the chest.
- 6 Hold, breathing slowly. The weight is on the shoulders and upper arms, not on the neck.
- 7 To come down, lower the legs slightly over the head, place the hands on the floor, and roll down vertebra by vertebra, using the abdominal muscles, keeping the head on the floor. In the classical sequence, move from here into plough.

#### 04 Breathing

Slow, quiet abdominal breathing. The chest is compressed towards the chin, so breathing will feel restricted; if it becomes laboured or the face flushes strongly, come down.

#### 05 Alignment

- ◆ Shoulders on the folded blankets, head on the floor: the cervical spine is not pressed flat into the floor.
- ◆ Elbows shoulder-width, hands supporting the back as high as possible.
- ◆ Hips over shoulders, legs over hips (full pose); or hips supported by the hands at an angle (half pose).
- ◆ Head and eyes still, gaze towards the chest or feet; never turn the head.

#### 06 Major joints and their positions

| Joint                | Position / movement                              |
|----------------------|--------------------------------------------------|
| Cervical spine       | Strongly flexed (reduced by blankets)            |
| Upper thoracic spine | Flexed                                           |
| Shoulders            | Extended, with scapulae retracted; elbows flexed |
| Hips                 | Extended (full pose) or flexed (half pose)       |
| Knees and ankles     | Extended; ankles relaxed                         |

#### 07 Major muscles and what they are doing

| Muscle(s)                             | Action in this pose       | Role                                                                  |
|---------------------------------------|---------------------------|-----------------------------------------------------------------------|
| Neck extensors, upper trapezius       | Lengthening under load    | Accommodate cervical flexion; the most vulnerable tissues in the pose |
| Back extensors, gluteals, hamstrings  | Isometric                 | Keep the trunk and legs lifted and vertical                           |
| Triceps, posterior deltoid, rhomboids | Isometric                 | Press the upper arms down and support the back                        |
| Abdominals                            | Concentric then eccentric | Lift into and lower out of the pose                                   |



## 08 Spinal action and planes of movement

Strong flexion of the lower cervical and upper thoracic spine, with the rest of the spine near neutral. The blankets allow the neck to flex less than it would with the shoulders flat on the floor.

Sagittal plane: cervical flexion, shoulder extension, hip extension.

## 09 Biomechanics: how the pose loads the body

Body weight rests on the shoulders, upper arms and the back of the head, while the neck approaches the end of its flexion range. The load on the neck depends on how vertical the body is, how much weight the arms take, and how much the blankets reduce flexion. Pushing the trunk further towards vertical by forcing the chin down transfers load to the cervical spine.

### COMMON MISTAKES

- ◆ Practising flat on the floor, with the neck pressed into full flexion.
- ◆ Elbows splaying wide, dropping the weight onto the neck.
- ◆ Turning the head to look around or to talk.
- ◆ Forcing the body vertical by pushing the chin into the chest.
- ◆ Rolling down fast, or lifting the head to come down.

### COMMON STUDENT LIMITATIONS

- ◆ **Stiff neck or upper back:** use more blanket height; stay in half shoulderstand.
- ◆ **Weak abdominals:** difficulty lifting; bend the knees to come up, or start from the wall.
- ◆ **Tight shoulders:** elbows splay; use a strap around the upper arms.
- ◆ **Large chest or abdomen:** breathing restricted; practise half shoulderstand or legs up the wall.

### ⚠ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

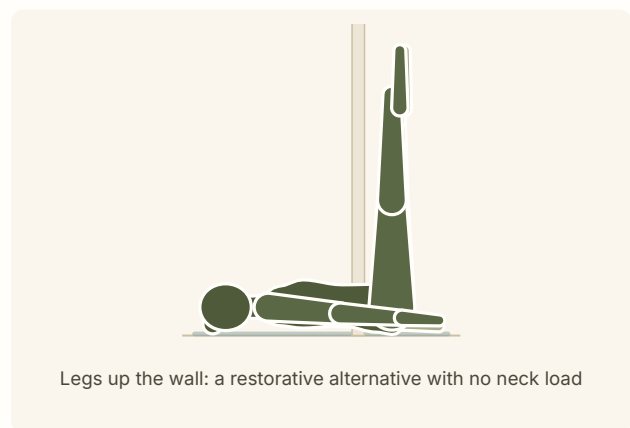
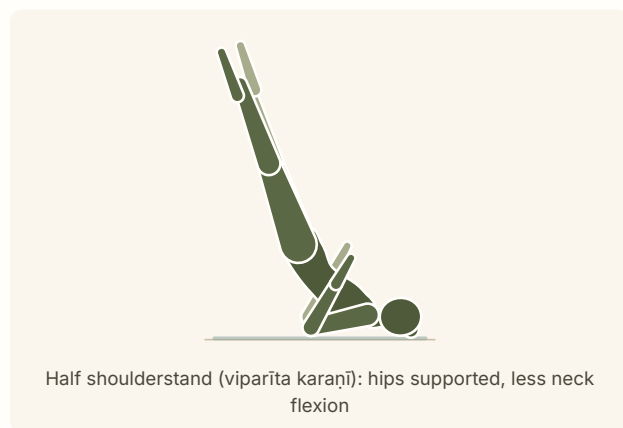
- ◆ Neck injury or neck pain; cervical disc problems
- ◆ Uncontrolled high blood pressure; heart disease
- ◆ Glaucoma and serious eye conditions (Jasien et al., 2015)
- ◆ Osteoporosis of the spine
- ◆ Pregnancy (except experienced practitioners with medical approval); hiatus hernia or reflux
- ◆ Ear or sinus infection; headache at the time

**Precautions:**

- ◆ Always use folded blankets; never use pillows or soft cushions.
- ◆ Build time gradually; never hold through neck discomfort.
- ◆ Many traditional teachers advise avoiding inversions during menstruation; the choice can be left to the student.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



### MODIFICATIONS

- ◆ Half shoulderstand (viparīta karaṇī), hips supported by the hands, body at an angle.
- ◆ Legs up the wall, for those who should not invert the trunk.
- ◆ Shoulderstand with the feet against a wall for support.

### BEGINNER VARIATION

Half shoulderstand on blankets, hips in the hands, legs angled over the head, held for 30 seconds to 1 minute. When this is comfortable and the breath steady, begin to straighten the body towards vertical.

### ADVANCED VARIATION

Unsupported shoulderstand (*nirālamba sarvāṅgāsana*, arms alongside the body or overhead); leg variations such as *eka pāda* (one leg to the floor) and lotus in shoulderstand — only after years of stable practice.

### USE OF PROPS

- ◆ Two or three firmly folded blankets under the shoulders (head off the blankets).
- ◆ A strap around the upper arms.
- ◆ A wall for the feet or for legs up the wall.

## 11 Teaching the pose

### SAFE TEACHING CUES

- ◆ “Shoulders on the blankets, head on the floor, eyes looking up.”
- ◆ “Walk your hands higher up the back.”
- ◆ “Bring the chest to the chin, not the chin to the chest.”
- ◆ “Keep your head still — no turning.”
- ◆ “Roll down slowly, one vertebra at a time.”

### HANDS-ON GUIDANCE

Setting the blankets precisely is the most important “adjustment”. With consent, a teacher can kneel behind the student and steady the legs once they are up, or support the back of the pelvis with a knee or hands while the student walks the hands higher. Never press down on the feet or legs, which drives load into the neck.

## 12 Transitions and counterpose

|                 |                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------|
| Into the pose   | From lying on the back after rest; lift through double leg raise.                             |
| Out of the pose | In the classical sequence, lower the legs directly into plough; otherwise roll down and rest. |
| Counterpose     | Matsyāsana (fish), after plough, for about half the time spent in shoulderstand.              |

### ⬇ EFFECTS TRADITIONALLY ATTRIBUTED

The tradition teaches that shoulderstand benefits the whole body, stimulates and regulates the thyroid gland through pressure on the throat, rests the heart, and calms the nervous system (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

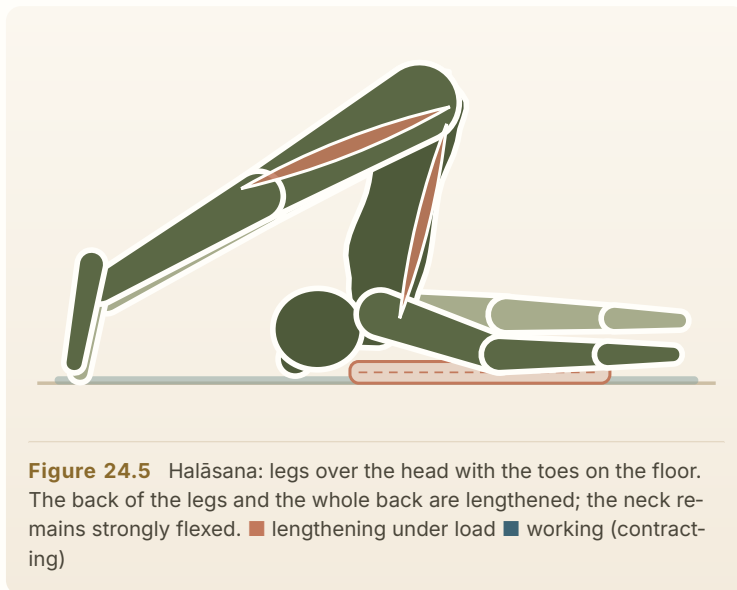
No controlled trials have examined shoulderstand's effects. There is no evidence that it regulates thyroid function. Inverted positions raise intraocular pressure; in a study of four common positions, pressure rose within minutes in legs-up-the-wall and plough as well as other poses (Jasien et al., 2015). Neck injuries associated with shoulderstand are described in case reports (Cramer et al., 2013). The posture's value lies in strength, body awareness and — for many practitioners — its calming effect; the specific health claims remain traditional.

# 3 Halāsana

PLOUGH

हलासन

ha-LAH-sa-na

**FAMILY**

Inversion; forward bend

**LEVEL**

Beginner (supported) to intermediate

**TRADITIONAL HOLD**

Build from 20 seconds to 1–3 minutes

**MAIN PLANES**

Sagittal (spinal and hip flexion)

**COUNTERPOSE**

Matsyāsana (fish)

*Hala* means “plough”: the body takes the shape of the traditional Indian plough.

## 01 Purpose

To flex the whole spine and lengthen the back of the body; in the classical sequence it continues from shoulderstand and prepares the spine for the forward bend.

## 02 Preparation

- ◆ Folded blankets under the shoulders, as for shoulderstand.
- ◆ Sufficient hamstring flexibility to bring the feet towards the floor, or a chair/blocks behind the head.
- ◆ Practice of shoulderstand.

## 03 Technique, step by step

- 1 From shoulderstand, exhale and slowly lower the legs over the head, keeping them straight.
- 2 Rest the toes on the floor (or on a chair or blocks) behind the head. The hands may stay supporting the back or be placed on the floor behind the back, palms down or with fingers interlaced.
- 3 Lift the hips and lengthen the spine; keep the head still and the throat soft.
- 4 Hold, breathing slowly.
- 5 To come out, place the hands on the floor and roll down slowly, keeping the legs straight or bending the knees, and the head on the floor.

## 04 Breathing

Slow, shallow breathing — the abdomen is compressed. If breathing becomes difficult, bend the knees or support the feet higher.

## 05 Alignment

- ◆ Weight on the shoulders and upper arms, never on the neck.
- ◆ Hips high, spine lengthened rather than collapsed.
- ◆ Toes resting on the floor; the feet may be flexed (toes tucked) or pointed.
- ◆ Head still, gaze towards the chest.

## 06 Major joints and their positions

| Joint                     | Position / movement                  |
|---------------------------|--------------------------------------|
| Cervical spine            | Strongly flexed                      |
| Thoracic and lumbar spine | Flexed                               |
| Hips                      | Strongly flexed                      |
| Knees                     | Extended (or flexed in modification) |
| Shoulders                 | Extended; hands on back or floor     |

## 07 Major muscles and what they are doing

| Muscle(s)                      | Action in this pose      | Role                                             |
|--------------------------------|--------------------------|--------------------------------------------------|
| Hamstrings, gastrocnemius      | Lengthening              | The main limit to bringing the feet to the floor |
| Back extensors, neck extensors | Lengthening              | Accommodate spinal flexion                       |
| Abdominals, hip flexors        | Eccentric then isometric | Lower the legs with control and hold             |
| Triceps, posterior deltoids    | Isometric                | Press the arms into the floor for stability      |

## 08 Spinal action and planes of movement

The whole spine is flexed, most strongly at the neck and upper back. If the hamstrings are short, the feet are pulled away from the floor and more flexion is demanded of the spine.

Sagittal plane: flexion of spine and hips.

## 09 Biomechanics: how the pose loads the body

Plough combines the neck flexion of shoulderstand with a strong forward bend. When the feet cannot reach the floor, gravity pulls the legs down and drags the spine into more flexion — especially the neck. Supporting the feet on a chair or blocks removes this pull.

### COMMON MISTAKES

- ◆ Dropping the legs quickly, jolting the neck.
- ◆ Collapsing the hips so that the weight rolls onto the neck.
- ◆ Forcing the feet to the floor with rounded back and strained breath.
- ◆ Turning the head.

### COMMON STUDENT LIMITATIONS

- ◆ **Short hamstrings:** feet do not reach; support the feet or bend the knees.
- ◆ **Stiff upper back or neck:** more blanket height; half plough.
- ◆ **Large abdomen or chest:** breathing restricted; support feet higher, keep time short.

### ⚠️ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

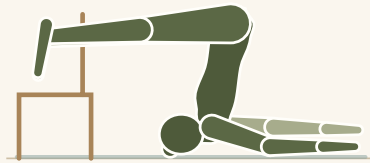
- ◆ Neck injury or pain; cervical disc problems
- ◆ Osteoporosis (loaded spinal flexion) (Sinaki, 2013)
- ◆ Glaucoma and serious eye conditions (Jasien et al., 2015)
- ◆ Uncontrolled high blood pressure; heart disease
- ◆ Pregnancy; hiatus hernia or reflux
- ◆ Acute low-back pain with sensitivity to flexion; sciatica

**Precautions:**

- ◆ Always use blankets under the shoulders.
- ◆ Lower the legs slowly; support them if they do not reach the floor.
- ◆ Come out at once if there is pain or tingling in the neck, arms or legs.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Feet supported on a chair: less spinal and neck flexion

### MODIFICATIONS

- ◆ Feet on a chair, a sofa edge or stacked blocks.
- ◆ Knees bent, resting near the forehead.
- ◆ Half plough: legs parallel to the floor, hands supporting the back.

### BEGINNER VARIATION

Feet on a chair placed behind the head; hands supporting the back; 5–10 breaths.

### ADVANCED VARIATION

*Karṇapīḍāsana* (ear-pressure pose): knees bend to rest beside the ears, arms wrapped around the backs of the knees — only for students with a very comfortable, pain-free plough.

### USE OF PROPS

- ◆ Blankets under the shoulders.
- ◆ A chair, blocks or bolster for the feet.

## 11 Teaching the pose

### SAFE TEACHING CUES

- ◆ “Lower the legs slowly — let the feet find the support.”
- ◆ “Lift the hips; let the spine lengthen.”
- ◆ “Breathe softly into the back.”

◆ “Head still; weight on the shoulders.”

### HANDS-ON GUIDANCE

Place a chair or blocks for the feet. With consent, a teacher may steady the hips from behind. Never press the legs down or push the feet towards the floor.

## 12 Transitions and counterpose

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| Into the pose   | From shoulderstand, lowering the legs over the head.                     |
| Out of the pose | Roll down slowly, head on the floor; rest in śavāsana for a few breaths. |
| Counterpose     | Matsyāsana (fish), which extends the neck and upper back.                |

### 🕉 EFFECTS TRADITIONALLY ATTRIBUTED

Plough is said to make the spine supple, stimulate the spinal nerves, massage the abdominal organs and relieve constipation, and to strengthen the back, shoulders and arms (Iyengar, 1966; Satyananda, 1969)

TRADITIONAL .

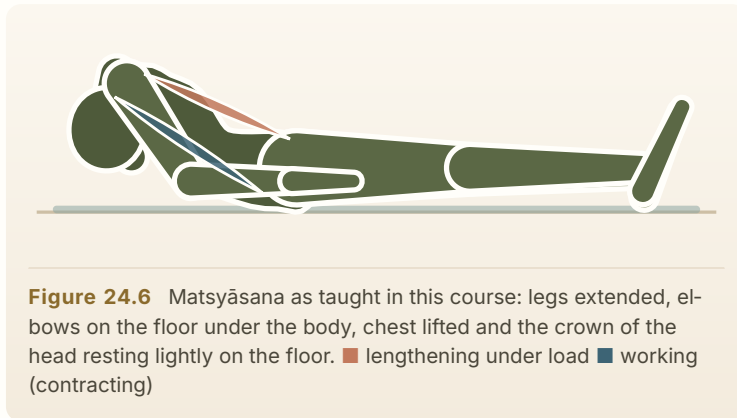
### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

No controlled trials address plough specifically. Plough raises intraocular pressure while held (Jasien et al., 2015). Loaded spinal flexion has been associated with vertebral compression fractures in people with osteoporosis (Sinaki, 2013). Claims about spinal nerves and digestion are traditional and untested.

# 4 Matsyāsana

FISH

मत्स्यासन  
mat-SYAH-sa-na

**FAMILY**

Backbend; counterpose

**LEVEL**

Beginner (supported) to intermediate

**TRADITIONAL HOLD**

About half the time spent in shoulderstand

**MAIN PLANES**

Sagittal (spinal extension)

**COUNTERPOSE**

Rest in śavāsana; gentle knees-to-chest

*Matsya* means “fish”. A traditional explanation is that with the legs in lotus one could float in water in this position **TRADITIONAL**.

## 01 Purpose

To extend the neck and upper back after the flexion of shoulderstand and plough; to open the chest and encourage full breathing.

## 02 Preparation

- ◆ Shoulderstand and plough completed.
- ◆ Comfortable neck extension; if not, use a supported version.
- ◆ Blocks or a bolster for supported fish.

## 03 Technique, step by step

- 1 Lie on the back, legs together and straight, arms beside the body.
- 2 Slide the hands, palms down, under the thighs or buttocks, bringing the elbows as close to each other under the back as comfortable.
- 3 Press the elbows and forearms down and, inhaling, lift the chest high, arching the upper back.
- 4 Tilt the head back and rest the crown lightly on the floor. Most of the weight remains on the elbows — the head carries very little.
- 5 Hold, breathing deeply into the chest.
- 6 To come out, press the elbows down, lift the head slightly, lower the back of the head and then the back to the floor; release the arms and rest.

## 04 Breathing

Deep, full breathing into the expanded chest; the tradition regards this as one of the best postures for full breathing.



## 05 Alignment

- ◆ Weight mainly on the elbows and forearms.
- ◆ The arch comes mostly from the upper back; the chest lifts.
- ◆ Crown of the head touches lightly; the neck is not compressed.
- ◆ Legs relaxed and together (or in lotus in the classical form).

## 06 Major joints and their positions

| Joint          | Position / movement                                      |
|----------------|----------------------------------------------------------|
| Cervical spine | Extended (moderately, with little load)                  |
| Thoracic spine | Extended                                                 |
| Lumbar spine   | Mildly extended                                          |
| Shoulders      | Extended, scapulae retracted; elbows flexed on the floor |
| Hips and knees | Extended                                                 |

## 07 Major muscles and what they are doing

| Muscle(s)                                        | Action in this pose       | Role                                     |
|--------------------------------------------------|---------------------------|------------------------------------------|
| Back extensors, rhomboids, middle trapezius      | Concentric then isometric | Lift and hold the arch of the upper back |
| Triceps, posterior deltoids                      | Isometric                 | Press the elbows into the floor          |
| Pectorals, intercostals, rectus abdominis        | Lengthening               | Open the front of the chest and trunk    |
| Neck flexors (sternocleidomastoid, deep flexors) | Lengthening / eccentric   | Control the tilting back of the head     |

## 08 Spinal action and planes of movement

Extension of the thoracic spine (and to a lesser extent the lumbar and cervical spines) — the opposite of the flexion of shoulderstand and plough.

Sagittal plane: spinal extension.

## 09 Biomechanics: how the pose loads the body

The elbows and forearms form the base of support. If they do not bear the weight, the head does, and the neck is loaded in extension — the principal risk of the pose. The thoracic spine, often stiff in extension, benefits most.

**COMMON MISTAKES**

- ◆ Placing weight on the head.
- ◆ Arching mainly from the lower back while the chest stays flat.
- ◆ Dropping the head back quickly.
- ◆ Holding the breath.

**COMMON STUDENT LIMITATIONS**

- ◆ **Stiff upper back:** the arch comes from the lower back; use a block under the upper back.
- ◆ **Neck discomfort in extension:** keep the head on a block or cushion, or keep the chin slightly towards the chest.
- ◆ **Shoulder stiffness:** the elbows cannot come under the back; keep the arms beside the body.

**⚠ CONTRAINDICATIONS AND PRECAUTIONS**

**Avoid, or practise only with individual guidance, when:**

- ◆ Neck injury or neck pain; cervical instability
- ◆ History of vertebral artery problems or dizziness with neck extension
- ◆ Acute low-back pain aggravated by extension
- ◆ Uncontrolled high blood pressure (head below heart is not an issue here, but strain is)

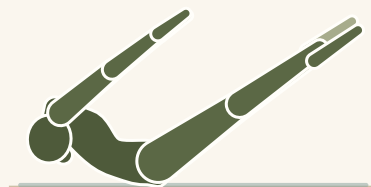
**Precautions:**

- ◆ Keep the weight on the elbows; the head rests lightly.
- ◆ Enter and exit slowly; come out if there is dizziness, pain or tingling.
- ◆ Prefer the supported fish for beginners and older students.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

**10 Modifications, variations and props**

Supported fish: blocks or bolster under the upper back and head



Advanced: *uttāna pādāsana* — legs and arms lifted

**MODIFICATIONS**

- ◆ Supported fish on a bolster or two blocks (one under the shoulder blades, one under the head).
- ◆ Head on the floor with a cushion, without the crown touching.
- ◆ Arms beside the body instead of under it.

**BEGINNER VARIATION**

Supported fish: lie back over a bolster or blocks placed under the upper back, head supported, arms relaxed; breathe into the chest for 1–3 minutes.

**ADVANCED VARIATION**

Classical fish with the legs in lotus (*padma matsyāsana*), hands holding the feet; or *uttāna pādāsana*, with legs and arms lifted at about 45° — both demand strong control of the neck and trunk.

**USE OF PROPS**

- ◆ Bolster, blocks or a rolled blanket under the upper back.

- ◆ Cushion or block under the head.

## 11 Teaching the pose

### SAFE TEACHING CUES

- ◆ “Press the elbows down and lift the chest to the sky.”
- ◆ “Let the crown of the head just kiss the floor.”
- ◆ “Breathe into the whole chest.”
- ◆ “To come out, lift the head a little first, then lower down.”

### HANDS-ON GUIDANCE

Mostly verbal. With consent, a teacher may slide a block or bolster under the upper back. Never press on the chest or pull the head back.

## 12 Transitions and counterpose

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| Into the pose   | From plough, roll down, rest briefly, then enter fish.                                  |
| Out of the pose | Lift the head slightly, lower the head and back, release the arms and rest in śavāsana. |
| Counterpose     | Rest in śavāsana; knees-to-chest if the back feels tight.                               |

### ⚠ EFFECTS TRADITIONALLY ATTRIBUTED

Fish is the counterpose to shoulderstand; it is said to relieve stiffness of the neck and shoulders, to expand the chest and aid deep breathing, and to tone the nerves of the neck and back (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

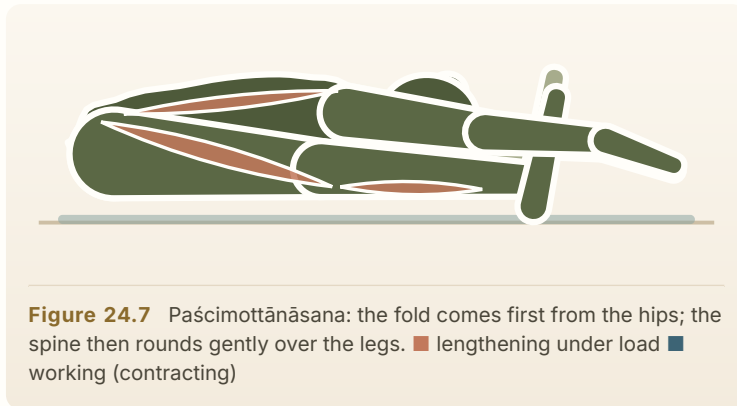
No trials specifically study fish. Case reports link extreme or loaded neck extension in yoga with rare vascular and cervical injuries (Cramer et al., 2013), supporting the practice of keeping the weight on the elbows and the head lightly supported. The role of fish as a counter-movement to sustained neck flexion is biomechanically sensible; its specific effects are untested.

# 5 Paścimottānāsana

## SEATED FORWARD BEND

पश्चिमोत्तानासन

PASH-chee-moh-tah-NAH-sa-na



**Figure 24.7** Paścimottānāsana: the fold comes first from the hips; the spine then rounds gently over the legs. ■ lengthening under load ■ working (contracting)

**FAMILY**

Seated forward bend

**LEVEL**

Beginner (with modifications)

**TRADITIONAL HOLD**

1–3 minutes, or longer in a gentle class

**MAIN PLANES**

Sagittal (hip and spinal flexion)

**COUNTERPOSE**

Cobra (next in the sequence); gentle backbend or tabletop

*Paścima* means “west” and, by extension, the back of the body (the traditional practitioner faces east, towards the sunrise); *uttāna* means “intense stretch”. It is one of the fifteen postures of the Haṭha Yoga Pradīpikā, which praises it highly (HYP 1.28–29) (Akers, 2002).

### 01 Purpose

To lengthen the whole back of the body — calves, hamstrings and back — and to cultivate patience and inward attention; in the classical sequence it follows the inversions and precedes the backbends.

### 02 Preparation

- ◆ Warm hamstrings (sun salutations, leg raises).
- ◆ A folded blanket or cushion under the sitting bones if the lower back rounds when sitting upright.
- ◆ A strap for students who cannot comfortably reach the feet.

### 03 Technique, step by step

- 1 Sit with the legs together and extended, feet flexed (toes towards the shins). Sit on a blanket if needed.
- 2 Inhale and raise the arms overhead, lengthening the spine.
- 3 Exhale and fold forward from the hips, keeping the spine long, leading with the chest.
- 4 Take hold of whatever you can reach comfortably — thighs, shins, ankles or feet — or use a strap around the feet.
- 5 With each inhalation, lengthen the spine a little; with each exhalation, relax a little further forward. Let the head and neck relax.
- 6 Hold for several breaths or minutes, softening rather than pulling.
- 7 To come out, inhale and lift the chest and arms with a long spine; exhale and lower the arms.

## 04 Breathing

Inhale to lengthen; exhale to release. The breath should be slow and unforced; many students find the abdomen presses against the thighs, so breathing moves into the back of the ribs.

## 05 Alignment

- ◆ The pelvis tilts forward over the thighs — the fold begins at the hips.
- ◆ Legs active but not locked; kneecaps drawn up gently; no hyperextension at the knees.
- ◆ Feet flexed or relaxed; knees bent as much as needed.
- ◆ Shoulders relaxed away from the ears; elbows soft; no pulling with the arms.

## 06 Major joints and their positions

| Joint                     | Position / movement                          |
|---------------------------|----------------------------------------------|
| Hips                      | Flexed                                       |
| Lumbar and thoracic spine | Flexed (moderately in a well-organised fold) |
| Cervical spine            | Relaxed flexion                              |
| Knees                     | Extended (or flexed in modification)         |
| Ankles                    | Dorsiflexed or relaxed                       |

## 07 Major muscles and what they are doing

| Muscle(s)                       | Action in this pose            | Role                                                          |
|---------------------------------|--------------------------------|---------------------------------------------------------------|
| Hamstrings                      | Lengthening                    | The main limit; tension felt at the back of the thighs        |
| Gastrocnemius                   | Lengthening                    | Tension behind the knees and calves when the feet are flexed  |
| Erector spinae, gluteus maximus | Lengthening                    | Allow the spine and pelvis to fold                            |
| Quadriceps                      | Light isometric                | Straighten the knees; reciprocal inhibition of the hamstrings |
| Hip flexors, abdominals         | Concentric early, then relaxed | Draw the trunk forward at the start of the fold               |

## 08 Spinal action and planes of movement

Flexion of the whole spine after the hips have flexed. If the hamstrings hold the pelvis back (posterior tilt), the lumbar spine flexes more to compensate.

Sagittal plane: hip flexion and spinal flexion.

## 09 Biomechanics: how the pose loads the body

The two-joint hamstrings limit the pose when the knees are straight (passive insufficiency). Bending the knees immediately allows more hip flexion. Pulling with the arms adds force at end range — load that the hamstring attachment at the sitting bones and the lumbar discs may not tolerate. Gravity assists only slightly in a seated fold, so the pose is gentle when practised passively.

**COMMON MISTAKES**

- ◆ Rounding the back from the start instead of folding from the hips.
- ◆ Pulling with the arms to go deeper.
- ◆ Locking or hyperextending the knees.
- ◆ Holding the breath; straining the neck to bring the head to the knees.
- ◆ Bouncing.

**COMMON STUDENT LIMITATIONS**

- ◆ **Short hamstrings:** pelvis tilts back and the lower back rounds; sit on a blanket, bend the knees, use a strap.
- ◆ **Stiff lower back:** fold less, emphasise length.
- ◆ **Nerve sensitivity:** tingling down the leg; bend the knees and reduce the stretch.
- ◆ **Large abdomen:** separate the legs slightly to make room.

**⚠ CONTRAINDICATIONS AND PRECAUTIONS**

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute low-back pain or disc problems aggravated by flexion; sciatica with leg symptoms
- ◆ Osteoporosis of the spine (avoid loaded rounding of the spine) (Sinaki, 2013)
- ◆ Proximal hamstring tendinopathy (pain at the sitting bone) (Goom et al., 2016)
- ◆ Later pregnancy (fold with legs apart instead)

**Precautions:**

- ◆ Never pull or bounce; move by breath and time.
- ◆ Bend the knees at the first sign of strain behind the knees or at the sitting bones.
- ◆ Move slowly out of the pose.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

**10 Modifications, variations and props**

Beginner: sitting tall with a strap around the feet, knees soft



Advanced: long spine, chest resting on the thighs

**MODIFICATIONS**

- ◆ Sit on a folded blanket or cushion.
- ◆ Bend the knees generously; rest the chest on the thighs with bent knees for a restorative version.
- ◆ Use a strap around the feet; keep the spine long rather than folding deeply.
- ◆ Bolster across the thighs to support the chest and head (restorative).

**BEGINNER VARIATION**

Sitting on a blanket, knees bent, strap around the feet, spine long; stay upright or fold a few degrees from the hips for 5–10 breaths.

**ADVANCED VARIATION**

Holding the big toes or wrapping the hands around the soles of the feet, chest resting on the thighs, elbows on the floor; long holds of 3–5 minutes in stillness.

**USE OF PROPS**

- ◆ Folded blanket under the sitting bones.
- ◆ Strap around the feet.
- ◆ Bolster or cushion on the thighs to support the chest.

**11 Teaching the pose****SAFE TEACHING CUES**

- ◆ “Sit on the front of your sitting bones.”
- ◆ “Inhale, grow tall; exhale, fold from the hips.”
- ◆ “Let the breath, not the arms, take you forward.”
- ◆ “Bend your knees as much as you need.”
- ◆ “Soften the face, the shoulders and the belly.”

**HANDS-ON GUIDANCE**

Most adjustment is verbal and with props. With consent, a light hand on the lower back can encourage the pelvis to tilt forward, or a hand between the shoulder blades can invite length. **Never push a student down into a forward bend** — this is one of the most common causes of hamstring and back injury in yoga.

**12 Transitions and counterpose**

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Into the pose   | From sitting (daṇḍāsana), after rest following fish.                                     |
| Out of the pose | Inhale and rise with a long spine; rest in śavāsana, then turn onto the front for cobra. |
| Counterpose     | The backbends that follow (cobra); or tabletop and a gentle cow.                         |

**⬇ EFFECTS TRADITIONALLY ATTRIBUTED**

The Haṭha Yoga Pradīpikā says paścimottānāsana makes the breath flow through the central channel, stokes the digestive fire, slims the waist and removes disease (HYP 1.29) (Akers, 2002). The tradition taught here adds that it stretches the whole back, massages the abdominal organs and calms the mind (Satyananda, 1969) **TRADITIONAL**.

**🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT**

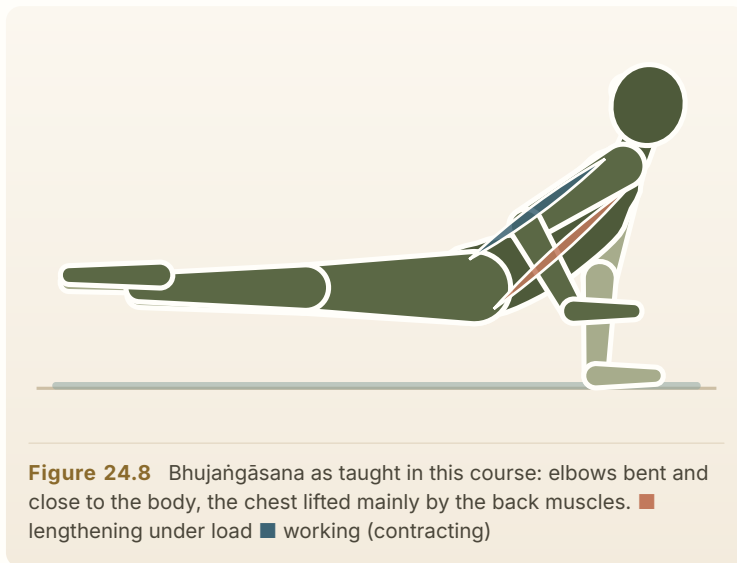
Regular stretching increases hamstring range of motion, mostly through increased stretch tolerance in the short term (Weppeler & Magnusson, 2010). No trials test the digestive or metabolic claims. Loaded spinal flexion has been associated with vertebral fractures in people with osteoporosis (Sinaki, 2013), and aggressive stretching can irritate the proximal hamstring tendon (Goom et al., 2016). As a calming, gentle forward bend practised without force, it is low-risk for most healthy students.

# 6 Bhujāṅgāsana

COBRA

भुजङ्गासन

boo-jahn-GAH-sa-na

**FAMILY**

Prone backbend

**LEVEL**

Beginner

**TRADITIONAL HOLD**

3 rounds of 5–10 breaths, or one longer hold

**MAIN PLANES**

Sagittal (spinal extension)

**COUNTERPOSE**

Child's pose; rest in makarāsana between backbends

*Bhujāṅga* means “serpent”, particularly the cobra, which lifts its head and hood while the body remains on the ground.

## 01 Purpose

To extend the spine gently and strengthen the back muscles; the first of the three prone backbends, preparing for locust and bow.

## 02 Preparation

- ◆ Sun salutation (position 7 is a cobra).
- ◆ Rest in makarāsana (crocodile) before and after.

## 03 Technique, step by step

- 1 Lie on the front, legs together, tops of the feet on the floor, forehead down.
- 2 Place the hands under the shoulders, fingers pointing forward, elbows bent and close to the sides.
- 3 Inhale and slowly lift the forehead, nose, chin and then the chest, as if the spine were rolling up one vertebra at a time. Use mainly the back muscles; the hands guide and support.
- 4 Keep the elbows bent and drawn in, the shoulders down and back, and the lower ribs and navel near or on the floor.
- 5 Hold, breathing evenly.
- 6 Exhale and roll down slowly — chest, chin, nose, forehead. Rest with the cheek on the hands.



## 04 Breathing

Inhale to rise, breathe steadily in the pose, exhale to lower. In the classical method the pose may be repeated three times, holding a little longer each time.

## 05 Alignment

- ◆ Pubic bone and legs remain on the floor; the legs relaxed or gently active.
- ◆ Extension distributed along the whole spine, especially the upper back — not concentrated in the lower back.
- ◆ Shoulders away from the ears; shoulder blades drawn gently back.
- ◆ Neck long, continuing the curve of the spine; gaze forward or slightly up.

## 06 Major joints and their positions

| Joint                     | Position / movement                        |
|---------------------------|--------------------------------------------|
| Thoracic and lumbar spine | Extended                                   |
| Cervical spine            | Mildly extended                            |
| Hips                      | Extended (neutral to slight extension)     |
| Shoulders                 | Extended, scapulae retracted and depressed |
| Elbows                    | Flexed                                     |

## 07 Major muscles and what they are doing

| Muscle(s)                             | Action in this pose       | Role                                        |
|---------------------------------------|---------------------------|---------------------------------------------|
| Erector spinae, multifidus            | Concentric then isometric | Lift and hold the trunk                     |
| Rhomboids, middle and lower trapezius | Isometric                 | Draw the shoulder blades back and down      |
| Triceps                               | Light isometric           | Support without pushing into the lower back |
| Rectus abdominis, hip flexors         | Lengthening               | Front of the trunk opens                    |
| Gluteals, hamstrings                  | Light isometric           | Stabilise the pelvis and legs               |

## 08 Spinal action and planes of movement

Extension of the thoracic and lumbar spine, with gentle cervical extension. The lumbar spine is the most mobile region in extension, so cues aim to spread the movement upward.

Sagittal plane: spinal extension.

## 09 Biomechanics: how the pose loads the body

When the lift comes from the back muscles, the height is limited to what the muscles can produce — a self-limiting and safe range. When the arms push hard, they lever the spine into extension beyond its active range, compressing the posterior lumbar structures.

**COMMON MISTAKES**

- ◆ Pushing up with straight arms and hanging into the lower back.
- ◆ Shrugging the shoulders to the ears.
- ◆ Throwing the head back.
- ◆ Lifting the pelvis off the floor.

**COMMON STUDENT LIMITATIONS**

- ◆ **Stiff upper back:** extension concentrates in the lower back; lift less, lengthen more.
- ◆ **Low-back sensitivity to extension:** practise sphinx or a very low cobra.
- ◆ **Weak back muscles:** lift less and hold for fewer breaths.

**⚠ CONTRAINDICATIONS AND PRECAUTIONS**

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute low-back pain aggravated by extension; spondylolisthesis or facet irritation aggravated by extension
- ◆ Pregnancy (after the first trimester, lying on the abdomen is uncomfortable and generally avoided)
- ◆ Recent abdominal surgery; hernia

**Precautions:**

- ◆ Keep the elbows bent; lift only as high as the back muscles allow.
- ◆ Stop if there is pinching or pain in the lower back.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

**10 Modifications, variations and props**

Beginner: sphinx — forearms on the floor, gentle extension

**MODIFICATIONS**

- ◆ **Sphinx:** forearms on the floor under the shoulders.
- ◆ **Baby cobra:** lift only the chest a few centimetres, hands barely pressing.
- ◆ Legs slightly apart to reduce compression in the lower back.

**BEGINNER VARIATION**

Sphinx or baby cobra, 5 breaths, repeated 2–3 times with rest.

**ADVANCED VARIATION**

Cobra with the hands lifted briefly off the floor (strength), or *rāja kapotāsana* preparations with the knees bent and feet reaching towards the head — only with a very supple, pain-free spine.

**USE OF PROPS**

- ◆ Folded blanket under the pelvis if the pubic bone is sensitive.

**11 Teaching the pose****SAFE TEACHING CUES**

- ◆ *“Peel the chest up, as if drawn forward and up by the sternum.”*

- ◆ “Elbows in; shoulders away from the ears.”
- ◆ “Let the back muscles do the work — the hands just rest.”
- ◆ “Lengthen the tailbone towards the heels.”
- ◆ “Roll down slowly, one vertebra at a time.”

### HANDS-ON GUIDANCE

Verbal and demonstration first. With consent, a teacher may place a hand lightly between the shoulder blades to invite the upper back to lift, or on the lower back to encourage length. Never press a student’s pelvis down while they push up.

## 12 Transitions and counterpose

|                 |                                                                       |
|-----------------|-----------------------------------------------------------------------|
| Into the pose   | From makarāsana (crocodile), after the forward bend and a brief rest. |
| Out of the pose | Roll down, turn the cheek to one side, rest in makarāsana.            |
| Counterpose     | Child’s pose after the series of backbends.                           |

### 🕉 EFFECTS TRADITIONALLY ATTRIBUTED

Cobra is said to make the spine supple and strong, tone the back muscles and spinal nerves, massage the abdominal organs and relieve menstrual irregularities and constipation (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

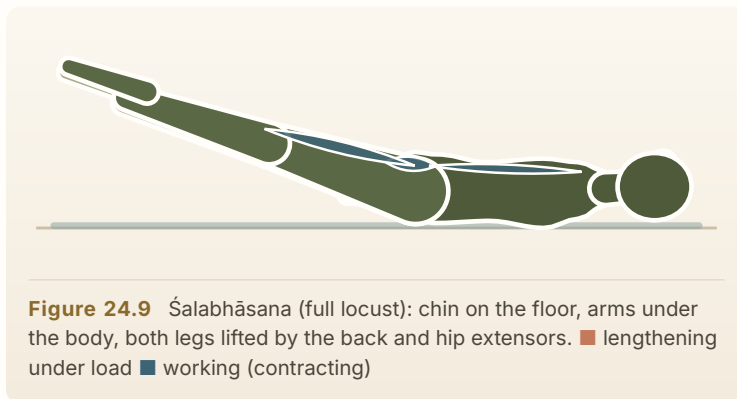
There are no trials of cobra alone. Yoga programmes that include gentle backbends improve pain and function in chronic low back pain overall (Wieland et al., 2022), but the contribution of individual postures is unknown. Claims about menstruation and digestion are traditional and untested.

# 7 Śalabhāsana

LOCUST

शलभासन

sha-la-BAH-sa-na

**FAMILY**

Prone backbend; strength

**LEVEL**

Beginner (half locust) to intermediate (full)

**TRADITIONAL HOLD**

Half locust 3–5 breaths each side; full locust 3–5 breaths, repeated

**MAIN PLANES**

Sagittal (hip and spinal extension)

**COUNTERPOSE**

Rest in makarāsana; child's pose after bow

*Śalabha* means “locust” or “grasshopper”, whose tail lifts when it rests.

## 01 Purpose

To strengthen the back muscles, gluteals and hamstrings; the most strengthening of the three prone backbends.

## 02 Preparation

- ◆ Cobra and rest.
- ◆ Half locust before full locust.

## 03 Technique, step by step

- 1 Lie on the front with the chin on the floor. Place the arms under the body, palms down or hands in fists, the arms close together and the elbows as straight as comfortable.
- 2 Half locust: inhale and lift one leg, keeping it straight and the hips on the floor; hold a few breaths; exhale and lower. Repeat with the other leg.
- 3 Full locust: inhale and, pressing the arms into the floor, lift both legs together as high as possible, keeping them straight.
- 4 Hold for a few breaths, then exhale and lower slowly.
- 5 Release the arms and rest with the cheek on the hands.

## 04 Breathing

In the traditional form the breath is held briefly during the lift of full locust; beginners and anyone with high blood pressure should instead breathe steadily throughout.

## 05 Alignment

- ◆ Chin (or forehead) on the floor; the neck long, not strained.

- ◆ Legs straight and together (or hip-width in modification); lift from the hips rather than bending the knees.
- ◆ Hips stay as level as possible in half locust.

## 06 Major joints and their positions

| Joint          | Position / movement               |
|----------------|-----------------------------------|
| Hips           | Extended                          |
| Lumbar spine   | Extended                          |
| Knees          | Extended                          |
| Shoulders      | Extended; arms beneath the body   |
| Cervical spine | Slightly extended (chin on floor) |

## 07 Major muscles and what they are doing

| Muscle(s)                                      | Action in this pose       | Role                                                     |
|------------------------------------------------|---------------------------|----------------------------------------------------------|
| Gluteus maximus, hamstrings                    | Concentric then isometric | Extend the hips to lift the legs                         |
| Erector spinae, multifidus, quadratus lumborum | Isometric                 | Extend and stabilise the lumbar spine                    |
| Triceps, latissimus dorsi, posterior deltoid   | Isometric                 | Press the arms down to lever the pelvis up (full locust) |
| Hip flexors                                    | Lengthening               | Front of the hips opens                                  |

## 08 Spinal action and planes of movement

Extension of the lumbar spine as the legs rise; the thoracic spine remains relatively neutral.

Sagittal plane: hip and spinal extension.

## 09 Biomechanics: how the pose loads the body

The legs are long levers lifted against gravity; the hip extensors and lower back muscles work hard. Compression of the posterior lumbar spine increases with height. Holding the breath raises intra-abdominal and blood pressure (a Valsalva-type effect), which helps lift but is inappropriate for some students.

### COMMON MISTAKES

- ◆ Bending the knees to lift higher.
- ◆ Straining the neck or lifting the chin.
- ◆ Twisting the pelvis in half locust.
- ◆ Holding the breath with a red, strained face.

### COMMON STUDENT LIMITATIONS

- ◆ **Weak hip extensors or back:** practise half locust only.
- ◆ **Arm discomfort under the body:** place the arms beside the body instead.
- ◆ **Low-back sensitivity:** lift less; keep the legs slightly apart.

### ⚠️ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Uncontrolled high blood pressure or heart disease (for the breath-holding version)
- ◆ Pregnancy
- ◆ Acute low-back pain aggravated by extension
- ◆ Recent abdominal surgery; hernia

**Precautions:**

- ◆ Teach half locust first.
- ◆ Breathe freely unless the student is experienced and healthy.
- ◆ Lower slowly, controlling the descent.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Beginner: half locust — one leg at a time

### MODIFICATIONS

- ◆ Half locust, one leg at a time.
- ◆ Arms beside the body, palms up or down.
- ◆ Lift the chest and arms instead of the legs (a form of locust with arms forward) for variety.

### BEGINNER VARIATION

Half locust, each leg three times for 3–5 breaths, breathing freely.

### ADVANCED VARIATION

Full locust held longer, and locust with arms extended forward and chest lifted together; the very advanced “full locust” with the legs rising over the head is a specialist practice not taught in general classes.

### USE OF PROPS

- ◆ Folded blanket under the pelvis.

## 11 Teaching the pose

### SAFE TEACHING CUES

- ◆ “Reach the leg long, then lift it.”
- ◆ “Keep both hips heavy on the floor.”
- ◆ “Lengthen the back of the neck.”
- ◆ “Lower slowly — feel the muscles lengthen as you come down.”

### HANDS-ON GUIDANCE

**Verbal:** most students need permission to lift less and reach more. With consent, a teacher may place a hand on the sacrum to help keep the pelvis level in half locust. Do not lift a student's legs.

## 12 Transitions and counterpose

|                 |                                   |
|-----------------|-----------------------------------|
| Into the pose   | From makarāsana, after cobra.     |
| Out of the pose | Lower slowly; rest in makarāsana. |
| Counterpose     | Rest; after bow, child's pose.    |

### 🕉 EFFECTS TRADITIONALLY ATTRIBUTED

Locust is said to strengthen the lower back, abdomen and legs, massage the abdominal organs, and improve digestion; with breath retention it is said to strengthen the lungs (Iyengar, 1966; Satyananda, 1969)

**TRADITIONAL** .

### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

No trials test locust alone. It is a well-recognised back-extensor and hip-extensor strengthening exercise. Breath-holding during effort raises blood pressure transiently, which is the reason for the precautions. Digestive claims are traditional.

## 8

## Dhanurāsana

BOW

धनुरासन

dha-noo-RAH-sa-na

**FAMILY**

Prone backbend

**LEVEL**

Intermediate

**TRADITIONAL HOLD**

3–5 breaths, repeated 2–3 times

**MAIN PLANES**

Sagittal (spinal and hip extension)

**COUNTERPOSE**

Child's pose

*Dhanus* means “bow”: the trunk and legs form the bow, the arms the bowstring. Dhanurāsana is one of the fifteen postures of the *Haṭha Yoga Pradīpikā* (HYP 1.25) (Akers, 2002).

## 01 Purpose

To extend the whole spine and open the front of the body while the back and legs work; the deepest of the three prone backbends in the sequence.

## 02 Preparation

- ◆ Cobra and locust.
- ◆ Quadriceps and hip-flexor mobility (lunge, half bow).

## 03 Technique, step by step

- 1 Lie on the front, forehead down. Bend the knees and take hold of the ankles (outside of the ankles) with the hands.
- 2 Inhale and press the feet back and up into the hands; this lifts the thighs and the chest at the same time.
- 3 Keep the arms straight — they act like ropes, not levers. Lift the chest and gaze forward.
- 4 Hold, breathing steadily; the body may rock slightly with the breath.
- 5 Exhale and lower slowly — thighs, chest, forehead — then release the ankles and rest.

## 04 Breathing

Inhale to lift; steady, often shallow breathing in the pose (the abdomen is compressed); exhale to lower.



## 05 Alignment

- ◆ Knees no wider than the hips (hip-width apart is fine for beginners).
- ◆ The lift comes from the legs pressing back, not from pulling with the arms.
- ◆ Extension spread along the spine; chest broad.
- ◆ Neck long, not thrown back.

## 06 Major joints and their positions

| Joint     | Position / movement            |
|-----------|--------------------------------|
| Spine     | Extended (thoracic and lumbar) |
| Hips      | Extended                       |
| Knees     | Flexed                         |
| Shoulders | Extended; arms straight behind |
| Ankles    | Plantarflexed (held)           |

## 07 Major muscles and what they are doing

| Muscle(s)                                      | Action in this pose | Role                                                |
|------------------------------------------------|---------------------|-----------------------------------------------------|
| Hamstrings, gluteus maximus                    | Isometric           | Press the feet back into the hands; extend the hips |
| Erector spinae, rhomboids                      | Isometric           | Support spinal extension and open the chest         |
| Quadriceps, hip flexors                        | Lengthening         | The main limit at the front of the thighs           |
| Rectus abdominis, pectorals, anterior deltoids | Lengthening         | Front of the trunk and shoulders opens              |

## 08 Spinal action and planes of movement

Extension of the whole spine, typically greatest in the lower lumbar and thoracolumbar region; cues aim to lift the chest to share the movement.

Sagittal plane.

## 09 Biomechanics: how the pose loads the body

Bow is a closed loop: the legs pushing against the hands generate the extension. This means the student can easily move beyond the range their back muscles could hold on their own, because the legs supply force. Compression of the lower lumbar spine and stretch on the knees (deep flexion with the ankles held) are the main loads.

### COMMON MISTAKES

- ◆ Pulling with bent arms instead of pressing with the legs.
- ◆ Knees splaying wide, turning the thighs out.
- ◆ Throwing the head back.
- ◆ Compressing into the lower back with pain.

### COMMON STUDENT LIMITATIONS

- ◆ **Tight quadriceps or hip flexors:** cannot reach the ankles; use a strap.
- ◆ **Knee discomfort:** hold a strap around the ankles to reduce knee flexion.
- ◆ **Stiff upper back:** extension concentrates low; practise half bow.

### ⚠️ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute low-back pain or conditions aggravated by extension
- ◆ Pregnancy
- ◆ Recent abdominal surgery, hernia
- ◆ Uncontrolled high blood pressure (if straining)
- ◆ Knee injury limiting deep flexion

**Precautions:**

- ◆ Practise cobra and locust first.
- ◆ Stop at any pinching in the lower back or pain in the knees.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Beginner: half bow — one side at a time, supported on the forearm

### MODIFICATIONS

- ◆ Half bow, one leg at a time, supported on the opposite forearm.
- ◆ Strap around the ankles, holding the strap.
- ◆ Lift only the legs or only the chest, keeping the other half down.

### BEGINNER VARIATION

Half bow, each side 3–5 breaths, or bow with a strap and minimal lift.

### ADVANCED VARIATION

Rocking bow (rocking forward and back with the breath) and *pūrṇa dhanurāsana* (full bow, holding the toes with the elbows pointing up) — only with a supple, pain-free spine.

### USE OF PROPS

- ◆ Strap around the ankles.
- ◆ Folded blanket under the pelvis.

## 11 Teaching the pose

### SAFE TEACHING CUES

- ◆ “Kick the feet back into the hands.”
- ◆ “Let the arms be long, like the string of a bow.”
- ◆ “Lift the chest forward and up.”
- ◆ “Breathe — let the breath rock you gently.”

**HANDS-ON GUIDANCE**

With consent, a teacher may help a student find the ankles or place a strap. Hands-on lifting of the legs or chest is not recommended.

**12 Transitions and counterpose**

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| Into the pose   | From rest after locust.                                       |
| Out of the pose | Lower slowly; release; rest in makarāsana, then child’s pose. |
| Counterpose     | Child’s pose; then the twist.                                 |

**🕉 EFFECTS TRADITIONALLY ATTRIBUTED**

Bow is said to combine the benefits of cobra and locust, keep the spine elastic, massage the abdominal organs, and improve digestion and appetite (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

**🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT**

No trials study bow alone. Its demands — deep lumbar extension driven by the legs — are the reason for its place late in the backbend sequence and the modifications above. Traditional claims about digestion are untested.

## 9

## Ardha Matsyendrāsana

HALF SPINAL TWIST

अर्धमत्स्येन्द्रासन

AR-dha mat-syen-DRAH-sa-na



**Figure 24.11** Ardha Matsyendrāsana, front view (schematic): the right knee is raised with the foot outside the left knee; the left arm presses against the right knee; the trunk turns to the right from the rib cage.

**FAMILY**

Seated twist

**LEVEL**

Beginner (easy twist) to intermediate

**TRADITIONAL HOLD**

30 seconds to 1 minute each side

**MAIN PLANES**

Transverse (spinal rotation)

**COUNTERPOSE**

Twist to the other side; a gentle forward bend

*Ardha* means “half”; *Matsyendra*, “lord of the fishes”, is the legendary founder of Haṭha Yoga, to whom the Haṭha Yoga Pradīpikā attributes the full twist, *matsyendrāsana* (HYP 1.26–27) (Akers, 2002).

**01 Purpose**

To rotate the spine in both directions after the backbends; to cultivate a lengthened, upright spine; and, in the traditional view, to tone the spinal nerves and abdominal organs.

**02 Preparation**

- ◆ Seated forward bend and backbends completed.
- ◆ A cushion under the sitting bones if the spine cannot remain upright.
- ◆ Simple seated twists (*sukhāsana* twist).

**03 Technique, step by step**

- 1 Sit on the heels (or with legs extended). Move the hips to the floor to the right of the feet, so that the left leg is folded with the left heel beside the right hip (or keep it straight).
- 2 Bring the right foot over the left knee and place it flat on the floor outside the left knee; the right knee points up.

- 3 Inhale and raise the left arm; exhale and bring it to the outside of the right knee, taking hold of the right foot or ankle, or simply pressing the upper arm against the knee.
- 4 Place the right hand on the floor behind the back, close to the spine, fingers pointing away.
- 5 Inhale and lengthen the spine upward; exhale and turn the trunk to the right, from the waist and rib cage; turn the head last to look over the right shoulder.
- 6 Hold, lengthening with each inhalation and turning a little with each exhalation.
- 7 Exhale and unwind slowly, head first. Repeat on the other side.

#### 04 Breathing

Inhale to lengthen, exhale to rotate. Breathing into the lower ribs is restricted on one side and expanded on the other.

#### 05 Alignment

- ◆ Both sitting bones grounded; spine upright and lengthened before rotating.
- ◆ Rotation from the thoracic spine and rib cage; the lower back turns only a little.
- ◆ The back hand supports the spine upright; it does not lever the twist.
- ◆ Head turns last and moderately; shoulders level and relaxed.

#### 06 Major joints and their positions

| Joint               | Position / movement                            |
|---------------------|------------------------------------------------|
| Thoracic spine      | Rotated (most of the twist)                    |
| Lumbar spine        | Slightly rotated; neutral to slightly extended |
| Cervical spine      | Rotated                                        |
| Top hip             | Flexed, adducted, internally rotated           |
| Bottom hip and knee | Flexed (or extended)                           |

#### 07 Major muscles and what they are doing

| Muscle(s)                                                         | Action in this pose       | Role                                     |
|-------------------------------------------------------------------|---------------------------|------------------------------------------|
| External oblique (opposite side) and internal oblique (same side) | Concentric then isometric | Rotate the trunk towards the raised knee |
| Erector spinae, multifidus                                        | Isometric                 | Keep the spine upright                   |
| Obliques of the other side, back muscles                          | Lengthening               | Allow rotation                           |
| Gluteus medius/minimus and deep rotators (top leg)                | Lengthening               | The top hip is drawn across the body     |

#### 08 Spinal action and planes of movement

Rotation, mainly thoracic, with the cervical spine completing the turn. The lumbar spine, whose facet joints limit rotation, contributes only a few degrees; forcing further rotation there should be avoided.

Transverse plane: spinal rotation; hip adduction occurs in the frontal plane.

## 09 Biomechanics: how the pose loads the body

The arm pressing against the knee provides a lever. Used gently, it helps the trunk rotate; used forcefully, it can push the spine past its active range. Combining rotation with a slumped, flexed spine increases load on the discs (McGill, 2016); sitting tall reduces it.

### COMMON MISTAKES

- ◆ Slumping the lower back (often because the hips are too tight — sit on a cushion).
- ◆ Levering hard with the arms to force the twist.
- ◆ Turning the head first and furthest.
- ◆ Lifting one sitting bone high off the floor.

### COMMON STUDENT LIMITATIONS

- ◆ **Tight hips:** the top knee cannot be crossed comfortably; keep the bottom leg straight or sit on a cushion.
- ◆ **Rounding of the lower back:** sit higher.
- ◆ **Shoulder stiffness:** hug the knee with the arm instead of binding.

### ⚠ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute disc problems or low-back pain aggravated by rotation
- ◆ Osteoporosis of the spine (avoid forced or deep twisting with flexion) (Sinaki, 2013)
- ◆ Later pregnancy (closed twists compress the abdomen; open twists turning away from the bent knee are used instead)
- ◆ Recent abdominal or spinal surgery

**Precautions:**

- ◆ Lengthen first, twist second.
- ◆ Unwind slowly and evenly; always twist both sides.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Beginner: easy seated twist in a cross-legged position

### MODIFICATIONS

- ◆ Easy twist sitting cross-legged or on a chair.
- ◆ Keep the bottom leg straight.
- ◆ Hug the raised knee with the arm rather than hooking the elbow outside it.

- ◆ Sit on a cushion or folded blanket.

#### BEGINNER VARIATION

Easy seated twist in sukhāsana or on a chair, 5 breaths each side, turning from the rib cage with a tall spine.

#### ADVANCED VARIATION

Bind (the lower arm wrapping under the raised knee to clasp the hand behind the back) or the full *matsyendrāsana* with the bottom foot in half-lotus — only with open hips and a relaxed, upright spine.

#### USE OF PROPS

- ◆ Cushion or folded blanket under the sitting bones.
- ◆ Chair for a seated twist.

### 11 Teaching the pose

#### SAFE TEACHING CUES

- ◆ “Inhale, grow tall; exhale, turn.”
- ◆ “Turn from the ribs, not the hands.”
- ◆ “Both sitting bones heavy.”
- ◆ “Let the head follow last, softly.”

#### HANDS-ON GUIDANCE

With consent: one hand on the shoulder in front and one behind can invite the rib cage to turn, but the student leads the movement and the teacher never adds force at the end of the range. Alternatively, a hand on the crown can invite the spine to lengthen.

### 12 Transitions and counterpose

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| Into the pose   | From kneeling or <i>danḍāsana</i> after the backbends.               |
| Out of the pose | Unwind on an exhalation, head first; change sides; rest.             |
| Counterpose     | The same twist to the other side; then a short forward bend or rest. |

#### ⚠ EFFECTS TRADITIONALLY ATTRIBUTED

The twist is said to keep the spine elastic and the spinal nerves toned, massage the abdominal organs, relieve digestive problems and help to balance the nervous system (Iyengar, 1966; Satyananda, 1969). The *Haṭha Yoga Pradīpikā* says the full *matsyendrāsana* kindles the digestive fire and awakens *kuṇḍalīnī* (HYP 1.27) (Akers, 2002) **TRADITIONAL**.

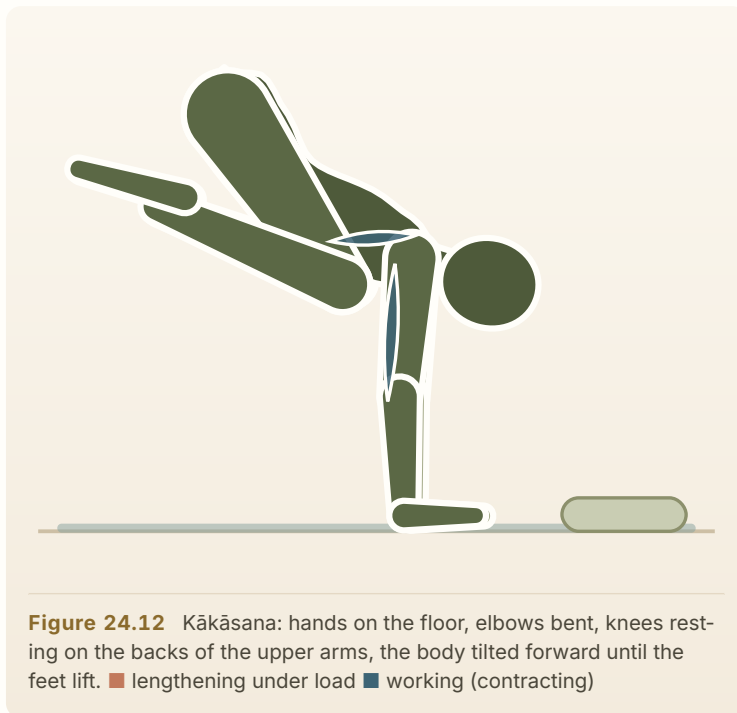
#### 🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT

No trials study the half spinal twist alone. There is no evidence that twisting “detoxifies” or “wrings out” the organs. Twisting and flexion under load are associated with vertebral fractures in people with osteoporosis (Sinaki, 2013). For healthy students, gentle, upright twisting is a normal spinal movement practised through a comfortable range.

# 10 Kākāsana

CROW

काकासन  
kah-KAH-sa-na

**FAMILY**

Arm balance

**LEVEL**

Intermediate

**TRADITIONAL HOLD**

Build from a few seconds to 30 seconds or more

**MAIN PLANES**

Balance over the hands (sagittal)

**COUNTERPOSE**

Wrist release; standing forward bend follows

*Kāka* means “crow”. The tradition taught here offers the peacock (*mayūrāsana*, one of the Haṭha Yoga Pradīpikā’s fifteen postures) as an alternative for advanced students.

## 01 Purpose

To build strength in the arms, shoulders, wrists and trunk; to develop balance, concentration and confidence.

## 02 Preparation

- ◆ Plank, downward dog and wrist-strengthening to prepare the hands.
- ◆ A squat (*malāsana*) to find the hip position.
- ◆ A cushion in front of the hands.

## 03 Technique, step by step

- 1 Squat with the feet together or hip-width, heels lifted if needed. Place the hands flat on the floor shoulder-width apart, fingers spread and pointing forward.
- 2 Bend the elbows slightly back (not outward) to make a shelf of the upper arms.
- 3 Lift the hips and place the knees on the backs of the upper arms, as close to the armpits as possible.
- 4 Look forward, not down, and slowly shift the weight forward into the hands until one foot, then the other, lifts.
- 5 Balance, drawing the feet towards the buttocks and the abdomen in.



- 6 To come out, lower the feet slowly to the floor and rest back into a squat or child's pose; release the wrists.

#### 04 Breathing

Steady breathing; the breath often wants to stop at the moment of lifting — keep it moving.

#### 05 Alignment

- ♦ Hands firmly pressing, especially through the knuckles of the index finger and thumb; fingers gripping lightly.
- ♦ Elbows pointing back and in, over the wrists.
- ♦ Shoulder blades protracted (“push the floor away”).
- ♦ Gaze forward at a point on the floor ahead.

#### 06 Major joints and their positions

| Joint          | Position / movement          |
|----------------|------------------------------|
| Wrists         | Strongly extended under load |
| Elbows         | Flexed (bent-arm crow)       |
| Shoulders      | Flexed, scapulae protracted  |
| Spine          | Flexed (rounded)             |
| Hips and knees | Strongly flexed              |

#### 07 Major muscles and what they are doing

| Muscle(s)                               | Action in this pose | Role                                        |
|-----------------------------------------|---------------------|---------------------------------------------|
| Serratus anterior, pectorals            | Isometric           | Protract the scapulae and support the trunk |
| Triceps, anterior deltoid, rotator cuff | Isometric           | Hold the bent-elbow shelf                   |
| Forearm flexors                         | Isometric           | Grip the floor and take load off the wrist  |
| Abdominals, hip flexors                 | Isometric           | Draw the knees up and the feet in           |

#### 08 Spinal action and planes of movement

The spine is flexed, drawing the knees high onto the arms; the neck is in slight extension to look forward.

Balance is controlled mainly in the sagittal plane — forward and back over the hands.

#### 09 Biomechanics: how the pose loads the body

The base of support is the two hands; the centre of mass must be brought forward over them, which requires the courage to lean. The wrists carry almost the whole body weight near their end range of extension (Figure 17.8), so wrist preparation and gradual progression are essential.

**COMMON MISTAKES**

- ◆ Elbows splaying outward.
- ◆ Looking down at the hands (the head drops and the body tips forward).
- ◆ Knees too low on the arms.
- ◆ Jumping into the pose; collapsing out of it.

**COMMON STUDENT LIMITATIONS**

- ◆ **Wrist discomfort:** build tolerance gradually; practise on fists or blocks, or substitute plank and squat work.
- ◆ **Weak upper body:** practise with one foot at a time or with the feet on a block.
- ◆ **Fear of falling forward:** cushion in front; practise lifting one foot only.

**⚠ CONTRAINDICATIONS AND PRECAUTIONS**

**Avoid, or practise only with individual guidance, when:**

- ◆ Wrist injury, carpal tunnel syndrome, or acute wrist pain
- ◆ Shoulder or elbow injury
- ◆ Pregnancy
- ◆ Uncontrolled high blood pressure (if straining)

**Precautions:**

- ◆ Place a cushion in front of the head.
- ◆ Warm the wrists; limit repetitions for beginners.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

**10 Modifications, variations and props**

Preparation: squat (malāsana) to find the hip position



Advanced: straight-arm crane (bakāsana)



Advanced alternative: peacock (mayūrāsana)

**MODIFICATIONS**

- ◆ Feet on a block, making the lift easier.
- ◆ One foot lifted at a time.
- ◆ Forehead resting on a block as a “third point” of support.

**BEGINNER VARIATION**

From a squat with feet on a block, place the knees on the arms and lean forward until the weight is in the hands; lift one foot, then the other, for a breath or two.

**ADVANCED VARIATION**

Straight-arm crane (*bakāsana*); side crow; peacock (*mayūrāsana*), in which the elbows press into the abdomen and the body is held horizontal.

**USE OF PROPS**

- ◆ Cushion or bolster in front of the hands.
- ◆ Block under the feet or forehead.

**11 Teaching the pose****SAFE TEACHING CUES**

- ◆ *“Spread the fingers and grip the floor.”*
- ◆ *“Knees high on the arms, close to the armpits.”*
- ◆ *“Look forward, not down.”*
- ◆ *“Lean — the feet lift by themselves.”*
- ◆ *“Push the floor away and round the back.”*

**HANDS-ON GUIDANCE**

Spotting from the side, with consent, a hand ready near the shoulder or upper back to prevent a forward fall. Do not lift the student’s hips or feet.

**12 Transitions and counterpose**

|                 |                                                                         |
|-----------------|-------------------------------------------------------------------------|
| Into the pose   | From a squat.                                                           |
| Out of the pose | Lower the feet slowly into a squat; release the wrists; rest.           |
| Counterpose     | Wrist circles and gentle wrist flexion; then the standing forward bend. |

**⚠ EFFECTS TRADITIONALLY ATTRIBUTED**

Crow is said to strengthen the arms, wrists and shoulders, increase concentration and balance, and develop mental poise and confidence (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

**🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT**

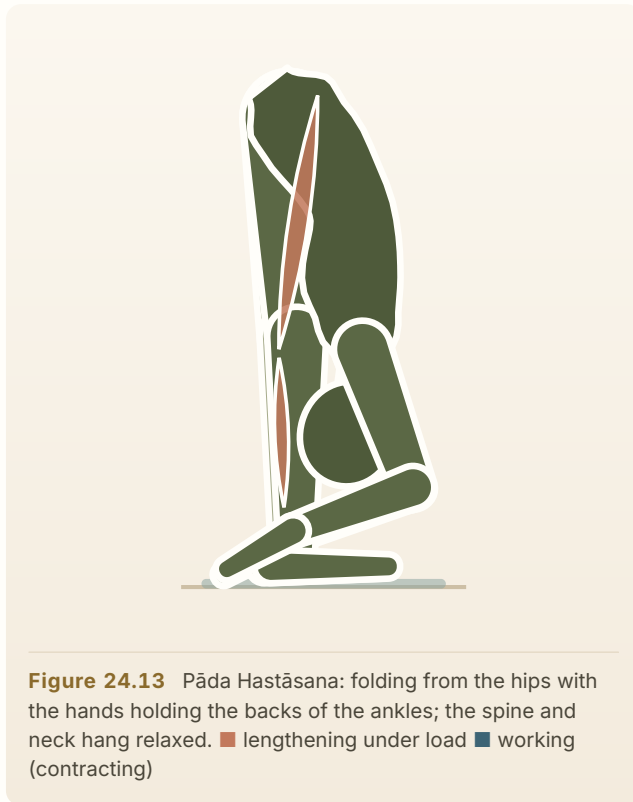
No specific trials. Crow is a demanding closed-chain strength and balance exercise for the upper body. The main documented risk area for arm-balancing practice is the wrist; gradual loading reflects general principles of tissue adaptation (Soligard et al., 2016).

# 11 Pāda Hastāsana

## STANDING FORWARD BEND

पादहस्तासन

PAH-da has-TAH-sa-na



**Figure 24.13** Pāda Hastāsana: folding from the hips with the hands holding the backs of the ankles; the spine and neck hang relaxed. ■ lengthening under load ■ working (contracting)

**FAMILY**

Standing forward bend; mild inversion

**LEVEL**

Beginner (with bent knees)

**TRADITIONAL HOLD**

30 seconds to 2 minutes

**MAIN PLANES**

Sagittal (hip and spinal flexion)

**COUNTERPOSE**

Standing upright; triangle follows

*Pāda* means “foot” and *hasta* “hand”: in the classical form the hands are placed flat under the feet, palms up.

### 01 Purpose

To lengthen the back of the legs and the spine with the help of gravity; to bring the head below the heart in a gentle inversion.

### 02 Preparation

- ◆ Sun salutation (positions 3 and 10).
- ◆ Seated forward bend earlier in the sequence.

### 03 Technique, step by step

- 1 Stand with the feet together (or hip-width), weight even.
- 2 Inhale and raise the arms overhead, lengthening the spine.
- 3 Exhale and fold forward from the hips, keeping the spine long at first.
- 4 Bring the hands to the floor beside the feet, hold the backs of the ankles, or slide the hands under the feet, palms up — bending the knees as needed.
- 5 Let the head and neck hang; relax the shoulders.

- 6 Hold, breathing slowly; with each exhalation release a little further.
- 7 To come up, bend the knees, place the hands on the thighs if needed, and inhale to rise with a long spine, head coming up last.

#### 04 Breathing

Exhale to fold; slow breathing into the back; inhale to rise.

#### 05 Alignment

- ◆ Weight slightly forward over the balls of the feet, hips over (not behind) the heels.
- ◆ The fold begins at the hips.
- ◆ Knees soft or bent; never locked.
- ◆ Neck relaxed.

#### 06 Major joints and their positions

| Joint  | Position / movement     |
|--------|-------------------------|
| Hips   | Strongly flexed         |
| Spine  | Flexed (hanging)        |
| Knees  | Extended or flexed      |
| Ankles | Slightly dorsiflexed    |
| Elbows | Flexed (holding ankles) |

#### 07 Major muscles and what they are doing

| Muscle(s)                       | Action in this pose                        | Role                                |
|---------------------------------|--------------------------------------------|-------------------------------------|
| Hamstrings, gastrocnemius       | Eccentric during descent, then lengthening | Control the fold; the main limit    |
| Erector spinae, gluteus maximus | Eccentric during descent, then relaxed     | Lower the trunk under gravity       |
| Quadriceps                      | Light isometric                            | Keep the knees extended if straight |
| Back extensors, hip extensors   | Concentric                                 | Lift the trunk back to standing     |

#### 08 Spinal action and planes of movement

Flexion under the pull of gravity. When hanging relaxed, the back extensors may go quiet (the flexion-relaxation phenomenon), leaving the passive tissues of the back to carry the load; rising requires them to work again.

Sagittal plane.

## 09 Biomechanics: how the pose loads the body

Unlike the seated forward bend, gravity pulls the trunk down, so the hamstrings work eccentrically to control the descent and the lower back is loaded in flexion. Coming up is the moment of greatest load on the back; bending the knees reduces it. The head below the heart raises pressure in the head and eyes (Jasien et al., 2015).

### COMMON MISTAKES

- ◆ Locking the knees and hanging from the lower back.
- ◆ Coming up quickly with straight legs and rounded back.
- ◆ Pulling on the ankles to force the head to the knees.
- ◆ Weight falling back into the heels.

### COMMON STUDENT LIMITATIONS

- ◆ **Short hamstrings:** bend the knees; hands on blocks or shins.
- ◆ **Low-back sensitivity:** half forward bend (ardha uttānāsana) with hands on thighs.
- ◆ **Dizziness with head below heart:** come up slowly; keep the head higher.

### ⚠ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute low-back pain or disc problems aggravated by flexion
- ◆ Osteoporosis (loaded flexion) (Sinaki, 2013)
- ◆ Glaucoma or serious eye conditions (Jasien et al., 2015); uncontrolled high blood pressure; vertigo
- ◆ Later pregnancy (legs apart, hands on a chair)

**Precautions:**

- ◆ Always bend the knees to come up.
- ◆ Rise slowly to avoid light-headedness.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Beginner: knees bent, hands to the floor



Half forward bend: flat back, hands on the shins

### MODIFICATIONS

- ◆ Knees generously bent.
- ◆ Hands on blocks, a chair or the thighs.
- ◆ Feet hip-width apart.

**BEGINNER VARIATION**

Knees bent, hands to the floor or blocks, head relaxed, 5–8 breaths; rise with bent knees.

**ADVANCED VARIATION**

The classical form: palms under the feet, legs straight, forehead to the shins; and *pādāṅguṣṭhāsana* (holding the big toes).

**USE OF PROPS**

- ◆ Blocks under the hands.
- ◆ Chair for the hands in later pregnancy or with low-back sensitivity.

**11 Teaching the pose****SAFE TEACHING CUES**

- ◆ “Fold from the hips like a hinge.”
- ◆ “Bend your knees as much as you like.”
- ◆ “Let the head hang heavy.”
- ◆ “Bend the knees, then roll or rise slowly, head last.”

**HANDS-ON GUIDANCE**

With consent, a light hand on the sacrum can encourage the weight forward over the feet. Never press down on the back.

**12 Transitions and counterpose**

|                 |                                                                     |
|-----------------|---------------------------------------------------------------------|
| Into the pose   | From standing, after crow and rest.                                 |
| Out of the pose | Bend the knees and rise slowly; pause standing.                     |
| Counterpose     | Standing upright with a gentle lift of the chest; triangle follows. |

🕉 **EFFECTS TRADITIONALLY ATTRIBUTED**

Standing forward bend is said to stretch the whole back of the body, increase blood supply to the brain, relieve digestive complaints and make the spine elastic (Iyengar, 1966; Satyananda, 1969) **TRADITIONAL**.

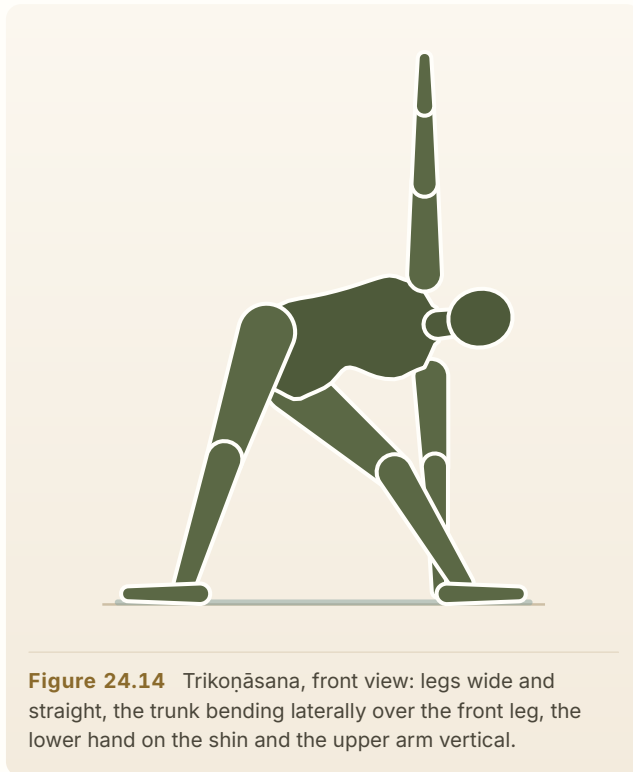
🔍 **WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT**

No trials of *pāda hastāsana* alone. Head-below-heart positions raise intraocular pressure within minutes; standing forward bend was one of the four positions shown to do so (Jasien et al., 2015). The claim of increased brain blood supply is not supported by physiology (cerebral blood flow is autoregulated) (Hall & Hall, 2021). As a hamstring stretch with a mild inversion, it is suitable for most healthy students when knees are bent freely.

# 12 Trikoṇāsana

TRIANGLE

त्रिकोणासन  
tri-ko-NAH-sa-na

**FAMILY**

Standing lateral bend

**LEVEL**

Beginner

**TRADITIONAL HOLD**

30 seconds to 1 minute each side

**MAIN PLANES**

Frontal (lateral flexion, hip abduction)

**COUNTERPOSE**

Standing; final relaxation follows

*Tri* means “three” and *koṇa* “angle”: the legs and floor, the trunk and legs, and the arms form triangles.

## 01 Purpose

To bend the spine laterally, stretch the sides of the trunk and legs, strengthen the legs and improve balance; the last posture of the twelve before final relaxation.

## 02 Preparation

- ◆ Standing postures in the sun salutation.
- ◆ Wide-legged stance practice; side bends.

## 03 Technique, step by step

- 1 Stand with the feet about one leg-length apart (roughly 1 metre, adjusted to proportions).
- 2 Turn the right foot out 90° and the left foot in slightly; align the right heel with the arch of the left foot.
- 3 Inhale and raise the arms to shoulder height, palms down.
- 4 Exhale and bend sideways to the right from the hip and waist, sliding the right hand down the right leg to the shin, ankle or a block; the left arm rises vertically.



- 5 Keep both legs straight but not locked, the trunk in the same plane as the legs; turn the head to look up at the left hand if the neck is comfortable.
- 6 Hold, breathing evenly.
- 7 Inhale to come up with the arms extended; repeat on the other side.

#### 04 Breathing

Exhale to bend; breathe freely — the upper side of the rib cage expands with each inhalation; inhale to return.

#### 05 Alignment

- ◆ Front knee in line with the second and third toes; knee cap lifted, not hyperextended.
- ◆ Trunk in the frontal plane (not collapsing forward); both sides of the waist long.
- ◆ Lower hand rests lightly — the legs and trunk hold the pose.
- ◆ Neck in line; looking down or forward if turning up strains it.

#### 06 Major joints and their positions

| Joint          | Position / movement                                         |
|----------------|-------------------------------------------------------------|
| Spine          | Laterally flexed towards the front leg (with some rotation) |
| Front hip      | Flexed, abducted and externally rotated                     |
| Back hip       | Abducted, relatively neutral                                |
| Knees          | Extended                                                    |
| Shoulders      | Abducted about 90° (lower arm) and 180° (upper arm)         |
| Cervical spine | Rotated to look up                                          |

#### 07 Major muscles and what they are doing

| Muscle(s)                                             | Action in this pose      | Role                                         |
|-------------------------------------------------------|--------------------------|----------------------------------------------|
| Obliques, quadratus lumborum (lower side)             | Eccentric then isometric | Control the side bend against gravity        |
| Obliques, quadratus lumborum, latissimus (upper side) | Lengthening              | Side of the trunk opens                      |
| Hamstrings and adductors (front leg)                  | Lengthening              | Back and inner thigh of the front leg open   |
| Quadriceps, gluteus medius                            | Isometric                | Keep the legs straight and the pelvis stable |
| Deltoids, trapezius                                   | Isometric                | Hold the arms                                |

#### 08 Spinal action and planes of movement

Lateral flexion of the lumbar and thoracic spine towards the front leg, usually with some rotation of the chest upward.

Mainly the frontal plane (lateral flexion, hip abduction); the front hip also flexes (sagittal) and the chest rotates (transverse).

## 09 Biomechanics: how the pose loads the body

The wide base makes triangle stable. The trunk is a lever held out sideways, controlled eccentrically by the muscles of the lower side of the waist. Hamstring length of the front leg determines how far the pelvis can tilt; forcing the hand to the floor causes the trunk to rotate forward or the front knee to hyperextend.

### COMMON MISTAKES

- ◆ Collapsing the trunk forward to reach the floor.
- ◆ Hyperextending (locking) the front knee.
- ◆ Leaning on the lower hand.
- ◆ Straining the neck to look up.

### COMMON STUDENT LIMITATIONS

- ◆ **Short hamstrings or adductors:** hand higher on the leg or on a block.
- ◆ **Neck discomfort:** look forward or down.
- ◆ **Balance difficulty:** back against a wall; shorter stance.

### ⚠ CONTRAINDICATIONS AND PRECAUTIONS

**Avoid, or practise only with individual guidance, when:**

- ◆ Acute low-back pain or sacroiliac pain aggravated by the stance
- ◆ Recent hip or knee injury
- ◆ Severe dizziness or balance disorders (practise at a wall)

### **Precautions:**

- ◆ Keep a micro-bend in the front knee if it tends to hyperextend.
- ◆ Come up on an inhalation with the arms extended, using the legs.

Contraindications in yoga rest largely on expert consensus, physiology and case reports rather than trials. When in doubt, the student's own healthcare professional has the final word.

## 10 Modifications, variations and props



Beginner: lower hand on a block, trunk higher



Advanced: hand on the floor behind the front foot, trunk parallel to the floor

### MODIFICATIONS

- ◆ Lower hand on a block or higher on the shin.
- ◆ Back against a wall for balance and alignment.
- ◆ Upper hand on the hip.

### BEGINNER VARIATION

Shorter stance, lower hand on the shin or a block, upper hand on the hip, gaze forward; 5 breaths each side.

**ADVANCED VARIATION**

Hand on the floor behind the front foot with the trunk parallel to the floor; revolved triangle (*parivṛtta trikoṇāsana*), which combines a twist with the forward-bending stance.

**USE OF PROPS**

- ◆ Block for the lower hand.
- ◆ Wall behind the back.

**11 Teaching the pose****SAFE TEACHING CUES**

- ◆ “Reach out long before you tip.”
- ◆ “Keep both sides of the waist long.”
- ◆ “Stand firmly on both feet.”
- ◆ “Let the hand rest; the legs hold you.”
- ◆ “Open the chest to the side wall.”

**HANDS-ON GUIDANCE**

With consent, a teacher standing behind can place a hand on the back of the pelvis and one on the upper shoulder to help the student feel the trunk in the frontal plane. Avoid pulling the upper arm.

**12 Transitions and counterpose**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Into the pose   | From standing after the standing forward bend.                                                                |
| Out of the pose | Inhale up; turn the feet parallel; step the feet together; stand quietly, then lie down for final relaxation. |
| Counterpose     | Standing, then final relaxation.                                                                              |

**📌 EFFECTS TRADITIONALLY ATTRIBUTED**

Triangle is said to make the spine flexible, tone the spinal nerves and abdominal organs, improve the appetite and help with constipation, and to reduce fat at the waist (Iyengar, 1966; Satyananda, 1969)

**TRADITIONAL** .

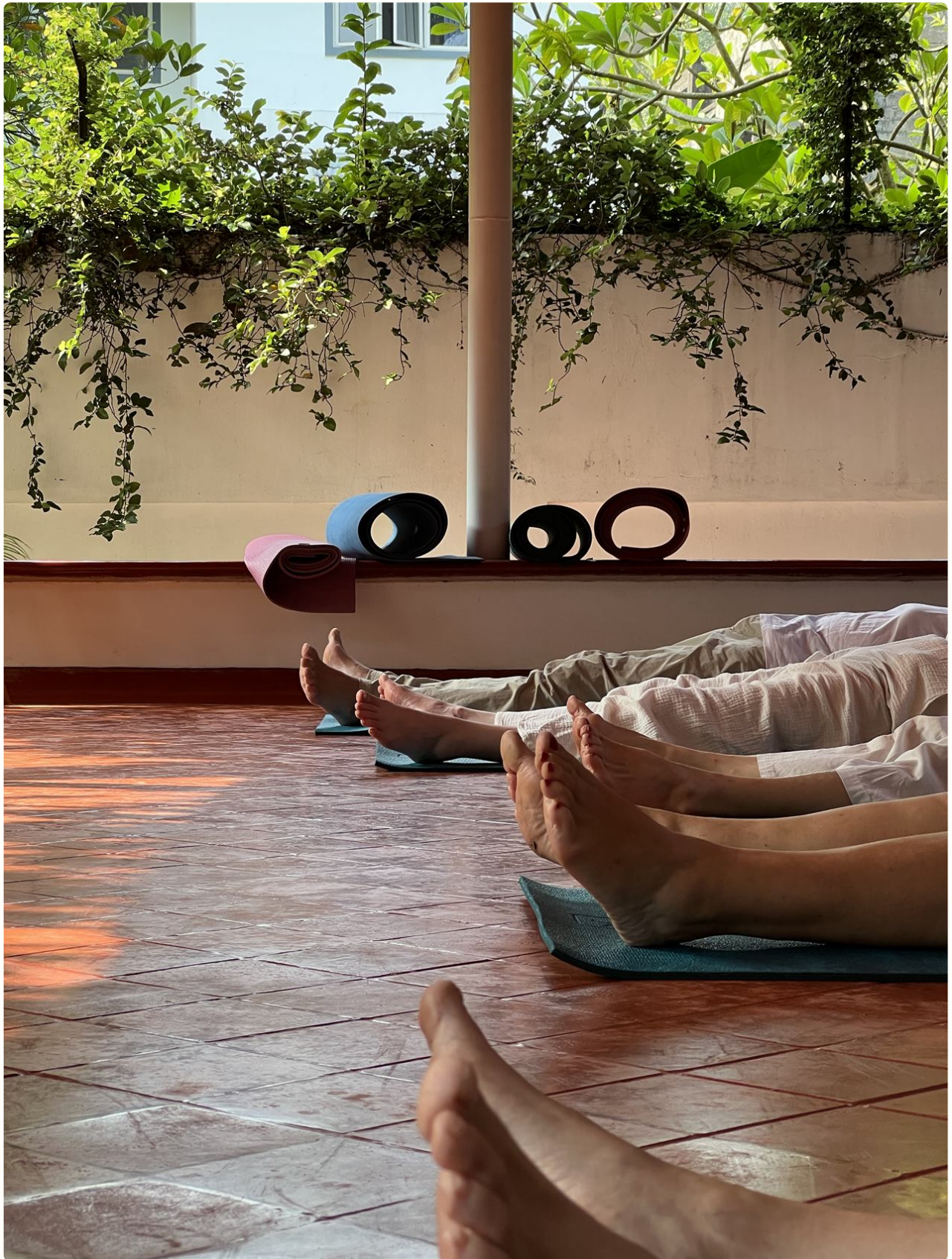
**🔍 WHAT MODERN EVIDENCE DOES AND DOES NOT SUPPORT**

No trials of triangle alone. Standing yoga postures challenge balance and leg strength, and yoga-based exercise improves balance and mobility in older adults (Youkhana et al., 2016). Claims about digestion and waist fat are traditional and untested.

**24.3 Completing the sequence: final relaxation**

After triangle, students lie down for the final relaxation described in **Chapter 23** — tensing and relaxing the body, autosuggestion from the feet to the head, and relaxation of the mind — for ten to fifteen minutes. This is not an optional extra. It is the stage in which the effects of the practice are absorbed, and for many students it is the part of the class they value most.





**Figure 24.2** Final relaxation in the garden pavilion at Santhi Yoga School, Fort Kochi.

## CHAPTER REVIEW

## 🍃 KEY TAKEAWAYS

- ◆ The twelve basic postures follow a fixed order: inversions (headstand, shoulderstand, plough) with fish as counterpose; forward bend; three backbends of increasing depth (cobra, locust, bow); twist; arm balance; standing forward bend; triangle.
- ◆ Each posture should be taught with preparation, clear technique, breath, alignment and a planned way in and out.
- ◆ Every posture has modifications; preparations are not lesser versions but the right versions for many students.
- ◆ Inversions demand screening for contraindications; the neck is protected by forearm strength in headstand and blankets in shoulderstand and plough.
- ◆ Traditional effects should be taught as traditional; modern evidence for individual postures is limited, and some traditional claims (such as increased blood flow to the brain) are not supported by physiology.

## 🍃 REFLECTION QUESTIONS

1. Which of the twelve is most challenging for you? Is the limit strength, mobility, bone shape, fear or understanding?
2. Which posture's traditional description moves you most, and how will you share it honestly?
3. What would a class of the twelve look like for a group of students in their seventies?

## 🙏 TEACHER-PRACTICE EXERCISES

1. Teach each of the twelve postures to a partner using only the “safe teaching cues” in its monograph plus one cue of your own.
2. For three postures, demonstrate the main posture, the beginner variation and one prop modification in sequence, explaining when you would offer each.
3. In groups of three (teacher, student, observer), practise spotting headstand at a wall with consent; the observer gives feedback on the teacher's position and communication.

## ☑️ KNOWLEDGE CHECK

1. Which posture immediately follows plough in the classical sequence? (a) cobra (b) fish (c) forward bend (d) headstand
2. In headstand, most of the weight should be carried by: (a) the head (b) the forearms and shoulders (c) the neck (d) the hands clasped behind the head
3. Why are folded blankets used under the shoulders in shoulderstand? (a) for comfort only (b) to reduce neck flexion (c) to increase the stretch (d) to make the pose higher
4. In cobra as taught in this course, the elbows are: (a) straight (b) bent and close to the body (c) wide (d) off the floor
5. Which claim is *not* supported by physiology? (a) inversions raise intraocular pressure (b) headstand greatly increases blood flow to the brain (c) crow loads the wrists in extension (d) bending the knees makes forward bends easier for short hamstrings

Answers 1 b · 2 b · 3 b · 4 b · 5 b

 **TEACHING ASSIGNMENT**

Prepare and teach a 60-minute class of the twelve basic postures for mixed-level beginners, with sun salutation and leg raises, relaxation between postures and a 10-minute final relaxation. Hand in a written plan listing, for each posture, the version you will teach, the modifications you will offer, and the contraindications you will screen for. This class forms part of your practicum portfolio (Chapter 40).

## CHAPTER 25

## Asana–Anatomy Mapping Tables

*“Observe the body as it is, not as the picture says it should be.” — a working principle of this manual*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ read and use an asana–anatomy map: joints, movements, muscles, types of contraction, loading, limitations and modifications;
- ◆ compare postures across families by the demands they place on the body;
- ◆ use the maps to choose preparatory postures, counterposes and modifications when planning a class;
- ◆ construct a map for any posture not included here.

## 25.1 How to read the maps

Each row of the tables that follow summarises one posture in eight columns:

- ◆ **Joints** — the joints doing the most significant work or reaching the greatest range.
- ◆ **Movement** — what those joints are doing, in anatomical terms (Chapter 16).
- ◆ **Key muscles** — the principal muscles involved (Chapter 19).
- ◆ **Muscle action** — C = concentric, E = eccentric, I = isometric, L = lengthening under load (Chapter 18).
- ◆ **Loading** — where the posture concentrates load, and any structure that deserves care (Chapter 20).
- ◆ **Common limitations** — what most often limits students.
- ◆ **Modifications** — first-line options for those limitations.

The maps are deliberately simplified. Every posture involves more muscles than are listed, and individuals perform the same posture in different ways. Use the maps as a planning tool and a memory aid, not as a description of what must happen in every body.

 USING THE MAPS TO PLAN

To prepare for a posture, look at its **joints** and **key muscles** and choose earlier postures that mobilise those joints and warm those muscles. To choose a counterpose, look at its **movement** and choose a posture that moves the main joints gently in the opposite direction. To prepare modifications, read the **limitations** column before class and have the props ready.

The mapping tables are printed on landscape pages.



**Table 25.1** The twelve basic postures

| Asana            |                                                                                     | Joints                                             | Movement                                            | Key muscles                                                                               | Muscle action                                           | Loading                                                                  | Common limitations                                | Modifications                                  |
|------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------|
| Śīrṣāsana        |    | Shoulders, elbows, cervical spine, trunk           | Shoulder flexion; spine neutral, inverted           | Serratus anterior, trapezius, deltoids, rotator cuff, triceps; abdominals, back extensors | I (shoulders, trunk); C/E hip flexors on entry and exit | Axial load through fore-arms, shoulders and partly the head/neck         | Shoulder strength and flexion; fear; neck comfort | Dolphin; walking in; tuck at a wall            |
| Sarvāṅgāsana     |    | Cervical and upper thoracic spine, shoulders, hips | Cervical flexion; shoulder extension; hip extension | Back extensors, gluteals; triceps, posterior deltoid; neck extensors (L)                  | I trunk and legs; L neck extensors                      | Body weight on shoulders and upper back with neck near end-range flexion | Neck and upper-back stiffness; abdominal strength | Blankets; half shoulderstand; legs up the wall |
| Halāsana         |  | Whole spine, hips, shoulders                       | Spinal and hip flexion                              | Hamstrings, calves, back extensors (L); abdominals (E/I)                                  | L posterior chain; E lowering the legs                  | Loaded spinal flexion, especially cervical                               | Hamstring length; upper-back stiffness            | Feet on a chair or blocks; knees bent          |
| Matsyāsana       |  | Thoracic and cervical spine, shoulders, elbows     | Spinal extension; shoulder extension                | Back extensors, rhomboids, triceps; pectorals, rectus abdominis, neck flexors (L)         | C then I back extensors; L front of trunk               | Weight on elbows; light load on crown of head                            | Thoracic stiffness; neck discomfort in extension  | Supported fish on bolster or blocks            |
| Paścimottānāsana |  | Hips, spine, knees, ankles                         | Hip and spinal flexion                              | Hamstrings, gastrocnemius, back extensors (L); quadriceps (I)                             | L hamstrings; light I quadriceps                        | Tension on hamstrings and proximal tendon; flexion of lumbar spine       | Hamstring length; pelvic tilt                     | Sit on blanket; bend knees; strap              |
| Bhujāṅgāsana     |  | Thoracic and lumbar spine, shoulders               | Spinal extension                                    | Erector spinae, rhomboids, trapezius; rectus abdominis (L)                                | C then I back extensors                                 | Compression of posterior lumbar elements                                 | Upper-back stiffness; low-back sensitivity        | Sphinx; baby cobra                             |
| Śalabhāsana      |  | Hips, lumbar spine                                 | Hip and lumbar extension                            | Gluteus maximus, hamstrings, erector spinae; triceps                                      | C then I                                                | Long-lever hip extension; lumbar compression                             | Hip-extensor strength                             | Half locust; arms beside body                  |

| Asana                |  | Joints                                    | Movement                                                                   | Key muscles                                                                                                    | Muscle action                       | Loading                                          | Common limitations                                 | Modifications                            |
|----------------------|--|-------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------|----------------------------------------------------|------------------------------------------|
| Dhanurāsana          |  | Spine, hips, knees, shoulders             | Spinal and hip extension; knee flexion; shoulder extension                 | Hamstrings, gluteals, back extensors (I); quadriceps, hip flexors, pectorals (L)                               | I posterior chain; L anterior chain | Deep lumbar extension driven by the legs         | Quadriceps/hip-flexor length; low-back sensitivity | Half bow; strap                          |
| Ardha Matsyendrāsana |  | Thoracic, lumbar and cervical spine; hips | Spinal rotation; hip flexion and adduction (top leg)                       | Obliques (C/I); back extensors (I); gluteals and deep rotators of top leg (L)                                  | C then I rotation                   | Rotation; with slumping, rotation plus flexion   | Hip tightness; lower back rounding                 | Easy twist; bottom leg straight; cushion |
| Kākāsana             |  | Wrists, elbows, shoulders, hips, knees    | Wrist extension; elbow flexion; scapular protraction; hip and knee flexion | Serratus anterior, pectorals, triceps, forearm flexors; abdominals, hip flexors                                | I                                   | Body weight on hands; wrist extension under load | Wrist tolerance; upper-body strength; fear         | Feet on block; one foot at a time        |
| Pāda Hastāsana       |  | Hips, spine, knees                        | Hip and spinal flexion under gravity                                       | Hamstrings and back extensors (E then L); calves (L)                                                           | E descent; L hold; C to rise        | Loaded flexion; head below heart                 | Hamstring length; low-back sensitivity             | Bent knees; blocks; half forward bend    |
| Trikoṇāsana          |  | Spine, hips, knees, shoulders             | Lateral flexion; hip abduction and flexion (front leg)                     | Obliques and QL of lower side (E/I); hamstrings and adductors of front leg (L); quadriceps, gluteus medius (I) | E then I                            | Trunk as a lateral lever; front knee             | Hamstring/adductor length; balance                 | Block; wall; shorter stance              |

Simplified summary; see the monographs in Chapter 25 for detail. Contraction types: C = concentric, E = eccentric, I = isometric, L = lengthening.

**Table 25.2** Standing postures

| Asana      |                                                                                     | Joints                                   | Movement                                                            | Key muscles                                                                              | Muscle action             | Loading                       | Common limitations                    | Modifications                                           |
|------------|-------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------|-------------------------------|---------------------------------------|---------------------------------------------------------|
| Tāḍāsana   |    | Ankles, knees, hips, spine               | Upright neutral stance                                              | Postural muscles: calves, quadriceps, gluteals, deep trunk                               | I (low-level, continuous) | Standing load; balance        | Habitual posture; knee hyperextension | Feet hip-width; micro-bend in knees                     |
| Vṛkṣāsana  |   | Standing ankle, knee and hip; lifted hip | Single-leg stance; lifted hip flexion, abduction, external rotation | Gluteus medius/minimus, foot and ankle muscles (standing leg); adductors (L, lifted leg) | I                         | Single-leg balance            | Balance; hip external rotation        | Foot on ankle or calf (never press into the knee); wall |
| Utkāṣāsana |  | Knees, hips, ankles, shoulders           | Knee and hip flexion; ankle dorsiflexion; shoulder flexion          | Quadriceps, gluteals, erector spinae; deltoids                                           | I                         | High quadriceps and knee load | Knee comfort; ankle dorsiflexion      | Less depth; hands at chest                              |

| Asana             |                                                                                   | Joints                                         | Movement                                                            | Key muscles                                                             | Muscle action | Loading                                                    | Common limitations                                | Modifications                    |
|-------------------|-----------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|---------------|------------------------------------------------------------|---------------------------------------------------|----------------------------------|
| Vīrabhadrāsana I  |  | Front knee and hip; back hip; spine; shoulders | Front hip and knee flexion; back hip extension; shoulder flexion    | Quadriceps and gluteals (front leg); hip flexors and calf (back leg, L) | I / L         | Front knee; lumbar extension if back hip flexors are short | Hip-flexor length; ankle dorsiflexion of back leg | Shorter stance; back heel lifted |
| Vīrabhadrāsana II |  | Hips, knees, shoulders                         | Front hip flexion, abduction, external rotation; shoulder abduction | Quadriceps, gluteals, hip rotators; adductors (L); deltoids             | I             | Front knee must track over the foot                        | Hip external rotation; endurance                  | Shorter stance; less depth       |
| Ardha Uttānāsana  |  | Hips, spine                                    | Hip flexion with neutral spine                                      | Back extensors (I); hamstrings (L)                                      | I / L         | Back extensors hold trunk horizontal                       | Hamstring length                                  | Hands on thighs; knees bent      |

**Table 25.3** Seated postures and hip openers

| Asana           |                                                                                    | Joints              | Movement                                                                                                                         | Key muscles                                                                                                | Muscle action | Loading                                                  | Common limitations               | Modifications                                                   |
|-----------------|------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------|----------------------------------|-----------------------------------------------------------------|
| Daṇḍāsana       |   | Hips, spine, knees  | Hip flexion 90°; upright spine                                                                                                   | Back extensors, hip flexors (I); hamstrings (L)                                                            | I             | Posterior pelvic tilt if hamstrings are short            | Hamstring length                 | Sit on a blanket; bend knees                                    |
| Sukhāsana       |   | Hips, knees, spine  | Hip flexion, abduction, external rotation                                                                                        | Deep hip rotators, adductors (L); back extensors (I)                                                       | L / I         | Sustained sitting                                        | Hip mobility; knees above hips   | Cushion; support under knees                                    |
| Vajrāsana       |   | Knees, ankles       | Full knee flexion; ankle plantarflexion                                                                                          | Quadriceps, tibialis anterior (L)                                                                          | L             | Deep knee flexion under body weight                      | Knee and ankle comfort           | Cushion between heels and buttocks; rolled blanket under ankles |
| Baddha Koṇāsana |   | Hips, knees         | Hip flexion, abduction, external rotation                                                                                        | Adductors (L)                                                                                              | L             | Adductor tension; knees                                  | Hip shape; adductor length       | Sit high; blocks under knees                                    |
| Malāsana        |  | Hips, knees, ankles | Deep hip and knee flexion; ankle dorsiflexion                                                                                    | Soleus, gluteals (L); tibialis anterior (I)                                                                | L / I         | Deep knee and hip flexion                                | Ankle dorsiflexion; knee comfort | Heels on a folded blanket; sit on a block                       |
| Gomukhāsana     |                                                                                    | Hips, shoulders     | Hip flexion, adduction, external rotation; one shoulder flexed and externally rotated, the other extended and internally rotated | Gluteals, deep rotators (L); triceps, latissimus (top arm, L); pectorals, anterior deltoid (bottom arm, L) | L             | Hips and shoulders at end range                          | Hip and shoulder mobility        | Sit on a cushion; strap between hands                           |
| Padmāsana       |                                                                                    | Hips, knees, ankles | Extreme hip external rotation, flexion and abduction                                                                             | Adductors, internal-rotator structures of the hip (L)                                                      | L             | Rotational load passes to the knee if the hip is limited | Hip shape; knee health           | Half lotus; easy pose — never force                             |

**Table 25.4** Backbends, core and arm-support postures

| Asana                          |                                                                                   | Joints                                 | Movement                                                          | Key muscles                                                                                                     | Muscle action      | Loading                                                                        | Common limitations                                    | Modifications                                     |
|--------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------|
| Setu Bandha Sarvāṅgāsana       |  | Hips, spine, shoulders, cervical spine | Hip extension; spinal extension; shoulder extension; neck flexion | Gluteals, hamstrings, back extensors (C/l); hip flexors (L)                                                     | C then l           | Weight on shoulders and feet; neck flexion                                     | Hip-flexor length; neck comfort                       | Block under sacrum (supported bridge)             |
| Uṣṭrāsana                      |  | Spine, hips, shoulders                 | Spinal and hip extension; shoulder extension                      | Back extensors, gluteals (l); quadriceps, hip flexors, abdominals, pectorals (L)                                | E descent; l hold  | Deep lumbar and cervical extension                                             | Thoracic and hip-flexor mobility                      | Hands on lower back; toes tucked; blocks by heels |
| Ūrdhva Dhanurāsana / Cakrāsana |  | Shoulders, wrists, spine, hips         | Full shoulder flexion; spinal and hip extension; wrist extension  | Triceps, deltoids, serratus, gluteals, back extensors (C/l); hip flexors, abdominals, latissimus, pectorals (L) | C then l           | High demand on shoulders, wrists and lumbar spine                              | Shoulder flexion; thoracic extension; wrist tolerance | Bridge; wheel at a wall; blocks under hands       |
| Nāvāsana                       |  | Hips, spine                            | Hip flexion; trunk balanced on sitting bones                      | Hip flexors, rectus abdominis, obliques                                                                         | l                  | Hip-flexor pull on lumbar spine                                                | Abdominal strength; hamstring length                  | Knees bent; hands behind thighs                   |
| Chaturāṅga Daṇḍāsana           |                                                                                   | Shoulders, elbows, wrists, trunk       | Elbow flexion to about 90° with neutral trunk                     | Triceps, pectorals, anterior deltoid, serratus anterior, rotator cuff, abdominals                               | E lowering; l hold | High load on the front of the shoulder when the shoulders dip below the elbows | Upper-body strength                                   | Knees down; lower halfway only                    |
| Sphinx                         |  | Spine, shoulders                       | Gentle spinal extension on forearms                               | Back extensors (l); abdominals (L)                                                                              | l / L              | Low                                                                            | Low-back sensitivity                                  | Forearms further forward                          |

**Table 25.5** Kneeling, supine and restorative postures

| Asana                     |  | Joints                     | Movement                                 | Key muscles                                  | Muscle action   | Loading                                   | Common limitations                | Modifications                                                        |
|---------------------------|--|----------------------------|------------------------------------------|----------------------------------------------|-----------------|-------------------------------------------|-----------------------------------|----------------------------------------------------------------------|
| Śavāsana                  |  | All at rest                | Supine neutral                           | —                                            | Relaxation      | Minimal                                   | Low-back discomfort; restlessness | Bolster under knees; side-lying; eyes open                           |
| Bālāsana                  |  | Knees, hips, ankles, spine | Knee and hip flexion; spinal flexion     | Back extensors, gluteals (L); quadriceps (L) | L               | Deep knee flexion                         | Knee or ankle discomfort          | Knees apart; cushion between heels and buttocks; forehead on a block |
| Bitilāsana / Mārjaryāsana |  | Spine, shoulders, hips     | Alternating spinal extension and flexion | Back extensors / abdominals alternately      | C / E, rhythmic | Low; weight on hands and knees            | Wrist or knee discomfort          | Forearms down; padding under knees                                   |
| Supta Pādāṅguṣṭhāsana     |  | Hip, knee                  | Hip flexion with straight knee           | Hamstrings, gastrocnemius (L)                | L               | Low; spine supported                      | Hamstring length                  | Strap; opposite knee bent                                            |
| Viparīta Karaṇī           |  | Hips, cervical spine       | Inversion with hips supported            | Back extensors, gluteals (I)                 | I (low)         | Less neck flexion than full shoulderstand | Neck comfort                      | Legs up the wall; block under sacrum                                 |
| Legs up the wall          |  | Hips                       | Hip flexion 90°, legs vertical           | Hamstrings (mild L)                          | L (mild)        | Very low                                  | Hamstring length                  | Move hips further from the wall                                      |

The sun-salutation positions are mapped in Chapter 23.

## 25.2 Mapping a new posture

### ✂ BUILD YOUR OWN MAP

Choose a posture not included in these tables — for example *pārsvottānāsana* (intense side stretch), *ardha candrāsana* (half moon) or *gomukhāsana* (cow face) arms. Complete all eight columns, then check your map with a partner: do you agree on the main movement, the key muscles and where the load goes? Finally, test your “modifications” column by teaching the posture to someone with the limitation you listed.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ An asana-anatomy map summarises joints, movements, muscles, contraction types, loading, limitations and modifications for a posture.
- ◆ Maps help in choosing preparations (same joints and muscles), counterposes (opposite movements) and modifications (from the limitations).
- ◆ The maps simplify: bodies differ, and so do their ways of making a shape.
- ◆ Any posture can be mapped with the same eight questions.

### 🍃 REFLECTION QUESTIONS

1. Which column of the maps is most useful to you as a new teacher? Why?
2. Which posture in the tables surprised you with its loading? How will that change how you teach it?
3. How could you use a simplified version of a map to help a student understand their own practice?

### ✂ TEACHER-PRACTICE EXERCISES

1. Using the tables, plan a sequence of five preparatory postures for wheel (ūrdhva dhanurāsana).
2. Choose a counterpose for each of the three prone backbends using the “movement” column, and justify each.
3. Map two postures you teach often that are not in the tables, and present them to your practice group.



☑ **KNOWLEDGE CHECK**

1. In the maps, “E” stands for: (a) extension (b) eccentric contraction (c) external rotation (d) end range
2. Which column would you consult first to choose a counterpose? (a) muscles (b) movement (c) loading (d) limitations
3. According to the maps, which posture loads the wrist in extension with the most body weight? (a) cat-cow (b) crow (c) cobra (d) bridge
4. Supported bridge places a block: (a) under the head (b) under the sacrum (c) under the feet (d) between the knees
5. The main limitation listed for pāścimottānāsana is: (a) wrist tolerance (b) hamstring length (c) shoulder flexion (d) neck extension

Answers 1 b · 2 b · 3 b · 4 b · 5 b

📄 **TEACHING ASSIGNMENT**

Choose a peak posture for a 45-minute class. Using the maps, write a one-page sequence plan showing: three preparatory postures (with the joints and muscles each prepares), the peak with two modifications, and two counterposes with the movements they reverse. Teach the class in your practice group.

## CHAPTER 26

# Injury Prevention and Teacher Safety

*“Ahimsā — in the presence of one established in non-harming, hostility is abandoned.” — Yoga Sūtra 2.35*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ apply the principles of safe teaching, including screening, informed choice and gradual progression;
- ♦ explain load management and distinguish useful discomfort from pain that means stop;
- ♦ summarise what research shows about the frequency and pattern of yoga-related injuries;
- ♦ recognise common problems of the shoulders, wrists, knees, hips, hamstrings, lower back, neck and sacroiliac region, and modify practice appropriately;
- ♦ know when and how to refer a student to a healthcare professional, and recognise emergencies;
- ♦ teach safely in inversions, deep backbends, deep twists, extreme hip positions and weight-bearing on the hands, and for pregnant students, older adults, hypermobile students and those with previous injuries.

## 26.1 How safe is yoga?

Yoga is, on the whole, a safe form of exercise. A systematic review of randomised trials found that adverse events in yoga were no more common than with usual care or other exercise, and serious adverse events were rare (Cramer et al., 2015). But “safe on average” is not “safe for everyone in every class”. Surveys and cohort studies of recreational practitioners find that a meaningful minority experience new musculoskeletal pain each year, and that practice sometimes aggravates existing problems (Campo et al., 2018; Penman et al., 2012). Emergency-department data from the United States suggest that yoga-related injuries rose as participation grew, particularly among older adults (Swain & McGwin, 2016). Case reports describe rarer serious events, including neck injuries and vascular problems associated with extreme neck positions and inversions (Cramer et al., 2013).

Surveys of practitioners and teachers consistently name the same regions: the lower back, neck, shoulders, wrists, knees and hamstrings (Wiese et al., 2019; Fishman et al., 2009; Mikkonen et al., 2008). Common threads in the reported mechanisms are **forcing** (by the student or by a teacher's adjustment), **end-range loading** that the tissues are not prepared for, **repetition** without variation or rest, and **ignoring pain**.

#### ◆ THE TEACHER'S SHARE OF SAFETY

You cannot guarantee that no student will ever be hurt; you can make injury much less likely by screening, offering options, progressing gradually, observing closely, never forcing, and creating a culture in which modifying and resting are respected rather than seen as failure.

## 26.2 Principles of safe teaching

1. **Know your students.** Use a brief health questionnaire for new students (Appendix I) and ask the class, discreetly, about injuries and conditions at the start. Invite private disclosure.
2. **Informed choice.** Explain options clearly and let students choose. Adults have the right to make decisions about their own practice; your job is to make those decisions well informed.
3. **Progress gradually.** New postures, longer holds, more repetitions and higher intensity are introduced step by step, over weeks, not in one class.
4. **Offer, don't impose.** Offer variations to the whole room ("you might choose... or...") rather than singling out individuals.
5. **Watch the breath and the face.** A held breath, a grimace or a flushed face mean the student is beyond their present capacity.
6. **Never force.** Neither students nor teachers should force a body into a shape. Hands-on assistance follows strict principles of consent and restraint (Chapter 36).
7. **Prepare the environment.** Adequate space between mats, non-slip surfaces, props available, a clear exit, ventilation and — in Kerala — shade, air movement and drinking water.
8. **Teach the exit.** For every challenging posture, teach how to come out safely before teaching how to go deeper.

## 26.3 Load management

Tissues adapt to the loads they meet regularly, and become irritated when load increases faster than they can adapt (Figure 20.3) (Soligard et al., 2016; Cook & Purdam, 2009). For a teacher, managing load means thinking about **how much, how often, how intensely and how suddenly**:

- ◆ Increase only one variable at a time — duration *or* depth *or* repetitions.
- ◆ Be especially careful with **repetition**: fifty chaturaṅgas a week are a different load from five.
- ◆ Plan lighter sessions and rest; on an intensive course, the second week is when overuse problems typically appear.
- ◆ Return to practice gradually after a break, illness or injury.

## 26.4 Pain and discomfort

Pain is a protective output of the nervous system, influenced by context and expectation, not a direct measure of damage (Raja et al., 2020; Butler & Moseley, 2013). Discomfort is a normal part of learning postures — stretching, muscular effort, fatigue, the unfamiliar. Pain that signals stop has different qualities.

| GREEN · continue                                                                                              | AMBER · modify                                                                                      | RED · stop                                                                                                         |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| warmth, muscular effort, trembling from work, a broad even stretch sensation, breath steady; fades soon after | unfamiliar or one-sided sensation, sensation that builds with each breath, breath becoming strained | sharp, stabbing, burning or electric pain; numbness or tingling; joint pain; dizziness, chest pain, breathlessness |
| Keep going; stay attentive.                                                                                   | Back off, modify, add support; reassess next practice.                                              | Come out slowly; do not resume that practice; refer if it persists.                                                |

**Figure 26.1** A traffic-light guide to sensation during practice. It is a teaching tool, not a diagnostic instrument; pain is a protective output of the nervous system and does not map neatly onto tissue damage (Raja et al., 2020; Butler & Moseley, 2013).

### A CLINICAL EXAMPLE: PAIN MONITORING

In some rehabilitation programmes, physiotherapists allow patients to exercise with mild pain as long as it settles quickly and does not worsen the next morning — the “pain-monitoring model” used, for example, in tendon rehabilitation (Silbernagel et al., 2007). This shows that some pain during activity is not necessarily harmful. It is a clinical tool used under professional supervision; in a group class, the more cautious traffic-light approach is appropriate.

### When a student should stop

A student should stop the posture or practice immediately, and not resume it in that class, if they experience:

- ◆ sharp, stabbing, burning, electric or shooting pain;
- ◆ numbness, tingling or weakness spreading into the arms or legs;
- ◆ pain in a joint (rather than a stretch in a muscle), or a “pop” or giving way;
- ◆ dizziness, light-headedness, visual disturbance or severe headache;
- ◆ chest pain, pressure or palpitations, or unusual breathlessness;
- ◆ nausea, confusion or feeling faint;
- ◆ in pregnancy: bleeding, fluid leakage, contractions, calf pain or swelling, or any of the above (ACOG, 2020).

### ⚠ EMERGENCIES

Call for emergency medical help immediately if a student shows signs of a **heart attack** (chest pain or pressure, pain spreading to the arm, jaw or back, breathlessness, sweating, nausea), a **stroke** (face drooping, arm weakness, speech difficulty — act fast), **loss of consciousness**, a **severe allergic reaction**, **heat stroke** (confusion, hot skin, collapse) or a **serious injury**. Know the local emergency number, where the nearest hospital is, and where the first-aid kit is kept. Every teacher should hold a current first-aid certificate.

## Heat and hydration

Kerala is hot and humid for much of the year. Exercise in heat raises core temperature and fluid loss, and humidity reduces the cooling effect of sweating (Hall & Hall, 2021). Schedule demanding practice in the cooler hours, ensure shade and air movement, encourage students to drink water before and after class, allow rests, and recognise **heat exhaustion** (heavy sweating, dizziness, headache, nausea, weakness): stop, move to a cooler place, lie down with legs raised, sip water, and seek help if symptoms do not improve quickly.

## 26.5 What teachers can address — and when to refer

**Table 26.1** Scope of the yoga teacher with regard to pain and injury

| Teachers can                                             | Teachers should refer when                                                                                       |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| ask about injuries and conditions, and adapt practice    | pain is new, severe, worsening, or not settling within a few days                                                |
| offer modifications, props and alternative postures      | pain wakes the student at night or is present at rest                                                            |
| reduce range, load, repetitions or duration              | there are neurological symptoms: numbness, tingling, weakness, changes in coordination                           |
| encourage gradual return to practice after clearance     | a joint is swollen, hot, locked or giving way                                                                    |
| give general information about evidence on yoga and pain | there is unexplained weight loss, fever, or a history of cancer with new back pain                               |
| suggest seeing a doctor or physiotherapist               | there are bladder or bowel changes, or numbness in the saddle area (a medical emergency) (Finucane et al., 2020) |
| work alongside a student's healthcare professional       | the student is pregnant with complications, or has a serious medical condition                                   |

A yoga teacher does not diagnose. “I can’t tell what is causing that, but it would be worth having it checked” is always an acceptable sentence (Chapter 15).

## 26.6 Region by region

**Table 26.2** Common problems by region, what teachers may notice, and how to respond

| Region                   | What you may notice                                                                                 | Practice patterns involved                                                                                                              | What to do in class                                                                                                        | Refer if                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Shoulders</b>         | pain at the front or side of the shoulder with arms overhead or in chaturaṅga; clicking with pain   | repeated chaturaṅga with the shoulders dipping below the elbows; collapsing in plank and down dog; forced binds                         | knees down; lower halfway only; strengthen the shoulder girdle; reduce repetitions; avoid binds                            | pain at night, weakness lifting the arm, pain after a fall, symptoms persisting over two weeks |
| <b>Wrists</b>            | pain on the back of the wrist in plank, down dog, crow; tingling in the fingers                     | sudden increase in time on the hands; sinking into the heel of the hand                                                                 | spread the load ( <b>Figure 17.8</b> ); fists, forearms or blocks; wedge under the heel of the hand; build gradually       | numbness or tingling in the hand (possible nerve compression), swelling, pain after a fall     |
| <b>Knees</b>             | pain inside or behind the knee in lotus, hero, deep squats; clicking with pain; swelling            | forcing lotus; twisting the knee when the hip is limited; hyperextension in standing poses                                              | never force the knee; cushion between heels and buttocks; half lotus or easy pose; micro-bend in standing poses            | swelling, locking, giving way, pain after a twisting incident                                  |
| <b>Hips</b>              | groin pain or pinching at the front of the hip in deep flexion; pain with pigeon or lotus           | forcing external rotation or deep flexion into compression                                                                              | reduce depth; change the angle; sit higher; respect bony limits (Griffin et al., 2016)                                     | persistent groin pain, catching or locking, pain walking                                       |
| <b>Ham-strings</b>       | deep pain at the sitting bone, worse sitting and in forward bends                                   | aggressive forward bending; pulling with the arms; long holds on a sensitised tendon                                                    | bend the knees; reduce depth; avoid pulling; strengthen (bridge, slow lowering) (Goom et al., 2016)                        | pain persisting for weeks, bruising after a sudden “pop”                                       |
| <b>Lower back</b>        | pain in forward bends or backbends; stiffness; pain after class                                     | loaded flexion (standing folds with straight legs), deep backbends concentrated in the lumbar spine, twisting while slumped             | bend the knees; lengthen before bending; smaller range; gentle movement usually helps more than rest (Qaseem et al., 2017) | red flags (see <b>Table 26.1</b> ), leg symptoms, pain after trauma, severe or worsening pain  |
| <b>Neck</b>              | pain in shoulderstand, plough, headstand or fish; headache after inversions                         | weight on the head in headstand; neck pressed flat in shoulderstand; dropping the head back under load; turning the head while inverted | forearm strength first; blankets; supported fish; no head turning                                                          | pain or tingling into the arms, dizziness, visual symptoms, severe headache                    |
| <b>Sacroiliac region</b> | one-sided pain at the back of the pelvis, over the SI joint, in twists or asymmetric standing poses | forcing the pelvis “square” in asymmetric poses; deep twists with the pelvis fixed; hypermobility                                       | let the pelvis move with the twist; shorter stances; symmetrical strengthening (Vleeming et al., 2012)                     | pain persisting, pain in pregnancy, pain after a fall                                          |

## 26.7 Special situations

### Inversions

Screen for neck problems, uncontrolled high blood pressure, heart disease, glaucoma and other serious eye conditions, recent head injury, dizziness and vertigo, and pregnancy in students not already practising inversions (Baskaran et al., 2006; Jasien et al., 2015). Build strength first; use blankets for shoulderstand and plough; teach headstand at a wall with a controlled entry (Hector & Jensen, 2015); never allow turning of the head; keep holds short for beginners.

### Deep backbends

Build through cobra and locust before bow, camel and wheel; distribute extension along the spine; keep the neck in line; practise with fresh (not fatigued) muscles; counterpose gently; avoid for students with acute back pain aggravated by extension.

### Deep twists

Sit tall; turn from the rib cage; avoid levering with the arms; let the pelvis move; avoid combining deep rotation and flexion in students with osteoporosis or acute disc problems (Sinaki, 2013).

### Extreme hip mobility

Lotus, splits and deep hip openers are where bone shape most clearly sets limits. Never press on a student's knees or pelvis; teach that the knee is protected by keeping the foot flexed and by sitting high; accept that some students will never do full lotus — and do not need to.

### Weight-bearing through the hands

Plank, down dog, crow and wheel load the wrists in extension (**Figure 17.8**). Warm the wrists, distribute load through the whole hand, increase time on the hands gradually over weeks, and offer forearm, fist or block variations.

### Pregnancy

For women with uncomplicated pregnancies, physical activity including modified yoga is recommended by obstetric guidelines (ACOG, 2020; Mottola et al., 2018), and reviews of prenatal yoga report no increase in adverse outcomes, though most trials are small (Babbar et al., 2012). A study of healthy women in late pregnancy found 26 yoga postures, including supine and some inverted positions, were well tolerated without changes in fetal heart rate (Polis et al., 2015). General precautions **CLINICAL CAUTION** :

- ◆ Obtain the student's confirmation that her pregnancy is uncomplicated and that she has discussed exercise with her midwife or doctor.
- ◆ Avoid overheating (no hot yoga), deep twists that compress the abdomen, lying on the front, strong abdominal work, breath retention and forceful breathing (kapālabhāti, bhastrikā), and deep backbends.
- ◆ After the first trimester, prolonged lying flat on the back may cause dizziness in some women; offer side-lying or an inclined position.
- ◆ Avoid postures with a risk of falling; use a wall for balances.
- ◆ Joint laxity increases in pregnancy: avoid pushing to end range.

- ♦ Stop and seek care for warning signs such as vaginal bleeding, fluid leakage, regular painful contractions, dizziness, chest pain, headache, calf pain or swelling (ACOG, 2020).

Teaching pregnant students beyond these general principles requires specialist prenatal training.

## Older adults

Yoga-based exercise improves balance and mobility in adults over 60 (Youkhana et al., 2016), and exercise that challenges balance reduces falls (Sherrington et al., 2019). But older students are more likely to have osteoporosis, osteoarthritis, high blood pressure and reduced balance. In spinal **osteoporosis**, repeated or loaded spinal flexion (e.g. forward bends, plough) has been associated with vertebral compression fractures (Sinaki, 2013): emphasise upright postures, gentle extension, standing balance with support, and leg strength. (A widely reported claim that a short daily yoga routine reverses bone loss rests on an uncontrolled study and should not be repeated as fact (Lu et al., 2016).) Offer chairs and walls, allow extra time for transitions, and help students up and down from the floor safely. For knee and hip **osteoarthritis**, exercise is a core recommended treatment, and yoga is conditionally recommended for the knee (Kolasinski et al., 2020): keep weight-bearing postures within a comfortable range and avoid sustained deep knee flexion under load. For **high blood pressure**, reviews suggest modest reductions with yoga that includes breathing and meditation (Hagins et al., 2013); avoid breath retention and long inversions unless it is well controlled.

## Hypermobility

Some students have joints that move well beyond the typical range; in a small number this is part of a hereditary connective-tissue disorder such as hypermobile Ehlers–Danlos syndrome (Malfait et al., 2017). Hypermobile students often look “advanced” but may have poor end-range control, joint pain, and a tendency to hang into ligaments. Teach them to work **away from end range**: micro-bends in elbows and knees, strength and stability rather than further stretch, and awareness of joint position. Praise control, not depth.

## Previous injuries

After an injury or surgery, students should follow their clinician’s advice. In class: ask what they have been told they can and cannot do; begin below their previous level; increase load gradually; and pay special attention to the injured region and its neighbours. Pain that increases over successive classes is a reason to reduce load and refer.

### Q THE HEALTH CONVERSATION

At the start of every class with new students: “Is there anything I should know — an injury, pregnancy, a health condition? You can tell me now, or quietly after I finish speaking.” Then listen, adapt, and follow up after class.



## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ Yoga is generally safe, but injuries occur, most often in the lower back, neck, shoulders, wrists, knees and hamstrings, commonly through forcing, end-range loading, repetition and ignoring pain.
- ◆ Safe teaching rests on screening, informed choice, gradual progression, offering options, observation, a safe environment and teaching the exit.
- ◆ Load should increase gradually; discomfort is normal, but sharp, neurological, joint or systemic symptoms mean stop.
- ◆ Teachers adapt and refer; they do not diagnose. Know the red flags and emergency responses.
- ◆ Inversions, deep backbends and twists, extreme hip positions and weight-bearing on the hands need preparation and options; pregnancy, older age, hypermobility and previous injury each need specific adaptations.

 REFLECTION QUESTIONS

1. Have you ever been injured in yoga? Looking back, which of the common mechanisms was involved?
2. How do you personally distinguish discomfort from pain? How will you help students do the same?
3. Which special population in this chapter do you feel least prepared to teach? What will you do about it?

 TEACHER-PRACTICE EXERCISES

1. Role-play the health conversation at the start of class with three different students: one with knee surgery last year, one in her second trimester, one with high blood pressure.
2. For each region in Table 26.2, demonstrate one modification in a relevant posture.
3. Walk through your school's emergency plan: phone numbers, first-aid kit, nearest hospital, how you would manage the rest of the class during an emergency.

 KNOWLEDGE CHECK

1. According to a meta-analysis of randomised trials, adverse events in yoga are: (a) much more common than in other exercise (b) no more common than usual care or exercise (c) never reported (d) mainly serious
2. Which of these means a student should stop immediately? (a) muscular trembling (b) a broad stretching sensation (c) tingling spreading into the hand (d) sweating
3. Numbness in the saddle area with bladder changes is: (a) normal after forward bends (b) a medical emergency (c) a sign of tight hamstrings (d) managed by rest
4. For a student with spinal osteoporosis, which should be avoided? (a) standing balance with support (b) loaded spinal flexion such as plough (c) gentle extension (d) leg strengthening
5. Hypermobile students should generally be taught to: (a) go deeper (b) work away from end range with strength and control (c) avoid all yoga (d) lock their joints for stability

Answers 1 b · 2 c · 3 b · 4 b · 5 b

 TEACHING ASSIGNMENT

Write a one-page “safety plan” for a 20-student drop-in class in a Kerala pavilion. Include your health-screening method, the environment (space, heat, water, props), your standard offers and modifications, your approach to hands-on help, your emergency procedure, and your referral wording. Present it to your mentor for feedback.

## CHAPTER 27

## Prāṇāyāma — Theory and Practice

*“Just as lions, elephants and tigers are tamed gradually, so the breath is brought under control gradually; otherwise it harms the practitioner.” —*  
*Haṭha Yoga Pradīpikā 2.15*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ practise and teach breath awareness, diaphragmatic breathing, full yogic breathing, nāḍī śodhana (anuloma viloma), ujjāyī, bhrāmārī, kapālabhāti, bhastrīkā and the cooling breaths;
- ◆ explain breath retention (kumbhaka), traditional ratios and the use of bandhas with retention, and how to progress safely;
- ◆ state the contraindications and precautions for each practice;
- ◆ distinguish the effects traditionally attributed to each practice from what physiological research shows;
- ◆ plan a prāṇāyāma sequence for beginners and for experienced students.

## 27.1 The place of prāṇāyāma

Prāṇāyāma is the fourth limb of Patañjali’s yoga (Chapter 7) and a central practice of Haṭha Yoga (Chapter 9). In the classical class it comes near the beginning, after the opening relaxation and before the sun salutation, so that the breath and mind are steady before the body moves.

The traditional aim of prāṇāyāma is the regulation of *prāṇa* and, through it, of the mind — ultimately preparing for concentration and meditation (YS 2.52–53) **TRADITIONAL**. The physiological mechanisms are described in Chapter 21: breathing rate and depth influence blood gases, heart-rate rhythms and the autonomic nervous system, and attention to the breath is itself a form of concentration. Proposed mechanisms linking slow breathing to calm — for instance through pulmonary stretch receptors and vagal pathways — remain largely hypotheses (Jerath et al., 2006; Brown & Gerbarg, 2005) **INTERPRETIVE**.

## 27.2 General guidelines

- ◆ **Empty stomach:** at least two to three hours after a meal (longer for kapālabhāti, bhastrikā and bandhas).
- ◆ **Posture:** a steady, comfortable seated posture with the spine upright — cross-legged on a cushion, kneeling, or on a chair. Lying down is suitable only for breath awareness and relaxed abdominal breathing.
- ◆ **Nose:** breathe through the nose unless the technique specifies otherwise.
- ◆ **Place:** clean, quiet, well ventilated, neither hot nor draughty.
- ◆ **No strain, ever.** The breath should never be forced, and the practice should end with the breath and mind calmer than at the start. Dizziness, tingling, breathlessness, headache or anxiety mean stop and return to normal breathing.
- ◆ **Progress slowly:** weeks, not days, between steps.

### ⚠ GENERAL CONTRAINDICATIONS AND CAUTIONS FOR PRĀṆĀYĀMA

Gentle slow breathing (awareness, diaphragmatic, full yogic breath without strain, nāḍī śodhana without retention, bhrāmari) is suitable for almost everyone. **Rapid or forceful breathing** (kapālabhāti, bhastrikā) and **breath retention** (kumbhaka) carry more risk and are commonly advised against, or offered only in gentle forms, in **CLINICAL CAUTION**: pregnancy; uncontrolled high blood pressure; heart disease; history of stroke; epilepsy; glaucoma, retinal detachment or recent eye surgery; hernia and recent abdominal or chest surgery; lung disease or a history of pneumothorax; panic disorder or severe anxiety (where over-breathing can provoke symptoms); and during acute illness. **Never practise breath retention in water, while driving, or standing where a faint could cause a fall** (Craig, 1961).

## 27.3 1. Breath awareness

**Technique.** Sit or lie comfortably. Without changing anything, notice the breath: where you feel it (nostrils, chest, abdomen), its rhythm, the pauses between inhalation and exhalation, the temperature of the air at the nostrils.

**Purpose.** Establishes attention on the breath and reveals habitual patterns (holding, shallow chest breathing, a rushed exhalation). It is the foundation of every other practice, and a meditation in itself.

**Teaching.** “Let the breath breathe itself. Your only job is to notice.”

## 27.4 2. Diaphragmatic (abdominal) breathing

**Technique.** Lying on the back with the knees bent or in śavāsana, place one hand on the abdomen and one on the chest. Inhale through the nose, letting the abdomen rise under the hand while the chest stays relatively quiet; exhale and let the abdomen fall. Continue for 3–5 minutes at a slow, comfortable pace.

**Physiology.** The diaphragm contracts in every inhalation; this practice lets its movement show in the abdomen and reduces the use of the upper-chest accessory muscles (Chapter 21).

**Common difficulty.** “Reverse breathing” — the abdomen pulling in on inhalation. Slow down, relax the belly, and let the hand rise first.

### 27.5 3. Full yogic breath

**Technique.** Sitting upright (or lying), inhale slowly and let the breath fill the body in three overlapping waves: the abdomen expands, then the lower ribs widen, then the upper chest rises slightly. Exhale smoothly in reverse, or simply let all three fall together. Continue for 5–10 rounds at an unforced pace.

**Physiology.** Uses the inspiratory and expiratory reserve volumes (Figure 21.3). Because it moves more air, it must be *slow*: a fast full breath is over-breathing.

**Cue.** “Belly — ribs — chest. Then let it all go.”

### 27.6 4. Nāḍī śodhana / anuloma viloma — alternate-nostril breathing

In the tradition taught here, alternate-nostril breathing with retention is called *anuloma viloma*; the preparatory form without retention is often called *nāḍī śodhana*, “purification of the channels”. The *Haṭha Yoga Pradīpikā* describes it as the principal practice for purifying the nāḍīs (HYP 2.7–10) **TRADITIONAL**.

**Hand position.** Right hand in *Viṣṇu mudrā*: fold the index and middle fingers into the palm; the thumb closes the right nostril, the ring finger (with the little finger) closes the left. Left-handed students may use either hand.

**Technique (beginner, no retention).**

1. Close the right nostril; inhale through the left for a count of 4.
2. Close the left nostril; release the right and exhale through it for 4.
3. Inhale through the right for 4.
4. Close the right; exhale through the left for 4. This is one round.
5. Practise 5–10 rounds, then lengthen the exhalation (4 : 8) over the following weeks.

**Technique (classical anuloma viloma, with retention).** Inhale left for 4 counts; close both nostrils and retain for 16; exhale right for 8; inhale right for 4; retain 16; exhale left for 8 — one round at the classical 1 : 4 : 2 ratio taught in this course.

#### ONE ROUND OF NĀḌĪ ŚODHANA (ANULOMA VILOMA) AT A 1 : 4 : 2 RATIO — base count 4



Beginners: omit retention and breathe in and out for equal counts (e.g. 4 : 4), then lengthen the exhalation (4 : 8).

Add retention only when the breath stays smooth and unhurried; the traditional 1 : 4 : 2 ratio is an advanced target, not a starting point.

Hand: *Viṣṇu mudrā* — right thumb closes the right nostril, ring (and little) finger the left.

**Figure 27.1** The pattern of alternate-nostril breathing as taught in this course. Retention (*kumbhaka*) is optional and is omitted for beginners and for anyone with contraindications.

### 🕉️ PROGRESSING ANULOMA VILOMA

Suggested progression over weeks: 4:4 (no retention) → 4:8 → 4:4:8 → 4:8:8 → 4:12:8 → 4:16:8. Only move on when the current ratio is comfortable for ten rounds with no gasping on the inhalation that follows retention. Many students never need retention at all to benefit.

**Physiology and evidence.** Alternate-nostril breathing is usually slow, and slow breathing has well-documented acute effects on heart-rate variability and baroreflex sensitivity (Russo et al., 2017; Zaccaro et al., 2018). In a study of healthy volunteers, several weeks of slow breathing practices including alternate-nostril breathing were associated with increased parasympathetic markers (Pal et al., 2004). Whether breathing through one nostril selectively stimulates one side of the brain or one branch of the autonomic system, as is sometimes claimed, has not been established (Raghuraj & Telles, 2008; Saoji et al., 2019).

## 27.7 5. Ujjāyī — the “victorious” breath

**Technique.** Sitting upright, breathe in and out through the nose while slightly narrowing the back of the throat (the glottis), as if whispering “haaa” with the mouth closed. The breath makes a soft, even sound, like distant waves. Keep the face and throat relaxed; the sound should be audible to you, barely to a neighbour.

**Uses.** Slows and smooths the breath; gives an audible anchor for attention; used in some postural styles throughout the practice. In the *Haṭha Yoga Pradīpikā* it is one of the eight kumbhakas (HYP 2.51–53) (Akers, 2002).

**Physiology and evidence.** Partial glottic closure increases airway resistance and slows breathing. In yoga beginners, slow breathing increased baroreflex sensitivity; adding ujjāyī did not improve on slow breathing alone (Mason et al., 2013).

**Common mistake.** Constricting the throat forcefully, producing a harsh sound and tension in the neck and jaw.

## 27.8 6. Bhrāmarī — the humming-bee breath

**Technique.** Sit upright. Close the ears gently with the thumbs (or index fingers on the ear flaps) and rest the other fingers over the closed eyes (*ṣaṇmukhī mudrā*) or in the lap. Inhale through the nose; exhale slowly while making a steady, low humming sound like a bee, with the lips closed and the teeth slightly apart. Feel the vibration in the head. Repeat 5–10 times.

**Physiology and evidence.** The long exhalation slows breathing; humming increases nasal nitric oxide many times over (Weitzberg & Lundberg, 2002). A systematic review of bhrāmarī reported short-term reductions in heart rate and blood pressure and improvements in some psychological measures, but the included studies were small and of limited quality (Kuppusamy et al., 2018).

**Cautions.** Very safe; avoid pressing hard on the eyes; not practised lying down.

## 27.9 7. Kapālabhāti — “skull-shining” breath

Traditionally one of the six cleansing practices (ṣaṭkarma) rather than a prāṇāyāma (HYP 2.35) (Akers, 2002), kapālabhāti is taught near the beginning of every class in this course.

### Technique.

1. Sit upright with the hands on the knees. Take two normal breaths, then inhale moderately.
2. Exhale sharply through the nose by briskly contracting the lower abdominal muscles, then let the abdomen relax: the inhalation happens passively.
3. Continue rhythmically — about one exhalation per second for beginners — for 15–30 exhalations. This is one round.
4. After the last exhalation, breathe normally for several breaths (in the traditional practice, the student then inhales fully and retains the breath briefly; beginners omit this).
5. Practise three rounds, with rest between. Over weeks, increase gradually to 60 or more exhalations per round, and the pace slightly.

**Common mistakes.** Forcing the inhalation; moving the shoulders and chest; pulling the abdomen in on the inhalation; going too fast or too long, causing dizziness.

**Physiology and evidence.** Kapālabhāti raises ventilation well above metabolic need, so carbon dioxide falls: dizziness and tingling are signs of over-breathing (Chapter 21). Acute studies of fast yogic breathing show mixed autonomic effects (Saoji et al., 2019). A case report describes pneumothorax following kapālabhāti in a young man (Johnson et al., 2004). The traditional claims — that it cleanses the frontal sinuses, energises and clears the mind — are traditional **TRADITIONAL**.

### ⚠ KAPĀLABHĀTI AND BHASTRIKĀ

Contraindicated or modified **CLINICAL CAUTION**: pregnancy; menstruation (many traditional teachers advise against it); high blood pressure; heart disease; epilepsy; hernia; recent abdominal or chest surgery; lung disease or history of pneumothorax; glaucoma; panic disorder. Teach slowly, stop at the first sign of dizziness, and never practise before driving or in water.

## 27.10 8. Bhastrikā — the bellows breath

**Technique.** Similar to kapālabhāti, but both inhalation and exhalation are active and forceful, like a blacksmith’s bellows. In the traditional form it is followed by a breath retention (HYP 2.59–67) (Akers, 2002).

**Teaching.** An intermediate practice. Teach only after kapālabhāti is well established, in short rounds (10–20 breaths) at a moderate pace, with full rest between. Contraindications as for kapālabhāti.

## 27.11 9. The cooling breaths: śītalī and sītkārī

**Śītalī.** Roll the tongue into a tube, protruding slightly from the lips; inhale slowly through the tongue; close the mouth and exhale through the nose. (Some people cannot roll the tongue; they practise sītkārī instead.)

**Sitkārī.** Teeth lightly together, lips parted; inhale slowly through the teeth with a hissing sound; close the mouth and exhale through the nose.

**Traditional use.** Cooling the body and calming the mind (HYP 2.54–58) **TRADITIONAL** — welcome on a hot afternoon in Kerala. Physiologically, inhaling through a moist surface cools the mouth and the inhaled air; the slow pace also slows breathing. Avoid in cold weather, and with sensitive teeth (sitkārī).

## 27.12 Breath retention (kumbhaka)

*Kumbhaka* means “pot” — the body held full (or empty) like a vessel.

- ♦ **Antara kumbhaka:** retention after inhalation.
- ♦ **Bāhya kumbhaka:** retention after exhalation.
- ♦ **Kevala kumbhaka:** spontaneous cessation of breath in deep meditation, described as the goal of prāṇāyāma (HYP 2.71–74) **TRADITIONAL**.

### Ratios

**Table 27.1** Common breathing ratios

| Ratio (in : hold : out : hold) | Name / use                | Suitable for                          |
|--------------------------------|---------------------------|---------------------------------------|
| 1 : 1                          | equal breathing           | beginners; calming and balancing      |
| 1 : 2                          | lengthened exhalation     | beginners; winding down, before sleep |
| 1 : 1 : 1 : 1                  | “box” or square breathing | beginners–intermediate; concentration |
| 1 : 2 : 2                      | mild retention            | intermediate                          |
| 1 : 4 : 2                      | classical anuloma viloma  | experienced practitioners             |

### Bandhas with retention

In classical practice, *jālandhara bandha* is applied during antara kumbhaka, and *uḍḍiyāna* and *mūla bandha* may be added in bāhya kumbhaka (Chapter 14). Bandhas with retention are advanced practices, taught individually to healthy students with an established foundation.

#### 🔍 BREATH RETENTION

Holding the breath causes carbon dioxide to rise and oxygen to fall; the urge to breathe comes mainly from the rising CO<sub>2</sub> (Chapter 21). Short retentions are harmless for healthy people. Long retentions, especially with straining against a closed glottis (a Valsalva-like manoeuvre), cause marked swings in blood pressure and heart rate — the reason for the cardiovascular contraindications. Over-breathing before retention can lead to loss of consciousness without warning (Craig, 1961). There is little research on the long-term effects of the traditional ratios (Saoji et al., 2019).



## 27.13 Traditional claims and modern evidence

**Table 27.2** Prāṇāyāma practices — traditional effects and physiological evidence

| Practice                              | Traditionally said to...<br><b>TRADITIONAL</b>    | What physiology and research show <b>SCIENTIFIC</b>                                                                                            | Strength of evidence                              |
|---------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Slow diaphragmatic and full breathing | calm the mind; improve vitality                   | slow breathing increases HRV and baroreflex sensitivity; associated with reduced anxiety (Zaccaro et al., 2018; Russo et al., 2017)            | moderate for acute effects                        |
| Nāḍī śodhana / anuloma viloma         | purify the nāḍīs; balance iḍā and piṅgalā         | slow-breathing effects; lateralised brain effects unproven (Pal et al., 2004; Raghuraj & Telles, 2008)                                         | moderate (slow breathing); weak (lateral effects) |
| Ujjāyī                                | heat the body; steady the mind; remove phlegm     | slows breathing; no added baroreflex benefit over slow breathing in beginners (Mason et al., 2013)                                             | limited                                           |
| Bhrāmarī                              | calm the mind; relieve anxiety; improve the voice | nasal nitric oxide rises greatly; short-term BP/HR reductions in small studies (Weitzberg & Lundberg, 2002; Kuppusamy et al., 2018)            | limited                                           |
| Kapālabhāti                           | cleanse the skull and sinuses; energise           | increases ventilation, lowers CO <sub>2</sub> ; mixed autonomic effects; rare serious harm reported (Saoji et al., 2019; Johnson et al., 2004) | limited                                           |
| Bhastrīkā                             | awaken kuṇḍalinī; remove disease                  | similar to kapālabhāti; little specific research                                                                                               | very limited                                      |
| Śītalī / sītkārī                      | cool the body; reduce thirst and hunger           | plausible local cooling; little research                                                                                                       | very limited                                      |
| Kumbhaka                              | steady prāṇa and mind; lead to kevala kumbhaka    | blood-gas and cardiovascular changes; little long-term research                                                                                | very limited                                      |

## 27.14 Teaching prāṇāyāma

- ◆ **Set up** everyone in a comfortable, upright seat first; offer cushions and chairs.
- ◆ **Demonstrate** the technique, then guide it with a calm, steady count — or let each student count at their own pace once learned.
- ◆ **Observe:** shoulders rising, faces straining, gasping after retention and swaying are signs to simplify.
- ◆ **Sequence:** calming practices (diaphragmatic, nāḍī śodhana, bhrāmarī) suit the end of the day; energising practices (kapālabhāti) the morning. In the classical class, kapālabhāti precedes anuloma viloma.
- ◆ **Duration** for beginners: 5–10 minutes in total, increasing over months.
- ◆ **Finish** with a few natural breaths and a moment of stillness to notice the effects.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ♦ Prāṇāyāma regulates the breath to steady the mind; it is practised seated, on an empty stomach, through the nose, without strain.
- ♦ Slow practices (awareness, diaphragmatic, full breath, nāḍī śodhana, ujjāyī, bhrāmari) suit almost everyone; fast practices (kapālabhāti, bhastrikā) and retention need screening and gradual progression.
- ♦ Anuloma viloma uses Viṣṇu mudrā and, in its classical form, a 1 : 4 : 2 ratio; beginners omit retention.
- ♦ Evidence best supports the acute autonomic effects of slow breathing; many specific traditional claims are untested.
- ♦ Never combine over-breathing with long retention, and never practise retention in water or while driving.

 REFLECTION QUESTIONS

1. Which practice calms you most? Which energises you? Does this match the tradition?
2. What happens to your mind during a breath retention?
3. How will you teach kapālabhāti to a class in which you do not know everyone's health history?

 TEACHER-PRACTICE EXERCISES

1. Practise and then teach each technique in this chapter to a partner, including the safety statement for kapālabhāti.
2. Record your breaths per minute at rest, then after five minutes of slow breathing and after three rounds of kapālabhāti; note how you feel after each.
3. Lead the prāṇāyāma section of a classical class: kapālabhāti (three rounds) and anuloma viloma (ten rounds), with a beginner option throughout.

 KNOWLEDGE CHECK

1. In Viṣṇu mudrā, the right nostril is closed with the: (a) index finger (b) thumb (c) ring finger (d) middle finger
2. The classical ratio of anuloma viloma is: (a) 1:1:1 (b) 1:2:1 (c) 1:4:2 (d) 2:4:1
3. In kapālabhāti, the active phase is: (a) inhalation (b) exhalation (c) retention (d) both
4. Tingling in the fingers during kapālabhāti suggests: (a) too much CO<sub>2</sub> (b) over-breathing and falling CO<sub>2</sub> (c) awakening of prāṇa (d) low blood sugar
5. Bhrāmari is associated with a large increase in: (a) nasal nitric oxide (b) blood CO<sub>2</sub> (c) heart rate (d) intraocular pressure

Answers 1b · 2c · 3b · 4b · 5a

 TEACHING ASSIGNMENT

Design a four-week prāṇāyāma course for complete beginners (two 20-minute sessions a week). Specify the practices, counts and durations for each week, the safety screening you will do in week 1, and how you will know a student is ready to progress. Teach the first session to your practice group.

## CHAPTER 28

## Meditation — Practice and Teaching

*“When the mind wanders, bring it back again and again.” — after  
Bhagavad Gītā 6.26*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ teach the fundamentals of a meditation practice: posture, place, time, attitude and duration;
- ◆ guide breath meditation, mantra meditation, trāṭaka, body scan and open-awareness (witnessing) practices;
- ◆ guide a structured meditation with appropriate language and silence;
- ◆ help students with common obstacles, including sleepiness, restlessness, doubt and difficult experiences;
- ◆ support students in establishing a daily practice, and teach meditation safely and ethically.

## 28.1 Fundamentals

Meditation practice rests on a few simple foundations, which the tradition taught here sums up as regularity of **place, time and practice** (Chapter 12).

### Posture

The aim is a posture in which the spine is upright and the body can remain still and comfortable for the whole session. Options, in order of increasing demand:

- ◆ sitting on a chair, feet flat, back away from the backrest or lightly supported;
- ◆ kneeling (vajrāsana) on a cushion or meditation bench;
- ◆ sitting cross-legged (sukhāsana) on a firm cushion high enough that the knees are at or below the hips;
- ◆ half lotus or siddhāsana; full lotus for those whose hips allow it without strain on the knees.

The hands rest on the knees (cin or jñāna mudrā) or in the lap (dhyāna mudrā). The head is balanced, chin slightly tucked; the eyes closed or softly lowered; the jaw, face and shoulders relaxed.

### Q NO ONE FAILS AT POSTURE

A student who meditates on a chair is not doing a lesser meditation. The *Bhagavad Gītā*'s instruction is a seat “neither too high nor too low” (6.11) and a body held steady (6.13); how a body achieves that varies. Discomfort that pulls attention every few seconds is the real obstacle — solve it with props.

## Time and duration

Five to ten minutes daily is a better beginning than an hour once a week. Mornings (before the day's activity) and evenings (before the evening meal) suit most people. The tradition regards the hours before dawn as especially favourable. Increase duration gradually — for example by five minutes a fortnight — towards twenty to thirty minutes.

## Attitude

- ♦ **Gentleness:** wandering is expected; return without self-criticism.
- ♦ **Consistency** over intensity.
- ♦ **Curiosity:** each session is different; there is no “good” or “bad” meditation.
- ♦ **Non-striving:** not chasing experiences, including calm.

## 28.2 A structure for a meditation session

A clear structure helps beginners. A 20-minute guided session might be:

**Table 28.1** A structure for a guided meditation session

| Minutes | Stage         | Guidance                                                                                                   |
|---------|---------------|------------------------------------------------------------------------------------------------------------|
| 0–2     | Arriving      | settle the posture; feel the contact with the floor or seat                                                |
| 2–4     | Relaxing      | scan and release the face, shoulders, hands, belly                                                         |
| 4–6     | Breath        | a few slow, deep breaths, then let the breath find its own rhythm                                          |
| 6–16    | Focus         | the main practice (breath, mantra, trāṭaka image or witnessing) with minimal instruction and long silences |
| 16–18   | Widen-<br>ing | expand awareness to the whole body and surroundings                                                        |
| 18–20   | Closing       | a few deeper breaths; move the fingers; open the eyes slowly; a moment of gratitude or dedication          |

## 28.3 Methods

### Breath meditation

**Technique.** Rest attention on the sensation of breathing at one place — the nostrils, the upper lip, the chest or the abdomen — without controlling the breath. When the mind wanders, notice where it went and return. Counting can help: count each exhalation from 1 to 10 and begin again at 1, or whenever the count is lost.

**Use.** The most universal method: requires no belief, is always available, and trains dhāraṇā directly.

## Mantra meditation (japa)

**Technique.** Repeat a mantra mentally, often coordinated with the breath — for example *Oṃ* on the exhalation, or *So* on the inhalation and *Ham* on the exhalation (*so’ham*, “I am That”). A *mālā* may be used to count repetitions (Chapter 14). When the mind wanders, return to the mantra.

**Use.** Central in the tradition taught here; gives a restless mind a steady, rhythmic object; can carry devotional meaning. Always translate and give context; offer a secular alternative (e.g. a word such as “peace”, or counting) for those who prefer it.

## Trāṭaka (steady gazing)

**Technique.** Place a candle at eye level about an arm’s length away in a dark, draught-free room. Gaze steadily at the flame, without straining, for 30 seconds to 2 minutes, blinking as little as comfortable. Then close the eyes and hold attention on the after-image at the space between the eyebrows until it fades. Repeat two or three times.

**Cautions.** Not for students with epilepsy triggered by flickering light, glaucoma, cataracts, recent eye surgery or eye infection, or those who find it distressing. Keep sessions short; never strain the eyes.

## Body scan and yoga nidrā

**Technique.** Move attention systematically through the body — toes to crown or the reverse — noticing sensation in each region without trying to change it. *Yoga nidrā* (“yogic sleep”), developed in the twentieth century by Swami Satyananda of the Bihar School, extends this into a structured practice with an intention (*saṅkalpa*), rotation of awareness through the body, breath awareness, opposite sensations and visualisation, practised lying down (Satyananda, 1976).

**Use.** Builds interoceptive awareness (Chapter 22); deeply relaxing; a doorway for students who cannot sit still. It overlaps with the classical final relaxation (Chapter 23).

## Witnessing (open awareness)

**Technique.** After settling with the breath, let go of any single object. Rest as awareness itself, noticing whatever arises — sounds, sensations, thoughts, emotions — without following or pushing away. Labelling can help: silently note “thinking”, “hearing”, “feeling”, and return to open awareness.

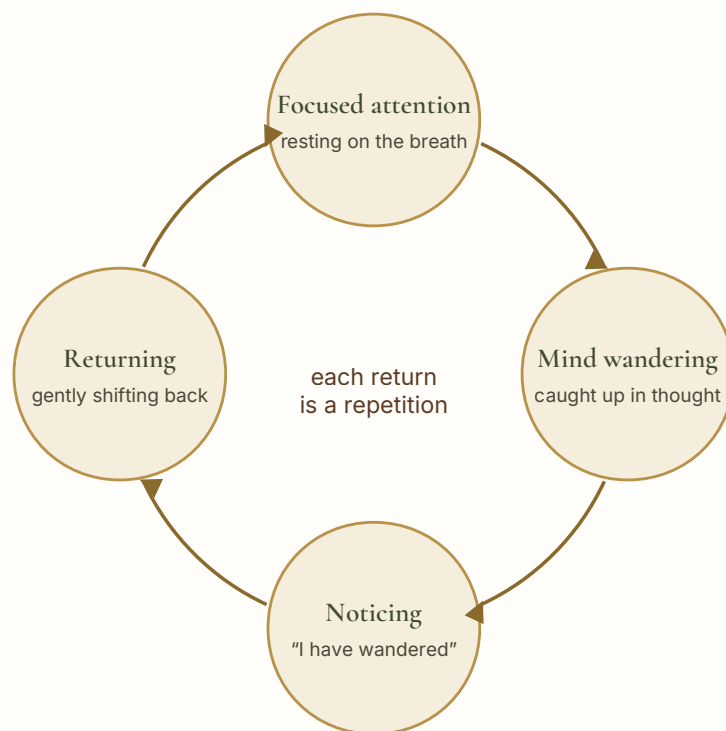
**Use.** Corresponds to the witness (*sākṣī*) of Vedānta (Chapter 10) and to open-monitoring practices in research (Lutz et al., 2008). Best introduced after some experience of concentration.

## 28.4 Working with obstacles

**Table 28.2** Common obstacles and practical responses

| Obstacle                       | Traditional name (YS 1.30) | What helps                                                                                                         |
|--------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Sleepiness, dullness           | styāna, ālasya             | sit more upright; open the eyes slightly; meditate earlier in the day or after <i>prāṇāyāma</i> ; shorter sessions |
| Restlessness, agitation        | vikṣepa (distraction)      | slow the breath; lengthen the exhalation; count breaths; walking meditation; mantra aloud                          |
| Doubt (“Am I doing it right?”) | saṁśaya                    | simple, clear instructions; remind that noticing wandering is the practice                                         |

| Obstacle                            | Traditional name (YS 1.30)          | What helps                                                                                                                |
|-------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Physical discomfort                 | vyādhi (illness)                    | adjust props; chair; shorter sessions; gentle movement before sitting                                                     |
| Craving experiences, disappointment | bhrānti-darśana, alabdha-bhūmikatva | drop expectations; practise for its own sake                                                                              |
| Irregular practice                  | anavasthitatva                      | same time, same place, small duration; practice journal                                                                   |
| Strong emotions or memories         | —                                   | open the eyes; feel the feet and hands; return to the breath or a neutral object; stop if needed and speak to the teacher |



**Figure 28.1** The cycle of attention in focused meditation, after Hasenkamp et al. (2012). Noticing that the mind has wandered is not failure; it is the moment in which the skill is trained.

#### ⚠ WHEN MEDITATION IS DIFFICULT

Most people find meditation calming, but a minority experience significant difficulties — anxiety, panic, dissociation, re-experiencing of traumatic memories, depersonalisation, disturbed sleep — especially with long or intensive practice (Lindahl et al., 2017; Britton et al., 2021). Teachers should:

- ◆ keep sessions short for beginners and offer eyes-open and movement options;
- ◆ invite students to stop at any time and ground themselves (feel the feet, look around the room, name objects, breathe slowly with a long exhalation);
- ◆ avoid intensive retreat-style practice for students with active mental-health conditions unless they are under appropriate care;
- ◆ follow up individually with anyone who seems distressed; and
- ◆ refer to a mental-health professional when difficulties persist. Meditation is not a substitute for treatment.

## 28.5 Establishing a daily practice

Help students with concrete structures:

1. **Anchor** practice to an existing habit (after brushing the teeth; before breakfast).
2. **Start small** — five minutes — and keep it daily.
3. **Prepare the place** — a cushion, a corner, perhaps a candle or picture that marks the space.
4. **Use a timer**, so that attention is not on the clock.
5. **Keep a journal** of practice (Appendix J).
6. **Practise with others** when possible — a weekly group sustains a daily habit.
7. **Expect ups and downs**; missing a day is not failure — resume the next.

## 28.6 Teaching meditation

### Language

- ◆ Use **invitational, present-tense** language: “Notice...”, “Let...”, “If you wish...”.
- ◆ Give **one instruction at a time**, then silence.
- ◆ Normalise wandering: “When you notice the mind has wandered, that noticing is a moment of awareness. Gently return.”
- ◆ Avoid promising experiences (“you will feel deep peace”) and avoid clinical claims (“this will cure your anxiety”).
- ◆ Keep your voice steady and warm, slower than normal speech.

### Silence

Beginning teachers talk too much. Plan silences and time them: 30 seconds, then a minute, then longer as the group settles. Silence is where the meditation happens.

### After the session

Invite brief sharing, especially with beginners: “What did you notice?” Receive all answers — boredom, sleepiness, restlessness, peace — with equal interest. Offer one practical tip, not a lecture.

#### ||| A FIVE-MINUTE BREATH MEDITATION

*Sit comfortably, with the spine upright and the body at ease... Let the eyes close or rest gently on the floor... Feel the weight of the body, the contact with the ground... Now bring your attention to the breath at the nostrils — the cool air coming in, the warmer air going out... You don't need to change the breath. Just feel it... (silence, 30 seconds)... When the mind wanders, as it will, notice where it has gone, and gently come back to the breath... (silence, 1–2 minutes)... Now let the attention widen to the whole body, breathing... (silence)... Take a slightly deeper breath... and when you are ready, open your eyes.*



### 🔍 MEDITATION RESEARCH: A BRIEF SUMMARY

Meta-analyses of meditation programmes find moderate improvements in anxiety, depression and pain relative to no treatment, and effects comparable to other active treatments (Goyal et al., 2014). Research identifies plausible mechanisms involving attention regulation, body awareness, emotion regulation and changes in self-perspective (Hölzel et al., 2011; Tang et al., 2015). Mind-wandering is common and associated with lower momentary happiness (Killingsworth & Gilbert, 2010). Researchers warn against overstating findings, and note that adverse effects are under-reported (Van Dam et al., 2018; Britton et al., 2021).

## CHAPTER REVIEW

### 📌 KEY TAKEAWAYS

- ◆ A stable practice rests on a comfortable upright posture, a regular place and time, a short daily duration increased gradually, and a gentle, non-striving attitude.
- ◆ Core methods: breath meditation, mantra (japa), trāṭaka, body scan and yoga nidrā, and open awareness (witnessing).
- ◆ Obstacles listed by Patañjali — dullness, distraction, doubt, irregularity — have practical remedies.
- ◆ A minority of meditators have difficult experiences; teachers offer grounding, options, short sessions and referral.
- ◆ Teaching meditation relies on clear, invitational language and generous silence; claims should be modest.

### 🙏 REFLECTION QUESTIONS

1. Which method suits your mind best right now? Which do you avoid?
2. What is your most frequent obstacle, and what have you learned about working with it?
3. How would you teach meditation to a group of teenagers? To a hospital staff group?

### 🙏 TEACHER-PRACTICE EXERCISES

1. Guide each of the five methods in this chapter to a partner for five minutes.
2. Record yourself guiding the five-minute breath meditation; time your silences and compare them with your intentions.
3. Role-play a student who becomes tearful and anxious during meditation; practise a calm, grounding response.

 **KNOWLEDGE CHECK**

1. For beginners, which is better? (a) one hour weekly (b) five to ten minutes daily (c) only on retreat (d) whenever inspired
2. In breath counting, if you lose count you should: (a) stop meditating (b) begin again at 1 (c) continue from any number (d) count faster
3. Trāṭaka should be avoided by students with: (a) tight hamstrings (b) photosensitive epilepsy or glaucoma (c) low blood pressure (d) knee pain
4. Yoga nidrā in its modern structured form is associated with: (a) Patañjali (b) Swami Satyananda (c) Śaṅkara (d) the Ṛgveda
5. A student reports distressing memories during meditation. A good first response is to: (a) tell them to push through (b) invite them to open the eyes, ground through the senses and breathe slowly (c) ignore it (d) increase the length of practice

Answers 1b · 2b · 3b · 4b · 5b

 **TEACHING ASSIGNMENT**

Design and teach a 20-minute guided meditation using the structure in Table 28.1 and one method of your choice. Afterwards, lead a two-minute sharing and write a one-page reflection on what you learned about your use of language and silence.





### SECTION III

# Teaching, Professional Practice & Graduation

Knowing yoga and teaching it are different skills. This section turns practitioners into teachers: the qualities and ethics of a good teacher, inclusive and trauma-aware teaching, the craft of cueing, demonstration and observation, sequencing and class planning, touch and consent, the management of a room, and the practical realities of a teaching life — ending with the practicum and the final teaching examination.

- 
- |    |                                                                           |    |                                                            |
|----|---------------------------------------------------------------------------|----|------------------------------------------------------------|
| 29 | What Makes a Good Yoga Teacher                                            | 36 | Hands-on Adjustments, Consent and Appropriate Touch        |
| 30 | Ethics, the Teacher–Student Relationship and Professional Boundaries      | 37 | Managing the Classroom and Handling Difficult Situations   |
| 31 | Inclusive and Trauma-Aware Teaching                                       | 38 | Private Classes, Workshops and Online Teaching             |
| 32 | Teaching Different Levels and Mixed-Level Classes                         | 39 | Professional Development and a Sustainable Teaching Career |
| 33 | The Teacher's Craft — Verbal Cueing, Demonstration, Observation and Voice | 40 | Practicum, Assessment and the Final Teaching Examination   |
| 34 | Sequencing — Warm-up, Peak Poses, Counterposes, Rest and Integration      |    |                                                            |
| 35 | Class Planning and Lesson Plans                                           |    |                                                            |

## CHAPTER 29

## What Makes a Good Yoga Teacher

*“The teacher is only the finger pointing at the moon.” — a saying common to many contemplative traditions*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ describe the qualities, knowledge and skills of an effective yoga teacher;
- ◆ explain how adults learn and what this implies for teaching yoga;
- ◆ identify your own strengths and areas for growth as a teacher;
- ◆ set personal development goals for the course and your first year of teaching.

## 29.1 Three things a teacher brings

Ask a group of experienced students what made their best yoga teachers good, and their answers fall into three groups:

1. **Who the teacher is** — presence, authenticity, humility, warmth, integrity, a living practice.
2. **What the teacher knows** — the practices, the tradition, the body, safety, how people learn.
3. **What the teacher can do** — see what is happening in the room, communicate clearly, adapt on the spot, hold a group’s attention, create a sense of safety.

A teacher-training course can give you the second and begin the third; the first grows from your own practice and life. None is optional. Knowledge without presence is a lecture; presence without knowledge is charisma, and can be dangerous.

**Table 29.1** Qualities of an effective yoga teacher

| Domain  | Qualities                                                                      | What it looks like in class                                   |
|---------|--------------------------------------------------------------------------------|---------------------------------------------------------------|
| Being   | presence, authenticity, humility, warmth, patience, equanimity                 | attention on students, not on self; calm when things go wrong |
| Knowing | technique, anatomy, safety, philosophy, pedagogy                               | accurate cues; appropriate modifications; honest answers      |
| Doing   | observation, clear communication, adaptation, classroom management, sequencing | teaching the students who are present, not the plan alone     |

| Domain     | Qualities                                     | What it looks like in class                  |
|------------|-----------------------------------------------|----------------------------------------------|
| Practising | a personal daily practice                     | teaching from experience; continued learning |
| Ethics     | honesty, boundaries, respect, confidentiality | safe, trustworthy relationships (Chapter 30) |

## 29.2 How adults learn

Most yoga students are adults, and adults learn differently from children. Theories of adult learning emphasise that adults (Knowles et al., 2015):

- ◆ **need to know why** they are learning something;
- ◆ **bring experience** that is a resource — and sometimes an obstacle;
- ◆ are **self-directed** and resent being treated as passive;
- ◆ learn best when the material is **relevant** to their lives and problems;
- ◆ are motivated mostly **internally** — by growth, health, meaning.

Motor-learning research adds that learners improve faster with **external-focus cues**, **choice** and a sense of **competence** (Chapter 22) (Wulf & Lewthwaite, 2016).

### ◆ TEACH THE PERSON IN FRONT OF YOU

Every principle above points the same way: good teaching starts from the student's experience, explains why, offers choice, and builds confidence. The fixed sequence of the classical class is a strength — but within it, every instruction is adapted to the people on the mats.

## 29.3 The teacher's own practice

A teacher's practice is not a private hobby; it is the source of their teaching. The tradition taught here expects its teachers to maintain a daily practice of āsana, prāṇāyāma and meditation, study and service. In practical terms, a teacher's practice:

- ◆ keeps the teacher's knowledge embodied — you understand a posture from the inside;
- ◆ reveals what students experience, especially when you practise as a beginner (modified postures, new styles);
- ◆ sustains the energy and steadiness teaching requires;
- ◆ keeps the teacher humble — there is always more to learn.

## 29.4 Common traps for new teachers

- ◆ **Teaching your own practice.** Sequencing the class you would enjoy rather than the class your students need.
- ◆ **Performing.** Demonstrating throughout the class; focusing on how you look rather than on what students need.
- ◆ **Over-talking.** Filling every silence with instruction or philosophy.
- ◆ **Correcting everything.** Giving five cues where one would do.



- ♦ **Imitating.** Adopting your teacher's voice and phrases rather than finding your own.
- ♦ **Needing approval.** Measuring success by compliments and numbers.

#### 🕒 YOUR FIRST HUNDRED CLASSES

Expect your first hundred classes to feel awkward. Plan carefully, teach simply, ask for feedback, and keep notes after each class: *What worked? What would I change? What did I notice about the students?* Improvement comes from reflection, not from repetition alone.

## 29.5 A self-assessment

### ✧ WHERE ARE YOU NOW?

Rate yourself from 1 (beginning) to 5 (confident) in each area: personal practice; knowledge of the twelve postures; anatomy; safety and modifications; philosophy; clear verbal cueing; demonstration; observation; voice; sequencing; managing a group; comfort with silence; ethical boundaries. Choose your two lowest scores and write one concrete action for each for the coming week. Repeat the assessment at the end of the course.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ♦ Good teachers bring being (presence, integrity), knowing (knowledge) and doing (skills), supported by practice and ethics.
- ♦ Adults learn best when they understand why, when their experience is respected, when they can choose and when they feel competent.
- ♦ A daily personal practice is the source of a teacher's authenticity and understanding.
- ♦ New teachers commonly teach their own practice, perform, over-talk and over-correct; reflection after each class is the remedy.

### 🕒 REFLECTION QUESTIONS

1. Describe your best yoga teacher in five words. Which of those qualities do you already have?
2. What does your teaching style risk becoming if you are not careful — performer, lecturer, drill sergeant, friend?
3. How will you protect your own practice when you are teaching many classes a week?

### ✧ TEACHER-PRACTICE EXERCISES

1. Teach a five-minute segment and ask your partner to note every time you gave an instruction without explaining why. Repeat, adding brief reasons.
2. Teach the same posture twice — once demonstrating throughout, once without demonstrating at all. Discuss the difference.

3. Interview an experienced teacher (at the school or elsewhere) about what they wish they had known in their first year.

☒ **KNOWLEDGE CHECK**

1. According to adult-learning theory, adults learn best when: (a) given no explanation (b) they understand why and can relate learning to their lives (c) treated as passive (d) corrected constantly
2. Which is a common trap for new teachers? (a) using silence (b) teaching the class they would enjoy themselves (c) offering choices (d) keeping notes
3. A daily personal practice matters for teachers because it: (a) impresses students (b) keeps teaching grounded in experience (c) is required by law (d) replaces study
4. Which cue supports learner autonomy? (a) "Do exactly as I do." (b) "Choose the variation that suits you today." (c) "Everyone must touch their toes." (d) "Don't move."

Answers 1 b · 2 b · 3 b · 4 b

 **TEACHING ASSIGNMENT**

Write your "teaching philosophy" in 300 words: why you teach, what you want students to experience, how you want to teach, and what you will do to keep learning. Share it with your mentor and revise it at the end of the course.



## CHAPTER 30

# Ethics, the Teacher–Student Relationship and Professional Boundaries

*“Truthfulness, non-harming, non-stealing... these are the great vow, universal.” — after Yoga Sūtra 2.30–31*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ apply the yamas and niyamas as a professional ethic for yoga teaching;
- ♦ explain the power dynamics of the teacher–student relationship, including projection and dependence;
- ♦ set and maintain clear physical, emotional, sexual, financial, time and digital boundaries;
- ♦ respond ethically to common dilemmas, including dual relationships, gifts, confidentiality and complaints;
- ♦ write a personal code of conduct.

## 30.1 Why ethics comes first

A yoga class places people in unusual positions: lying with eyes closed, in close physical proximity to a teacher, sometimes emotionally open, often trusting the teacher’s judgement about their bodies. That trust is the ground of good teaching — and the reason abuse of it is so damaging. The history of modern yoga includes prominent teachers who exploited students sexually, financially and psychologically, often protected by their communities’ reverence (Remski, 2019). Ethics is not an optional module; it is the foundation on which everything else in this section stands.

## 30.2 The yamas and niyamas as a professional code

**Table 30.1** A professional ethic drawn from the yamas and niyamas

| Principle         | In professional practice                                                                                      |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| Ahiṃsā            | put student safety and wellbeing first; no forcing; no humiliation; no discrimination                         |
| Satya             | honest representation of training, qualifications and lineage; no exaggerated health claims; admit mistakes   |
| Asteya            | respect students' time, money and ideas; credit sources; do not take students from other teachers dishonestly |
| Brahmacarya       | no sexual or romantic involvement with current students; appropriate touch only with consent                  |
| Aparigraha        | do not cultivate dependence; let students go; avoid possessiveness over students or methods                   |
| Śauca             | clean, safe environment; clear intentions; transparent business practice                                      |
| Santoṣa           | contentment with an honest living; no pressure-selling                                                        |
| Tapas             | commitment to preparation, continuing education and your own practice                                         |
| Svādhyāya         | reflective practice; seek supervision; learn from complaints and feedback                                     |
| Īśvara-praṇidhāna | humility; the teaching is not about you                                                                       |

## 30.3 The teacher–student relationship

### Power

Teachers hold power: of knowledge, of position at the front of the room, of physical proximity and touch, and of the spiritual authority students may project onto them. Power is not in itself a problem — a teacher needs authority to keep a class safe — but it must be recognised, used for students' benefit and never exploited (Farhi, 2006).

### Projection and dependence

Students sometimes see their teacher as wiser, kinder or more spiritually advanced than any human being is — a process psychologists call *projection* or *transference*. They may seek approval, personal advice or special attention, or develop romantic feelings. Teachers can also project onto students (seeing them as disciples, friends or admirers). Recognising this is the first step; maintaining steady, professional warmth is the second. A good teacher's aim is to make students **less** dependent on them over time — more able to practise, discern and care for themselves.

#### ◆ THE DIRECTION OF CARE

In a professional relationship, care flows from teacher to student. If you notice that you are relying on students for emotional support, friendship, admiration or romantic attention, that is a signal to seek support elsewhere — from peers, mentors or professionals.

## 30.4 Boundaries

**Table 30.2** Professional boundaries in yoga teaching

| Boundary               | Clear practice                                                                                                                                                                                                                                                                                                   |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sexual</b>          | No sexual or romantic relationships with current students. If a genuine mutual attraction arises, the teaching relationship ends first — the student is referred to another teacher, and time passes — and the power imbalance is honestly considered. Sexualised comments, jokes or touch are never acceptable. |
| <b>Physical</b>        | Touch only with consent, for a clear teaching purpose, on appropriate areas, and with the student free to decline at any time ( <b>Chapter 36</b> ).                                                                                                                                                             |
| <b>Emotional</b>       | Listen with care, but do not act as a therapist; refer when a student's needs exceed your role.                                                                                                                                                                                                                  |
| <b>Financial</b>       | Transparent prices; no pressure-selling; no borrowing from or lending to students; declare conflicts of interest (e.g. commissions on products you recommend).                                                                                                                                                   |
| <b>Time</b>            | Start and end on time; set clear availability for messages; protect your own rest.                                                                                                                                                                                                                               |
| <b>Digital</b>         | Professional social-media presence; no photographs or videos of students without explicit consent; careful private messaging.                                                                                                                                                                                    |
| <b>Confidentiality</b> | Health information and personal disclosures stay private, except where there is a risk of serious harm.                                                                                                                                                                                                          |

## Dual relationships

A *dual relationship* exists when a teacher has a second relationship with a student — friend, employer, landlord, business partner, family member. In small communities (and in Fort Kochi) these are often unavoidable. They are not forbidden, but they require extra care: keep roles distinct, be alert to conflicts of interest, and avoid situations in which a student could feel unable to say no.

## Gifts

Small gifts of gratitude are part of many cultures, including India's, and can be graciously received. Large or repeated gifts, or gifts that seem to carry expectations, should be politely declined or discussed.

## 30.5 Cultural respect

Yoga comes from Indian traditions and is being taught here in its land of origin to students from many countries. Respect means:

- ◆ learning and pronouncing Sanskrit carefully, and translating it;
- ◆ presenting practices with their context and lineage;
- ◆ not using sacred images, mantras or symbols as decoration or marketing;
- ◆ acknowledging the diversity of Indian traditions rather than presenting one as “the” yoga;
- ◆ welcoming students of all faiths without asking them to adopt a religious identity.

## 30.6 When things go wrong

Mistakes will happen: a cue that hurt someone, a careless comment, a boundary blurred. The ethical response is to **acknowledge, apologise, repair and learn**. Schools should have a clear, accessible way for students to raise concerns, and teachers should welcome feedback rather than defending against it. If

you observe another teacher behaving unethically — particularly anything involving sexual misconduct or abuse — it must be reported to the school's leadership, and where the law requires, to the appropriate authorities.

### ✧ ETHICAL DILEMMAS

In small groups, discuss what you would do:

1. A student you teach weekly invites you for dinner, alone.
2. A student tells you in confidence that she is being harmed at home.
3. A studio asks you to advertise that your class “detoxes the liver and cures back pain”.
4. A student gives you an expensive gift after you helped them through a difficult time.
5. You notice a colleague repeatedly adjusting one student in a way that seems unnecessary and intimate.
6. A student asks you to photograph them in an advanced posture for their social media, while other students are in the background.

For each, name the principles in tension and decide on a course of action.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ The trust students place in teachers creates power, which must be recognised and used only for their benefit.
- ◆ The yamas and niyamas translate directly into a professional code of conduct.
- ◆ Projection and dependence are normal; the teacher's aim is to make students more independent.
- ◆ Clear sexual, physical, emotional, financial, time, digital and confidentiality boundaries protect students and teachers.
- ◆ Mistakes are acknowledged, apologised for, repaired and learned from; misconduct by others is reported.

### 🕯 REFLECTION QUESTIONS

1. Where have you seen power used well by a teacher? Where have you seen it misused?
2. Which boundary do you expect to find hardest to hold? Why?
3. What does cultural respect mean for you personally, studying yoga in India?

### ✧ TEACHER-PRACTICE EXERCISES

1. Role-play saying “no” kindly and clearly to three boundary-crossing requests (a personal invitation, a request for medical advice, a request for a free private class in exchange for “exposure”).
2. Draft the consent and photography policy you would post in your studio.
3. Discuss the dilemmas in the exercise above and present one group's reasoning to the class.

 **KNOWLEDGE CHECK**

1. Which is always outside professional boundaries? (a) friendship with a former student (b) a sexual relationship with a current student (c) receiving a small thank-you gift (d) teaching a family member
2. A “dual relationship” means: (a) teaching two classes (b) having another relationship with a student besides teaching (c) co-teaching (d) teaching online and in person
3. Projection is when a student: (a) demonstrates a pose (b) attributes qualities or feelings to the teacher that belong to their own inner world (c) projects their voice (d) photographs the class
4. The best response to a mistake that hurt a student is to: (a) deny it (b) acknowledge, apologise, repair and learn (c) blame the student (d) ignore it
5. Which is an example of asteya in teaching? (a) starting on time (b) forcing a posture (c) exaggerating benefits (d) keeping secrets

Answers 1b · 2b · 3b · 4b · 5a

 **TEACHING ASSIGNMENT**

Revise the “teacher’s charter” you began in Chapter 15 into a formal personal code of conduct (one page), covering each boundary in Table 30.2, your approach to consent, confidentiality and complaints, and your commitments to continued learning. Sign and date it; it becomes part of your graduation portfolio.

## CHAPTER 31

# Inclusive and Trauma-Aware Teaching

*“Friendliness towards the happy, compassion towards the suffering...” —*  
*Yoga Sūtra 1.33*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ identify barriers that exclude people from yoga and ways to reduce them;
- ♦ use language, demonstration and props that include students of different bodies, ages, abilities, genders, cultures and languages;
- ♦ explain the principles of a trauma-informed approach and apply them to a yoga class;
- ♦ adapt classes for students with common disabilities and chronic conditions within your scope;
- ♦ recognise when a student needs specialist support.

### 31.1 Who is not in the room?

Look at the images used to advertise yoga and you will see a narrow range of bodies — young, slim, flexible, able-bodied. Many people who would benefit from yoga assume it is not for them: larger people, older people, disabled people, people with chronic illness, men who feel stiff, people who cannot afford studio prices, people who fear being judged or touched. Inclusive teaching begins by asking who is *not* in the room, and why.

### 31.2 Inclusive teaching in practice

#### Language

- ♦ **Offer options without hierarchy.** Say “option one... option two...” rather than “the full pose” and “the easy version”. Let students choose without feeling demoted.
- ♦ **Describe actions, not ideal shapes.** “Reach the arms up” works for everyone; “touch your toes” excludes many.

- ◆ **Avoid body-shaming or diet language:** no “burn off last night’s dessert”, no “work off the belly”.
- ◆ **Avoid assumptions** about gender, family, religion, ability or lifestyle. Use students’ names and pronouns as they give them; avoid gendered group terms (“ladies”, “guys”) in favour of “everyone”.
- ◆ **Speak plainly** for international students: short sentences, common words, Sanskrit translated, key words demonstrated.

## Demonstration and props

- ◆ Demonstrate modifications as normal options, not as concessions; sometimes demonstrate the modified version first.
- ◆ Make props available to everyone, and use them yourself.
- ◆ Invite students to adapt freely: “If your body says something different, listen to your body.”

## The environment

- ◆ Accessible entry, toilets and practice space where possible; chairs available.
- ◆ A range of class times and prices, including concessions where feasible.
- ◆ A clear, friendly welcome for newcomers; information in advance about what to expect.

### ◆ UNIVERSAL DESIGN

Rather than designing a class for an imagined “standard” student and then adapting it for others, **design every class from the start to include a range of bodies:** offer options, have props ready, give time for transitions. Adaptations then become part of the class for everyone rather than special treatment for a few.

## 31.3 Teaching students with disabilities and chronic conditions

A 200-hour training does not make you a yoga therapist, but you will meet students with a wide range of conditions. General principles:

- ◆ **Ask, don’t assume.** The student is the expert on their own body. “What works for you? What should I know?”
- ◆ **Work with their clinician’s advice,** within your scope.
- ◆ **Adapt the form, keep the purpose:** a chair-based sun salutation can serve the same aims as the standing one.
- ◆ **Know when to refer** to a specialist teacher or yoga therapist.

**Table 31.1** Examples of common adaptations

| Situation                          | Adaptations                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------|
| Wheelchair users, limited mobility | chair-based practice; upper-body and breath-focused sequences; transfers only when the student chooses |
| Visual impairment                  | precise verbal cues; describe the room; hands-on guidance only with consent; consistent layout         |
| Hearing impairment                 | position so the student can see you; demonstrate; visual signals; written notes                        |
| Chronic pain or fatigue            | shorter holds, more rest, gentler sequences; emphasise breath and relaxation                           |

| Situation           | Adaptations                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Arthritis           | gentle movement through comfortable range; padding under joints; avoid sustained end-range weight-bearing                        |
| High blood pressure | avoid breath-holding and straining; keep the head above the heart for longer holds; gentle inversions only with medical approval |
| Larger bodies       | wider stances; blocks and straps; space for the abdomen in forward bends (legs apart); side-lying options                        |

### 31.4 Trauma-aware teaching

Many people carry the effects of trauma — experiences that overwhelmed their capacity to cope, from accidents and violence to war, abuse or medical procedures. You will not know who they are. Yoga can support people living with trauma: a randomised trial found that ten weeks of trauma-sensitive yoga reduced symptoms of post-traumatic stress disorder in women with chronic, treatment-resistant PTSD more than a health-education control (van der Kolk et al., 2014). But yoga classes can also include triggers — being touched without warning, lying with eyes closed in a darkened room, a commanding voice, being watched from behind.

A **trauma-informed approach**, as described by the US Substance Abuse and Mental Health Services Administration, rests on six principles: safety; trustworthiness and transparency; peer support; collaboration and mutuality; empowerment, voice and choice; and attention to cultural, historical and gender issues (SAMHSA, 2014). Trauma-sensitive yoga, as developed by Emerson & Hopper (2011), applies these through invitational language, choice, predictable structure, an emphasis on noticing sensation, and very careful (often no) hands-on assistance.

**Table 31.2** Trauma-aware practices in a general class

| Principle                        | In class                                                                                                           |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Safety                           | predictable structure; explain what will happen; clear view of the door; students choose their place; no surprises |
| Trustworthiness and transparency | say what you are doing and why; ask before touching and respect the answer                                         |
| Choice                           | invitational language ("if you like", "you might..."); options for every posture; eyes open or closed              |
| Collaboration                    | the student is the authority on their own experience; invite feedback                                              |
| Empowerment                      | emphasise noticing and choosing rather than achieving a shape                                                      |
| Cultural and gender awareness    | awareness of how gender, culture and history shape students' experience of touch, space and authority              |

#### Q INVITATIONAL LANGUAGE

Compare: "Close your eyes. Relax completely." with "You might like to close your eyes, or let your gaze rest softly on the floor. Notice what it's like to rest here." The second offers choice, invites noticing, and does not command. It is also simply better teaching for everyone.



### ⚠ SCOPE AND TRAUMA

Trauma-aware teaching is a way of teaching everyone more safely. It is **not** trauma therapy. Do not invite students to recall or process traumatic experiences, do not interpret their reactions, and do not promise that yoga will heal trauma. If a student is distressed, help them feel grounded and safe (Chapter 28) and encourage them to seek professional support.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ Inclusion starts by asking who is not in the room and removing barriers of image, language, cost, access and assumption.
- ◆ Offer options without hierarchy; describe actions not ideal shapes; avoid body-shaming and assumptions; speak plainly.
- ◆ Design classes from the start for a range of bodies (universal design), with props and time for transitions.
- ◆ Ask students with disabilities or chronic conditions what works for them; adapt the form and keep the purpose; refer when needed.
- ◆ A trauma-informed approach emphasises safety, transparency, choice, collaboration and empowerment; it is good teaching, not therapy.

### 🍷 REFLECTION QUESTIONS

1. When have you felt excluded or judged in a yoga class? What would have helped?
2. Which of your habitual cues might exclude some students?
3. How will you create choice in a traditional class with a fixed sequence?

### ✋ TEACHER-PRACTICE EXERCISES

1. Rewrite five of your usual cues into invitational, non-hierarchical language, then teach them.
2. Teach a sun salutation entirely from chairs to your practice group.
3. With a partner, practise asking a new student about their needs in a way that is warm, private and non-intrusive.

 **KNOWLEDGE CHECK**

1. “Offering options without hierarchy” means: (a) naming the full pose and the easy version (b) presenting variations as equal choices (c) giving no options (d) teaching only advanced poses
2. Which is one of SAMHSA’s six trauma-informed principles? (a) competition (b) empowerment, voice and choice (c) silence (d) correction
3. In trauma-aware teaching, hands-on assistance should be: (a) routine (b) given without asking (c) used with great care and explicit consent, or not at all (d) given more often
4. Trauma-aware teaching is: (a) trauma therapy (b) a way of teaching everyone more safely (c) only for clinical settings (d) about recalling memories
5. The best first step with a student who uses a wheelchair is to: (a) assume what they can do (b) ask what works for them (c) exclude them from the class (d) insist on floor work

Answers 1b · 2b · 3c · 4b · 5b

 **TEACHING ASSIGNMENT**

Adapt a 45-minute classical-style class plan so that it is fully accessible to a mixed group including an older student with knee arthritis, a student who prefers not to be touched, and a student with limited English. Mark every adaptation on the plan and teach it to your practice group.

## CHAPTER 32

## Teaching Different Levels and Mixed-Level Classes

*“Practice becomes firmly grounded when cultivated for a long time, without interruption, and with devotion.” — Yoga Sūtra 1.14*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ describe what distinguishes beginner, intermediate and advanced students — beyond flexibility;
- ◆ teach an effective first class and beginners’ course;
- ◆ challenge experienced students safely and meaningfully;
- ◆ use tiered options, layering and group organisation to teach a mixed-level class well;
- ◆ decide what to teach when a class is more varied than planned.

### 32.1 What “level” means

Level is not flexibility. A flexible dancer in her first yoga class is a beginner; a stiff sixty-year-old who has practised daily for ten years is not. Level is better understood as a combination of:

- ◆ **familiarity** with the practices and the class structure;
- ◆ **body awareness** — knowing what one is doing and feeling;
- ◆ **self-regulation** — knowing when to rest, modify or stop;
- ◆ **strength, mobility and stamina** for the practices;
- ◆ **steadiness of breath and attention.**

**Table 32.1** Characteristics and needs at each level

|                    | Beginner                                                                    | Intermediate                                                  | Advanced                                                                |
|--------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------|
| Typical experience | first weeks to months                                                       | regular practice for a year or more                           | many years of steady practice                                           |
| Needs              | clear, simple instructions; safety; confidence; understanding the structure | refinement; consistency; deeper understanding; more challenge | subtlety; inner focus; variations; guidance in pranayama and meditation |

|                 | Beginner                                                     | Intermediate                                                       | Advanced                                                    |
|-----------------|--------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------|
| Common risks    | doing too much; copying neighbours; not knowing when to stop | plateau; habits; over-confidence in inversions and backbends       | overuse from repetition; ego; neglect of foundations        |
| Teacher's focus | foundation, breath, relaxation, the "why"                    | alignment within the student's body, holds, transitions, prāṇāyāma | stillness, longer holds, advanced variations, self-practice |

## 32.2 Teaching beginners

The first class shapes whether someone ever returns. Make it welcoming, clear and a little easier than you think necessary.

- ◆ **Before class:** greet each newcomer; ask about health, injuries and experience; show them the props and where to place their mat (near the middle, where they can see you and others).
- ◆ **Explain the structure** in a sentence: "We start lying down, then breathing, sun salutations and postures, with rest between, and a long relaxation at the end."
- ◆ **Teach fewer things, more clearly.** Two or three rounds of sun salutation, simplified; six or eight of the twelve basic postures in preparation form.
- ◆ **Demonstrate more; talk less** – and walk around to check understanding.
- ◆ **Normalise rest and modification** from the start.
- ◆ **Finish on time**, with a proper relaxation, and ask afterwards how it felt.

### Q A BEGINNERS' COURSE

Many schools teach beginners in a closed course of four to eight weeks, building the class piece by piece: week 1 relaxation and breathing; week 2 sun salutation; week 3 leg raises and inversions preparation; and so on, until the full class is familiar. Students who learn the structure first practise more safely in open classes later. Appendix G includes a beginners' class plan.

## 32.3 Teaching experienced students

Experienced students need challenge, but more depth rather than more difficulty:

- ◆ longer holds and slower transitions;
- ◆ advanced variations for those who are ready, with preparation built into the sequence;
- ◆ refinement of breath, bandhas and dṛṣṭi (gaze);
- ◆ more prāṇāyāma and meditation;
- ◆ the freedom to practise with fewer instructions.

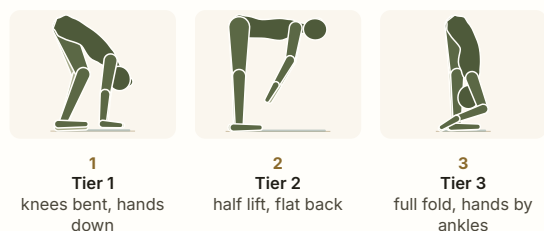
Watch for **overuse**: experienced students often have the most repetitive practices. Vary sequences and include strength and rest.

## 32.4 Mixed-level classes

Most public classes are mixed. The key is **one class, several paths** through it.

## Tiered options

Teach the foundation of each posture to everyone, then offer tiers: “Stay here — or, if it suits you today, take the next step...” The foundation remains valuable for everyone; advanced students are not waiting, and beginners are not rushed.



## Layering

Build a posture in layers, each one complete in itself: *feet* → *legs* → *pelvis* → *spine* → *arms* → *gaze* → *breath*. Students stop at the layer that is right for them.

## Organising the room

- ◆ Place beginners where they can see you and a steady student.
- ◆ Place experienced students where their practice will not distract beginners (e.g. at the back or sides) — and where you can still see them.
- ◆ Pair or group for partner work by experience when needed.

## Timing

Give the class a common rhythm — everyone enters and leaves postures together — so that no one is left behind or held up. Where variations take longer (e.g. headstand), give a clear “stay in child’s pose or dolphin until everyone comes down”.

### ◆ TEACH THE WHOLE ROOM

In a mixed class, the most important skill is not knowing advanced variations; it is **seeing** the whole room and **pacing** the class so that every student has a meaningful practice. When in doubt, teach to the least experienced student present, and offer the extras.

## 32.5 When the room surprises you

You planned an intermediate class and five newcomers walk in; or you planned gentle yoga and the room is full of athletes. Have a **flexible plan**: a core sequence that can be simplified or extended; spare postures in both directions; the confidence to drop your plan. The students in front of you are always more important than the plan in your notebook.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ Level combines familiarity, body awareness, self-regulation, capacity and steadiness — not flexibility alone.
- ◆ Beginners need a warm welcome, the class structure, few clear instructions, demonstration and permission to rest.
- ◆ Experienced students need depth, refinement, variety and care for overuse.
- ◆ Mixed classes are taught as one class with several paths: tiered options, layering, room organisation and a common rhythm.
- ◆ Always teach the students in front of you, not the plan.

 REFLECTION QUESTIONS

1. What level are you, by the definition in this chapter? Where are you a beginner?
2. Recall a class that was too hard or too easy for you. What could the teacher have done?
3. Which is harder for you to teach — beginners or experienced students? Why?

 TEACHER-PRACTICE EXERCISES

1. Teach triangle with three tiers to a mixed group of classmates.
2. Teach a posture by layering, stopping after each layer; ask students where they chose to stop.
3. Plan a first class for complete beginners; teach it to classmates asked to behave as beginners.

 KNOWLEDGE CHECK

1. Level in yoga is best described as: (a) flexibility (b) years of practice only (c) a combination of familiarity, awareness, self-regulation, capacity and steadiness (d) ability to do headstand
2. In a mixed-level class, “tiered options” means: (a) separate classes (b) a foundation for all with optional further steps (c) teaching only the advanced version (d) teaching only beginners
3. For a beginners’ first class, you should: (a) teach all twelve postures fully (b) teach fewer things, clearly, with permission to rest (c) skip relaxation (d) avoid demonstrating
4. A common risk for advanced students is: (a) doing too little (b) overuse from repetitive practice (c) not knowing the sequence (d) fear of relaxation

Answers 1 c · 2 b · 3 b · 4 b

 TEACHING ASSIGNMENT

Design a 60-minute mixed-level class based on the classical sequence, with a written three-tier option for every posture and a plan for pacing. Teach it to a group containing at least one beginner and one experienced practitioner, and ask each for feedback.

## CHAPTER 33

## The Teacher's Craft — Verbal Cueing, Demonstration, Observation and Voice

*“Speak so that the student can do it with their eyes closed.” — a teaching maxim*

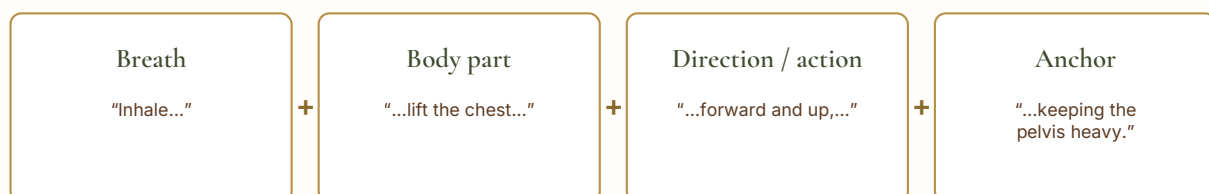
### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ construct clear, economical verbal cues and use different types of cue for different purposes;
- ◆ decide when and how to demonstrate, and demonstrate effectively;
- ◆ observe a class systematically, identify what matters most and give helpful feedback;
- ◆ use your voice with appropriate volume, pace, pitch, tone and silence, and care for it;
- ◆ communicate with presence and warmth, verbally and non-verbally.

### 33.1 Verbal cueing

A cue is any instruction that helps a student do or notice something. Good cues are **clear, brief, timed and purposeful**.



One cue = one breath + one body part + one clear direction (+ what stays still). Then pause and let students do it.

**Figure 33.1** The anatomy of a clear verbal cue.

## Building a cue

A reliable structure: **breath** → **body part** → **action/direction** → **(what stays still)**. “Inhale, lift the arms forward and up, keeping the shoulders soft.” Then **stop talking** and let students do it.

## Types of cue

**Table 33.1** Types of verbal cue

| Type                    | Purpose                                  | Example                                               |
|-------------------------|------------------------------------------|-------------------------------------------------------|
| Instructional           | move students into and out of postures   | “Step the right foot back into a lunge.”              |
| Breath                  | coordinate movement and breath           | “Inhale here... exhale, fold.”                        |
| Alignment               | organise the body safely and effectively | “Knee over the ankle.”                                |
| Safety / options        | protect and include                      | “If your wrists complain, come onto your forearms.”   |
| Action (external focus) | create an effective action with an image | “Press the floor away.”                               |
| Awareness / inquiry     | turn attention inward                    | “Where do you feel this most?”                        |
| Energetic / imagery     | evoke a quality                          | “Lengthen as if drawn up by a thread from the crown.” |
| Philosophical           | connect practice to meaning              | “Steady and easy — <i>sthira, sukha</i> .”            |

Balance your cues. Beginners need mostly instructional, breath and safety cues; experienced students benefit from awareness and inquiry cues. External-focus action cues are often more effective for learning movement than internal anatomical cues (Wulf, 2013).

## Principles of good cueing

- ◆ **One cue at a time**, then pause.
- ◆ **From the ground up**: feet, legs, pelvis, trunk, arms, head — the foundation first.
- ◆ **Positive phrasing**: say what to do (“keep the knee over the ankle”) rather than what not to do (“don’t let your knee collapse”).
- ◆ **Timed with the breath**: instruct the movement on the breath that performs it.
- ◆ **Specific**: “lift the chest” rather than “open up”.
- ◆ **Consistent language**: use the same words for the same action every time; students learn your vocabulary.
- ◆ **Economical**: remove filler words (“just”, “kind of”, “go ahead and”, “I want you to”).
- ◆ **Left and right**: say them clearly; if you demonstrate facing the class, mirror (use your left when you say “right”) and say so.

### Q CUE LESS, SEE MORE

Record yourself teaching a posture and count the words. Then teach it again with half as many. Most new teachers discover that the shorter version is clearer, and that the silences let them see the room.



### 33.2 Demonstration

Demonstration is powerful — people copy what they see more than what they hear — but it has costs: while demonstrating, you cannot observe; students may copy your body rather than find their own; and a teacher who demonstrates everything performs rather than teaches.

**When to demonstrate:** introducing a new posture or transition; showing a modification; clarifying a confusing instruction; with beginners and language differences.

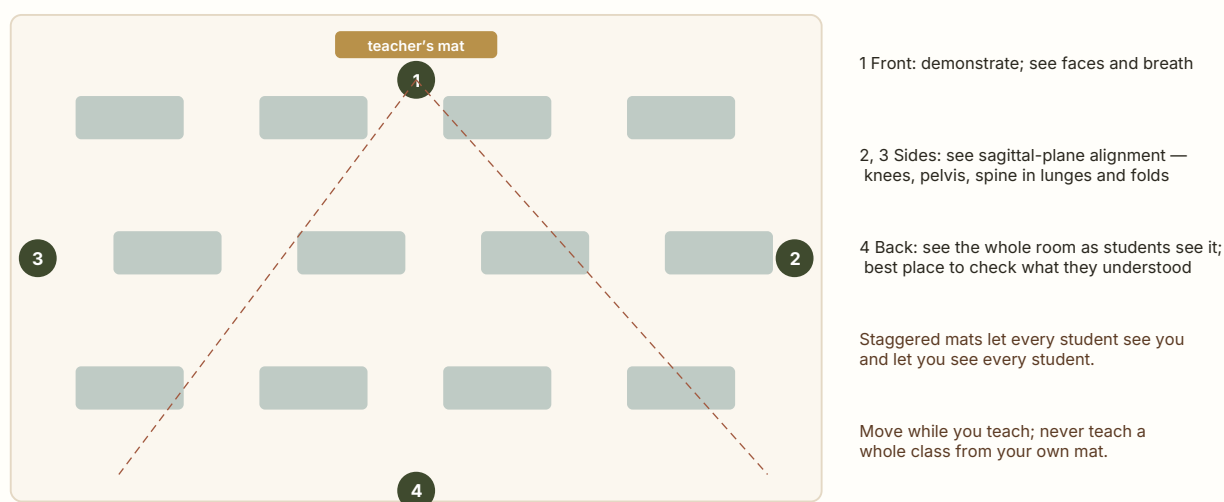
**When not to:** in postures students already know; during holds, when you should be observing; when your version would be discouraging or unsafe to copy.

**How to demonstrate well:**

- ◆ Stand where everyone can see — raise yourself (on a stage or at the front) and turn side-on for sagittal-plane postures.
- ◆ Demonstrate the **version you want most students to do**, often the modified one; show variations as equals.
- ◆ Demonstrate **partially** — just the feet, just the transition — rather than the whole pose.
- ◆ Use a **student demonstration** (with consent) to show how a pose looks in a different body.
- ◆ Demonstrate, then **teach verbally while you walk around**.

### 33.3 Observation

Observation is the skill that makes teaching responsive. It can be trained.



**Figure 33.2** Teaching positions and sight lines in a group class. Circulating between these positions is how you observe; staying on your own mat is how you perform.

#### How to observe

1. **Scan the whole room** first — is everyone breathing, following, safe?
2. **Look at foundations** — feet and hands, then knees and hips.

3. **Look for the pattern, not the individual** — if six people do the same thing, your instruction was unclear; re-cue the class.
4. **Then look at individuals** — especially beginners, anyone who has told you of an injury, and anyone struggling.
5. **Move around:** see the room from the sides and back; different planes are visible from different angles.
6. **Watch faces and breath** as closely as shapes.

The observation checklist in Table 20.1 applies to every posture.

### Giving feedback

- ◆ **Prioritise:** safety first, then the foundation, then refinements. Choose the one thing that matters most.
- ◆ **Address the group first** with a general cue; individualise only if needed.
- ◆ **Be specific and positive:** “Good — now let the knee move over the ankle,” rather than “Wrong”.
- ◆ **Be discreet:** go to the student and speak quietly, rather than calling out their name across the room.
- ◆ **Check the effect:** did the cue help? If not, try a different one, a demonstration or a prop.

## 33.4 Voice

Your voice is your main instrument. In a two-hour class, you may speak for an hour or more.

**Table 33.2** Qualities of the teaching voice

| Quality      | Guidance                                                                                                                                   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Volume       | loud enough for the back of the room without strain; lower for relaxation — but still audible                                              |
| Pace         | slower than conversation; slower still in relaxation and meditation                                                                        |
| Pitch        | use your natural pitch; avoid a “yoga voice” that sounds unnatural                                                                         |
| Tone         | warm, steady, confident; match the tone to the part of the class (clear and energetic in standing postures; soft and steady in relaxation) |
| Articulation | clear consonants, especially for international students                                                                                    |
| Silence      | planned pauses after each instruction; long silences in holds and relaxation                                                               |

### Caring for the voice

- ◆ Breathe from the diaphragm and support the voice with the breath rather than the throat.
- ◆ Warm up the voice before teaching (humming, bhrāmarī, gentle chanting).
- ◆ Drink water; avoid shouting over fans, traffic or music — reduce noise or use a microphone.
- ◆ Rest the voice between classes; seek professional help for persistent hoarseness.

## 33.5 Communication and presence

Much of teaching is non-verbal. Students read your posture, facial expression, movement around the room and attention. Presence — being fully with the class, not in your notes, your phone or your own practice — is felt immediately.

- ◆ **Learn names** and use them.

- ♦ **Make eye contact** at the start and end of class.
- ♦ **Listen** — before class, after class and in the pauses.
- ♦ **Use humour** lightly, and never at a student's expense.
- ♦ **Be yourself:** authenticity is more trustworthy than a borrowed persona.

#### ||| CUEING TĀḌĀSANA FROM THE GROUND UP

*Stand with the feet together or hip-width apart... Feel the weight in the heels, the base of the big toes and the base of the little toes... Let the legs be strong, the kneecaps lifting softly... Let the pelvis rest level... Lengthen up through the spine... Let the shoulders soften down, the arms hang heavy... The crown of the head rises... Breathe... Stand.*

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ♦ A clear cue combines breath, body part and direction, is given once, and is followed by silence.
- ♦ Use different types of cue for different purposes; cue from the ground up; phrase positively; be specific and economical.
- ♦ Demonstrate to introduce and clarify; do not demonstrate during holds when you should be observing.
- ♦ Observe systematically: the room, foundations, patterns, individuals, faces and breath; move around the room.
- ♦ Give prioritised, specific, positive and discreet feedback, and check its effect.
- ♦ Care for your voice; teach with presence and authenticity.

### 🕉 REFLECTION QUESTIONS

1. Which of your verbal habits (filler words, “just”, over-talking) do you notice most?
2. When do you demonstrate because students need it, and when because you want to be seen?
3. What does your voice sound like when you are nervous? What helps?

### ✧ TEACHER-PRACTICE EXERCISES

1. Teach tādāsana and trikoṇāsana using only one cue at a time with a full breath of silence after each.
2. In groups of three, one teaches, one practises, one observes the teacher and counts filler words, demonstrations and cues given while not looking at students.
3. Practise observing a line of five classmates in downward dog for 30 seconds; list what you saw, prioritised by importance.

☒ **KNOWLEDGE CHECK**

1. Which cue uses an external focus? (a) “Contract your serratus anterior.” (b) “Push the floor away.” (c) “Rotate the humerus.” (d) “Engage.”
2. When six students make the same error, the best response is to: (a) correct each individually (b) re-cue the whole class (c) ignore it (d) demonstrate the advanced version
3. During long holds, the teacher should mostly be: (a) demonstrating (b) observing (c) doing their own practice (d) checking notes
4. Positive phrasing means: (a) praising everything (b) saying what to do rather than what not to do (c) avoiding safety cues (d) speaking loudly
5. A good way to protect the voice is to: (a) shout over music (b) support the voice with the breath and warm up (c) whisper throughout (d) speak continuously

Answers 1 b · 2 b · 3 b · 4 b · 5 b

 **TEACHING ASSIGNMENT**

Record a 15-minute section of your teaching. Transcribe two minutes of it. Rewrite the transcript to be clearer and shorter, then teach the same section again and record it. Submit both transcripts with a short reflection on what changed.

## CHAPTER 34

## Sequencing — Warm-up, Peak Poses, Counterposes, Rest and Integration

*“Take the practice step by step, as a stairway.” — after Haṭha Yoga Pradīpikā 1.1*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ apply the core principles of sequencing: preparation, progression, balance, counterpose and integration;
- ◆ design warm-ups that prepare the specific joints and muscles a class requires;
- ◆ build a peak-pose sequence;
- ◆ choose appropriate counterposes;
- ◆ place rest and integration effectively, including the final relaxation.

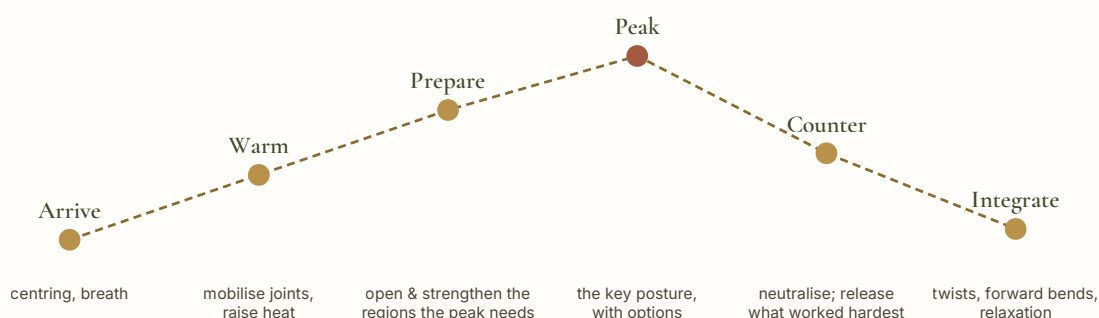
### 34.1 Why sequence matters

The same postures, in a different order, make a different class. Sequencing determines whether the body is ready for what is asked of it, whether the nervous system is stimulated or settled, whether the class feels coherent, and — often — whether it is safe. The classical sequence is a carefully tested answer to the sequencing question (Chapter 24). Understanding *why* it works lets you teach it well and design other sequences when needed.

### 34.2 Principles

1. **Preparation before demand.** Warm the body and prepare the specific regions a posture needs before asking for depth or load.
2. **Simple before complex; familiar before new.** Build from the known.
3. **Progression in intensity,** rising to a peak and then descending.

4. **Balance.** Move the spine in all its directions (flexion, extension, lateral flexion, rotation) and work both sides of the body equally.
5. **Counterpose.** After a strong movement in one direction, return the body gently towards neutral or the opposite.
6. **Rest and integration.** Short rests between postures; a long relaxation at the end.
7. **Fit to the students, the time of day and the setting** — a Kerala afternoon at 34 °C calls for a different class from a cool dawn.



**Figure 34.1** The arc of a peak-pose class. Everything before the peak prepares for it; everything after it restores balance.

### 34.3 Warm-up and preparation

A warm-up raises body temperature, increases blood flow to muscles, mobilises joints through gentle range, and turns attention to the body and breath. In the classical class, relaxation and *prāṇāyāma* settle the mind; *sūrya namaskār* warms the whole body; leg raises prepare the abdomen and hip flexors (Chapter 23).

A good warm-up is **specific** as well as general. Look at the joints and muscles the main postures will use (Chapter 25) and include movements that prepare them:

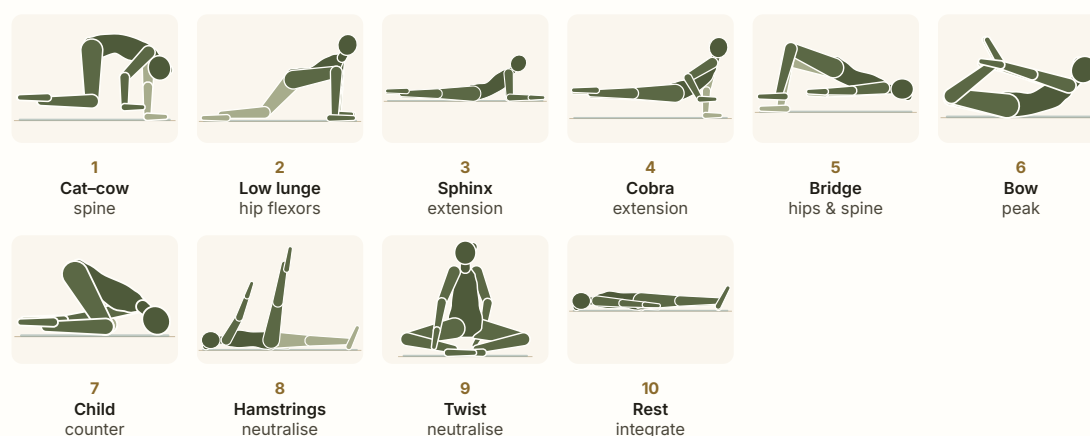
**Table 34.1** Specific preparation for posture families

| Main work         | Prepare with                                                                                                                                      |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Inversions        | shoulder mobility (arm circles, cow-face arms), shoulder strength (dolphin, plank), trunk strength (leg raises), neck mobility (gentle rotations) |
| Backbends         | spinal extension (cat-cow, sphinx, cobra), hip-flexor length (lunge), shoulder flexion (arms overhead), thoracic mobility                         |
| Forward bends     | hamstring warm-up (leg raises, supine hamstring stretch), hip mobility, gentle spinal flexion                                                     |
| Twists            | spinal mobility in all directions, hip mobility                                                                                                   |
| Arm balances      | wrist mobility and loading, plank, core strength, squats                                                                                          |
| Standing balances | foot and ankle mobility, leg strength, simpler balances                                                                                           |
| Hip openers       | hip circles, gentle external rotation, adductor warm-up                                                                                           |

### 34.4 Peak-pose sequencing

A **peak pose** is the most demanding posture of a class, towards which the whole sequence builds. Designing a peak-pose class:

1. Choose the peak suitable for the group.
2. Analyse it with the mapping table: joints, movements, muscles, loading.
3. List what it requires — strength, mobility, balance, confidence.
4. Build preparatory postures that develop each requirement, from simpler to closer approximations of the peak (“component poses”).
5. Offer the peak with options (Chapter 32).
6. Counterpose and integrate.



### 34.5 Counterposes

A counterpose returns the body towards balance after a strong action. It should be **gentler** than the pose it follows and should move the main joints **in the opposite direction** without jolting.

**Table 34.2** Common counterposes

| After                                   | Counterpose                                        | Why                                       |
|-----------------------------------------|----------------------------------------------------|-------------------------------------------|
| Shoulderstand and plough (neck flexion) | fish (neck and upper-back extension)               | reverses cervical and thoracic flexion    |
| Backbends                               | child's pose; gentle forward bend; knees to chest  | gentle spinal flexion                     |
| Deep forward bends                      | gentle backbend (cobra, bridge, tabletop with cow) | gentle spinal extension                   |
| Twists                                  | twist to the other side; neutral spine             | symmetry                                  |
| Headstand                               | child's pose, then rest                            | allows circulation and the neck to settle |
| Arm balances                            | wrist flexion and release; child's pose            | wrists and shoulders                      |
| Hip openers (external rotation)         | gentle internal rotation (windscreen-wiper knees)  | balance hip rotation                      |
| Standing sequences                      | rest lying down                                    | lowers heart rate                         |

Counterposes are not an absolute rule. A gentle backbend does not require an immediate forward bend; the need increases with the intensity and duration of the preceding posture.

### 34.6 Rest and integration

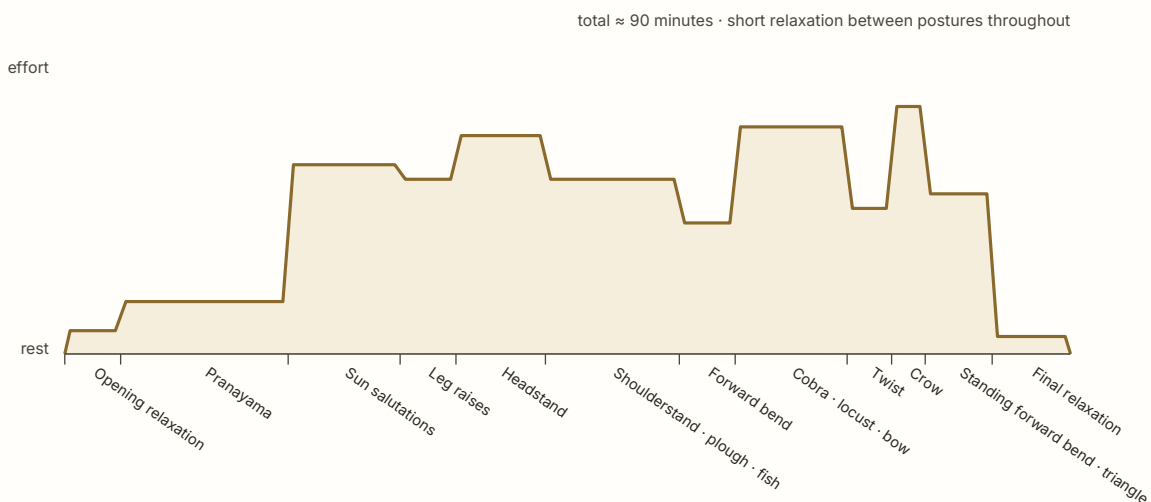
Rest is part of the practice, not a gap in it.

- ◆ **Between postures:** a few breaths in śavāsana, makarāsana or child's pose lets heart rate and breath settle and effects be felt — the hallmark of the classical method.
- ◆ **Final relaxation:** allow about 10–15 minutes in a 90-minute class, and at least 5 minutes in a 45–60-minute class. Never cut it short to fit in more postures.
- ◆ **After relaxation:** a transition to sitting, perhaps a short meditation, chant or silence, before students leave.

#### ◆ ENDING WELL

Students remember the end of a class more than the middle. A calm, unhurried ending — a long relaxation, a moment of silence, a closing word — integrates the practice and sends students out more steady than they came in.

### 34.7 The energetic shape of a class



**Figure 34.2** The shape of a traditional 90-minute classical class: effort rises and falls, with savasana between postures and a long final relaxation. Timings are indicative.

A well-sequenced class has a recognisable arc: arrival, warming, building, peak, descent and rest. The classical class places its demanding inversions early and its most restful posture at the end, with a rhythm of effort and rest throughout.



### 🕒 SEQUENCING FOR THE TIME OF DAY

Morning classes can include more stimulating practice (kapālabhāti, sun salutations, backbends); evening classes more calming practice (forward bends, longer holds, lengthened exhalations, bhrāmari). In the heat of a Kerala afternoon, reduce the number of fast rounds, lengthen rests, and add cooling breaths.

## CHAPTER REVIEW

### 📌 KEY TAKEAWAYS

- ◆ Sequencing determines readiness, safety, coherence and the effect of a class.
- ◆ Core principles: preparation, simple to complex, progression, balance of spinal directions and sides, counterpose, rest and integration, fit to context.
- ◆ Warm-ups should be general and specific to the postures to come.
- ◆ Peak-pose classes analyse the peak and build its requirements through component poses.
- ◆ Counterposes are gentler movements in the opposite direction; rest and a long final relaxation are essential.

### 📖 REFLECTION QUESTIONS

1. Which principle is most often broken in classes you have attended?
2. How does the classical order (inversions first) compare with sequences you know? What are the advantages of each?
3. What makes an ending memorable for you?

### 👤 TEACHER-PRACTICE EXERCISES

1. Design and teach a 30-minute peak-pose sequence towards crow, using component poses.
2. Take a random list of eight postures and put them into a safe, coherent order; explain your reasoning.
3. Teach a 5-minute warm-up specific to backbends.

### ✅ KNOWLEDGE CHECK

1. Which is the best counterpose to shoulderstand? (a) plough (b) fish (c) headstand (d) forward bend
2. A counterpose should be: (a) more intense than the pose before (b) gentler, moving in the opposite direction (c) always a forward bend (d) optional in all cases
3. Specific preparation for arm balances includes: (a) wrist mobility and loading (b) hamstring stretching only (c) neck rotation (d) fish
4. In a 90-minute class, final relaxation should last about: (a) 1–2 minutes (b) 10–15 minutes (c) 30 minutes (d) it is optional
5. In peak-pose sequencing, component poses are: (a) random warm-ups (b) postures that develop the requirements of the peak (c) counterposes (d) restorative poses

Answers 1b · 2b · 3a · 4b · 5b

 **TEACHING ASSIGNMENT**

Design two 60-minute classes with different peak poses (one backbend, one forward bend or twist). For each, provide a written sequence with timings, the purpose of each section, component poses, counterposes and options. Teach one of them in your practice group.

## CHAPTER 35

## Class Planning and Lesson Plans

*“Prepare carefully; then teach the students in front of you.” — a teaching maxim*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ plan a class from its intention and its students, not only from a list of postures;
- ◆ write a clear lesson plan with timings, sequence, cues, options and safety notes;
- ◆ plan a series of classes that progress over weeks;
- ◆ adapt a plan in the moment and reflect on it afterwards.

## 35.1 Planning begins with questions

Before choosing a single posture, answer five questions:

1. **Who** is coming? (Level, age, health, language, numbers.)
2. **What** is the setting? (Time of day, length, temperature, space, props, noise.)
3. **Why** — what is the intention of this class? (A foundation class; strength; calm; a peak pose; a philosophical theme.)
4. **What** will students do? (Sequence, breathing, relaxation, meditation.)
5. **How** will I know it worked? (What do I hope students will feel, understand or be able to do?)

## 35.2 The shape of a lesson plan

A lesson plan is a tool for thinking and a safety net while teaching. It should be clear enough to glance at mid-class.

**Table 35.1** Elements of a lesson plan

| Element   | Content                                             |
|-----------|-----------------------------------------------------|
| Header    | date, time, length, place, expected students, level |
| Intention | one sentence: the purpose of the class              |

| Element          | Content                                                              |
|------------------|----------------------------------------------------------------------|
| Theme (optional) | a teaching point, quality or idea woven through                      |
| Sequence         | sections with timings; postures in order; breath; holds; repetitions |
| Key cues         | the two or three most important cues for each main posture           |
| Options          | modifications and variations; props needed                           |
| Safety notes     | contraindications to screen for; postures needing particular care    |
| Transitions      | how students move between postures and sections                      |
| Closing          | relaxation script outline; closing words                             |
| Reflection       | space to write after class                                           |

### 35.3 A sample lesson-plan template

**LESSON PLAN**

Class

Date · Time · Length · Place

Students

Number · Level · Known health notes

Intention

Theme

Props

| Time  | Section                | Practice (postures, breath, holds) | Key cues | Options / safety |
|-------|------------------------|------------------------------------|----------|------------------|
| 0–5   | Arrival and relaxation |                                    |          |                  |
| 5–15  | Prāṇāyāma              |                                    |          |                  |
| 15–25 | Sun salutation         |                                    |          |                  |
| 25–30 | Leg raises             |                                    |          |                  |
| 30–70 | Postures               |                                    |          |                  |
| 70–85 | Final relaxation       |                                    |          |                  |
| 85–90 | Closing                |                                    |          |                  |

**Reflection after class:** What worked? What would I change? What did I notice about the students?

Appendix G contains complete sample plans for 60- and 90-minute classes, a classical class, and beginner, intermediate and mixed-level classes.

### 35.4 Planning a series

Students learn best when classes build on one another. A four- or six-week series might:

- ♦ introduce one element of the classical class each week (for beginners);

- ♦ work towards a peak pose over several weeks;
- ♦ follow a theme — the five points, the eight limbs, the koshas — one aspect per week.

Each class in a series should stand on its own for students who miss a week.

### 35.5 In the moment

No plan survives the first five minutes unchanged. Adapt when:

- ♦ the students differ from what you expected;
- ♦ the room is too hot, crowded or noisy;
- ♦ students look tired, rushed or restless;
- ♦ something is not working — a cue, a sequence, a pace.

Adaptation is a skill, not a failure. Have spare postures in both directions (easier and harder), and be ready to shorten sections rather than rush.

#### 🕒 TIME KEEPING

Put timings in your plan and a clock where you can see it without looking at your wrist. The most common timing error is running over in the postures and cutting the final relaxation. Protect the relaxation first.

### 35.6 After class: reflection

Write a few lines immediately after each class. Over months these notes become your most valuable teaching resource. Useful prompts:

- ♦ What was the best moment? Why?
- ♦ Where did students look confused, strained or bored?
- ♦ Which cue worked best? Which didn't?
- ♦ Did the timing work?
- ♦ What will I do differently next time?

## CHAPTER REVIEW

### 📌 KEY TAKEAWAYS

- ♦ Plan from the students, setting and intention, not only from postures.
- ♦ A lesson plan includes intention, sequence with timings, key cues, options, safety notes, transitions and closing.
- ♦ Series of classes should progress while each class stands alone.
- ♦ Adapting the plan in the moment is a skill; protect the final relaxation.
- ♦ Brief written reflection after every class accelerates improvement.

 **REFLECTION QUESTIONS**

1. What is the intention behind the class you most like to practise?
2. When have you seen a teacher adapt well to an unexpected class?
3. What will your reflection routine be after teaching?

 **TEACHER-PRACTICE EXERCISES**

1. Write a 60-minute lesson plan using the template, and exchange it with a partner for feedback.
2. Take your partner's plan and teach 15 minutes of it exactly as written; discuss what was unclear.
3. Adapt your plan in two minutes for a class in which half the students are complete beginners.

 **KNOWLEDGE CHECK**

1. The first question when planning a class is: (a) which peak pose? (b) who is coming? (c) what music? (d) how long is relaxation?
2. Which is *not* an essential element of a lesson plan? (a) timings (b) safety notes (c) a list of your personal achievements (d) key cues
3. The most common timing error is: (a) starting early (b) cutting the final relaxation (c) too much prāṇāyāma (d) too many rests
4. Each class in a series should: (a) depend entirely on the previous week (b) stand on its own for students who missed a week (c) be identical (d) have no theme

Answers 1 b · 2 c · 3 b · 4 b

 **TEACHING ASSIGNMENT**

Plan a four-week series for beginners introducing the classical class (one lesson plan per week). Teach week 1 in your practice group and submit all four plans with your reflection on week 1.

## CHAPTER 36

## Hands-on Adjustments, Consent and Appropriate Touch

*“First, do no harm.” — a principle of medical ethics*

### ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ explain the purposes and risks of hands-on assistance in yoga;
- ◆ obtain, check and respect consent for touch at every stage;
- ◆ distinguish verbal, visual, prop-based and hands-on adjustments and choose between them;
- ◆ give safe, clear, appropriate hands-on assistance in the basic postures, using good body mechanics;
- ◆ recognise touch that is never appropriate.

### 36.1 Why touch — and why caution

Touch can be a powerful teaching tool. A light hand can show a student where to lengthen, reassure them in an unfamiliar position, or help them feel a part of the body they cannot sense. Touch is also one of the most frequently reported causes of yoga injury — especially when a teacher pushes a student deeper into a stretch (Fishman et al., 2009; Wiese et al., 2019) — and, when misused, a source of profound harm and violation (Remski, 2019). Modern teaching has therefore moved from routine adjusting towards **assisting sparingly, with explicit consent, for a clear purpose.**

### ◆ ADJUST OR ASSIST?

The word *adjustment* suggests the student is wrong and the teacher will fix them. *Assist* suggests the teacher is helping the student do what they are trying to do. The difference is not only in words: an assist follows the student’s movement and intention; it never overrides them.

### 36.2 The hierarchy of adjustments

Most corrections do not need touch. Work through this hierarchy, and use hands-on assistance only when the others will not do:

1. **Verbal cue to the group** (“everyone, knee over ankle”).
2. **Verbal cue to the individual**, quietly.
3. **Visual cue**: demonstrate, or point to where you want movement.
4. **Prop**: a block, strap, blanket or wall often solves the problem better than a hand.
5. **Hands-on assist**, with consent.

### 36.3 Consent

Consent must be **informed, freely given, specific, ongoing and revocable**.



**Figure 36.1** A consent process for hands-on assistance. Consent given at the start of class does not cover every touch; it is re-checked moment to moment.

- ◆ **Ask at the start of class.** Explain that you sometimes offer hands-on assists, and offer an easy way to opt in or out: consent cards or chips placed at the front of the mat (turned one way for yes, the other for no), or a raised hand with eyes closed. Make “no” easy and without explanation.
- ◆ **Ask again before touching:** “May I place a hand on your back?” — and wait for a clear yes. Silence is not consent.
- ◆ **Tell them what you will do and why.**
- ◆ **Check during:** “How is that?” Stop immediately if the student says so, stiffens, holds their breath or seems uncertain.
- ◆ **Respect changes:** a student who said yes at the start may say no later, and vice versa.
- ◆ **Never assume** that previous consent or a friendly relationship implies consent now.

### 36.4 Principles of safe touch

- ◆ **Purpose:** every touch has a clear teaching purpose that you could explain.
- ◆ **Areas:** use broad, neutral areas — shoulders, upper back, sacrum (with a flat hand), outer thighs, feet, hands. **Never touch** the chest, breasts, groin, buttocks or inner thighs.
- ◆ **Quality:** clear, steady, confident — not tentative, not heavy; use the flat of the hand; no stroking or lingering.
- ◆ **Direction:** guide (show direction) rather than push; never add force at the end of a student’s range.



- ♦ **Timing:** enter with the student's breath; release slowly and clearly.
- ♦ **Your body:** stable stance, bent knees, spine neutral — you must be stable before you can steady anyone else.
- ♦ **Brevity:** a few breaths at most.

#### ⚠ NEVER

- ♦ Never push a student deeper into a stretch, especially forward bends, twists and hip openers.
- ♦ Never adjust the neck or head in inversions, or lift a student into or out of headstand.
- ♦ Never use your body weight on a student's back, pelvis or legs.
- ♦ Never touch without consent, or in a way you would not be comfortable explaining to the student, a colleague or a family member.
- ♦ Never adjust a student who has told you of an injury in that area, or a pregnant student, unless you are specifically trained.

## 36.5 Safe assists for the basic postures

**Table 36.1** Examples of simple, safe assists (only with consent)

| Posture               | Purpose                 | Assist                                                                               | Avoid                                                |
|-----------------------|-------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------|
| Tāḍāsana              | grounding               | hands lightly on the shoulders, drawing attention down                               | pressing down on the head                            |
| Child's pose          | release                 | flat hands on the lower back, following the exhalation, very light                   | pressing down with weight                            |
| Downward dog          | length of the spine     | hands on the sides of the pelvis guiding it up and back                              | pulling hard; pushing on the spine                   |
| Cobra                 | upper-back extension    | a light hand between the shoulder blades inviting lift                               | pressing the pelvis down while the student pushes up |
| Seated forward bend   | pelvic tilt             | a hand on the lower back encouraging the pelvis to tilt forward                      | pushing the back down                                |
| Half spinal twist     | rotation from the ribs  | one hand in front of a shoulder, one behind the other, following the student's turn  | adding force at end range                            |
| Triangle              | frontal-plane alignment | hands on the back of the pelvis and upper shoulder                                   | pulling the upper arm                                |
| Headstand (at a wall) | safety                  | standing beside the student, hands ready near the hips or legs                       | lifting; standing in front                           |
| Śavāsana              | release                 | adjusting a blanket; gently lengthening the legs from the heels (with prior consent) | touching the head or face                            |

## 36.6 Hands-on in practice training

Learn assists by giving and receiving them in training, with feedback from the receiver. Always practise:

- ♦ asking and waiting for consent;
- ♦ stance and body mechanics;
- ♦ the lightest touch that communicates the intention;

- ♦ asking the receiver: *What did you feel? Was it clear? Was it comfortable?*

### ✂ THE LIGHTEST TOUCH

In pairs, one partner holds downward dog. The other, with consent, places the flat hands on the sides of the pelvis and guides it up and back — first with firm pressure, then with half the pressure, then with the lightest possible touch. Ask the receiver which communicated the direction most clearly. Most people choose the lightest.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ♦ Touch can teach powerfully but is a common cause of injury and, when misused, of serious harm.
- ♦ Use the hierarchy: group cue, individual cue, visual cue, prop — hands-on only when needed.
- ♦ Consent is informed, freely given, specific, ongoing and revocable; ask at the start and before each touch.
- ♦ Safe touch has a clear purpose, uses neutral areas, guides rather than pushes, and never adds force at end range.
- ♦ Certain adjustments are never appropriate: pushing into stretches, adjusting the neck in inversions, using body weight, touching intimate areas.

### 🍃 REFLECTION QUESTIONS

1. Recall an adjustment you received that helped you, and one that did not. What was the difference?
2. How comfortable are you with touch — giving and receiving? How will this affect your teaching?
3. How will you make it genuinely easy for students to say “no”?

### ✂ TEACHER-PRACTICE EXERCISES

1. Practise the consent process at the start of a class, including consent cards.
2. Practise each assist in Table 36.1 with a partner, asking for feedback after each.
3. Role-play a student who withdraws consent mid-assist; practise stopping gracefully.

☑ **KNOWLEDGE CHECK**

1. Which should be tried *before* a hands-on assist? (a) nothing (b) verbal, visual and prop-based cues (c) pushing (d) a louder voice
2. Consent given at the start of class: (a) covers every touch (b) can be withdrawn at any time and should be re-checked (c) is unnecessary (d) is permanent
3. Which area should never be touched in an assist? (a) shoulders (b) outer thighs (c) inner thighs (d) hands
4. In a seated forward bend, a safe assist is to: (a) push the back down (b) encourage the pelvis to tilt forward with a light hand on the lower back (c) sit on the student (d) pull the arms
5. In headstand, the teacher should: (a) lift the student up (b) adjust the neck (c) stand beside the student ready to steady the hips or legs (d) stand in front

Answers 1b · 2b · 3c · 4b · 5c

📄 **TEACHING ASSIGNMENT**

Write your studio's touch-and-consent policy (under 200 words) and the exact words you will use at the beginning of class to explain it. Demonstrate three assists from Table 36.1 to your mentor, including the consent process.

## CHAPTER 37

# Managing the Classroom and Handling Difficult Situations

*“Equanimity towards the non-virtuous.” — Yoga Sūtra 1.33*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ set up and manage a practice space, before, during and after class;
- ◆ establish clear, welcoming class agreements;
- ◆ respond calmly to common difficult situations: lateness, phones, talking, competitiveness, disruptive or distressed students, unwanted attention and medical events;
- ◆ reflect on difficult situations and seek support.

### 37.1 Before class

- ◆ **Arrive early.** Prepare the room: clean floor, ventilation, lighting, temperature, props laid out, water available, first-aid kit accessible.
- ◆ **Arrange the mats** so that you can see everyone and everyone can see you (Figure 33.2), with space to walk between them.
- ◆ **Greet students** at the door; learn names; ask newcomers about health and experience.
- ◆ **Set the tone:** calm, unhurried, welcoming.

### 37.2 Class agreements

Students relax when they know what to expect. State the basics briefly — or post them — and follow them yourself:

- ◆ arrive a few minutes early; if late, enter quietly (many schools do not admit students after the opening relaxation, for safety and for others’ concentration);
- ◆ phones silent and away;
- ◆ tell the teacher about injuries, conditions or pregnancy;

- ♦ rest whenever you need to; follow your own body;
- ♦ stay for the final relaxation — or, if you must leave early, tell the teacher and leave before it begins.

### 37.3 Common situations

**Table 37.1** Responding to common difficult situations

| Situation                                                         | Response                                                                                                                                                                                                              |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Late arrival                                                      | welcome quietly; guide to a space; after class, explain the reason for arriving on time (safety, warm-up, others' focus). If late arrivals would miss the warm-up before strong postures, offer a gentler alternative |
| Phone rings                                                       | a neutral reminder to the whole room at the start of the next class, not public embarrassment                                                                                                                         |
| Talking during class                                              | a general, friendly reminder ("let's keep a quiet room for everyone"); speak privately afterwards if it continues                                                                                                     |
| Competitive or over-striving student                              | offer options to the whole class; praise steadiness and breath rather than depth; speak privately about safety if needed                                                                                              |
| Student ignores safety instructions                               | calm, private, clear explanation of the risk; if they persist in something dangerous, ask them to stop — safety of the student and others comes first                                                                 |
| Student is distressed or tearful                                  | allow space; offer quiet reassurance; invite rest, eyes open, grounding; check in privately afterwards; refer if appropriate                                                                                          |
| Student gives unsolicited advice to others                        | thank them, and gently reclaim the teaching role ("let's let everyone follow their own practice today")                                                                                                               |
| Unwanted personal attention towards the teacher or other students | clear, calm boundary-setting; involve the school leadership; never ignore harassment                                                                                                                                  |
| A student feels faint or unwell                                   | help them to lie down (or sit with head low), legs raised if faint; stay with them; seek medical help if symptoms do not settle or are serious ( <b>Chapter 26</b> )                                                  |
| Emergency                                                         | follow the emergency plan: call for help, give first aid within your training, assign another adult to manage the rest of the class                                                                                   |
| You make a mistake                                                | acknowledge it simply, correct it, continue; learning from mistakes models what yoga teaches                                                                                                                          |

#### ♀ PRIVATE, NOT PUBLIC

Almost every correction of behaviour is better given privately, after class, than publicly during it. Public correction embarrasses the student and unsettles the room; private conversation preserves dignity and relationship.

### 37.4 Hot, crowded and noisy rooms

Fort Kochi is not a silent monastery. Traffic, temple loudspeakers, crows, rain on the roof and the heat of the afternoon are part of practice here. Teach students to include sounds in their awareness rather than resent them (**Chapter 7**). Adjust the practice for heat, and reduce numbers rather than crowd mats.

## 37.5 Looking after yourself

Difficult situations are stressful. Afterwards:

- ◆ **write a brief note** of what happened and what you did (especially for any incident involving injury, illness or complaint);
- ◆ **talk to a colleague or mentor**;
- ◆ **reflect** without self-blame: what would you do the same, what differently?

### CHAPTER REVIEW

#### KEY TAKEAWAYS

- ◆ Prepare the room and yourself before class; greet students and learn names.
- ◆ Clear, welcoming agreements reduce problems; model them yourself.
- ◆ Respond to difficulties calmly, generally first and privately after; safety always comes first.
- ◆ Manage fainting, distress and emergencies with a clear plan.
- ◆ Record incidents, talk to a mentor and reflect afterwards.

#### REFLECTION QUESTIONS

1. Which situation in Table 37.1 would you find hardest? Why?
2. How do you react when you feel challenged or disrespected? How does that affect your teaching?
3. How can the yogic attitudes of YS 1.33 guide you with a difficult student?

#### TEACHER-PRACTICE EXERCISES

1. Role-play five situations from Table 37.1 with a partner playing the student.
2. Write and rehearse your class agreements in under 60 seconds of speaking.
3. Walk through your response to a student fainting in standing forward bend.

#### KNOWLEDGE CHECK

1. The best response to repeated talking during class is usually: (a) naming the student publicly (b) a general reminder, then a private conversation (c) ignoring it (d) ending the class
2. If a student feels faint, you should: (a) ask them to stand up quickly (b) help them lie down or sit with the head low, stay with them, and seek help if needed (c) continue teaching (d) give them kapālabhāti
3. Corrections of behaviour are generally best given: (a) publicly during class (b) privately after class (c) by email only (d) never
4. After a difficult incident you should: (a) forget it (b) record it, talk to a mentor and reflect (c) post about it online (d) blame the student

Answers 1 b · 2 b · 3 b · 4 b

#### TEACHING ASSIGNMENT

Write a one-page “room and rules” guide for your future classes: set-up, greeting, agreements, what you will say about late arrival and phones, and your emergency plan. Present it to your practice group.

## CHAPTER 38

# Private Classes, Workshops and Online Teaching

*“Teach according to the student’s capacity.” — the principle of adhikāra*

## ◎ LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ◆ conduct an initial consultation and plan a private class or series;
- ◆ design and deliver a workshop with clear learning outcomes;
- ◆ teach live online classes safely and effectively;
- ◆ recognise the particular ethical and safety considerations of each format.

## 38.1 Private classes

A private class is the most individual form of teaching — close to the traditional one-to-one relationship between teacher and student.

### The first session

1. **Consultation (10–15 minutes).** Ask about goals, experience, health, injuries, medication relevant to exercise, pregnancy, lifestyle and what the student enjoys. Use a written health questionnaire (Appendix I).
2. **Observation.** Watch the student in a few simple movements — standing, forward bend, a lunge, sitting, breathing — without correcting.
3. **Practice.** Teach a short, safe, appropriate practice.
4. **Home practice.** Give a simple routine (10–20 minutes) they can remember, with a written or drawn sheet.
5. **Agree next steps.** Frequency, goals, how progress will be recognised.



## Planning a series

Set realistic goals with the student — for example, a steady home practice, comfort sitting for meditation, preparation for headstand. Build progressively, review regularly and keep notes. Private students often have specific health goals; remember your scope (Chapter 15) and liaise with their healthcare professional where appropriate.

### ⚠️ PRIVATES AND BOUNDARIES

Private sessions happen in private spaces — the student's home, a hotel room, a small studio. Professional boundaries matter even more (Chapter 30): clear appointments and payment, a professional setting where possible, touch only with explicit consent, and no social or romantic entanglement. Tell someone where you are when teaching in unfamiliar private settings.

## 38.2 Workshops

A workshop (typically 2–3 hours, or a day) explores one topic in depth: inversions, backbends, prāṇāyāma, the Bhagavad Gītā, yoga for runners.

### Designing a workshop:

- ♦ **Title and outcomes.** Write two to four outcomes: “By the end, participants will be able to...”.
- ♦ **Structure.** Arrival and practice; teaching segments alternating with practice; partner or small-group work; a closing practice and questions.
- ♦ **Variety.** Mix explanation, demonstration, practice, discussion and reflection; change activity every 15–20 minutes.
- ♦ **Handout.** A one-page summary students can take home.
- ♦ **Prerequisites and safety.** State the level and any contraindications in advance (e.g. an inversion workshop).

## 38.3 Online teaching

Live online classes extend teaching beyond the room, but change it.

**Table 38.1** Adapting to online teaching

| Consideration | Good practice                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| Screening     | health questionnaire before the first class; emergency contact and address for each student; consent form       |
| Visibility    | ask students to place the camera to see the whole body; demonstrate side-on to the camera for sagittal postures |
| Cueing        | more verbal precision; say “right” and “left” clearly; describe what you are showing                            |
| Observation   | watch the gallery view; name individuals for feedback only with consent; unmute for questions                   |
| Safety        | teach less demanding postures than in person; no inversions for new students; props from household items        |
| Technology    | stable connection, good microphone, lighting in front of you; a backup plan                                     |
| Privacy       | no recording without explicit consent; secure meeting links; respect students' homes                            |
| Emergencies   | know each student's location; if a student is unwell, contact their emergency contact or local services         |

### 🔍 HOUSEHOLD PROPS

Books for blocks, a belt or scarf for a strap, a folded blanket or firm cushion, a sturdy chair, a wall. Name the substitutes in advance so students can prepare.

## CHAPTER REVIEW

### 📌 KEY TAKEAWAYS

- ◆ Private classes begin with consultation and observation, and progress through a planned series with home practice.
- ◆ Workshops have clear outcomes, varied activities, handouts and stated prerequisites.
- ◆ Online teaching needs screening, emergency information, precise verbal cues, careful observation, safer content and respect for privacy.
- ◆ Each format carries specific boundary and safety considerations.

### 📝 REFLECTION QUESTIONS

1. Which format appeals to you most? Which least? Why?
2. What would you do in a private class if the student asks for help with a medical condition?
3. What is lost in online teaching? What is gained?

### ✂️ TEACHER-PRACTICE EXERCISES

1. Conduct a mock first private session (consultation, observation, 20-minute practice, home routine) with a classmate.
2. Design a two-hour workshop outline on one topic from this manual.
3. Teach a 15-minute online segment to classmates (using phones or laptops in separate rooms) and gather feedback on visibility and cueing.

### ☑️ KNOWLEDGE CHECK

1. The first step in a first private class is: (a) a vigorous practice (b) consultation about goals, health and experience (c) an advanced posture (d) hands-on adjustment
2. A workshop should have: (a) no plan (b) clear outcomes and varied activities (c) only lecture (d) only practice
3. For online classes, you should have each student's: (a) photograph (b) emergency contact and location (c) passport number (d) nothing
4. Recording an online class requires: (a) nothing (b) explicit consent of participants (c) payment (d) a paid platform

Answers 1 b · 2 b · 3 b · 4 b

### 📅 TEACHING ASSIGNMENT

Design a six-week private-class programme for a fictional student (describe their goals and health), including the consultation questions, a plan for each week, home practice sheets and how you will review progress.

## CHAPTER 39

# Professional Development and a Sustainable Teaching Career

*“Yoga is skill in action.” — Bhagavad Gītā 2.50*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ plan your continuing professional development after this course;
- ♦ describe the practical foundations of a teaching career: credentials, insurance, first aid, record-keeping and legal compliance;
- ♦ market your teaching honestly and ethically;
- ♦ design a sustainable working life that protects your practice, body, voice and finances.

### 39.1 After the 200 hours

Completing a 200-hour training is the beginning of a teacher’s education, not its end. Most teachers find that the first year of teaching raises more questions than the training answered. Plan for:

- ♦ **Mentoring:** regular contact with an experienced teacher who observes you and whom you can consult.
- ♦ **Assisting:** assisting an experienced teacher’s classes is one of the best ways to learn observation and adjustment.
- ♦ **Continuing education:** workshops and further trainings in areas you need (anatomy, prenatal yoga, yoga for older adults, prāṇāyāma, meditation, philosophy). Advanced trainings (e.g. 300- or 500-hour programmes) deepen knowledge; yoga therapy is a separate, longer clinical training.
- ♦ **Study:** the reading list in Appendix M, and returning to this manual.
- ♦ **Your own practice** — the foundation of all the rest.

## 39.2 Credentials and recognition

Graduates of this course receive the **Santhi Yoga School 200-hour certificate**. The school is not a Yoga Alliance Registered Yoga School. Many studios and retreats accept certificates from established schools; some employers, particularly outside India, ask specifically for registration with a particular body. **Check the requirements of the places where you intend to teach**, and plan any further training accordingly. Represent your qualifications accurately at all times (*satya*).

## 39.3 Practical foundations

**Table 39.1** Practical foundations of a teaching practice

| Area             | What to do                                                                                                                                             |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| First aid        | hold a current first-aid certificate and renew it as required                                                                                          |
| Insurance        | obtain professional liability insurance where available and where you teach; studios may require it                                                    |
| Legal and tax    | find out the rules for teaching, business registration, tax and visas in each country where you teach; never teach on a visa that does not permit work |
| Health records   | keep student health questionnaires securely and confidentially; know how long records should be kept locally                                           |
| Incident records | record any injury, illness or complaint, with date, facts and actions taken                                                                            |
| Contracts        | written agreements with studios (fees, cancellation, cover, insurance, intellectual property)                                                          |
| Data and privacy | handle personal data carefully; follow local data-protection law                                                                                       |

## 39.4 Marketing with integrity

Marketing is simply telling people honestly what you offer, so that the right students can find you.

- ◆ **Describe what you teach**, your training and your lineage accurately.
- ◆ **Make no health promises** or unsupported claims (“cures back pain”, “detoxes the body”) — they breach *satya* and may breach consumer or advertising law.
- ◆ **Use images respectfully**: consent from anyone photographed; representative, diverse images; no sacred symbols as decoration.
- ◆ **Build relationships**, not just followers: word of mouth from students who feel well taught remains the strongest marketing.
- ◆ **Price fairly**: know your costs; consider concessions; avoid pressure-selling.

## 39.5 A sustainable working life

Teaching yoga full-time is physically, vocally and emotionally demanding, and often financially precarious. Sustainable teachers tend to:

- ◆ **limit the number of classes** per week to what they can teach well — often fewer than new teachers expect;

- ◆ **not practise every class they teach** — demonstrate selectively and keep separate time for their own practice;
- ◆ **diversify income**: group classes, privates, workshops, retreats, online, corporate and community teaching;
- ◆ **protect the voice and body** (Chapter 33);
- ◆ **take real rest**: days off and holidays;
- ◆ **stay connected** with a community of teachers;
- ◆ **seek support** early when tired, disillusioned or overwhelmed.

#### ◆ KARMA YOGA AND A LIVING

The Gītā's teaching of action without attachment to results (Chapter 4) is not a reason to be exploited or to undervalue your work. It is a way of doing your work — fully, well, and without letting praise, numbers or income define your worth — while still earning a fair living from it.

#### ✧ A ONE-YEAR PLAN

Write a one-year plan with four sections: **practice** (your daily practice commitment), **learning** (two areas of study, one mentor), **teaching** (where, how often, whom), and **sustainability** (rest, income, support). Review it with your mentor at the end of the course.

## CHAPTER REVIEW

### 🍃 KEY TAKEAWAYS

- ◆ A 200-hour training begins a teacher's education; plan mentoring, assisting, continuing education and practice.
- ◆ Know what your certificate is and is not; check the requirements where you intend to teach; represent qualifications truthfully.
- ◆ Establish practical foundations: first aid, insurance, legal and tax compliance, records, contracts, data protection.
- ◆ Market honestly, without health promises, with respectful images and fair pricing.
- ◆ Sustainability depends on limiting teaching load, diversifying income, protecting body and voice, rest and community.

### 🌺 REFLECTION QUESTIONS

1. What does a sustainable teaching life look like for you in five years?
2. What would tempt you to exaggerate in your marketing? How will you resist?
3. Who could be your mentor after the course?

### ✧ TEACHER-PRACTICE EXERCISES

1. Write an honest 100-word teacher biography for a studio website.
2. Draft a simple budget for teaching in your home town: costs, fees, number of classes needed.

3. Critique three yoga advertisements you find online for accuracy and integrity.

#### ☑ KNOWLEDGE CHECK

1. Graduates of this course receive: (a) Yoga Alliance registration (b) the Santhi Yoga School 200-hour certificate (c) a yoga therapy licence (d) a medical qualification
2. Which marketing claim is appropriate? (a) “Cures back pain” (b) “A traditional Hatha class with breathing, postures and deep relaxation” (c) “Detoxes your organs” (d) “Guaranteed weight loss”
3. Which is a practical foundation of a teaching practice? (a) professional liability insurance where available (b) teaching on any visa (c) keeping health forms in a public folder (d) avoiding contracts
4. Sustainable teachers tend to: (a) teach as many classes as possible (b) limit classes to what they can teach well and protect their own practice (c) practise fully in every class they teach (d) never take holidays

Answers 1 b · 2 b · 3 a · 4 b

#### 📄 TEACHING ASSIGNMENT

Prepare your “professional launch pack”: an honest biography, a one-page class description, your code of conduct (Chapter 30), your health questionnaire and your one-year plan. Present it to your mentor for review before graduation.

## CHAPTER 40

# Practicum, Assessment and the Final Teaching Examination

*“Knowledge is complete only when it is practised.” — a teaching maxim*

## 🕒 LEARNING OBJECTIVES

By the end of this chapter you will be able to:

- ♦ describe the components of the practicum and how you will be assessed;
- ♦ plan and log your practice teaching, observation and feedback;
- ♦ give and receive feedback constructively;
- ♦ prepare for the written and practical final examinations;
- ♦ complete the requirements for graduation.

## 40.1 The practicum

The **practicum** is where you learn to teach by teaching — to your fellow trainees, and in supervised settings. It runs through the whole course, growing in length and responsibility:

**Table 40.1** Practicum progression (indicative; your course leader will confirm the schedule)

| Week | Practicum task                                                                | Setting                    |
|------|-------------------------------------------------------------------------------|----------------------------|
| 1    | teach one posture to a partner; teach the opening relaxation                  | pairs                      |
| 2    | teach sūrya namaskār and two basic postures; observe a senior teacher's class | small groups               |
| 3    | teach a 30-minute class segment; give and receive structured feedback         | groups of 4–6              |
| 4    | teach a full class (45–60 minutes) as the final practical examination         | whole group / public class |

Throughout, you will keep a **practicum log** (Appendix I) of every teaching session: date, length, content, feedback received, and your reflection.



## 40.2 Observation

Observing experienced teachers — and fellow trainees — trains your eye. Use the **teacher observation sheet** (Appendix I) to note: structure and timing; cueing (clarity, economy, types); demonstration; observation and adjustment; use of voice and silence; inclusion and options; safety; atmosphere. Focus on what you can learn, not on judging.

## 40.3 Giving and receiving feedback

Feedback is a gift when it is specific, balanced and kind.

### Giving feedback:

- ◆ Start by asking the teacher how it felt to teach.
- ◆ Name **specific strengths** (“Your cue ‘press the floor away’ transformed my downward dog”).
- ◆ Offer **one or two specific suggestions**, phrased as possibilities (“You might try cueing the feet before the arms”).
- ◆ Base feedback on what you observed and experienced, not on personal preference.

### Receiving feedback:

- ◆ Listen without defending or explaining.
- ◆ Ask for examples if something is unclear.
- ◆ Thank the giver.
- ◆ Decide afterwards what to act on.

## 40.4 Assessment

You will be assessed on knowledge, practice, teaching and professional conduct. The components, used together to judge readiness to teach, are:

1. **Attendance and participation** in all sessions (absences agreed with the course leader and made up).
2. **Personal practice**: the ability to practise the twelve basic postures (with modifications appropriate to your body), sūrya namaskār, the prāṇāyāma taught and meditation.
3. **Assignments**: the teaching assignments at the end of each chapter form your portfolio.
4. **Practicum log and observation sheets**.
5. **Written examination**: philosophy, anatomy, safety, prāṇāyāma, meditation and teaching methodology.
6. **Practical teaching examination**: a full class taught to a group and assessed with the practicum assessment form.
7. **Professional conduct**: punctuality, ethics, care for others and for the school.

### What “ready to teach” means

The assessment does not ask for perfect postures or flawless classes. It asks whether you can teach a safe, clear, well-sequenced class to generally healthy beginners and mixed-level students; adapt it for common limitations; explain what you teach with honesty; and conduct yourself as a trustworthy professional.

## 40.5 The final teaching examination

The practical examination is a full class (45–60 minutes, as assigned) taught to fellow trainees and/or members of the public, observed by one or more senior teachers.

**Table 40.2** Practical examination criteria

| Criterion                         | What assessors look for                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------|
| Preparation                       | a written lesson plan with intention, timings, options and safety notes                         |
| Opening                           | welcome, health check, clear orientation, opening relaxation                                    |
| Sequencing                        | coherent, safe structure with preparation, counterposes and adequate rest                       |
| Instruction                       | clear, economical verbal cues timed with the breath; correct Sanskrit and English names         |
| Demonstration                     | purposeful and appropriate, not continuous                                                      |
| Observation and adaptation        | circulates, sees the room, responds to students, offers options                                 |
| Safety                            | screens for contraindications; modifications; no forcing; appropriate use of touch with consent |
| Breath, relaxation and meditation | prāṇāyāma taught correctly; relaxation guided with appropriate voice and silence                |
| Presence and voice                | calm, audible, warm, authentic                                                                  |
| Time management                   | starts and ends on time; relaxation protected                                                   |
| Reflection                        | honest self-assessment after the class                                                          |

The written examination and a sample of its questions are in Appendix K. The assessment forms are in Appendix I; the completion checklist is in Appendix L.

### 📌 PREPARING FOR THE PRACTICAL EXAMINATION

Teach your exam class at least twice beforehand to classmates. Time it. Memorise the structure, not a script. Have a plan for beginners and a plan for anyone with an injury. On the day, arrive early, set up the room, breathe — and teach the people in front of you.

## 40.6 Graduation

On successful completion of all components you receive the Santhi Yoga School 200-hour certificate, traditionally presented in a closing ceremony. Graduation is an ending and a beginning: of your life as a teacher, and of a lifetime of study.

## CHAPTER REVIEW

 KEY TAKEAWAYS

- ◆ The practicum builds from teaching a single posture to teaching a full class, supported by a log, observation and feedback.
- ◆ Specific, balanced, kind feedback — given and received well — accelerates learning.
- ◆ Assessment covers attendance, personal practice, assignments, practicum, written and practical examinations, and professional conduct.
- ◆ “Ready to teach” means safe, clear, well-sequenced, adaptable, honest and professional — not perfect.

 REFLECTION QUESTIONS

1. What kind of feedback helps you most? What kind do you find hardest to hear?
2. What would make you feel ready to teach your first public class?
3. What do you want your students to say about your final class?

 TEACHER-PRACTICE EXERCISES

1. Teach your planned exam class to a small group and ask each student to complete the assessment criteria in Table 40.2 for you.
2. Observe a fellow trainee’s class using the observation sheet, and give feedback using the structure above.
3. Take the sample written examination in Appendix K under timed conditions.

 KNOWLEDGE CHECK

1. The practicum log records: (a) your diet (b) each teaching session, feedback and reflection (c) attendance only (d) postures you can do
2. Good feedback is: (a) vague and positive (b) specific, balanced and kind (c) mostly critical (d) based on personal preference
3. “Ready to teach” means: (a) perfect postures (b) teaching safe, clear, adaptable classes with honesty and professionalism (c) knowing every posture (d) years of experience
4. When receiving feedback, you should first: (a) defend your choices (b) listen without defending (c) disagree (d) ignore it

Answers 1 b · 2 b · 3 b · 4 b

 TEACHING ASSIGNMENT

Submit your graduation portfolio: all chapter teaching assignments, your practicum log, three observation sheets, your lesson plan for the final examination, your code of conduct, and a one-page reflection on how your understanding of yoga has changed during the course.

## A

## Sanskrit Glossary

Sanskrit terms used in this manual, in the IAST transliteration (see the note on transliteration at the front of the manual), with Devanāgarī and a brief definition. Definitions give the sense used in this manual; many terms carry wider meanings in other texts and schools. Pose names are listed separately in Appendices C and D.

## A

**abhiniveśa** अभिनिवेश

Clinging to life; fear of death. The fifth kleśa (YS 2.9).

**abhyāsa** अभ्यास

Practice; steady, repeated effort (YS 1.12–14).

**adhikāra** अधिकार

Eligibility or readiness of a student for a particular teaching.

**adhyāsa** अध्यास

Superimposition; in Advaita Vedānta, the mistaken projection of the properties of one thing onto another.

**advaita** अद्वैत

Non-duality; the school of Vedānta taught by Śaṅkara.

**ahaṁkāra** अहंकार

The “I-maker”; the sense of individual identity and ownership.

**ahiṁsā** अहिंसा

Non-harming; the first yama.

**ājñā cakra** आज्ञा

The “command” centre between the eyebrows.

**anāhata cakra** अनाहत

The heart centre; “unstruck” (sound).

**ānandamaya kośa** आनन्दमय

The sheath of bliss; the causal body.

**annamaya kośa** अन्नमय

The food sheath; the physical body.

**antaḥkaraṇa** अन्तःकरण

The “inner instrument”: manas, buddhi, ahaṁkāra and citta.

**antaraṅga** अन्तरङ्ग

“Inner limb”; dhāraṇā, dhyāna and samādhi (YS 3.7).

**anuloma viloma** अनुलोम विलोम

Alternate-nostril breathing, in the tradition taught here usually with retention.

**apāna** अपान

The downward and outward vāyu, governing elimination.

**aparigraha** अपरिग्रह

Non-possessiveness, non-grasping; the fifth yama.

**āsana** आसन

Seat; posture. The third limb of Patañjali’s yoga.

**asmitā** अस्मिता

“I-am-ness”; identification of the seer with the instrument of seeing (YS 2.6).

**aṣṭāṅga** अष्टाङ्ग

“Eight-limbed”; Patañjali’s eight limbs of yoga.

**asteya** अस्तेय

Non-stealing; the third yama.

**ātman** आत्मन्

The Self; in Vedānta, identical with brahman.

**avidyā** अविद्या

Ignorance, misapprehension; the root kleśa (YS 2.4–5).

## B

**bandha** बन्ध

Lock; a sustained contraction used to direct prāṇa (mūla, uḍḍiyāna, jālandhara).

**bhakti** भक्ति

Devotion; the path of love for the divine.

**bhastrikā** भस्त्रिका

“Bellows”; a forceful breathing practice.

**bhrāmarī** भ्रामरी

“Bee”; the humming breath.

**bīja** बीज

Seed; a one-syllable seed mantra, e.g. lam, vaṃ, raṃ.

**brahmacarya** ब्रह्मचर्य

“Conduct in brahman”; sexual restraint and moderation; the fourth yama.

**brahman** ब्रह्मन्

The ultimate reality, ground of all being.

**buddhi** बुद्धि

Intellect; the discriminating, deciding faculty.

**C****cakra** चक्र

Wheel; a centre in the subtle body where nāḍis converge.

**cin mudrā** चिन्मुद्रा

Hand gesture with index finger and thumb joined, palms down.

**citta** चित्त

The mind-field; in the Yoga Sūtra, the whole of mental activity.

**D****darśana** दर्शन

“Seeing”; a school of Indian philosophy.

**dhāraṇā** धारणा

Concentration; binding the mind to one place (YS 3.1). The sixth limb.

**dharma** धर्म

Duty, righteousness, the right order of things; one's own path of duty (svadharma).

**dhyāna** ध्यान

Meditation; the uninterrupted flow of awareness to one object (YS 3.2). The seventh limb.

**dr̥ṣṭi** दृष्टि

Gaze; a point of visual focus.

**duḥkha** दुःख

Suffering, unsatisfactoriness.

**dveṣa** द्वेष

Aversion; the fourth kleśa (YS 2.8).

**G****guṇa** गुण

Quality or strand of nature: sattva, rajas, tamas.

**guru** गुरु

Teacher; traditionally, “dispeller of darkness”.

**H****haṭha** हठ

Force; the yoga that uses body, breath and prāṇa as its means.

**I****iḍā** इडा

The left, “lunar” nāḍī, associated with the left nostril.

**īśvara** ईश्वर

The Lord; in the Yoga Sūtra a special puruṣa untouched by affliction.

**īśvara-praṇidhāna** ईश्वरप्रणिधान

Devotion or surrender to Īśvara; the fifth niyama.

**J****jālandhara bandha** जालन्धर बन्ध

The chin lock.

**japa** जप

Repetition of a mantra.

**jñāna** ज्ञान

Knowledge; the path of knowledge and discrimination.

**jñāna mudrā** ज्ञानमुद्रा

Hand gesture with index finger and thumb joined, palms up.

**K****kaivalya** कैवल्य

“Aloneness”; liberation in Patañjali's system.

**kapālabhāti** कपालभाति

“Skull-shining”; rapid active exhalations. One of the ṣaṭkarmas.

**karma** कर्म

Action, and the law by which actions bear fruit.

**karma yoga** कर्मयोग

The yoga of selfless action.

**kīrtan** कीर्तन

Devotional singing, often call and response.

**kleśa** क्लेश

Affliction; the five roots of suffering (YS 2.3).

**kośa** कोश

Sheath; one of the five layers of the person (Taittiriya Upaniṣad).

**kriyā** क्रिया

Action; a practice, especially a cleansing practice.

**kriyā yoga** क्रियायोग

The yoga of action: tapas, svādhyāya and īśvara-praṇidhāna (YS 2.1).

kumbhaka कुम्भक

Breath retention (antara, after inhalation; bāhya, after exhalation; kevala, spontaneous).

kuṇḍalīni कुण्डलिनी

“The coiled one”; the dormant power said to rest at the base of the spine.

## M

mahāvākya महावाक्य

“Great saying”; the four Upaniṣadic statements of the identity of ātman and brahman.

mālā माला

String of beads (usually 108) for counting mantra repetitions.

manas मनस्

Mind; the faculty that receives impressions, doubts and deliberates.

maṇipūra cakra मणिपूर

The navel centre; “city of jewels”.

manomaya kośa मनोमय

The mental sheath.

mantra मन्त्र

Sacred sound, word or verse used in meditation and worship.

māyā माया

The power by which the one appears as many; appearance.

mokṣa मोक्ष

Liberation from the cycle of rebirth.

mudrā मुद्रा

Seal or gesture; hand gestures and Haṭha techniques.

mūla bandha मूलबन्ध

The root lock; contraction of the pelvic floor.

mūlādhāra cakra मूलाधार

The root centre at the base of the spine.

## N

nāda नाद

Sound; the inner sound meditated on in Haṭha Yoga.

nāḍī नाडी

Channel of prāṇa in the subtle body.

nāḍī śodhana नाडीशोधन

“Purification of the channels”; alternate-nostril breathing.

niṣkāma karma निष्काम कर्म

Action without craving for its fruits (Bhagavad Gītā).

niyama नियम

Observance; the second limb (śauca, santoṣa, tapas, svādhyāya, īśvara-praṇidhāna).

## O

Oṃ ॐ

The sacred syllable (praṇava), analysed as A-U-M and silence.

## P

paramparā परम्परा

Lineage; the succession of teachers.

piṅgalā पिङ्गला

The right, “solar” nāḍī, associated with the right nostril.

prakṛti प्रकृति

Nature, matter; in Sāṃkhya, all that can be experienced, including the mind.

prāṇa प्राण

Breath, vital energy; also the inward-moving vāyu of the chest.

prāṇamaya kośa प्राणमय

The vital sheath.

prāṇāyāma प्राणायाम

Regulation of the breath; the fourth limb.

pratipakṣa-bhāvanam प्रतिपक्षभावनम्

Cultivating the opposite (YS 2.33).

pratyāhāra प्रत्याहार

Withdrawal of the senses; the fifth limb.

puruṣa पुरुष

Pure consciousness, the seer; in Sāṃkhya, distinct from prakṛti.

## R

rāga राग

Attachment; the third kleśa (YS 2.7).

rāja yoga राजयोग

“Royal yoga”; the state of samādhi (Haṭha texts); the eight-limbed path (modern usage).

rajas रजस्

The guṇa of activity and passion.

## S

sādhana साधन

Spiritual practice.

sahasrāra सहस्रार

The thousand-petalled centre at the crown of the head.

sākṣī साक्षी

The witness; awareness observing all experience.

samādhi समाधि

Absorption; the eighth limb (YS 3.3).

samāna समान

The balancing vāyu of the navel region, governing digestion.

Sāṃkhya साङ्ख्य

“Enumeration”; the dualist philosophy underlying classical Yoga.

saṃskāra संस्कार

Latent impression left by experience.

saṃyama संयम

Dhāraṇā, dhyāna and samādhi practised together on one object (YS 3.4).

saṅkalpa संकल्प

Intention or resolve, as used in yoga nidrā.

śānti शान्ति

Peace. The name of the school, from Malayalam ശാന്തി.

santoṣa संतोष

Contentment; the second niyama.

ṣaṭkarma षट्कर्म

The six cleansing practices of Haṭha Yoga.

sattva सत्त्व

The guṇa of clarity, lightness and harmony.

satya सत्य

Truthfulness; the second yama.

śauca शौच

Cleanliness, purity; the first niyama.

siddhi सिद्धि

Attainment; an extraordinary power (YS chapter 3).

śraddhā श्रद्धा

Faith, trust.

sthira स्थिर

Steady, firm (YS 2.46).

sukha सुख

Ease, comfort, happiness (YS 2.46).

suṣumṇā सुषुम्णा

The central channel of the subtle body.

sūtra सूत्र

Thread; an aphorism.

svādhiṣṭhāna cakra स्वाधिष्ठान

The sacral centre; “one’s own seat”.

svādhyāya स्वाध्याय

Study; recitation; self-study. The fourth niyama.

## T

tamas तमस्

The guṇa of inertia and heaviness.

tapas तपस्

Heat; disciplined effort; austerity. The third niyama.

tattva तत्त्व

Principle; one of the 25 categories of Sāṃkhya.

trāṭaka त्राटक

Steady gazing, usually at a candle flame.

turiya तुरीय

“The fourth”; pure awareness beyond waking, dreaming and deep sleep.

## U

udāna उदान

The upward vāyu of throat and head.

uḍḍiyāna bandha उड्डीयान बन्ध

The abdominal lock, applied after exhalation.

ujjāyī उज्जायी

“Victorious”; breathing with slight closure of the glottis.

Upaniṣad उपनिषद्

Philosophical texts concluding the Vedas.

## V

vairāgya वैराग्य

Dispassion, detachment (YS 1.15).

vāsanā वासना

Latent tendency formed from accumulated impressions.

vāyu वायु

Wind; one of the five functions of prāṇa.

Veda वेद

“Knowledge”; the oldest sacred texts of the Hindu tradition.

Vedānta वेदान्त

“End of the Veda”; philosophy based on the Upaniṣads, Gitā and Brahma Sūtras.

vijñānamaya kośa विज्ञानमय

The sheath of discernment.

viparītakaraṇī विपरीतकरणी

“Reversing action”; inverted mudrā, ancestor of shoulderstand.

Viṣṇu mudrā विष्णुमुद्रा

Hand position for alternate-nostril breathing.

viśuddha cakra विशुद्ध

The throat centre; “purified”.

viveka विवेक

Discrimination between the real and the unreal.

vṛtti वृत्ति

Fluctuation or movement of the mind (YS 1.2, 1.5).

vyāna व्यान

The pervading vāyu, governing circulation and co-ordination.

## Y

yama यम

Restraint; the first limb (ahiṃsā, satya, asteya, brahmacharya, aparigraha).

yoga योग

From *yuj*, to yoke or to be absorbed; both the discipline and its goal.

yoga nidrā योगनिद्रा

“Yogic sleep”; a guided practice of deep relaxation and awareness.



## APPENDIX

## B

# Anatomy and Physiology Glossary

Anatomical and physiological terms used in Section II. Directional terms assume the anatomical position (Figure 16.1). For individual muscles see the muscle atlas (Chapter 19) and Appendix E.

## A

**abduction**

Movement away from the midline in the frontal plane.

**active insufficiency**

Inability of a two-joint muscle to shorten fully over both joints at once.

**active range of motion**

The range a person reaches and controls with their own muscles.

**adduction**

Movement towards the midline in the frontal plane.

**afferent**

Carrying signals towards the central nervous system (sensory).

**agonist**

The muscle chiefly responsible for a movement (prime mover).

**alveoli**

Tiny air sacs of the lungs where gas exchange occurs.

**anatomical position**

Standing upright, facing forward, feet together, arms by the sides, palms forward; the reference for all anatomical terms.

**annulus fibrosus**

The tough fibrocartilaginous outer ring of an intervertebral disc.

**antagonist**

A muscle opposing the action of the agonist.

**anterior**

Towards the front of the body.

**ASIS**

Anterior superior iliac spine; the front “hip point” of the pelvis.

**autonomic nervous system**

The part of the nervous system regulating organs, vessels and glands; sympathetic and parasympathetic branches.

## B

**baroreflex**

Reflex adjustment of heart rate and vessel tone in response to changes in blood pressure.

## C

**cervical**

Relating to the neck; the seven cervical vertebrae.

**circumduction**

Circular movement combining flexion, abduction, extension and adduction.

**closed chain**

Movement in which the end of the limb is fixed, e.g. against the floor.

**compression**

A load pressing tissues together; a movement limit felt on the side moved towards.

**concentric contraction**

Muscle shortening while producing force.

**contralateral**

On the opposite side.

**coronal plane**

See frontal plane.

## D

**diaphragm**

Dome-shaped muscle between chest and abdomen; the principal muscle of inspiration.

**distal**

Further from the trunk (on a limb).

**dorsiflexion**

Lifting the top of the foot towards the shin.

**E****eccentric contraction**

Muscle lengthening while producing force, controlling a movement.

**efferent**

Carrying signals away from the central nervous system (motor).

**end range**

The last part of a joint's range, where passive tissues become taut.

**eversion**

Turning the sole of the foot outward.

**extension**

Increasing the angle between segments; returning from flexion (sagittal plane).

**F****facet joint**

Zygapophyseal joint; the paired synovial joints between vertebral arches.

**fascia**

Sheets and wrappings of connective tissue around and within muscles and organs.

**flexion**

Decreasing the angle between segments (sagittal plane).

**frontal plane**

Vertical plane dividing front from back; abduction, adduction and lateral flexion occur in it.

**functional residual capacity**

Air remaining in the lungs after a quiet exhalation.

**G****glenohumeral joint**

Ball-and-socket joint between the humerus and the glenoid of the scapula.

**Golgi tendon organ**

Sensory receptor in tendons sensitive to tension.

**H****heart rate variability**

Variation in the interval between heartbeats; reflects autonomic regulation.

**hyperextension**

Extension beyond the neutral or anatomical position.

**hypermobility**

Joint range greater than typical; sometimes part of a connective-tissue disorder.

**hypocapnia**

Low carbon dioxide in the blood, caused by over-breathing.

**I****inferior**

Towards the feet.

**interoception**

Sense of the internal physiological condition of the body.

**intra-abdominal pressure**

Pressure within the abdominal cavity, raised by diaphragm, abdominal wall and pelvic floor.

**intraocular pressure**

Fluid pressure inside the eye; rises in inverted positions.

**inversion (foot)**

Turning the sole of the foot inward.

**ipsilateral**

On the same side.

**ischial tuberosity**

The "sitting bone"; origin of the hamstrings.

**isometric contraction**

Muscle producing force without changing length.

**K****kyphosis**

A curve convex posteriorly, normal in the thoracic spine.

**L****lateral**

Away from the midline.

**lateral flexion**

Side-bending of the spine in the frontal plane.

**ligament**

Dense connective tissue joining bone to bone.

**lordosis**

A curve concave posteriorly, normal in the cervical and lumbar spine.

**lumbar**

Relating to the lower back; the five lumbar vertebrae.

**M****medial**

Towards the midline.

**meniscus**

C-shaped fibrocartilage disc deepening the knee joint.

**motor unit**

A motor neuron and the muscle fibres it controls.

**muscle spindle**

Receptor within muscle sensitive to length and rate of stretch.

**N****nucleus pulposus**

The gel-like centre of an intervertebral disc.

**nutiation**

Small forward nodding movement of the sacrum between the ilia.

**O****open chain**

Movement in which the end of the limb is free.

**P****parasympathetic**

Autonomic branch supporting rest, digestion and repair.

**passive insufficiency**

Inability of a two-joint muscle to lengthen fully over both joints at once.

**passive range of motion**

Range reached with external help.

**pelvic tilt**

Forward (anterior) or backward (posterior) rotation of the pelvis on the femoral heads.

**plantarflexion**

Pointing the foot downward.

**posterior**

Towards the back of the body.

**pronation**

Turning the palm downward (forearm).

**proprioception**

Sense of the position and movement of the body.

**protraction**

Forward movement of the scapula around the rib cage.

**proximal**

Closer to the trunk (on a limb).

**PSIS**

Posterior superior iliac spine; the “dimples” at the low back.

**R****reciprocal inhibition**

Reflex relaxation of an antagonist when its agonist contracts.

**residual volume**

Air remaining in the lungs after maximal exhalation.

**respiratory sinus arrhythmia**

Rise in heart rate with inhalation and fall with exhalation.

**retraction**

Backward movement of the scapula towards the spine.

**rotation**

Turning about a long axis (transverse plane).

**rotator cuff**

Supraspinatus, infraspinatus, teres minor and subscapularis; stabilisers of the shoulder.

**S****sacroiliac joint**

Joint between the sacrum and ilium; strong, with small movement.

**sagittal plane**

Vertical plane dividing left from right; flexion and extension occur in it.

**scapulohumeral rhythm**

Coordinated movement of scapula and humerus as the arm rises.

**stabiliser**

A muscle holding a segment steady so that movement can occur elsewhere.

**superior**

Towards the head.

**supination**

Turning the palm upward (forearm).

**sympathetic**

Autonomic branch preparing the body for action.

**synergist**

A muscle assisting the agonist or neutralising unwanted actions.

**synovial joint**

A freely movable joint with a fluid-filled capsule.

**T****tendon**

Connective tissue joining muscle to bone.

**tension (tensile load)**

A load pulling tissues apart; a movement limit felt on the side being lengthened.

**thoracic**

Relating to the chest; the twelve thoracic vertebrae.

**tidal volume**

Air moved in a normal quiet breath (about half a litre).

**transverse plane**

Horizontal plane dividing upper from lower; rotation occurs in it.

**V**

---

**vagus nerve**

Cranial nerve X; carries most parasympathetic fibres, mostly sensory.

**vital capacity**

Maximum air exhaled after maximal inhalation.

## APPENDIX

## C

# Asana Index (English)

Postures in this manual, alphabetically by English name, with the Sanskrit name and a page reference.

★ = full illustrated monograph in **Chapter 24**; other page numbers point to the mapping tables in **Chapter 25** or to the chapter where the pose is taught.

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# Sanskrit-to-English Pose Index

Postures in this manual, alphabetically by Sanskrit name (ignoring diacritics), with English equivalents. Most Sanskrit posture names are compounds ending in *āsana*: *paścima* (west, the back of the body) + *ut-tāna* (intense stretch) + *āsana* gives *paścimottānāsana*, the “intense stretch of the back of the body”. Learning a few dozen common elements — *ardha* (half), *eka* (one), *pāda* (foot), *hasta* (hand), *supta* (reclining), *baddha* (bound), *parivṛtta* (revolved), *ūrdhva* (upward), *adho* (downward), *mukha* (face) — makes most names transparent.

★ = full illustrated monograph in **Chapter 24**; other page numbers point to the mapping tables in **Chapter 25** or to the chapter where the pose is taught.

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| Gomukhāsana — Cow face .....                      | 252   |
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| Legs up the wall — Legs up the wall .....         | 254   |
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| Uṣṭrāsana — Camel .....                                 | 253   |
| Utkaṭāsana — Chair .....                                | 250   |
| Vajrāsana — Thunderbolt (kneeling) .....                | 252   |
| Viparīta Karaṇī — Half shoulderstand / legs up .....    | 254   |
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## APPENDIX

## E

## Muscle Index

The muscles described in the muscle atlas (Chapter 19), alphabetically, with their region and the page of their atlas card. Each card gives origin, insertion, actions, role in yoga, whether the muscle more often needs stretching or strengthening, postures that use it and common misconceptions.

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|                                                              |     | Triceps brachii — Shoulder girdle and arm.....          | 153 |

# Teaching Cue Library

A working collection of cues for the practices in this manual, gathered from the chapters on the twelve postures, prāṇāyāma, relaxation and teaching skills. Use them as a starting point, not a script: the best cue is the one that helps the student in front of you, in your own voice (Chapter 33). Each entry gives the pose or moment, the cue, and why it works or what to watch for.

## Arriving, relaxing and closing

Speak slowly, leave silence, and lower the pitch slightly. Invite rather than command.

| Pose / moment            | Cue                                                                                         | Why it works / watch for                                                     |
|--------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Opening relaxation       | "Let the whole weight of the body be taken by the floor."                                   | Gives permission to release; avoids "relax!" as an order.                    |
| Opening relaxation       | "Notice the breath without changing it."                                                    | Establishes the witnessing attitude used throughout the class.               |
| Śavāsana                 | "Feet a little wider than the hips, arms away from the body, palms up."                     | Space between limbs lets the shoulders and legs roll outward and rest.       |
| Śavāsana                 | "If the low back is uncomfortable, bend the knees or place a bolster under them."           | Normalises options; relieves lumbar extension.                               |
| Final relaxation         | "Tense ... hold ... and let go completely."                                                 | Contrast teaches students to feel the difference between effort and release. |
| Coming out of relaxation | "Begin to deepen the breath. Move fingers and toes. Roll to the right side and rest there." | Gradual return protects blood pressure and the lower back.                   |

## Breath

| Pose / moment       | Cue                                                                                | Why it works / watch for                                                 |
|---------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Abdominal breathing | "Let the belly rise as you breathe in and fall as you breathe out."                | Simple image of diaphragmatic breathing; hand on abdomen gives feedback. |
| Full yogic breath   | "Fill from the bottom up — abdomen, ribs, upper chest — and empty gently."         | Watch for shoulders lifting to the ears.                                 |
| Kapālabhāti         | "Short, sharp exhalations from the belly; the inhalation happens by itself."       | The active phase is exhalation; stop if dizzy.                           |
| Anuloma viloma      | "Keep the breath smooth and silent; the count serves the breath, not the reverse." | Prevents strain and air hunger; retention is optional for beginners.     |
| Any posture         | "If the breath becomes short or held, you have gone too far — come back a little." | Breath is the most reliable guide to appropriate effort.                 |



## Sūrya namaskār

| Pose / moment     | Cue                                                                                     | Why it works / watch for                                       |
|-------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Prayer pose       | "Stand evenly on both feet; palms together at the chest."                               | A still, grounded beginning.                                   |
| Raised arms       | "Inhale, arms up; lift the chest more than you bend the back."                          | Keeps the extension spread through the thoracic spine.         |
| Hand to foot      | "Exhale, fold forward from the hips; bend the knees as much as you need."               | Protects the lumbar spine and hamstrings.                      |
| Lunge             | "Step the right foot back; front knee over the ankle; lower the back knee if you like." | Knee over ankle limits shear; knee-down option for all levels. |
| Eight-limbed pose | "Knees, chest, chin down; hips stay up."                                                | Beginners may lower the knees first, then chest.               |
| Cobra             | "Hands under shoulders; lift with the back muscles, elbows soft."                       | Avoids pushing into the lumbar spine with the arms.            |
| Downward dog      | "Press the floor away; lift the hips; heels towards the floor, not necessarily on it."  | Length in the spine is more important than straight legs.      |

## Headstand and its preparation

Teach headstand only to students who are ready, step by step, with a clear exit. See Chapter 24.

| Pose / moment    | Cue                                                                                          | Why it works / watch for                        |
|------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------|
| Setting the arms | "Grasp the elbows, then release the hands without moving the elbows; interlace the fingers." | Sets the forearm width to the shoulders.        |
| Head placement   | "Crown of the head on the floor, back of the head cradled in the hands."                     | Most of the weight should stay on the forearms. |
| Walking in       | "Walk the feet towards the face until the hips are over the shoulders."                      | Stack before lifting; do not jump.              |
| In the pose      | "Press the forearms down so the shoulders lift away from the floor."                         | Reduces load on the cervical spine.             |
| Coming down      | "Lower slowly with control, rest in child's pose for several breaths."                       | Allows blood pressure to normalise.             |

## Shoulderstand, plough and fish

| Pose / moment | Cue                                                                                                      | Why it works / watch for                                                        |
|---------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Sarvāṅgāsana  | "Hands support the back close to the shoulder blades; elbows no wider than the shoulders."               | Gives a stable base; a folded blanket under the shoulders reduces neck flexion. |
| Sarvāṅgāsana  | "Do not turn the head while in the pose."                                                                | Protects the cervical spine.                                                    |
| Halāsana      | "Feet to the floor only if they reach easily; otherwise rest them on a chair or keep hands on the back." | Limits end-range cervical flexion.                                              |
| Matsyāsana    | "Lift the chest from the elbows; the head rests lightly — the weight is on the elbows."                  | Avoids compressing the neck.                                                    |

## Forward bends

| Pose / moment        | Cue                                                                        | Why it works / watch for                                     |
|----------------------|----------------------------------------------------------------------------|--------------------------------------------------------------|
| Paścimottānāsana     | "Inhale, lengthen the spine; exhale, fold from the hips."                  | Hip-hinging protects the discs and lengthens the hamstrings. |
| Paścimottānāsana     | "Hold the legs where you comfortably can — the shins or a strap are fine." | Removes the pressure to reach the feet.                      |
| Seated forward bends | "Sit on a folded blanket if the lower back rounds."                        | Tilts the pelvis forward when the hamstrings are tight.      |
| Any forward bend     | "Let each exhalation release you a little further — never pull."           | Pulling provokes protective muscle guarding.                 |

## Backbends

| Pose / moment   | Cue                                                                         | Why it works / watch for                                 |
|-----------------|-----------------------------------------------------------------------------|----------------------------------------------------------|
| Bhujāṅgāsana    | "Lift the head and chest with the back muscles; hands light on the floor."  | Builds back strength; protects the low back.             |
| Bhujāṅgāsana    | "Keep the elbows bent and close to the body; shoulders away from the ears." | Avoids shrugging and over-extension.                     |
| Śalabhāsana     | "Lengthen the leg back before you lift it."                                 | Length before height spreads the load.                   |
| Dhanurāsana     | "Kick the feet into the hands to lift the chest."                           | The legs do the work; the arms stay straight like ropes. |
| After backbends | "Rest in child's pose, or lie on the front with head to one side."          | Neutral rest before further work.                        |

## Twists, standing postures and balance

| Pose / moment        | Cue                                                                              | Why it works / watch for                                        |
|----------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Ardha matsyendrāsana | "Inhale, grow tall; exhale, turn from the belly, then the chest, then the head." | Rotation distributed along the spine; lengthen before twisting. |
| Ardha matsyendrāsana | "Keep both sitting bones grounded."                                              | Stabilises the pelvis.                                          |
| Trikoṇāsana          | "Reach out along the side before you go down; hand on the shin, block or floor." | Lateral length before depth; props make the pose accessible.    |
| Trikoṇāsana          | "Both sides of the waist long."                                                  | Avoids collapsing into the lower side.                          |
| Vṛkṣāsana            | "Foot on the inner calf or thigh — not on the knee."                             | Avoids a sideways load on the knee joint.                       |
| Any balance          | "Fix the gaze on one still point at eye level."                                  | Visual fixation improves balance.                               |

## Arm balances

| Pose / moment | Cue                                                                             | Why it works / watch for       |
|---------------|---------------------------------------------------------------------------------|--------------------------------|
| Kākāsana      | "Spread the fingers; knees high on the backs of the upper arms."                | Wide hands and a stable shelf. |
| Kākāsana      | "Lean forward until the feet feel light — put a cushion in front of your face." | Reduces fear of falling.       |

| Pose / moment | Cue                                                                    | Why it works / watch for                         |
|---------------|------------------------------------------------------------------------|--------------------------------------------------|
| Mayūrāsana    | "Elbows into the abdomen, fingers pointing back; lean forward slowly." | Teach only with good wrist and core preparation. |

## Cues to avoid, and what to say instead

Some familiar cues are unclear, unsafe or not supported by anatomy. Replace them with specific, accurate language.

| Pose / moment                       | Cue                                                        | Why it works / watch for                                                           |
|-------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
| "Tuck the tailbone"                 | "Let the tailbone drop slightly towards the heels."        | Aggressive tucking flattens the lumbar curve; ask for a small, neutral adjustment. |
| "Suck the belly in"                 | "Let the belly be soft, with gentle tone below the navel." | Rigid gripping restricts diaphragmatic breathing.                                  |
| "Lock the knees"                    | "Straighten the legs without pushing the knees back."      | Avoids hyperextension, especially in hypermobile students.                         |
| "Squeeze the glutes" (in backbends) | "Let the buttocks work enough to support the lift."        | Maximal gluteal gripping often turns the thighs out and compresses the low back.   |
| "Breathe into your kidneys"         | "Feel the back of the ribs expand as you inhale."          | Accurate and still evocative.                                                      |
| "This detoxes the liver"            | "Many people find twists refreshing."                      | Avoid unsupported health claims (Chapter 30).                                      |
| "Everyone should be able to..."     | "Find the version that suits you today."                   | Inclusive, honest, and safer.                                                      |

# Sample Class Plans

Six complete class plans built on the principles of Chapter 34 and Chapter 35. Each is a model to learn from and adapt, not a fixed routine. Timings are approximate; protect the final relaxation whatever else you shorten. Before every class, ask about injuries, health conditions and pregnancy, and offer the options shown.

## G.1 The classical class (90 minutes)

The open class as taught at the school (Chapter 24). Its fixed order — relaxation, breath, sun salutation, leg raises, the twelve postures from inversions to standing poses, and deep relaxation — is itself a sequencing lesson: each posture prepares for or counterbalances its neighbours.



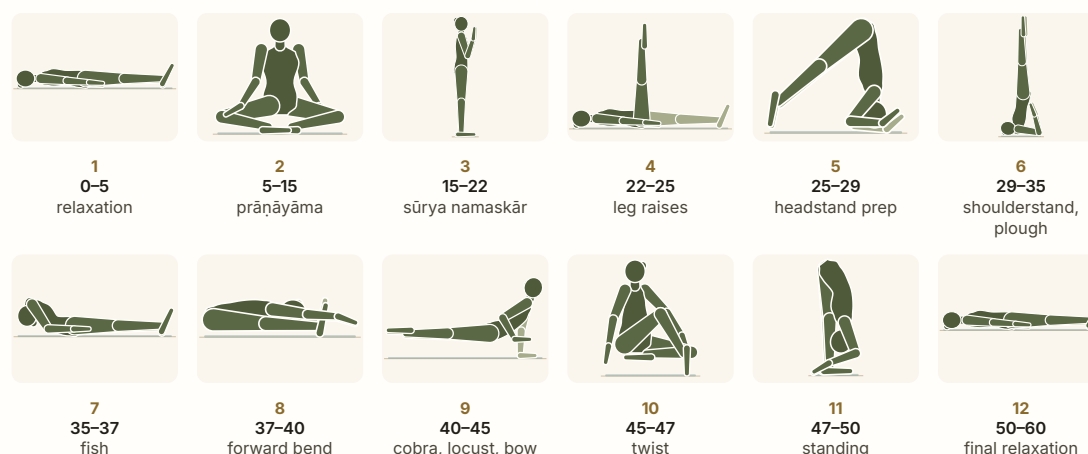
**Table G.1** The classical open class, 90 minutes

| Time  | Practice                                             | Notes and options                                            |
|-------|------------------------------------------------------|--------------------------------------------------------------|
| 0–5   | Opening chant; relaxation in śavāsana                | abdominal breathing; arrive and settle                       |
| 5–20  | Kapālabhāti (3 rounds); anuloma viloma (5–10 rounds) | beginners omit retention; seated on a cushion or chair       |
| 20–30 | Sūrya namaskār, 6–12 rounds                          | slow first rounds; knee-down option                          |
| 30–35 | Single and double leg raises; relaxation             | bend the knees; hands under the pelvis if the low back lifts |

| Time  | Practice                                                                               | Notes and options                                                   |
|-------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 35–43 | Śīrṣāsana or its preparation; child's pose                                             | dolphin for those not ready; never a first-class pose               |
| 43–53 | Sarvāṅgāsana, halāsana, matsyāsana                                                     | blanket under the shoulders; plough feet on a chair; supported fish |
| 53–58 | Paścimottānāsana; inclined plane                                                       | strap; blanket under the pelvis                                     |
| 58–66 | Bhujahṅgāsana, śalabhāsana, dhanurāsana; child's pose                                  | half locust, half bow; rest between                                 |
| 66–70 | Ardha matsyendrāsana                                                                   | easy seated twist                                                   |
| 70–73 | Kākāsana or mayūrāsana                                                                 | squat (mālāsana) as preparation                                     |
| 73–76 | Pāda hastāsana                                                                         | knees bent; hands on shins                                          |
| 76–79 | Trikoṇāsana                                                                            | block; hand on shin                                                 |
| 79–90 | Final relaxation: tense-and-release, then guided relaxation and silence; closing chant | a blanket for warmth; roll to the side to come up                   |

## G.2 A 60-minute general class

A compressed version of the classical order for studios and lunchtime classes. Fewer rounds, shorter holds, and alternatives to the full inversions — but the same arc (Figure 34.2).



**Table G.2** A 60-minute general class

| Time  | Practice                                      | Key cues                          | Options                         |
|-------|-----------------------------------------------|-----------------------------------|---------------------------------|
| 0–5   | Relaxation                                    | "Let the floor take your weight." | knees bent on a bolster         |
| 5–15  | Kapālabhāti 2 rounds; anuloma viloma 5 rounds | "Smooth and silent."              | no retention                    |
| 15–22 | Sūrya namaskār, 4–6 rounds                    | "Move with the breath."           | knee-down lunge                 |
| 22–25 | Leg raises                                    | "Lower slowly."                   | single leg only                 |
| 25–29 | Dolphin / headstand preparation               | "Press the forearms down."        | child's pose                    |
| 29–35 | Shoulderstand, plough                         | "Keep the head still."            | legs up the wall; feet on chair |
| 35–37 | Fish                                          | "Weight on the elbows."           | supported fish                  |
| 37–40 | Seated forward bend                           | "Fold from the hips."             | strap, blanket                  |
| 40–45 | Cobra, half locust, bow                       | "Lift with the back."             | sphinx; half bow                |

| Time  | Practice                        | Key cues                | Options                |
|-------|---------------------------------|-------------------------|------------------------|
| 45–47 | Half spinal twist               | "Grow tall, then turn." | easy twist             |
| 47–50 | Standing forward bend, triangle | "Reach out, then down." | block                  |
| 50–60 | Final relaxation                | "Nothing to do."        | eyes open if preferred |

### G.3 A 90-minute general class with a theme

**Theme:** *Sthira sukham āsanam* — steadiness and ease (Chapter 7). **Intention:** to notice where effort is needed and where it can be released. Revisit the theme in a sentence or two at the opening, once in the middle, and at the close; do not lecture.

**Table G.3** A 90-minute themed class

| Time  | Section              | Practice                                                         | Teaching focus                                 |
|-------|----------------------|------------------------------------------------------------------|------------------------------------------------|
| 0–8   | Arrival              | relaxation; introduce the theme in two sentences                 | voice, pacing, silence                         |
| 8–23  | Breath               | full yogic breath; kapālabhāti 3 rounds; anuloma viloma 8 rounds | steady breath as the model of sthira and sukha |
| 23–28 | Warm-up              | cat-cow, low lunge, gentle twists                                | mobilise spine and hips                        |
| 28–38 | Sūrya namaskār       | 3 slow rounds taught in detail, then 6 flowing                   | even breath throughout                         |
| 38–42 | Leg raises           | single, then double                                              | core steadiness; soft face                     |
| 42–52 | Inversions           | headstand tiers; child's pose                                    | steadiness first, then ease                    |
| 52–62 | Shoulderstand group  | shoulderstand, plough, fish                                      | neck safety; ease in the breath                |
| 62–68 | Forward bends        | paścimottānāsana; janu śīrṣāsana                                 | ease: waiting rather than pulling              |
| 68–76 | Backbends            | cobra, locust, bow                                               | steadiness: strength from the back             |
| 76–79 | Twist                | ardha matsyendrāsana                                             | length, then rotation                          |
| 79–82 | Balance              | crow preparation or vṛkṣāsana                                    | steady gaze                                    |
| 82–85 | Standing             | pāda hastāsana, trikoṇāsana                                      | even weight                                    |
| 85–90 | Relaxation and close | guided relaxation; one sentence on the theme; silence            | ending on time                                 |

### G.4 A beginners' class (60–75 minutes)

For a first course of four to six weeks. Teach less and explain more; introduce one new element each week; give every posture a simpler option and time to rest (Chapter 32).

**Table G.4** A beginners' class, week 2 of a 6-week course

| Time  | Practice                                                    | Why / teaching notes                               |
|-------|-------------------------------------------------------------|----------------------------------------------------|
| 0-10  | Relaxation; abdominal breathing with a hand on the belly    | teaches diaphragmatic breathing by feel            |
| 10-18 | Full yogic breath; nāḍī śodhana without retention, 5 rounds | introduce Viṣṇu mudrā; allow the free hand to rest |
| 18-22 | Cat-cow; low lunge                                          | warms the spine and hips before sūrya namaskār     |
| 22-32 | Sūrya namaskār taught in three parts, then 3 slow rounds    | demonstrate once; then teach from observation      |
| 32-35 | Single leg raises                                           | knees soft; lower back relaxed                     |
| 35-40 | Dolphin; child's pose                                       | builds shoulder strength for later headstand       |
| 40-46 | Viparīta karaṇī or legs up the wall; bridge; supported fish | introduces inversion and its counterpose safely    |
| 46-50 | Forward bend with strap                                     | "fold from the hips"                               |
| 50-56 | Sphinx; half locust; child's pose                           | back strength without load on the arms             |
| 56-59 | Easy twist                                                  | sitting on a cushion                               |
| 59-63 | Standing forward bend with bent knees; triangle with block  | stable, accessible standing poses                  |
| 63-75 | Final relaxation with tense-and-release                     | the practice students most often remember          |

## G.5 An intermediate class (90 minutes)

For students who practise regularly, hold the basic postures steadily, and have learned headstand with a teacher. Longer holds, retention in prāṇāyāma, and one or two variations per posture group.

**Table G.5** An intermediate class

| Time  | Practice                                                                              | Progression                                       |
|-------|---------------------------------------------------------------------------------------|---------------------------------------------------|
| 0–5   | Relaxation                                                                            | shorter; students settle quickly                  |
| 5–25  | Kapālabhāti 3 × 40–60; anuloma viloma with retention 1:4:2, 10 rounds                 | stop if strain appears; count adapted to capacity |
| 25–35 | Sūrya namaskār, 12 rounds                                                             | steady pace, continuous breath                    |
| 35–40 | Double leg raises                                                                     | lower slowly on the exhalation                    |
| 40–48 | Śīrṣāsana 3–5 min, with leg variations for those ready                                | child's pose after                                |
| 48–60 | Sarvāṅgāsana 3–5 min; halāsana; bridge from shoulderstand for those ready; matsyāsana | counterpose held half the time of shoulderstand   |
| 60–66 | Paścimottānāsana; nāvāsana                                                            | forward bend, then abdominal strength             |
| 66–74 | Bhujāṅgāsana, full śalabhāsana, dhanurāsana, uṣṭrāsana                                | child's pose between                              |
| 74–77 | Ardha matsyendrasana                                                                  | full arm bind if available                        |
| 77–80 | Mayūrāsana or kākāsana                                                                | wrist preparation first                           |
| 80–84 | Pāda hastāsana; trikoṇāsana; vṛkṣāsana                                                | balance with eyes closed for a few breaths        |
| 84–90 | Final relaxation                                                                      | silence for the last minutes                      |



## G.6 A mixed-level class (75 minutes)

Plan every posture in **tiers** (Chapter 32): demonstrate or describe the first tier, then offer the next. Name tiers neutrally (“option one, option two”), never “easy” and “advanced”. Keep inversions optional and give a clear alternative to everyone who does not do them.



**Table G.6** A mixed-level class — tiered options

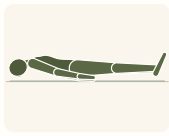
| Posture group  | Tier 1                                        | Tier 2                    | Tier 3          |
|----------------|-----------------------------------------------|---------------------------|-----------------|
| Prāṇāyāma      | abdominal breath, nāḍī śodhana                | anuloma viloma 1:2        | with retention  |
| Sūrya namaskār | knee-down lunge and plank                     | full, slow                | full, 12 rounds |
| Headstand      | dolphin                                       | knees tucked              | full pose       |
| Shoulderstand  | legs up the wall                              | half shoulderstand        | full pose       |
| Plough         | omitted or feet on chair                      | feet on floor             | —               |
| Fish           | supported on bolster                          | classic, weight on elbows | lotus legs      |
| Forward bend   | strap, knees bent                             | hands to shins or feet    | full fold       |
| Backbends      | sphinx, half locust                           | cobra, full locust        | bow             |
| Twist          | easy seated twist                             | half spinal twist         | full bind       |
| Arm balance    | squat                                         | crow, one foot lifted     | crow            |
| Standing       | forward bend, knees bent; triangle with block | classic forms             | hold longer     |

## G.7 A restorative interlude

For tired, stressed, unwell or recovering students — or as the second half of a workshop. Hold each pose 3–8 minutes with ample props; keep the room warm; talk little.



1  
Supported child's  
pose



2  
Supported fish



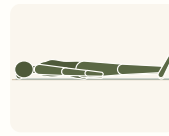
3  
Legs up the wall



4  
Bound angle  
supported



5  
Gentle twist



6  
Relaxation  
15 min

### ♀ ADAPTING ANY PLAN

Before class, ask: *Who is coming? What do they need today?* After class, write three lines in your practicum log (Appendix I): what worked, what did not, what you will change.

## APPENDIX

## H

# Sample Prāṇāyāma and Meditation Sessions

Two model sessions, with a guided script, that trainees teach during the practicum. The techniques, contraindications and evidence are in **Chapter 27** and **Chapter 28**. Read the scripts slowly, pausing where marked (...); the pauses are part of the teaching.

## H.1 A 30-minute prāṇāyāma session

**Suitable for:** generally healthy beginners and mixed groups. **Setting:** seated on a cushion, bench or chair, spine upright and comfortable; empty stomach (two to three hours after a meal). **Screening:** ask about pregnancy, high blood pressure, heart disease, recent surgery, glaucoma, epilepsy and panic or anxiety disorders; offer the gentle options below.

**Table H.1** A 30-minute prāṇāyāma session

| Time  | Practice                        | Instructions                                                       | Options and safety                                                                                          |
|-------|---------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 0–4   | Settling                        | sit comfortably; observe the natural breath                        | chair, wall support                                                                                         |
| 4–9   | Abdominal and full yogic breath | 10 slow breaths of each; hands on belly, then ribs                 | no forcing; shoulders relaxed                                                                               |
| 9–15  | Kapālabhāti                     | 3 rounds of 20–40 exhalations, each followed by 3–4 normal breaths | omit in pregnancy, hypertension, recent abdominal surgery, dizziness; replace with slow abdominal breathing |
| 15–25 | Anuloma viloma / nāḍī śodhana   | beginners: 4:8 (in:out), no retention; experienced: 1:4:2          | never strain in retention; omit retention in pregnancy, hypertension, heart disease, anxiety                |
| 25–28 | Bhrāmarī                        | 5–7 rounds of humming exhalation                                   | quiet version if self-conscious                                                                             |
| 28–30 | Silence                         | sit and observe; roll the shoulders; open the eyes slowly          | —                                                                                                           |

### SCRIPT: INTRODUCING NĀḌĪ ŚODHANA

Bring the right hand up and fold down the first two fingers, so that the thumb, ring finger and little finger are free ... This is Viṣṇu mudrā. Let the left hand rest on the thigh ... Close the right nostril with the thumb, and breathe out slowly through the left ... Breathe in through the left for a count of four ... Close the left nostril with the ring finger; release the thumb and breathe out through the right for a count of eight ... Breathe in through the right ... close ... and breathe out through the left ... That is one round. If the count feels long, shorten it: the breath should stay smooth and silent ... Let the arm rest whenever it tires.

### ⚠ STOP SIGNALS

Stop any prāṇāyāma and breathe normally if you feel dizzy, light-headed, breathless, tingling in the hands or face, anxious, or unwell. These are usually signs of over-breathing or strain (Chapter 27).

## H.2 A 25-minute guided meditation session

**Suitable for:** beginners and mixed groups. **Object:** the breath, with an optional mantra. **Structure:** posture → body scan → breath awareness → one-pointed attention → open awareness → close. Invite students to keep their eyes open or lowered if they prefer, and remind them that they can move or stop at any time (Chapter 31).

**Table H.2** A 25-minute meditation session

| Time  | Stage                 | Guidance                                                                                                            |
|-------|-----------------------|---------------------------------------------------------------------------------------------------------------------|
| 0–3   | Posture               | stable, upright, comfortable; hands resting; eyes closed or lowered                                                 |
| 3–7   | Body scan             | from the feet to the head, noticing and softening                                                                   |
| 7–12  | Breath awareness      | the breath at the nostrils or abdomen; counting 1–10 if helpful                                                     |
| 12–20 | One-pointed attention | resting on the breath (or mantra: so on inhalation, ham on exhalation, or Om); returning each time the mind wanders |
| 20–23 | Open awareness        | releasing the object; resting as the witness of sounds, sensations and thoughts                                     |
| 23–25 | Closing               | deepen the breath; move gently; open the eyes; a short sharing if appropriate                                       |

### SCRIPT: RETURNING TO THE BREATH

Rest your attention on the breath where you feel it most clearly ... at the nostrils, or the rise and fall of the belly ... There is no need to change it ... Sooner or later the mind will wander — into planning, remembering, imagining. This is not a mistake; it is what minds do ... The moment you notice that you have wandered is a moment of awareness. Gently, without judging, come back to the next breath ... and again ... and again. Each return is the practice.

### 🕒 AFTER THE SESSION

Allow a minute of silence before speaking. If you invite sharing, keep it optional and brief. Some students may feel emotional or unsettled; stay available after class and know when to suggest other support (Chapter 28, Chapter 36 on trauma-aware teaching).

## H.3 A 20-minute yoga nidrā outline

A simplified outline in the style popularised by the Bihar School (Satyananda (1976)). Students lie in śavāsana with a blanket; guide in a steady voice with long pauses.

1. **Settling** (2 min): body position, stillness, sounds from far to near.
2. **Saṅkalpa** (1 min): a short, positive resolve in the student's own words, repeated three times mentally.
3. **Rotation of consciousness** (6 min): naming body parts in sequence, right side then left, back, front, whole body.

4. **Breath awareness** (4 min): natural breath; counting backwards from 27 to 1.
5. **Sensations and images** (3 min): heaviness and lightness, warmth and coolness; a few simple, peaceful images (optional – some students prefer to omit visualisation).
6. **Saṅkalpa** (1 min): repeat the resolve.
7. **Return** (3 min): awareness of the breath, the body, the room; move slowly; roll to the side; sit up.

# Forms for Teaching and Assessment

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Photocopiable forms for use during the course and in your teaching. You may copy and adapt them for your own classes. Store completed health forms securely and confidentially (Chapter 39).

1. Student health questionnaire and consent
2. Teacher observation sheet
3. Practicum log
4. Practicum assessment form (final teaching examination)
5. Student feedback form
6. Attendance sheet

## Student health questionnaire and consent

|                                 |                                 |
|---------------------------------|---------------------------------|
| Name                            | Date of birth                   |
| Phone                           | Email                           |
| Emergency contact (name, phone) | Address / where you are staying |

**Experience.** How long have you practised yoga? Which styles?

**Health.** Do you have, or have you had, any of the following? Tick and give brief details.

|                                                                                                       |                                                                                      |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <input type="checkbox"/> High or low blood pressure                                                   | <input type="checkbox"/> Heart condition                                             |
| <input type="checkbox"/> Glaucoma, detached retina, eye surgery                                       | <input type="checkbox"/> Epilepsy or seizures                                        |
| <input type="checkbox"/> Asthma or breathing difficulties                                             | <input type="checkbox"/> Diabetes                                                    |
| <input type="checkbox"/> Back or neck problems                                                        | <input type="checkbox"/> Osteoporosis or osteopenia                                  |
| <input type="checkbox"/> Joint problems (shoulder, wrist, hip, knee, ankle)                           | <input type="checkbox"/> Hernia                                                      |
| <input type="checkbox"/> Recent surgery or injury (last 12 months)                                    | <input type="checkbox"/> Hypermobility                                               |
| <input type="checkbox"/> Pregnancy or recent birth                                                    | <input type="checkbox"/> Anxiety, depression, trauma or other mental-health concerns |
| Any other condition, medication relevant to exercise, or anything you would like the teacher to know: |                                                                                      |
|                                                                                                       |                                                                                      |

**Touch.** In class, teachers may offer hands-on adjustments. ☐ I welcome hands-on adjustments ☐ I prefer verbal guidance only. *You may change your mind at any time, and you can use the consent card in class.*

**Declaration.** I have answered these questions truthfully and will inform the teacher of any change in my health. I understand that yoga involves physical activity; that I should practise within my own limits and stop if I feel pain, dizziness or discomfort; and that yoga is not a substitute for medical care. Where I have a medical condition, I have consulted, or will consult, my doctor about taking part.

Signature

Date

## Teacher observation sheet

|          |                  |                      |
|----------|------------------|----------------------|
| Observer | Teacher observed | Date / class / level |
|----------|------------------|----------------------|

Use this sheet to learn, not to judge. Record specific examples: what the teacher said or did, and its effect.

| Area                                            | What I observed (examples) | What I can learn / use |
|-------------------------------------------------|----------------------------|------------------------|
| Arrival, welcome, health check                  |                            |                        |
| Structure and timing                            |                            |                        |
| Verbal cueing (clarity, economy, breath timing) |                            |                        |
| Demonstration                                   |                            |                        |
| Observation of students; circulation            |                            |                        |
| Options, modifications, props                   |                            |                        |
| Adjustments and consent                         |                            |                        |
| Safety                                          |                            |                        |
| Voice, silence, presence                        |                            |                        |
| Prāṇāyāma, relaxation, meditation               |                            |                        |
| Atmosphere and inclusion                        |                            |                        |

**The one thing I will try in my own teaching:**

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# Practicum log

Record every teaching session during the course: to a partner, a group or a public class. Bring the log to your mentor meetings.

[illegible]

## Practicum assessment form — final teaching examination

|              |                    |       |
|--------------|--------------------|-------|
| Trainee      | Assessor(s)        | Date  |
| Class length | Number of students | Level |

Rate each criterion: 4 confident and consistent · 3 competent · 2 developing — needs further practice · 1 not yet safe or not demonstrated. A rating of 1 in *Safety* requires re-examination.

| Criterion                                           | 4 | 3 | 2 | 1 | Comments |
|-----------------------------------------------------|---|---|---|---|----------|
| Preparation and lesson plan                         |   |   |   |   |          |
| Opening: welcome, health check, orientation         |   |   |   |   |          |
| Sequencing and counterposes                         |   |   |   |   |          |
| Verbal instruction and breath cues                  |   |   |   |   |          |
| Sanskrit and English names                          |   |   |   |   |          |
| Demonstration                                       |   |   |   |   |          |
| Observation and adaptation; options                 |   |   |   |   |          |
| <b>Safety:</b> screening, modifications, no forcing |   |   |   |   |          |
| Touch and consent (if used)                         |   |   |   |   |          |
| Prāṇāyāma taught correctly                          |   |   |   |   |          |
| Relaxation / meditation guidance                    |   |   |   |   |          |
| Voice, presence, atmosphere                         |   |   |   |   |          |
| Time management                                     |   |   |   |   |          |
| Self-reflection after the class                     |   |   |   |   |          |

**Strengths:**

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**Areas to develop:**

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**Result:** ☐ Pass ☐ Pass with recommendations ☐ Re-examination required

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Assessor signature

---

Trainee signature

---

Date

## Student feedback form

Completed anonymously by students in a trainee's class. Thank you for helping our trainee teachers learn.

| Please circle                                      | Not at all |   |   |   |   | Very much |
|----------------------------------------------------|------------|---|---|---|---|-----------|
| The instructions were clear                        | 1          | 2 | 3 | 4 | 5 |           |
| I felt safe and was offered options that suited me | 1          | 2 | 3 | 4 | 5 |           |
| The pace was right for me                          | 1          | 2 | 3 | 4 | 5 |           |
| The teacher noticed and helped me when needed      | 1          | 2 | 3 | 4 | 5 |           |
| The breathing and relaxation were well guided      | 1          | 2 | 3 | 4 | 5 |           |
| I felt welcome and respected                       | 1          | 2 | 3 | 4 | 5 |           |
| I would come to this teacher's class again         | 1          | 2 | 3 | 4 | 5 |           |

**What did the teacher do especially well?**

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**What one thing could make the class better?**

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**Any other comments:**

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## Attendance sheet

|                |               |
|----------------|---------------|
| Course / group | Course leader |
|----------------|---------------|

[illegible]

Key: ✓ present · A absent (agreed) · L late · M made up

APPENDIX

# J Daily Practice Journal

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A journal turns practice into learning. Spend five minutes each evening of the course completing one page; copy the template for the weeks after the course. Over time, the journal shows you patterns — what helps, what disturbs, how the mind and body change — that no teacher could show you. It is private: write honestly.

🕒 HOW TO USE THE JOURNAL

Write briefly and specifically. “Headstand: 2 min, steadier; neck fine” is more useful than “good practice”. Note one thing you noticed about your mind, not only your body. Once a week, read back over the week and write a short summary.

## Daily page

|      |               |                |
|------|---------------|----------------|
| Date | Day of course | Hours of sleep |
|------|---------------|----------------|

| Practice             | Time / rounds | Observations (body, breath, mind) |
|----------------------|---------------|-----------------------------------|
| Meditation           |               |                                   |
| Prāṇāyāma            |               |                                   |
| Āsana                |               |                                   |
| Relaxation           |               |                                   |
| Study / svādhyāya    |               |                                   |
| Karma yoga / service |               |                                   |

**Energy today** (circle): low · steady · restless · bright    **Mood:** \_\_\_\_\_

**One thing I learned today** (from teaching, study or practice):

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**A yama or niyama I noticed in action** (or its absence):

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**A question to bring to my teacher:**

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**Gratitude:**

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## Weekly review

**What has changed in my practice this week?**

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**What did I find difficult? What helped?**

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**What have I understood about teaching?**

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**Which text or idea stayed with me?**

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**My intention for next week:**

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## APPENDIX

## K

# The Final Examination

The final examination has two parts: a **written paper** (2 hours 30 minutes, closed book) and a **practical teaching examination** (Chapter 40). This appendix gives the structure of both and a complete sample written paper for practice. The paper set on the course follows the same pattern with different questions.

## K.1 The written paper

**Table K.1** Structure of the written paper

| Part  | Format                                                      | Marks | Suggested time |
|-------|-------------------------------------------------------------|-------|----------------|
| A     | 30 multiple-choice questions, one mark each                 | 30    | 30 min         |
| B     | 12 short-answer questions (3–5 sentences), three marks each | 36    | 60 min         |
| C     | 2 extended answers from a choice of 6, 17 marks each        | 34    | 60 min         |
| Total |                                                             | 100   | 2 h 30 min     |

The pass mark is 70, with at least 50 per cent in each part. A candidate who does not pass may re-sit once.

## K.2 Sample paper — Part A: multiple choice

Choose the single best answer.

### Philosophy and tradition

1. Yoga Sūtra 1.2 defines yoga as: (a) balance of sun and moon (b) the stilling of the fluctuations of the mind (c) union with God (d) skill in action
2. The five kleśas, in the order given by Patañjali, begin with: (a) asmitā (b) avidyā (c) dveṣa (d) rāga
3. Which is **not** one of the yamas? (a) ahimsā (b) asteya (c) aparigraha (d) santoṣa
4. The Bhagavad Gītā is part of the: (a) Mahābhārata (b) Ṛg Veda (c) Haṭha Yoga Pradīpikā (d) Rāmāyaṇa
5. Most scholars date the Yoga Sūtra to approximately: (a) the 15th century CE (b) 325–425 CE (c) 1500 BCE (d) 800 BCE
6. The *Haṭha Yoga Pradīpikā* was composed by: (a) Śaṅkara (b) Patañjali (c) Svātmārāma (d) Vyāsa
7. In the *Haṭha Yoga Pradīpikā*, the practice that comes first is: (a) prāṇāyāma (b) mudrā (c) nāḍānusandhāna (d) āsana
8. The kośa associated with the breath and vital energy is the: (a) prāṇamaya (b) vijñānamaya (c) manomaya (d) annamaya
9. The guṇa of clarity and harmony is: (a) prakṛti (b) rajas (c) sattva (d) tamas

10. Karma yoga is best described as: (a) action performed without attachment to its results (b) yoga of knowledge (c) yoga of devotion (d) meditation on the chakras

### Anatomy, physiology and safety

11. Flexion and extension occur in the: (a) transverse plane (b) coronal plane (c) sagittal plane (d) frontal plane
12. The principal muscle of quiet inhalation is the: (a) pectoralis major (b) rectus abdominis (c) diaphragm (d) sternocleidomastoid
13. The hamstrings originate on the: (a) sacrum (b) ischial tuberosity (c) greater trochanter (d) ASIS
14. In pascimottānāsana, the hamstrings are mainly: (a) contracting concentrically (b) unchanged (c) shortening (d) lengthening
15. Iliopsoas is primarily a: (a) hip flexor (b) shoulder adductor (c) hip extensor (d) knee extensor
16. The rotator cuff includes all of these **except**: (a) supraspinatus (b) infraspinatus (c) deltoid (d) subscapularis
17. The normal curve of the lumbar spine is a: (a) lordosis (b) flat back (c) scoliosis (d) kyphosis
18. Respiratory sinus arrhythmia describes: (a) an irregular breathing pattern (b) heart rate rising with inhalation and falling with exhalation (c) a heart disease (d) hyperventilation
19. Prolonged exhalation is generally associated with: (a) sympathetic dominance (b) hypocapnia (c) increased parasympathetic (vagal) influence (d) raised blood pressure
20. A student feels tingling in the hands and light-headedness during fast breathing. The most likely cause is: (a) a heart attack (b) low oxygen (c) low carbon dioxide from over-breathing (d) high blood pressure
21. Which is a contraindication for full inversions without medical advice? (a) tight hamstrings (b) flat feet (c) being a beginner at meditation (d) uncontrolled high blood pressure or glaucoma
22. In a mixed class, a student with osteoporosis should avoid: (a) breathing exercises (b) all practice (c) standing postures (d) loaded spinal flexion such as deep rounded forward bends
23. Radiating pain, numbness or weakness down a leg during practice means: (a) switch sides (b) stop and refer to a health professional (c) hold longer (d) stretch further

### Teaching methodology

24. The best time to cue an upward movement is usually on the: (a) exhalation (b) retention (c) inhalation (d) it makes no difference
25. A counterpose for shoulderstand in the classical sequence is: (a) fish (b) plough (c) headstand (d) cobra
26. When adjusting a student, the teacher must first: (a) check the time (b) obtain consent (c) demonstrate (d) apply firm pressure
27. Offering tiered options in a mixed-level class aims to: (a) speed the class (b) separate students by ability (c) avoid demonstration (d) let each student find an appropriate version safely
28. Trauma-sensitive teaching includes: (a) closing all doors and dimming lights without explanation (b) surprising students with adjustments (c) offering choice and invitational language (d) requiring eyes closed
29. Which wording is appropriate for a class description? (a) “A calm class of postures, breathing and relaxation” (b) “Detoxifies the liver” (c) “Cures anxiety” (d) “Guaranteed to fix back pain”
30. A student asks for advice about stopping their medication. You should: (a) ignore the question (b) refer them to their doctor (c) tell them to stop (d) advise them based on yoga philosophy

### K.3 Sample paper — Part B: short answers

Answer each question in three to five sentences.

1. Distinguish between *abhyāsa* and *vairāgya* and explain how they work together (YS 1.12–16).
2. Name the eight limbs of Patañjali's yoga and explain why the yamas and niyamas come first.
3. Explain, in plain language, the teaching of *niṣkāma karma* in the Bhagavad Gītā and give an example from a yoga teacher's life.
4. What is the difference between a historical claim, a traditional teaching and a scientific finding? Give one example of each from this course.
5. Describe the five kośas and suggest how a single yoga class might address each.
6. Describe the actions of the diaphragm during inhalation and exhalation, and one practical implication for teaching breath.
7. Explain why tight hamstrings can lead to lumbar flexion in seated forward bends, and give two modifications.
8. List four precautions for teaching sarvāṅgāsana, and describe one safe alternative.
9. What are the signs that a student should stop a prāṇāyāma practice, and what should they do?
10. Describe three ways of making a class accessible to a student who cannot get down to the floor.
11. Explain the stages of motor learning and how they should shape the way you teach a new posture.
12. What is informed consent for touch, and how would you obtain it in a group class?

### K.4 Sample paper — Part C: extended answers

Choose **two** questions. Plan your answer; aim for a clear structure of about one and a half to two pages each. Credit is given for accuracy, use of sources, balance between tradition and evidence, and practical application to teaching.

1. "Yoga is the stilling of the fluctuations of the mind." Explain this definition with reference to the *vṛttis*, the *kleśas* and the practices Patañjali recommends. How is it relevant to a modern āsana class?
2. Compare the paths of karma, bhakti, jñāna and rāja yoga. How can they be integrated in one practice, and how might a teacher reflect it in a 90-minute class?
3. Choose one of the twelve basic postures. Describe its technique, the joint actions and principal muscles involved, common mistakes, contraindications, modifications and a suitable counterpose. Explain what the evidence does and does not tell us about its effects.
4. Discuss the evidence for yoga's effects on the autonomic nervous system and stress. Distinguish what is well supported, what is plausible, and what is overclaimed, and explain how you would speak about it to students.
5. Plan a 75-minute class for a mixed group including a beginner, a student in the second trimester of pregnancy, and a regular practitioner with a history of low-back pain. Justify your sequencing, options and safety measures.
6. What makes a teacher trustworthy? Discuss the ethics of the teacher–student relationship, including power, boundaries, touch and scope of practice, with reference to the yamas and niyamas and to documented problems in modern yoga communities.

## K.5 The practical teaching examination

**Task.** Teach a 45- to 60-minute class (as assigned) to a group of fellow trainees and/or members of the public, following the structure of the classical class or a plan of your own approved in advance.

**Submit** your written lesson plan (Chapter 35) to the assessors 24 hours before the examination.

**Assessment** uses the practicum assessment form in Appendix I against the criteria in Table 40.2. Assessors look for safe, clear, well-sequenced teaching adapted to the people present — not for a perfect performance.

**Reflection.** Immediately after the class, you will have ten minutes to discuss with the assessors what went well, what you would change, and why.

### Q PREPARING FOR THE WRITTEN PAPER

Revise from the Key Takeaways and Knowledge Checks at the end of each chapter; test yourself with the glossaries (Appendices A and B); and practise explaining one idea from each chapter aloud to a classmate in two minutes. If you can explain it simply, you understand it.

## APPENDIX

## L

# Completion Checklist

Use this checklist to track your progress through the course. Your course leader will sign each section when it is complete. All items must be signed before the certificate is awarded (Chapter 40).

| Requirement                                                | Done                     | Signed / date |
|------------------------------------------------------------|--------------------------|---------------|
| <b>1. Attendance and participation</b>                     |                          |               |
| Attendance at all sessions (absences agreed and made up)   | <input type="checkbox"/> |               |
| Daily meditation, chanting and āsana classes               | <input type="checkbox"/> |               |
| Karma yoga (selfless service) as assigned                  | <input type="checkbox"/> |               |
| <b>2. Personal practice</b>                                |                          |               |
| Sūrya namaskār (12 rounds) with correct breathing          | <input type="checkbox"/> |               |
| The twelve basic postures, or appropriate modifications    | <input type="checkbox"/> |               |
| Kapālabhāti, anuloma viloma, bhrāmarī and ujjāyī as taught | <input type="checkbox"/> |               |
| Daily practice journal kept (Appendix J)                   | <input type="checkbox"/> |               |
| <b>3. Portfolio</b>                                        |                          |               |
| Teaching assignments, Chapters 1–15                        | <input type="checkbox"/> |               |
| Teaching assignments, Chapters 16–28                       | <input type="checkbox"/> |               |
| Teaching assignments, Chapters 29–40                       | <input type="checkbox"/> |               |
| Personal code of conduct (Chapter 30)                      | <input type="checkbox"/> |               |
| Health questionnaire prepared for your own classes         | <input type="checkbox"/> |               |
| <b>4. Practicum</b>                                        |                          |               |

|                                                         |                          |  |
|---------------------------------------------------------|--------------------------|--|
| Practicum log complete (Appendix I)                     | <input type="checkbox"/> |  |
| Three teacher observation sheets                        | <input type="checkbox"/> |  |
| Taught a posture, sūrya namaskār, a 30 - minute segment | <input type="checkbox"/> |  |
| Taught a prāṇāyāma session and a guided relaxation      | <input type="checkbox"/> |  |
| <b>5. Examinations</b>                                  |                          |  |
| Written examination passed                              | <input type="checkbox"/> |  |
| Lesson plan for the practical examination submitted     | <input type="checkbox"/> |  |
| Practical teaching examination passed                   | <input type="checkbox"/> |  |
| <b>6. Professional</b>                                  |                          |  |
| Professional conduct throughout the course              | <input type="checkbox"/> |  |
| One - year professional development plan (Chapter 39)   | <input type="checkbox"/> |  |
| Final reflection submitted                              | <input type="checkbox"/> |  |

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 Trainee

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 Course leader

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 Date of completion

## APPENDIX

## M

# Recommended Reading

A short list for continuing study, chosen for reliability and usefulness to a new teacher. Full details of these and all other works cited are in the References. Start with one book from each group; return to the primary texts throughout your life.

## Primary texts in translation

- ◆ **Yoga Sūtra.** Bryant (2009) — a scholarly translation with extensive traditional commentary; Satchidananda (1978) — a practical, accessible commentary.
- ◆ **Bhagavad Gītā.** Easwaran (2007) for readability; Sargeant (2009) for word-by-word Sanskrit.
- ◆ **Upaniṣads.** Olivelle (1996) — an accurate, annotated translation of the principal Upaniṣads.
- ◆ **Haṭha Yoga Pradīpikā.** Muktibodhananda (1998), with practical commentary; Akers (2002) for a plain translation.
- ◆ **Vedānta.** Śaṅkara's *Vivekacūḍāmaṇi* (Madhavananda, 1921).

## Other lineages

- ◆ Iyengar (1966) and Iyengar (1981) — detailed technique of posture and breath from a closely related modern lineage.
- ◆ Satyananda (1969) — the Bihar School's comprehensive practice manual, including bandhas and mudrās.
- ◆ Desikachar (1995) — adapting practice to the individual, from the Krishnamacharya tradition.

## History and scholarship

- ◆ Mallinson & Singleton (2017) — primary sources on haṭha and early yoga, translated and introduced; the best single book for understanding where the practices come from.
- ◆ Singleton (2010) and De Michelis (2004) — how modern posture practice took shape.
- ◆ White (2014) — the life of the Yoga Sūtra through history.
- ◆ Flood (1996) — the religious and cultural context.
- ◆ Feuerstein (2008) — a broad survey of the tradition; readable, though some of its historical claims have been revised by later scholarship.

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## Anatomy, physiology and movement

- ◆ Kaminoff & Matthews (2021) — clear, yoga-specific anatomy with excellent illustrations.
- ◆ Neumann (2017) — the standard kinesiology text, for depth.
- ◆ Calais-Germain (2006) — the anatomy of breathing for teachers.
- ◆ Clark (2016) — individual variation in the skeleton and why poses look different in different bodies.
- ◆ Coulter (2001) and Keil (2014) — further yoga-specific anatomy and kinesiology.
- ◆ Kendall et al. (2005) — muscle testing and posture, for the detail of individual muscles.
- ◆ Butler & Moseley (2013) — a readable introduction to modern pain science.

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## Meditation and the mind

- ◆ Gunaratana (2011) and Thich Nhat Hanh (1975) — practical guides to mindfulness.
- ◆ Kabat-Zinn (1990) — the foundation of mindfulness-based stress reduction.

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## Teaching

- ◆ Farhi (2006) — the teacher–student relationship and ethics.
- ◆ Stephens (2010) — practical teaching methods.
- ◆ Emerson & Hopper (2011) — trauma-sensitive yoga; van der Kolk (2014) for the wider context of trauma and the body, read with awareness that some of its claims are debated.
- ◆ Remski (2019) — a sobering account of abuse in yoga communities, and how to prevent it.

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## Evidence and health

- ◆ Khalsa et al. (2016) — yoga in health care: research, principles and practice.
- ◆ Broad (2012) — a journalist's account of the risks and benefits; read critically, alongside the research reviews cited in **Chapter 26**.



## APPENDIX

## N

## References

Every reference cited in this manual is listed below. Citation keys were checked at build time: the manual cites nothing that is not listed here. Where a traditional text exists in many translations, the translation consulted is listed; verse numbering of the *Hatha Yoga Pradīpikā* in particular differs slightly between editions.

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